VII. **Identity (unit) matrix.** A scalar matrix in which each diagonal elements are all one. For example

Example $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are identity matrix.

VIII. **Upper triangular matrix.** A square matrix in which all the elements below the diagonal elements are zero

Example, $\begin{pmatrix} 2 & 3 \\ 0 & 5 \end{pmatrix}, \begin{pmatrix} 5 & 7 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$ are upper triangular.

- **Lower triangular matrix.** A square matrix in which all the elements above the diagonal elements are zero.

Example: $\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 \\ 5 & 1 & 0 \\ 3 & 4 & 6 \end{pmatrix}$ are lower triangular matrix.

- **Triangular matrix.** Either upper triangular or lower triangular.

Equality of matrices

Two matrices $A_{m\times n}$ and $B_{m\times n}$ are said to be equal, written

i) $A = B$ if and only if they have the same order and their corresponding elements are equal ($a_{ij} = b_{ij}$ for $1 \le i \le m$ and $1 \le j \le n$

Illustrative Example

2. Find x, y, and z if the matrices

$A = \begin{pmatrix} -5 & 5x-8 & 4 \\ 3 & 8 & y-3 \\ 10 & 3z+5 & 12 \end{pmatrix}$ *and* $B = \begin{pmatrix} -5 & 2x+3 & 4 \\ 3 & 8 & 5y-6 \\ 10 & -3z+5 & 12 \end{pmatrix}$

are equal

Solution: If A = B, then corresponding element must be equal

$5x - 8 = 2x + 3$

$5y - 6 = y - 3$ and $3z + 5 = -3z + 5$

Solving, $5x - 8 = 2x + 3 \Rightarrow 3x = 11 \Rightarrow x = \frac{11}{3}$

$5y - 6 = y - 3 \Rightarrow 3y = 3 \Rightarrow y = 1$

$3z + 5 = -3z + 5 \Rightarrow 6z = 0 \Rightarrow z = 0$

Addition and subtraction of matrices

If A and B are two matrices of the same order then the sum of A and B denoted by **A + B** obtained by adding the corresponding element of A and B

Let $A = a_{ij}$, $B = b_{ij}$, and $C = c_{ij}$ be m × n matrices.

Then $C = A + B$ if and only if $c_{ij} = a_{ij} + b_{ij}$

And $C = A - B$ if and only if $c_{ij} = a_{ij} - b_{ij}$

Properties of matrix addition

i. $A + B = B + A$ (commutative property)

ii. $(A + B) + C = A + (B + C)$ (Associative)

iii. $A + 0 = 0 + A = A$ (Existence of additive identity.)

iv. $A + (-A) = 0$ (Existence of additive inverse)

Multiplication of a matrix by a scalar

The product of a matrix $\mathbf{A} = \mathbf{a_{ij}}$ by a scalar k is denoted by KA and is obtained by multiplying every element of A by K

Thus $KA = (Ka_{ij})$

If $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$, then $KA = \begin{bmatrix} ka_1 & ka_2 & ka_3 \\ kb_1 & kb_2 & kb_3 \end{bmatrix}$

Illustrative Example

3. Let $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}$

Find a) 2A + 3B b) 2A − B

Solution: a) $2\begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} + 3\begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 2 & 6 \end{pmatrix} + \begin{pmatrix} -3 & 9 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 7 \\ 2 & 12 \end{pmatrix}$

b) $2\begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} - \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 2 & 6 \end{pmatrix} + \begin{pmatrix} 1 & -3 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 5 & -5 \\ 2 & 4 \end{pmatrix}$

4. Let $A = \begin{pmatrix} 9 & 1 \\ 4 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 5 \\ 7 & 12 \end{pmatrix}$, find the matrix x such that $3A + 2B + \frac{1}{2}x = 0$

Solution: $3A + 2B + \frac{1}{2}x = 0 \Rightarrow x = -2[3A + 2B]$

$$\Rightarrow x = -2\left[3\begin{pmatrix} 9 & 1 \\ 4 & 3 \end{pmatrix} + 2\begin{pmatrix} 1 & 5 \\ 7 & 12 \end{pmatrix}\right] = -2\left[\begin{pmatrix} 27 & 3 \\ 12 & 9 \end{pmatrix} + \begin{pmatrix} 2 & 10 \\ 14 & 24 \end{pmatrix}\right]$$

$$\Rightarrow x = -2\begin{bmatrix} 29 & 13 \\ 26 & 33 \end{bmatrix} = \begin{bmatrix} -58 & -26 \\ -52 & -66 \end{bmatrix}$$

5. Find x, y, z and w if $3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$

Solution: $\begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix} = \begin{bmatrix} x+4 & 6+x+y \\ -1+z+w & 2w+3 \end{bmatrix}$

Equating of the corresponding element

$\Rightarrow 3x = x + 4 \Rightarrow 2x = 4 \Rightarrow x = 2$

$\Rightarrow 3w = 2w+3 \Rightarrow w = 3$

$\Rightarrow 3z = -1 + z + w \Rightarrow 2z = -1 + 3 = 2 \Rightarrow z = 1$

$\Rightarrow 3y = 6 + x + y \Rightarrow 2y = 6 + 2 \Rightarrow y = 4$

6. Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 7 & 8 & 9 \\ -3 & 0 & 4 \end{bmatrix}$ find matrix B such that

a) A + B is a scalar matrix

b) A + B is an identity matrix

Solution: Let $B = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

Now, $A + B = \begin{bmatrix} 1 & -2 & 3 \\ 7 & 8 & 9 \\ -3 & 0 & 4 \end{bmatrix} + \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$ is scalar

So that $\begin{bmatrix} 1+a_1 & -2+a_2 & 3+a_3 \\ 7+b_1 & 8+b_2 & 9+b_3 \\ -3+c_1 & 0+c_2 & 4+c_3 \end{bmatrix} = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$

$\Leftrightarrow 1 + a_1 = k \Rightarrow a_1 = k - 1$ and $-2 + a_2 = 0 \Rightarrow a_2 = 2$

$3 + a_3 = 0 \Rightarrow a_3 = -3$ and $7 + b_1 = 0 \Rightarrow b_1 = -7$, $8 + b_2 = k$

$\Rightarrow b_2 = k - 8$, $9 + b_3 = 0 \Rightarrow b_3 = 3$ again $-3 + c_1 = 0 \Rightarrow c_1 = 3$, $0 + c_2 = 0 \Rightarrow c_2 = 0$, and $4 + c_3 = k \Rightarrow c_3 = k - 4$

$$\therefore B = \begin{pmatrix} k-1 & 2 & 3 \\ -7 & k-8 & -9 \\ 3 & 0 & k-4 \end{pmatrix}$$

b) $\begin{pmatrix} 1 & -2 & 3 \\ 7 & 8 & 9 \\ -3 & 0 & 4 \end{pmatrix} + \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ identity matrix

$\Rightarrow a_1 + 1 = 1 \Rightarrow a_1 = 0$, $-2 + a_2 = 0 \Rightarrow a_2 = 2$, $a_3 = -3$

$\Rightarrow 7 + b_1 = 0 \Rightarrow b_1 = -7$, $8 + b_2 = 1 \Rightarrow b_2 = -7$, $b_3 = -9$

$-3 + c_1 = 0 \Rightarrow c_1 = 3$, $0 + c_2 = 0 \Rightarrow c_2 = 0$, $c_3 + 4 = 1$, $c_3 = -3$

$$\therefore B = \begin{pmatrix} 0 & 2 & -3 \\ -7 & -7 & -9 \\ 3 & 0 & -3 \end{pmatrix}$$

7. For matrices $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, C = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$

$D = \begin{bmatrix} 3 & 3 \\ 0 & -2 \end{bmatrix}$ for what values of a, b and c the matrix D is linear combination (aA + bB + cC) of A, B and C.

A. 1, -2, 1 B. 2, 2, -2 C. 1, 1, -1 D. 2, 1, -1

Solution:

$$= \begin{bmatrix} a & a \\ a & -a \end{bmatrix} + \begin{bmatrix} 0 & b \\ b & b \end{bmatrix} + \begin{bmatrix} -c & b \\ c & c \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 0 & -2 \end{bmatrix}$$

$\Rightarrow a + 0 + (-c) = 3$ and $a + b + 0 + 3$

$\Rightarrow 0 + b + c = 0 \Rightarrow b = -c$

$\therefore a = 2, b = 1, c = 1$

Answer: D

8. Construct a 3×2 matrix if $A = (a_{ij})$ where $a_{ij} = 3i - 4j$

Solution: $a_{11} = 3 - 4 = -1$

$a_{12} = 3 - 4(2) = -5$

$a_{21} = 3(2) - 4 = 2$

$a_{22} = 3(2) - 4(2) = 6 - 8 = -2$

$a_{31} = 3(3) - 4(1) = 9 - 4 = 5$

$a_{32} = 3(3) - 4(2) = 9 - 8 = 1$

$$\therefore A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} = \begin{pmatrix} -1 & -5 \\ 2 & -2 \\ 5 & 1 \end{pmatrix}$$

Matrix Multiplication

Two matrices A and B are said to be conformable for the product AB (in this every order of A and B) if the number of columns in A is equal to the number of rows in B.

Thus, if the orders of A and B are $m \times n$ and $p \times q$ respectively then

i) AB is defined if number of columns in A = number of rows in B i.e. if $n = p$

ii) BA is defined if number of columns in B equal number of rows in A i.e. if $q = m$

The rule for multiplication of two conformable matrices is called row – by – column method

Consider $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3\times2}$

Orders of A and B are 3×3 respectively. AB is defined and is of order 3×2.

$$= \begin{pmatrix} 3 & 14 \\ -3 & -7 \end{pmatrix}$$

$$BA = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ -1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 0 & 1 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 0 & 1 & -3 \\ 3 & -2 & -1 & -8 \\ 8 & 4 & 9 & -5 \end{bmatrix}$$

$\therefore AB \neq BA$

Properties of Multiplication of Matrices

- $A(BC) = (AB)C$ (Associative)
- $A(B+C) = AB + AC$ (Distributive)
- $AB \neq BA$ (Not commutative)
- If A is square matrix, then $A^2 = A.A$, $A^3 = A.A.A$,....

Note: The multiplication of diagonal matrices is commutative.

Example, Let $A = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ then

Find a) AB b) BA

Solution: a) $AB = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 3 \end{pmatrix}$

b) $BA = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 3 \end{pmatrix}$

$\therefore AB = BA \longleftarrow$ diagonal matrix are commutative

9 If $A = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$, $B = \begin{pmatrix} a & 1 \\ b & -1 \end{pmatrix}$ *and* $(A+B)^2 = A^2 + B^2$

find a and b.

Solution: $A + B = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} + \begin{pmatrix} a & 1 \\ b & -1 \end{pmatrix} = \begin{pmatrix} a+1 & 0 \\ b+2 & -2 \end{pmatrix}$

- $(A+B)^2 = \begin{pmatrix} a+1 & 0 \\ b+2 & -2 \end{pmatrix}\begin{pmatrix} a+1 & 0 \\ b+2 & -2 \end{pmatrix} = \begin{pmatrix} a^2+2a+1 & 0 \\ ab-b+2a-2 & 4 \end{pmatrix}$
- $A^2 = A \cdot A = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

- $B^2 = B \cdot B = \begin{pmatrix} a & 1 \\ b & -1 \end{pmatrix}\begin{pmatrix} a & 1 \\ b & -1 \end{pmatrix} = \begin{pmatrix} a^2+b & a-1 \\ ab-b & b+1 \end{pmatrix}$
- $A^2 + B^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} a^2+b & a-1 \\ ab-b & b+1 \end{pmatrix} = \begin{pmatrix} a^2+b-1 & a-1 \\ ab-b & b \end{pmatrix}$

Now, $(A + B)^2 = A^2 + B^2$

$$\Rightarrow \begin{pmatrix} a^2+2a+1 & 0 \\ ab-b+2a-2 & 4 \end{pmatrix} = \begin{pmatrix} a^2+b-1 & a-1 \\ ab-b & b \end{pmatrix}$$

Thus, $a - 1 = 0 \Rightarrow a = 1$ and $b = 4$, $\therefore a = 1, b = 4$

10. If $f(x) = x^2 - 5x$ and $A = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}$ then find f(A).

Solution: Here, Note, $A^2 = A \cdot A$

$$f(A) = A^2 - 5A = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix} - 5\begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}$$

$$\Rightarrow A^2 = A \cdot A = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1-4 & -2+2 \\ 2-2 & -4+1 \end{pmatrix} = \begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}$$

$$\therefore f(A) = A^2 - 5A = \begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix} - 5\begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -8 & 10 \\ -10 & 2 \end{pmatrix}$$

11. If $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, then find a) A^2 b) A^3 C) A^4 D) A^n

Solution: Here

a) $A^2 = A \cdot A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1+1 & 1+1 \\ 1+1 & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$$\therefore A^2 = 2A = 2\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

b) $A^3 = A^2 \cdot A = 2A \cdot A = 2A^2 = 2\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 4\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 4A$

c) $A^4 = A^3 \cdot A = 4A \cdot A = 4A^2 = 4(2A) = 8A$

19. Given, $A=\begin{bmatrix}4 & -3 & 1\\ -5 & 2 & -1\end{bmatrix}$ *and* $B=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$

Let C_{ij} denote the element in the i^{th} row and j^{th} column of the product $C = AB$ then find a) C_{23} b) C_{21} c) C_{13}

Solution: a) $C_{23} = A_2B^3$ ⟵ Product of 2^{nd} row of A and 3^{rd} column of B

$$\therefore C_{23}=(-5 \quad 2 \quad -2)\begin{pmatrix}3\\6\\9\end{pmatrix}=-15+12-18=-21$$

b) $$C_{21}=A_2B^1=(-5\ 2\ -2)\begin{pmatrix}1\\4\\7\end{pmatrix}=-5+8-14=-11$$

c) $$C_{13}=A_1B^3=(4\ -3\quad 1)\begin{pmatrix}3\\6\\9\end{pmatrix}=12-18+9=3$$

Transpose of Matrix

If we interchange row in to column and column in to row of **m×n** matrix A we get **n ×m** matrix, denoted by A^t which is called the transpose of A.

Example: a) If $A=\begin{bmatrix}1 & 2 & 3\\ 5 & 6 & 7\end{bmatrix}_{2\times3}$ *then* $A^t=\begin{pmatrix}1 & 5\\ 2 & 6\\ 3 & 7\end{pmatrix}_{3\times2}$

b) If $A=\begin{pmatrix}3 & 4\\ 5 & 6\end{pmatrix}_{2\times2}$ *then* $A^t=\begin{pmatrix}3 & 5\\ 4 & 6\end{pmatrix}_{2\times2}$

Properties of transpose matrix

i) $(A^t)^t = A$ ii) $(A+B)^t = A^t + B^t$ iii) $(AB)^t = B^tA^t$

Symmetric and skew symmetric matrices

Definition: i) A square matrix A is called **symmetric** matrix if $A^t=A$

Definition: ii) A square matrix A is called **skew symmetric** matrix if A^t = -A or $A^t + A = 0$

In skew symmetric matrices A

- $A = [a_{ij}]$, $a_{ij} = -a_{ji}$
- $a_{11} = a_{22} = a_{33} = \ldots = 0 \longleftarrow$ all diagonal element 0
- $a_{12} = -a_{21}$, $a_{13} = -a_{31}$ .....
- all main diagonal element of skew symmetry matrix is zero.

To show, put i = j

$\Rightarrow a_{ji} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0$ for a_{11}, i = j

20. Let $A = \begin{pmatrix} 3 & 4 \\ 4 & 6 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 3 & 5 \\ 3 & -1 & 4 \\ 5 & 4 & 8 \end{pmatrix}$ then show that

a) $A^t = A$ b) $B^t = B$

Solution: a) $A^t = \begin{pmatrix} 3 & 4 \\ 4 & 6 \end{pmatrix} \Rightarrow A^t = A \Rightarrow A$ is symmetric

b) $B^t = \begin{pmatrix} -2 & 3 & 5 \\ 3 & -1 & 4 \\ 5 & 4 & 8 \end{pmatrix} \Rightarrow B^t = B \Rightarrow B$ is symmetric

21. Let $A = \begin{bmatrix} 0 & 4 & -6 \\ -4 & 0 & 2 \\ 6 & -2 & 0 \end{bmatrix}$ then show that A is skew symmetric

Solution: • $a_{11} = a_{22} = a_{33} = 0$

- $a_{12} = 4$ and $a_{21} = -4 \Rightarrow a_{12} = -a_{21}$
- $a_{13} = -6$ and $a_{31} = 6 \Rightarrow a_{13} = -a_{31}$

$\therefore$ A is skew symmetric.

Note: If A is a square matrix then

i) $A + A^t$ is symmetric matrix

ii) $A - A^t$ is skew symmetric

iii) $\left(\frac{A + A^t}{2}\right) + \left(\frac{A - A^t}{2}\right) = A$

22. Let $A = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$ then

a) find symmetric matrix from A

c) $\begin{vmatrix} x & x+2 \\ x+1 & 2x \end{vmatrix} = -2$ d) $\begin{vmatrix} 2x-1 & x+1 \\ x+2 & x-2 \end{vmatrix} = 9$

Solution:

a) $\begin{vmatrix} x-3 & 8 \\ 2 & x+3 \end{vmatrix} = 0 \Leftrightarrow x^2 - 9 - 16 = 0 \Rightarrow x^2 - 25 = 0 \Rightarrow x$

b) $\begin{vmatrix} 4 & 2x \\ 5x & 3 \end{vmatrix} = -8 \Leftrightarrow 12 - 10x^2 = -8 \Rightarrow -10x^2 = -20 \Rightarrow x^2 =$

$\therefore x = \pm\sqrt{2}$

c) $\begin{vmatrix} x & x+2 \\ x+1 & 2x \end{vmatrix} = -2 \Leftrightarrow 2x^2 - (x+2)(x+1) = -2$

$\Leftrightarrow 2x^2 - (x^2 + 3x + 2) = -2$

$\Rightarrow x^2 - 3x - 2 = -2 \Rightarrow x^2 - 3x = 0 \therefore x = 0$ or $x = 3$

d) $\begin{vmatrix} 2x-1 & x+1 \\ x+2 & x-2 \end{vmatrix} = 9$ ⟵ exercise left for you (Tip!)

Determinant of order 3 × 3 matrices

In 3 × 3 matrix, the determinants are defined as follows

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

Illustrative Example

26. $\begin{vmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \\ 5 & -2 & 6 \end{vmatrix} = 2\begin{vmatrix} 4 & 2 \\ -2 & 6 \end{vmatrix} - 1\begin{vmatrix} 1 & 5 \\ 5 & 6 \end{vmatrix} + 3\begin{vmatrix} 1 & 4 \\ 5 & -2 \end{vmatrix} = 5 + 4 - 66 = -6$

Consider the following illustration of **minors and cofactor.**

- Let the minors of the element: a_{11}, a_{12} and a_{23} be denoted by M_{11}, M_{12} and M_{23} respectively

Let the cofactor of the element a_{11}, a_{12} and a_{23} be denoted by A_{11}, A_{12} and A_{23} respectively. Then:

Matrix	Minor	Cofactor
$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$	$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$ $= a_{22}a_{33} - a_{32}a_{23}$	$A_{11} = (-1)^{1+1}M_{11} = M_{11}$
$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$	$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$	$A_{12} = (-1)^{1+2}M_{12} = -M_{12}$
$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$	$M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$	$A_{23} = (-1)^{2+3}M_{23} = -M_{23}$

Minors and Cofactor

Definition: Let $A = a_{ij}$ be a square matrix of order $n > 1$

i) The **minor** M_{ij} of the element a_{ij} is the determinant of the matrix of order $n - 1$ obtained by deleting the **i^{th} row** and **j^{th} column** of A.

ii) The **cofactor** A_{ij} of the element a_{ij} is $A_{ij} = (-1)^{i+j}M_{ij}$.

Illustrative Example

27. Find all the minors and the cofactors of the elements in the matrix $(a_{11}, a_{12}, a_{21}, a_{22})$

a) $\begin{bmatrix} 6 & -4 \\ 3 & 2 \end{bmatrix}$ b) $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Solution: a) $M_{11} = 2$ ⟵ deleting 1st row and 1st column

$M_{12} = 3$ ⟵ minor of -4

$M_{21} = -4$ and $M_{22} = 6$

$$\therefore \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -4 & 6 \end{bmatrix}$$

2 -3

- Minor of 6 is 2
- Minor of -4 is 3
- Minor of 3 is -4

Cofactor $(A_{ij} = (-1)^{i+j}(M_{ij})$

Thus, $A_{11} = (-1)^{1+1}(M_{11}) = (1)(2) = 2 \leftarrow$ cofactor of 6
$A_{12} = (-1)^{1+2}(M_{12}) = (-1)(3) = -3 \leftarrow$ cofactor of 4
$A_{21} = (-1)^{2+1}M_{21} = (-1)(-4) = 4 \leftarrow$ cofactor of 3
$A_{22} = (-1)^{2+2}M_{22} = (1)(6) = 6 \leftarrow$ cofactor of 2

$$\therefore \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 4 & 6 \end{bmatrix}$$

b) $\therefore \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} = \begin{bmatrix} d & c \\ b & a \end{bmatrix} \leftarrow$ minor of entry $a_{11}, a_{12}, a_{21}, a_{2}$

Cofactor, of $a_{11}, a_{12}, a_{21}, a_{22}$ $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix} \leftarrow$ use $\begin{bmatrix} + & \\ & \end{bmatrix}$

28. If $A = \begin{pmatrix} 1 & -3 & 3 \\ 4 & 2 & 0 \\ -2 & -7 & 5 \end{pmatrix}$, then find

a) the minor of a_{11}, a_{21}, a_{23}
b) the cofactor of a_{11}, a_{21}, a_{23}

Solution: Deleting the right row and column of A.

a) Minor of a_{11} is $M_{11} = \begin{vmatrix} 2 & 0 \\ -7 & 5 \end{vmatrix} = 10 \leftarrow$ deleting 1st row and 1st column.

Minor of a_{21} is $M_{21} = \begin{vmatrix} -3 & 3 \\ -7 & 5 \end{vmatrix} = -15 + 21 = 6 \leftarrow$ deleting 2nd row and 1st column

Minor of a_{23} is $M_{23} = \begin{vmatrix} 1 & -3 \\ -2 & -7 \end{vmatrix} = -7 - 6 = -13 \leftarrow$ deleting 2nd row and 3rd column

b) Cofactor of a_{11} is $A_{11} = (-1)^{1+1}M_{11} = (1)(10) = 10$
Cofactor of a_{21} is $A_{21} = (-1)^{2+1}M_{21} = (-1)(6) = -6$
Cofactor of a_{23} is $A_{23} = (-1)^{2+3}M_{23} = (-1)(-13) = 13$

Finding determinant of 3 × 3 matrix by expanding by cofactors.

$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ is given by expanding

- Along Row I: $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$.
- Along Row III: $|A| = a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33}$.
- Along Column II: $|A| = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32}$.

29. Let $A = \begin{pmatrix} 3 & -2 & 2 \\ 6 & 1 & -1 \\ -2 & -3 & 2 \end{pmatrix}$ then determinant |A| by expanding along:

a) first row b) first column c) 2nd row

Solution: a) along first row, $|A| = 3A_{11} - 2A_{12} + 2A_{13}$

$$= 3(-1)^2\begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} + (-2)(-1)^3\begin{vmatrix} 6 & -1 \\ -2 & 2 \end{vmatrix} + 2(-1)^2\begin{vmatrix} 6 & 1 \\ -2 & -3 \end{vmatrix}$$

$= 3(2-3) + 2(12-2) + 2(-18+2) = -15$

$\therefore |A| = -15$

b) Along first column: $|A| = 3A_{11} + 6A_{21} + (-2)A_{31}$

$$= 3(-1)^2\begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} + 6(-1)^3\begin{vmatrix} -2 & 2 \\ -3 & 2 \end{vmatrix} + (-2)(-1)^4\begin{vmatrix} -2 & 2 \\ 1 & -1 \end{vmatrix}$$

$= 3(2-3) + -6(-4+6) + -2(2-2) = -15 \quad \therefore |A| = -15$

c) Exercise left for you

30. Solve each of the following equation:

a) $\begin{vmatrix} -1 & 3 & -6 \\ x & 3 & 0 \\ 1 & -2 & 1 \end{vmatrix} = 1$ b) $\begin{vmatrix} x+1 & 2 & 3 \\ x & 1 & x \\ 1 & 1 & 1 \end{vmatrix} = 0$

Solution:

Since the second row contains a zero, we shall expand the 2nd row.

$\Rightarrow xA_{21} + 3A_{22} + 0A_{23} = 1$

$$\Rightarrow (-1)^3(x)\begin{vmatrix}3 & -6\\ -2 & 1\end{vmatrix}+3(-1)^4\begin{vmatrix}-1 & -6\\ 1 & 1\end{vmatrix}+0(-1)^5\begin{vmatrix}-1 & 3\\ 1 & -2\end{vmatrix}=1$$

$$-x(3-12)+3(-1+6)+0=1$$

$$9x+15=1$$

$$9x=-14$$

$$\therefore x=\frac{-14}{9}$$

b) $(x+1)A_{11}+2A_{12}+3A_{13}=0$ (by expanding 1st row)

$$\Rightarrow (x+1)(-1)^2\begin{vmatrix}1 & x\\ 1 & 1\end{vmatrix}+2(-1)^3\begin{vmatrix}x & x\\ 1 & 1\end{vmatrix}+3(-1)^4\begin{vmatrix}x & 1\\ 1 & 1\end{vmatrix}=0$$

$$\Rightarrow (x+1)(1-x)-2(x-x)+3(x-1)=0$$

$$\Rightarrow 1-x^2+3x-3=0 \Rightarrow x^2-3x+2=0 \Leftrightarrow (x-2)(x-1)=0$$

$$\therefore x=2 \text{ or } x=1$$

Determinant of triangular matrix form

- Let $A=\begin{pmatrix}a_{11} & a_{12} & a_{13}\\ 0 & a_{22} & a_{23}\\ 0 & 0 & a_{33}\end{pmatrix}$ then $|A| = a_{11}\, a_{22}\, a_{33}$

- Let $A=\begin{pmatrix}a_{11} & 0 & 0 & 0\\ a_{21} & a_{22} & 0 & 0\\ a_{31} & a_{32} & a_{33} & 0\\ a_{41} & a_{42} & a_{43} & a_{44}\end{pmatrix}$ = then $|A| = a_{11}\, a_{22}\, a_{33}\, a_{44}$

- *If A be squar matrix of order **n** in triangular form then determinant of A is the product of the elements on the main diagonal.*

$$|A| = a_{11}\, a_{22}\, A_{33} \ldots A_{nn}.$$

31. Compute each of the following determinant

a) $\begin{vmatrix}3 & 4\\ 0 & 6\end{vmatrix}$

b) $\begin{vmatrix}2 & 0 & 0\\ 6 & 7 & 0\\ 5 & 4 & 1\end{vmatrix}$

c) $\begin{vmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{vmatrix}$ d) $\begin{vmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{vmatrix}$

Solution: a) $3 \times 6 = 18$ c) $(2)(-2)(1)(3) = -12$

b) $2 \times 7 \times 1 = 14$ d) $(5)(2)(-3) = -30$

Properties of Determinant

1. *If A is square matrix then* $|A| = |A^t|$, *Example,* $\begin{vmatrix} 3 & 5 \\ 4 & 6 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix}$

2. Inter **change** of any two rows or column **change** the sign of the determinant

Example:

a) $\begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} = -\begin{vmatrix} 5 & 6 \\ 3 & 4 \end{vmatrix}$ b) $\begin{vmatrix} 2 & 0 & 1 \\ 6 & 4 & 3 \\ 0 & 3 & 5 \end{vmatrix} = -\begin{vmatrix} 6 & 4 & 3 \\ 2 & 0 & 1 \\ 0 & 3 & 5 \end{vmatrix}$

3. If the elements one row (or column) of a determinant are multiplied by scalar k, then the new determinant multiplied by $|B| = k|A|$

Example:

- $\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = \begin{vmatrix} ka & b \\ kc & d \end{vmatrix} = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$
- $\begin{vmatrix} ka & kb \\ mc & md \end{vmatrix} = (km)\begin{vmatrix} a & b \\ c & d \end{vmatrix}$
- $\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = (k)^2\begin{vmatrix} a & b \\ c & d \end{vmatrix}$
- $\begin{vmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \\ ka_3 & kb_3 & kc_3 \end{vmatrix} = k^3\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Example: $\begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} = -2, \begin{vmatrix} 6 & 4 \\ 10 & 6 \end{vmatrix} = 2\begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} = 2(-2) = -4$

35. If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 4$ then which of the following is not true

A. $\begin{vmatrix} 2a_1 & c_1 & 3b_1 \\ 2a_2 & c_2 & 3b_2 \\ 2a_3 & c_3 & 3b_3 \end{vmatrix} = 24$ C. $\begin{vmatrix} 3a_3 & 3c_3 & 6b_3 \\ a_2 & c_2 & 2b_2 \\ a_1 & c_1 & 2b_1 \end{vmatrix} = -24$

B. $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 + 4a_2 & b_3 + 4b_2 & c_3 + 4c_2 \end{vmatrix} = 4$ D. $\begin{vmatrix} \frac{1}{2}a_1 & \frac{1}{2}b_1 & \frac{1}{2} \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Solution:

A. $-(2)(3)(4) = -24$ (2^{nd} column and 3^{rd} column are inter changed and factoring)

B. $4 \longleftarrow$ Adding multiple of row to another row, the determinant value is the same.

C. $(-(3)(2)(4) = -24 \longleftarrow$ Inter change 1^{st} row and 3^{rd} row and factoring 3 and 2

D. $\frac{1}{2} \cdot 0 = 0 \longleftarrow$ factoring $\frac{1}{2}$ and two identical row.

36. If $\begin{vmatrix} x & t+w \\ y & S+U \end{vmatrix} = m$, *and* $\begin{vmatrix} x & t \\ y & s \end{vmatrix} = n$, then $\begin{vmatrix} x & w \\ y & u \end{vmatrix}$ is equal

A. m + n B. m − n C. mn D. m

Solution: $\begin{vmatrix} x & t+w \\ y & S+U \end{vmatrix} = \begin{vmatrix} x & t \\ y & s \end{vmatrix} + \begin{vmatrix} x & w \\ y & u \end{vmatrix}$

$\Rightarrow m = n + \begin{vmatrix} x & w \\ y & u \end{vmatrix} \Rightarrow \begin{vmatrix} x & w \\ y & u \end{vmatrix} = m - n$

Answ

37. a) Use the properties of determinant to show that the equation of straight line through the points (x_1, y_1) and (x_2, y_2) is given by

$$\begin{vmatrix} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{vmatrix} = 0$$

b) Find the equation of the line passing through the points (2, 3) and (-1, 4) (using #37a)

Solution: a) Putting x_1 for x and y_1 for y.

$$\Rightarrow \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_1 & y_1 \\ 1 & x_2 & x_2 \end{vmatrix} = 0 \leftarrow \text{we get two identical row}$$

$\therefore (x_1, y_1)$ and (x_2, y_2) are on the same line

b) putting $x_1 = 2$, $y_1 = 3$ and $x_2 = -1$, $y_2 = 4$

$$\Rightarrow \begin{vmatrix} 1 & x & y \\ 1 & 2 & 3 \\ 1 & -1 & 4 \end{vmatrix} = 1\begin{vmatrix} 2 & 3 \\ -1 & 4 \end{vmatrix} + (-1)^3 x \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} + y\begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = 0$$

$\Rightarrow 8 + 3 - x(4 - 3) + y(-1 - 2) = 0$

$\Rightarrow -x - 3y = -11 \Rightarrow x + 3y = 11$

$\therefore$ The equation of line is $x + 3y = 11$

Removing common factors from Rows and Column.

38. Find the factor and determinant value of each of the following

a) $\begin{vmatrix} a & b \\ a^2 & b^2 \end{vmatrix}$ b) $\begin{vmatrix} 14 & -6 & 4 \\ 4 & -5 & 12 \\ -21 & 9 & -6 \end{vmatrix}$ c) $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$

Solution: Here

a) $\begin{vmatrix} a & b \\ a^2 & b^2 \end{vmatrix} = ab\begin{vmatrix} 1 & 1 \\ a & b \end{vmatrix} = ab(b - a) \longrightarrow$ Removing factor a, b from 1^{st} and 2^{nd} column.

b) $\begin{vmatrix} 14 & -6 & 4 \\ 4 & -5 & 12 \\ -21 & 9 & -6 \end{vmatrix} = (2)(-3)\begin{vmatrix} 7 & -3 & 2 \\ 4 & -5 & 12 \\ 7 & -3 & 2 \end{vmatrix} = (-6)(0) = 0$

- 2 is common factor of row 1
- -3 is common factor of row 3
- Row 1 and Row 3 are identical

c) $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = abc\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ ← Removing factor a, b, c

$= abc\begin{vmatrix} 1-1 & 1-1 & 1 \\ a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \end{vmatrix}$ ← from 1st, 2nd and 3rd column

- $C_1 = C_1 - C_2$
- $C_2 = C_2 - C_3$ (by property four

$= abc\begin{vmatrix} 0 & 0 & 1 \\ a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \end{vmatrix} = abc\begin{vmatrix} a-b & b-c \\ a^2-b^2 & b^2-c^2 \end{vmatrix}$ ←

Determinant by expanding 1st row

- Factor 1st and 2nd column of a – b and b – c gives:

$= abc(a-b)(b-c)\begin{vmatrix} 1 & 1 \\ a+b & b+c \end{vmatrix}$

$\therefore \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (abc)(a-b)(b-c)(c-a)$

Adjoint of a square matrix

Definition: The adjoint of a square matrix A is the transpose of the matrix obtained by replacing each element of A by its co-factor in A denoted by **adj A= $(A_{ij})^T$**

If $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ then *cofactor of* $A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$

$\therefore$ adj A = (cofactor of A)^t $= \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{32} & A_{33} \end{pmatrix}$

Example a) Find adj.A if $A = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$

Solution: First, find the cofactor, A_{11}, A_{12}, A_{21}, A_{22}

$A_{11} = (-1)^{1+1}|w| = w$ and $A_{12} = (-1)^{1+2}|z| = -z$

$A_{21} = (-1)^{2+1}|y| = -y$ and $A_{22} = (-1)^{2+2}|x| = x$

Then. cofactor $= \begin{pmatrix} w & -z \\ -y & x \end{pmatrix}$, adjoint is transpose of cofactor

$$\therefore adjA = \begin{pmatrix} w & -y \\ -z & x \end{pmatrix}$$

Thus. adj $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ adj $\begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 6 & -4 \\ -5 & 3 \end{pmatrix}$

Singular and non-singular matrices

Definition: A square matrix A is called

i) **Singular** if $|A| = 0$

ii) **Non-singular** if $|A| \neq 0$

Example: Find |A| if (i) $A = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}$ (ii) $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$

Solution: i) $|A| = (3)(8) - (6)(4) = 24 - 24 = 0 \leftarrow$ A is singular

ii) $|A| = 3.6 - 4.5 = 18 - 20 = -2 \leftarrow$ non singular (Invertible)

39. For what value of x the given matrix A is singular

a) $A = \begin{pmatrix} x+1 & x \\ 2x & -x \end{pmatrix}$ b) $A = \begin{pmatrix} x+2 & 6 \\ 2 & x-2 \end{pmatrix}$

Solution: a) $|A| = (x+1)(-x) - 2x^2 = -x^2 - x - 2x^2 = -3x^2 - x$

$\Rightarrow -3x^2 - x = 0 \Rightarrow x(3x+1) = 0$

$\therefore x = 0$ or $x = -\frac{1}{3}$ ⟵ singular

b) $|A| = 0 \Rightarrow (x+2)(x-2) - 12 = 0 \Rightarrow x^2 - 4 - 12 = 0$
$\Rightarrow x^2 - 16 = 0$ $\therefore x = 4$ or $x = -4$

40. For what values set of x the matrix A is non singular (Invertible)

a) $A = \begin{pmatrix} x & 3 \\ 4 & x-4 \end{pmatrix}$ b) $A = \begin{pmatrix} x-3 & -2 \\ 4 & x \end{pmatrix}$

Solution:

a) $|A| \neq 0 \Rightarrow x^2 - 4x - 12 \neq 0 \Rightarrow (x-6)(x+2) \neq 0$
$\Rightarrow x \neq 6$ and $x \neq -2$.

$\therefore$ S.S= R\{6,-2}→ Invertible for all real number except 6,-2

b) $|A| \neq 0 \Rightarrow x^2 - 3x + 8 \neq 0$, for all x, $|A| \neq 0$

$\therefore$ S.S= R ← Invertible for every x

Inverse of a square matrix

Definition: A square matrix A said to be **Invertible (non-singular)** if and only if there is square matrix B such that $AB = BA = I$ where I is the identity matrix.

I. The inverse of A is denoted by A^{-1} i.e $B = A^{-1}$ so that $AA^{-1} = I_n$

II. $|AA^{-1}| = |A||A^{-1}| = |I_n| = 1$ $\therefore \left|A^{-1}\right| = \frac{1}{|A|}$

III. $A^{-1} = \frac{1}{|A|} adj.A$ IV. $(AB)^{-1} = B^{-1}A^{-1}$ V) $(KA)^{-1} = \frac{1}{k}A^{-1}$

41. Which of the following is an invertible matrix?

A. $\begin{bmatrix} 2 & -5 \\ 4 & -10 \end{bmatrix}$ C. $\begin{bmatrix} -2 & 3 \\ 4 & -6 \end{bmatrix}$

B. $\begin{bmatrix} 2 & -8 \\ -3 & 12 \end{bmatrix}$ D. $\begin{bmatrix} 4 & -3 \\ 6 & 9 \end{bmatrix}$

Solution:

A. $\begin{vmatrix} 2 & -5 \\ 4 & -10 \end{vmatrix} = -20 + 20 = 0$ ⟵ Not invertible

B. $\begin{vmatrix} 2 & -8 \\ -3 & 12 \end{vmatrix} = 24 - 24 = 0 \longleftarrow$ Not invertible

C. $\begin{vmatrix} -2 & 3 \\ 4 & -6 \end{vmatrix} = 12 - 12 = 0 \longleftarrow$ Not invertible

D. $\begin{vmatrix} 4 & -3 \\ 6 & 9 \end{vmatrix} = 36 + 18 = 54 \neq 0 \longleftarrow$ Invertible **Answer: D**

42. Using adjoint-cofactor method find the inverse of each of the following matrices

A. $A = \begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix}$ B. $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ C. $A = \begin{bmatrix} 2 & 5 \\ -1 & 3 \end{bmatrix}$

***Solution*:** To find the inverse:

i) determine, |A|
ii) determine cofactor of A
iii) determine adj.A
iv) write $A^{-1} = \frac{1}{|A|} adj.A$

Explanation
adj $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

a) $|A| = \begin{vmatrix} 1 & 3 \\ 4 & 13 \end{vmatrix} = 13 - 12 = 1$

cofactor of A: $A_{11} = 13$, $A_{12} = (-1)^3(4) = -4$, $A_{21} = (-1)^3(3) = -3$, $A_{22} = 1$

$$\therefore Adj.A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^t = \begin{pmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{pmatrix} = \begin{pmatrix} 13 & -3 \\ -4 & 1 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{|A|} adj.A = \frac{1}{1}\begin{pmatrix} 13 & -3 \\ -4 & 1 \end{pmatrix} = \begin{pmatrix} 13 & -3 \\ -4 & 1 \end{pmatrix}$$

b) $|A| = \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix} = 10 - 3 = 7$, adj.A $= \begin{pmatrix} 5 & -3 \\ -1 & 2 \end{pmatrix}$

$$\therefore A^{-1} = \frac{1}{|A|} adj.A = \frac{1}{7}\begin{pmatrix} 5 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{7} & \frac{-3}{7} \\ \frac{-1}{7} & \frac{2}{7} \end{pmatrix}$$

c) $|A| = \begin{vmatrix} 2 & 5 \\ -1 & 3 \end{vmatrix} = 6 + 5 = 11$, adj.A $= \begin{pmatrix} 3 & -5 \\ 1 & 2 \end{pmatrix}$

$$\therefore A^{-1} = \frac{1}{|A|} adj.A = \frac{1}{11}\begin{pmatrix} 3 & -5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{11} & \frac{-5}{11} \\ \frac{1}{11} & \frac{2}{11} \end{pmatrix}$$

43. Find the inverse of each matrix:

a) $A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$ b) $A = \begin{bmatrix} k & 1 \\ 1 & 1 \end{bmatrix}, k \neq 1$

c) $A = \begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix}, k \neq \pm 1$

Solution:

a) $A = \begin{vmatrix} 1 & 0 \\ 0 & k \end{vmatrix} = \text{k}, \; adj.A = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$

$$\therefore \text{A}^{-1} = \frac{1}{k}\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{k} \end{pmatrix}$$

b) $A = \begin{vmatrix} k & 1 \\ 1 & 1 \end{vmatrix} = \text{k}, \; adj.A = \begin{pmatrix} 1 & -1 \\ -1 & k \end{pmatrix}$

$$\therefore \text{A}^{-1} = \frac{1}{|A|} adj.A = \frac{1}{k-1}\begin{pmatrix} 1 & -1 \\ -1 & k \end{pmatrix} = \begin{pmatrix} \frac{1}{k-1} & \frac{-1}{k-1} \\ \frac{-1}{k-1} & \frac{k}{k-1} \end{pmatrix}$$

c) $A = \begin{vmatrix} k & 1 \\ 1 & k \end{vmatrix} = \text{k}^2 - 1, \; adj.A = \begin{pmatrix} k & -1 \\ -1 & k \end{pmatrix}$

$$\therefore \text{A}^{-1} = \frac{1}{|A|} adj.A = \frac{1}{k^2-1}\begin{pmatrix} k & -1 \\ -1 & k \end{pmatrix} = \begin{pmatrix} \frac{k}{k^2-1} & \frac{-1}{k^2-1} \\ \frac{-1}{k^2-1} & \frac{k}{k^2-1} \end{pmatrix}$$

44. Find the inverse of each of the following

a) $A=\begin{pmatrix}1 & 2 & -1\\ 3 & 1 & 1\\ 1 & -1 & 2\end{pmatrix}$ b) $A=\begin{pmatrix}1 & 0 & 2\\ 2 & -1 & 3\\ 4 & 1 & 8\end{pmatrix}$

Solution: First find the cofactor, then adjA

a) $A_{11}=(-1)^{1+1}\begin{vmatrix}1 & 1\\ -1 & 2\end{vmatrix}=3$ $A_{12}=(-1)^{1+2}\begin{vmatrix}3 & 1\\ 1 & 2\end{vmatrix}=-5$

$A_{13}=(-1)^{1+3}\begin{vmatrix}3 & 1\\ 1 & -1\end{vmatrix}=-4$

$A_{21}=(-1)^{2+1}\begin{vmatrix}2 & -1\\ -1 & 2\end{vmatrix}=-3$ $A_{22}=(-1)^{4}\begin{vmatrix}1 & -1\\ 1 & 2\end{vmatrix}=3$

$A_{23}=(-1)^{2+3}\begin{vmatrix}1 & 2\\ 1 & -1\end{vmatrix}=3$

$A_{31}=(-1)^{3+1}\begin{vmatrix}2 & -1\\ 1 & 1\end{vmatrix}=3$ $A_{32}=(-1)^{3+2}\begin{vmatrix}1 & -1\\ 1 & 2\end{vmatrix}=-3$

$A_{33}=(-1)^{6}\begin{vmatrix}1 & 2\\ 3 & 1\end{vmatrix}=-5$

$\therefore \text{cof.A}=\begin{pmatrix}3 & -5 & -4\\ -3 & 3 & 3\\ 3 & -3 & -5\end{pmatrix}$, $\therefore$ adjA $=\begin{pmatrix}3 & -3 & 3\\ -5 & 3 & -3\\ -4 & 3 & -5\end{pmatrix}$

And, $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = 1.3 + 2.)-5) + (-1)(-4) = -3$

$\therefore\ A^{-1}=\frac{1}{|A|}adj.A=\frac{1}{-3}\begin{pmatrix}3 & -3 & 3\\ -5 & 3 & -3\\ -4 & 3 & -5\end{pmatrix}=\begin{pmatrix}-1 & 1 & -1\\ \frac{5}{3} & -1 & 1\\ \frac{4}{3} & -1 & \frac{5}{3}\end{pmatrix}$

b) $A_{11} = (-1)^2 \begin{vmatrix} -1 & 3 \\ 1 & 8 \end{vmatrix} = -11$ $A_{12} = (-1)^3 \begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix} = -4$,

$A_{13} = (-1)^4 \begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix} = 6$

Similarly, $A_{21} = 2$, $A_{22} = 0$, $A_{23} = -1$

$A_{31} = 2$, $A_{32} = 1$, $A_{33} = -1$

And $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = 1(-11)+0.(-4)+2.6 = 1$

$$\therefore A^{-1} = \frac{1}{|A|} adj.A = \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix}$$

45. Which of the following pair of matrices are not inverses of e other?

A. $\begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix}$ *and* $\begin{bmatrix} 13 & -3 \\ -4 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ *and* $\begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ *and* $\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 2 & 6 \\ 6 & 5 \end{bmatrix}$ *and* $\begin{bmatrix} \frac{5}{8} & \frac{-3}{4} \\ \frac{-3}{8} & \frac{1}{4} \end{bmatrix}$

Solution: Multiply the pair of matrices, if you get the identi matrices then, the pair of matrices are inverse of each other, that,

$$\begin{bmatrix} 1 & 3 \\ 4 & 13 \end{bmatrix} \cdot \begin{bmatrix} 13 & -3 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \cdot \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

But D. $\begin{bmatrix} 2 & 6 \\ 6 & 5 \end{bmatrix} \cdot \begin{bmatrix} \frac{5}{8} & \frac{-3}{4} \\ \frac{-3}{8} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ \frac{15}{8} & \frac{-13}{4} \end{bmatrix}$ ← Not identity

that not inverse of each other.

Answer:

46. If the matrices $\begin{bmatrix} 3 & x+2 \\ x & 3 \end{bmatrix}$ *and* $\begin{bmatrix} x+1 & -4 \\ -x & x+1 \end{bmatrix}$ are inverses of each other, then the possible value of x is

A. 0 B. -5 C. -1 D. 2

Solution: $\begin{pmatrix} 3 & x+2 \\ x & 3 \end{pmatrix} \cdot \begin{pmatrix} x+1 & -4 \\ -x & x+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} 3x+3-x^2-2x & -12+x^2+3x+2 \\ x^2+x-3x & -4x+3x+3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow C_{11} = -x^2 + x + 3 = 1 \Leftrightarrow x^2 - x - 2 = 0$ .....(*)

$\Rightarrow (x-2)(x+1) = 0 \Leftrightarrow x = 2, x = -1$

$\Rightarrow C_{12} = 0 \Rightarrow x^2 + 3x - 10 = 0 \Leftrightarrow (x-2)(x+5) = 0 \Leftrightarrow x = 2, x = -5$..(**)

$\Rightarrow C_{21} = 0 \Rightarrow x^2 - 2x = 0 \Leftrightarrow x(x-2) = 0 \Leftrightarrow x = 0, x = 2$.... (***)

$\Rightarrow C_{22} = 1 \Rightarrow -4x + 3x + 3 = 1 \Leftrightarrow -x + 3 = 1 \Leftrightarrow x = 2$ .....(****)

Thus, the common value from (*), (**), (***) and (****) is 2.

Answer: D

47. Let $A = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ then

Find a) A^t b) A^{-1} c) $A^t.A^{-1}$

Solution: a) $A^t = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) to find A^{-1}

$$\text{adj. } A = \begin{bmatrix} \begin{vmatrix} \cos\theta & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} \sin\theta & 0 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} \sin\theta & \cos\theta \\ 0 & 0 \end{vmatrix} \\ \begin{vmatrix} -\sin\theta & 0 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} \cos\theta & 0 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} \cos\theta & -\sin\theta \\ 0 & 0 \end{vmatrix} \\ \begin{vmatrix} -\sin\theta & 0 \\ \cos\theta & 0 \end{vmatrix} & \begin{vmatrix} \sin\theta & 0 \\ \cos\theta & 0 \end{vmatrix} & \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} \end{bmatrix} \cdots \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\therefore \text{adj. } A = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$= \cos\theta\cos\theta + \sin\theta\sin\theta + 0$$

$$\therefore |A| = \cos^2\theta + \sin^2\theta = 1$$

$$A^{-1} = \frac{1}{|A|} adj.\ A = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \longleftarrow A^t = A^{-1}$$

c. $A^{-1}A^t \longrightarrow$ Exercise left for students

A system of equation can be written as matrix equation

System **matrix equation**

$a_1x + b_1y = C_1$

$a_2x + b_2y = C_2$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$$

$$A \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$$ can be solved using the inverse of A

$$A^{-1}A\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}\begin{bmatrix} C_1 \\ C_2 \end{bmatrix}$$

Find the value of x, y, for each of the following matrix equation

a) $\begin{bmatrix} 3 & 4 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ b) $\begin{pmatrix} -5 & 3 \\ 4 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

Solution: Multiplying each side of matrix equation by the inverse

a) Let $A = \begin{bmatrix} 3 & 4 \\ 3 & 5 \end{bmatrix}, adj.A = \begin{pmatrix} 5 & -4 \\ -3 & 3 \end{pmatrix}$, $|A| = 15 - 12 = 3$

Thus,

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 5 & -4 \\ -3 & 3 \end{pmatrix}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{3} & \frac{-4}{3} \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

$\therefore$ x = -3 and y = 2

b) Let $A = \begin{pmatrix} -5 & 3 \\ 4 & -1 \end{pmatrix}, adj.A = \begin{pmatrix} -1 & -3 \\ -4 & -5 \end{pmatrix}, |A| = 5 - 12 = -7$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}\begin{pmatrix} 5 \\ 3 \end{pmatrix} = \frac{1}{-7}\begin{pmatrix} -1 & -3 \\ -4 & -5 \end{pmatrix}\begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{7} & \frac{3}{7} \\ \frac{4}{7} & \frac{5}{7} \end{pmatrix}\begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} \Rightarrow x=2,\ y=5$$

Systems of equations with two or three variables

In this section, describes, how to solve the system of equations by matrix method.

Consider, a matrix created from a system of equation.

As an example, we start with the system $\begin{cases} x - 2y + 3z = 4 \\ 2x + y - 4z = 3 \\ -3x + 4y - z = -2 \end{cases}$

Now, using only the coefficients of the variable and constants of this system, we can write the 3 × 4 matrix

$$\begin{array}{ll} x - 2y + 3z = 4 \rightarrow & \\ 2x + y - 4z = 3 \rightarrow & \\ -3x + 4y - z = -2 \rightarrow & \end{array}\begin{pmatrix} 1 & -2 & 3 & 4 \\ 2 & 1 & -4 & 3 \\ -3 & 4 & -1 & -2 \end{pmatrix} \leftarrow \text{this called the augmented matrix.}$$

The matrix formed by the coefficient of the variable is called **coefficient matrix**

$$\begin{pmatrix} 1 & -2 & 3 \\ 2 & 1 & -4 \\ -3 & 4 & -1 \end{pmatrix} \longleftarrow \text{coefficient matrix}$$

$$\begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} \longleftarrow \text{constant matrix (formed from the constant)}$$

Elementary Row operations

1. Interchange two rows.
   Symbolic representation: $R_i \longleftrightarrow R_j$
2. Multiply all the elements in a row by the same non zero number
   Symbolically: $kR_j \rightarrow R_i$
3. Replace a row by the sum of that row and a multiple of any other row. **$(kR_i + R_j \rightarrow R_j)$**

The goal of row operations is transforming the matrix to upper **triangular matrix** with main diagonal element 1 and 0's to the left of the 1's in all rows except the first. This called the **echelon form** of the matrix. Examples of echelon form are:

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 1 & -1 \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & -3 & -1 & -6 \\ 0 & 1 & -2 & 7 \\ 0 & 0 & 1 & -2 \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 2 & -3 & 5 \\ 0 & 1 & -4 & 6 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

The order in which the elements in the 2×3 matrix were changed is important.

1. Change a_{11} to 1
2. Change a_{21} to zero
3. Change a_{22} to 1

$$\left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{array}\right] = \left[\begin{array}{ccc} 1 & a_{12} & a_{13} \\ 0 & 1 & a_{23} \end{array}\right]$$

Illustrative Example

49. Solve by using a matrix, row operations.

a) $\begin{cases} 2x + 3y = 6 \\ 4x + 5y = 8 \end{cases}$ b) $\begin{cases} 3x - 5y = -7 \\ 5x - 3y = -1 \end{cases}$

Solution: Write the system in matrix.

$$\begin{cases} 2x + 3y = 6 \longrightarrow \\ 4x + 5y = 8 \longrightarrow \end{cases} \left[\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 5 & 8 \end{array}\right]$$

i. Change element a_{11} to 1.

$\Rightarrow$ Multiply row 1 by $\frac{1}{2}$

$$\left[\begin{array}{cc:c} 1 & \frac{3}{2} & 3 \\ 4 & 5 & 8 \end{array}\right]$$

$$R_1 \rightarrow \frac{1}{2}R_1$$

ii. Change element a_{21} to zero.

$\Rightarrow$Multiply row 1 by -4, and add it to row 2. Replace row 2 by the sum.

$(-4R_1 + R_2 \rightarrow R_2)$

$$\left[\begin{array}{cc:c} 1 & \frac{3}{2} & 3 \\ 0 & -1 & -4 \end{array}\right]$$

iii. Change element a_{22} to 1.

$\Rightarrow$Multiply row 2 by -1.

The matrix is now in **echelon form.**

$$\left[\begin{array}{cc:c} 1 & \frac{3}{2} & 3 \\ 0 & 1 & 4 \end{array}\right]$$

Write the system of equations represented by the matrix.

$$\begin{cases} x + \frac{3}{2}y = 3 \Rightarrow x + \frac{3}{2}(4) = 3 \Rightarrow x = 3 - 6 = -3 \\ y = 4 \end{cases}$$

$\Rightarrow x = -3, y = 4 \therefore$ The solution is (-3, 4).

b) $\begin{cases} 3x - 5y = -7 \longrightarrow \\ 5x - 3y = -1 \longrightarrow \end{cases} \left[\begin{array}{ccc} 3 & -5 & -7 \\ 5 & -3 & -1 \end{array}\right]$

i. Change a_{11} to 1.

$$\left(R_1 \longrightarrow \frac{1}{3}R_1\right)$$

$\Rightarrow$Multiply row 1 by $\frac{1}{3}$

$$\left[\begin{array}{ccc} 1 & \frac{-5}{3} & -\frac{7}{3} \\ 5 & -3 & -1 \end{array}\right]$$

ii. Change a_{21} to zero

$\Rightarrow$Multiply row 1 by (-5), and add it to row 2. Put the sum in row 2. $(-5R_1 + R_2 \rightarrow R_2)$

$$\left[\begin{array}{cc|c} 1 & \frac{5}{3} & \frac{7}{3} \\ 0 & \frac{16}{3} & \frac{32}{3} \end{array}\right]$$

iii. Change a_{22} to 1.

$$\frac{3}{16}R_2 \longrightarrow R_2$$

$$\Rightarrow \text{Multiply row 2 by } \frac{3}{16} \left(\begin{array}{cc|c} 1 & -\frac{5}{3} & -\frac{7}{3} \\ 0 & 1 & 2 \end{array}\right)$$

Write the system of equation represented in matrix form

$$\Rightarrow \begin{cases} x - \frac{5}{3}y = -\frac{7}{3} \\ y = 2 \end{cases} \Rightarrow \begin{cases} x - \frac{5}{3}(2) = -\frac{7}{3} \\ \Rightarrow x = -\frac{7}{3} + \frac{10}{3} = \frac{3}{3} = 1 \end{cases}$$

$\Rightarrow x = 1, y = 2$

The order in which the elements in a 3×4 matrix are changed is:

1. Change a_{11} to 1
2. Change a_{21} and a_{31} to zero.
3. Change a_{22} and a_{33} to 1.
4. Change a_{32} to zero.

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right]$$

Illustrative Example

Solve by using a matrix row operation.

a) $\begin{cases} 3x + 2y + 3z = 2 \\ 2x - 3y + 4z = 5 \\ x + 4y + 2z = 8 \end{cases}$ b) $\begin{cases} 3x - 2y - 3z = 5 \\ x + 3y - 2z = -4 \\ 2x + 6y + 3z = b \end{cases}$

Solution:

i) Write the system in matrix form.

$$\left[\begin{array}{ccc|c} 3 & 2 & 3 & 2 \\ 2 & -3 & 4 & 5 \\ 1 & 4 & 2 & 8 \end{array}\right]$$

ii. Change a_{11} to 1

$\Rightarrow$Inter change row 1 and row 3. $(R_1 \longleftrightarrow R_3)$

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 8 \\ 2 & -3 & 4 & 5 \\ 3 & 2 & 3 & 2 \end{array}\right]$$

iii. Change a_{21} and a_{31} to zero.

Multiply row 1 by (-2) and add to row 1.

Multiply row 1 by (-3) and add to row 1.

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 8 \\ 0 & -11 & 0 & -11 \\ 0 & -10 & -3 & -22 \end{array}\right]$$

iv. Change a_{22} to 1.

$\Rightarrow$ Multiply row 2 by

v. Change a_{32} to zero.

$\Rightarrow$Multiply row 2 by 10 and add to row 3.

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & -10 & -3 & -22 \end{array}\right]$$

vi. Change a_{33} to 1.

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -3 & -12 \end{array}\right].$$

Multiply row 3 by $\left(-\frac{1}{3}\right)$.

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

Write the system of equations represented by the matrix.

$\Rightarrow x + 4y + 2z = 8$

$y = 1$

$z = 4$

Substitute y = 1 and z = 4 into the equation: x + 4y + 2z = 8 and solve for x.

$\Rightarrow x + 4(1) + 2(4) = 8 \Rightarrow x = 8 - 4 - 8 = -4$

$\therefore x = -4, y = 1, z = 4$

b) **Exercise left for you. (Ans. (3, -1, 2)**

Cramer's Rule

Cramer's rule is a method of solving a system of linear equations of

A) Two variable

Let $\begin{cases} a_1x + b_1y = k_1 \\ a_2x + b_2y = k_2 \end{cases}$ the Solution of the system is given by

i) $x = \dfrac{D_x}{D} = \dfrac{\begin{vmatrix} k_1 & b_1 \\ k_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_1 & b_2 \end{vmatrix}}$, ii) $y = \dfrac{D_y}{D} = \dfrac{\begin{vmatrix} a_1 & k_1 \\ a_2 & k_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_1 & b_2 \end{vmatrix}}$

$\therefore$ S.S $= \left\{\left(\dfrac{D_x}{D}, \dfrac{D_y}{D}\right)\right\}$ where $D \neq 0$

B) Three variable

Let $\begin{cases} a_1x + b_1y + c_1z = k_1 \\ a_2x + b_2y + c_2z = k_2 \\ a_3x + b_3y + c_3z = k_3 \end{cases}$ the Solution of the system is given by

i) $x = \dfrac{D_x}{D} = \dfrac{\begin{vmatrix} k_1 & b_1 & c_1 \\ k_2 & b_2 & c_2 \\ k_3 & b_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$, ii) $y = \dfrac{D_y}{D} = \dfrac{\begin{vmatrix} a_1 & k_1 & c_1 \\ a_2 & k_2 & c_2 \\ a_3 & k_3 & c_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$

iii) $$z=\frac{D_z}{D}=\frac{\begin{vmatrix} a_1 & b_1 & k_1 \\ a_2 & b_2 & k_2 \\ a_3 & b_3 & k_3 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}$$

$$\therefore S.S=\{(x,y,z)\}=\left\{\left(\frac{D_x}{D},\ \frac{D_y}{D},\frac{D_z}{D}\right)\right\} \text{ where } D\neq 0$$

The three possibility of Solution

1. Case **I.** If $D \neq 0$, then the system has **exactly** one Solution (consistent)
2. Case **II.** If $D = 0$, $D_x = D_y = D_z = 0$ then the system has an **infinite number** of Solution (dependent)
3. Case **III.** If $D = 0$, $D_x \neq 0$, $D_y \neq 0$ $D_z \neq 0$, then the system has **no Solution** (inconsistent)

Illustrative Example

51. Solve by cramer's rule

a) $\begin{cases} 3x-y=-8 \\ 2x+6y=3 \end{cases}$ b) $\begin{cases} 3x+2y=4 \\ 5x+5y=9 \end{cases}$ c) $\begin{cases} x-2y=-1 \\ 2x+5y=16 \end{cases}$

Solution:

a) $\begin{cases} 3x-y=-8 \\ 2x+6y=3 \end{cases}$, $$x=\frac{D_x}{D}=\frac{\begin{vmatrix} -8 & -1 \\ 3 & 6 \end{vmatrix}}{\begin{vmatrix} 3 & -1 \\ 2 & 6 \end{vmatrix}}=\frac{-48+3}{18+2}=\frac{-9}{4},$$

$$y=\frac{D_y}{D}=\frac{\begin{vmatrix} 3 & 4 \\ 5 & 9 \end{vmatrix}}{\begin{vmatrix} 3 & -1 \\ 2 & 6 \end{vmatrix}}=\frac{25}{20}=\frac{5}{4}$$

b) $\begin{cases} 3x+2y=4 \\ 5x+5y=9 \end{cases}, x=\dfrac{D_x}{D}=\dfrac{\begin{vmatrix} 4 & 2 \\ 9 & 5 \end{vmatrix}}{\begin{vmatrix} 3 & 2 \\ 5 & 5 \end{vmatrix}}=\dfrac{2}{5}, y=\dfrac{D_y}{D}=\dfrac{\begin{vmatrix} 3 & 4 \\ 5 & 9 \end{vmatrix}}{\begin{vmatrix} 3 & 2 \\ 5 & 5 \end{vmatrix}}$

$\therefore$ S.S $=\left\{\left(\dfrac{2}{5},\dfrac{7}{5}\right)\right\}$

c) $\begin{cases} x-2y=-1 \\ 2x+5y=16 \end{cases}, x=\dfrac{D_x}{D}=\dfrac{\begin{vmatrix} -1 & -2 \\ 16 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 2 & 5 \end{vmatrix}}=\dfrac{27}{9}=3,$

$y=\dfrac{D_y}{D}=\dfrac{\begin{vmatrix} 1 & -1 \\ 2 & 16 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 2 & 5 \end{vmatrix}}=\dfrac{18}{9}=2 \ \therefore$ S.S $=\{(3,2)\}$

52. Solve the following systems of equations

a) $\begin{cases} 3x+2y+z=-5 \\ x-y+3z=-5 \\ 2x+3y+z=0 \end{cases}$ b) $\begin{cases} 2x+y-z=9 \\ 3x-4y+z=3 \\ x-3y+2z=-4 \end{cases}$

Solution:

a) $D=\begin{vmatrix} 3 & 2 & 1 \\ 1 & -1 & 3 \\ 2 & 3 & 1 \end{vmatrix}=3\begin{vmatrix} -1 & 3 \\ 3 & 2 \end{vmatrix}-2\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}+\begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix}=-15$

$D_x=\begin{vmatrix} -5 & 2 & 1 \\ -5 & -1 & 3 \\ 0 & 3 & 1 \end{vmatrix}=45, D_y=\begin{vmatrix} 3 & -5 & 1 \\ 1 & -5 & 3 \\ 2 & 0 & 1 \end{vmatrix}=-30,$

$$D_z = \begin{vmatrix} 3 & 2 & -5 \\ 1 & -1 & -5 \\ 2 & 3 & 0 \end{vmatrix} = 0,$$

$$S.S = \{(x,y,z)\} = \left\{\left(\frac{D_x}{D}, \frac{D_y}{D}, \frac{D_z}{D}\right)\right\} = \left\{\left(\frac{45}{-15}, \frac{-30}{-15}, \frac{0}{-15}\right)\right\}$$

$$= \{(-3,2,0)\}$$

$$D = \begin{vmatrix} 2 & 1 & -1 \\ 3 & -4 & 1 \\ 1 & -3 & 2 \end{vmatrix} = -10, \quad D_x = \begin{vmatrix} 9 & 1 & -1 \\ 3 & -4 & 1 \\ -4 & -3 & 2 \end{vmatrix} = -30$$

$$D_y = \begin{vmatrix} 2 & 9 & -1 \\ 3 & 3 & 1 \\ 1 & -4 & 2 \end{vmatrix} = -10, \quad D_z = \begin{vmatrix} 2 & 1 & 9 \\ 3 & -4 & 3 \\ 1 & -3 & -4 \end{vmatrix} = 20$$

$$\therefore x = \frac{D_x}{D} = \frac{-30}{-10} = 3, \; y = \frac{D_y}{D} = \frac{-10}{-10} = 1, \; z = \frac{D_z}{D} = \frac{20}{-10} = -2$$

$\therefore$ S.S = {(3,1,-2)}

3. Identify, the system has: unique solution, no solution, infinite many solution.

a) $\begin{cases} x + 3y = 6 \\ 2x + 6y = -18 \end{cases}$ b) $\begin{cases} 2x - 5y = 11 \\ 3x + y = 8 \end{cases}$ c) $\begin{cases} 8x - 4y = 16 \\ 2x - y = 4 \end{cases}$

Solution:

a) $D = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 6 - 6 = 0, \; D_x = \begin{vmatrix} 6 & 3 \\ -18 & 6 \end{vmatrix} = 36 + 54 \neq 0,$

Since $D = 0$, $D_x \neq 0$, $D_y \neq 0$ $\therefore$ No Solution (inconsistence)

b) $D = \begin{vmatrix} 2 & -5 \\ 3 & 1 \end{vmatrix} = 2 + 15 = 17$

Since $D \neq 0$, $\therefore$ It has one Solution (consistence)

c) $D = \begin{vmatrix} 8 & -4 \\ 2 & -1 \end{vmatrix} = -8 + 8 = 0, \; D_x = \begin{vmatrix} 16 & -4 \\ 4 & -1 \end{vmatrix} = -16 + 16 = 0$

Since $D = 0, D_x = 0, D_y = 0,$

$\therefore$ It has infinite Solution (dependent)

54. Determine the value of a and b for which the system $\begin{cases} 3x - ay = 1 \\ 6x + 4y = b \end{cases}$ have

a) unique solution b) no solution c) infinite solu

Solution: First $D = \begin{vmatrix} 3 & -a \\ 6 & 4 \end{vmatrix} = 12 + 6a$

$$D_x = \begin{vmatrix} 1 & -a \\ b & 4 \end{vmatrix} = 4 + ab$$

$$D_y = \begin{vmatrix} 3 & 1 \\ 6 & b \end{vmatrix} = 3b - 6$$

i) System has unique solution, if $D \neq 0 \Rightarrow 12 + 6a \neq 0$

$\therefore a \neq -2, b \in R$

ii) has no solution if $D = 0, D_x = 0, D_y \neq 0$

$\Rightarrow a = -2, ab + 4 \neq 0 \Rightarrow ab \neq -4, \Rightarrow b \neq 2$

$\therefore a = -2$ and $b \neq 2$

55. For what values of k does the system $\begin{cases} kx + 4y = 5 \\ 9x + ky = 2 \end{cases}$ have

a) Unique Solution

b) No Solution

c) Infinite number of Solution

Solution: First, $D = \begin{vmatrix} k & 4 \\ 9 & k \end{vmatrix} = k^2 - 36$

$$D_x = \begin{vmatrix} 5 & 4 \\ 2 & k \end{vmatrix} = 5k - 8, \quad D_y = \begin{vmatrix} k & 5 \\ 9 & 2 \end{vmatrix} = 2k - 45$$

The system have:

a) Unique Solution if $D \neq 0 \Rightarrow k^2 - 36 \neq 0 \Rightarrow k \neq 6$

b) No Solution if $D = 0, D_x \neq 0$ and $D_y \neq 0$

$\Rightarrow k = 6$ or $k = -6$, $5k - 8 \neq 0 \Rightarrow 5k \neq 8 \Rightarrow k \neq \frac{8}{5}$

$\therefore k = 6$ or $k = -6$, $k \neq \frac{8}{5}$, $k \neq \frac{45}{2}$

c) Infinite Solution if $D = 0$, $D_x = 0$, $D_y = 0$

$\therefore k = 6$ or $k = -6$, $k = \frac{8}{5}$, $k = \frac{45}{2}$

56. Determine the value of a and b for which the system

a)
$$\begin{cases} 3x - 2y + z = b \\ 5z - 8y + 9z = 3 \\ 2x + y + az = -1 \end{cases}$$

i) unique Solution ii) No Solution iii) infinite many solve

Solution: First we find determinant of coefficient matrix (D)

$$D = \begin{vmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{vmatrix} = 3\begin{vmatrix} -8 & 9 \\ 1 & a \end{vmatrix} - (2)\begin{vmatrix} 5 & 9 \\ 2 & a \end{vmatrix} + 1\begin{vmatrix} 5 & -8 \\ 2 & 1 \end{vmatrix} = -14a - 42$$

$$D_x = \begin{vmatrix} b & -2 & 1 \\ 3 & -8 & 9 \\ -1 & 1 & a \end{vmatrix} = b\begin{vmatrix} -8 & 9 \\ 1 & a \end{vmatrix} - (-2)\begin{vmatrix} 3 & 9 \\ -1 & a \end{vmatrix} + 1\begin{vmatrix} 3 & -8 \\ -1 & 1 \end{vmatrix}$$

$= -8ab - 9b + 6a + 13$

The system have:

a) unique Solution if $D \neq 0 \Rightarrow -14a \neq 42$ $\therefore a \neq -3$

b) No Solution if $D = 0$, $D_x \neq 0$, $D_y \neq 0$, $D_z \neq 0$

$\Rightarrow a = -3$ and $D_x \neq 0 \Rightarrow -8(-3)b - 9b + -18 + 13 \neq 0$

$\Rightarrow 15b - 5 \neq 0$ $\quad b \neq \frac{5}{15}$, $b \neq \frac{1}{3}$

$\therefore a = -3$ and $b \neq \frac{1}{3}$

c) Infinite many Solution if $D = 0$, $D_x = D_y = D_z = 0$

$\therefore a = -3$ and $b = \frac{1}{3}$

Homogeneous linear equation

The system $\begin{cases} a_1x + b_1y = 0 \\ a_2x + b_2y = 0 \end{cases}$ is called homogeneous linear equation

- A system of homogeneous equation always has the trivial Solution
  $(x = 0, y = 0, z = 0), (D \neq 0, D_x = 0, D_y = 0)$
- Some time Non-trivial (non zero) Solution exist (infinitely many Solution) $(D = 0)\ D_x = 0, D_y = 0, D_z = 0)$

Illustrative Example

57. Solve each of the following homogeneous system of equation (identify type of Solution)
    i) trivial ii) non-trivial

a) $\begin{cases} 3x + 5y = 0 \\ 2x - 4y = 0 \end{cases}$ b) $\begin{cases} 2x - 6y = 0 \\ -x + 3y = 0 \end{cases}$

Solution: a) $D = \begin{vmatrix} 3 & 5 \\ 2 & -4 \end{vmatrix} = -12 - 10 = -22 \neq 0$

$$D_x = \begin{vmatrix} 0 & 5 \\ 0 & -4 \end{vmatrix} = 0, \qquad D_y = \begin{vmatrix} 3 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

$$\therefore\ x = \frac{D_x}{D} = 0,\ y = 0$$

$\therefore$ S.S = {(0, 0)} ⟵ trivial Solution

b) $D = \begin{vmatrix} 2 & -6 \\ -1 & 3 \end{vmatrix} = 6 - 6 = 0$ ⟵ Non trivial

$\therefore$ S.S = {(x, y): x = 3y} ⇔ {(3a, a)}

58. For what values of k does the homogeneous system has
    i) Trival Solution
    ii) non-zero Solution (non trival Solution)

a) $\begin{cases} (k-1)x + y = 0 \\ 3x + (k+1)y = 0 \end{cases}$ b) $\begin{cases} x + 2y + kz = 0 \\ 2x + ky + 2z = 0 \\ 3x + y + z = 0 \end{cases}$

Solution;

i) trivial Solution (x=0, y= 0, z = 0) is obtained when D = 0

ii) non-zero Solution are obtained when $D \neq 0$

a) $D = \begin{vmatrix} k-1 & 1 \\ 3 & k+1 \end{vmatrix} = k^2 - 1 - 3 \Rightarrow k^2 - 4$

i) trivial Solution, $k^2 - 4 \neq 0 \Rightarrow k \neq \pm 2$

ii) non trivial Solution, when $D = 0 \Rightarrow k^2 - 4 = 0 \Rightarrow k = 2$ or $k = -2$

b) $D = \begin{vmatrix} 1 & 2 & k \\ 2 & k & 2 \\ 3 & 1 & 1 \end{vmatrix} = 3k^2 - 3k - 6$

i) trivial Solution: $D \neq 0 \Rightarrow 3k^2 - 3k - 6 \Leftrightarrow (k+1)(k-2) \neq 0$
$\therefore k \neq -1$ or $k \neq 2$

ii) Non-zero Solution, $k = -1$ or $k = 2$

Conceptual Example

59. If the determinant $|A| = 3$ and $|B| = 2$, find $|2(AB)^{-1}|$ for 4×4 matrices A and B.

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{8}{3}$

Solution:

Recall that $|A^{-1}| = \frac{1}{|A|}$

$$|2(AB)^{-1}| = 2^4 \times \frac{1}{|A||B|} = \frac{16}{(3)(2)} = \frac{8}{3}$$

Answer: D

60. For what values of x and y, $\begin{pmatrix} 2 & 1 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3x \\ 5y+8 \end{pmatrix}$

Solution: $\begin{pmatrix} 2x+y \\ 3x-4y \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3x \\ 5y+8 \end{pmatrix}$

$$\Rightarrow \begin{cases} 2x - 3x + y = -4 \\ 3x - 4y + 2 - 5y = 8 \end{cases} \Rightarrow 3\begin{cases} -x + y = -4 \\ 3x - 9y = 6 \end{cases}$$

$$\Rightarrow \begin{cases} -3x+3y=-12 \\ +\ 3x-9y=6 \end{cases}$$

$$-6y=-6, \qquad \therefore y=1,\ x=5$$

61. If $A=\begin{pmatrix}2 & 7\\ 1 & 3\end{pmatrix}$ and $B^{-1}=\begin{pmatrix}1 & 1\\ 0 & 2\end{pmatrix}$ then What is the value of $(2AB)^{-1}$?

Solution:

Recall that $(2AB)^{-1}=\frac{1}{2}B^{-1}A^{-1}$

If $A=\begin{pmatrix}a & c\\ b & d\end{pmatrix}$ then $A^{-1}=\frac{1}{|A|}\begin{pmatrix}d & -c\\ -b & a\end{pmatrix}$

If $A=\begin{pmatrix}2 & 7\\ 1 & 3\end{pmatrix}$ then $A^{-1}=\frac{1}{-1}\begin{pmatrix}3 & -7\\ -1 & 2\end{pmatrix}=\begin{pmatrix}-3 & 7\\ 1 & -2\end{pmatrix}$

$$\therefore (2AB)^{-1}=\frac{1}{2}B^{-1}A^{-1}=\frac{1}{2}\begin{pmatrix}1 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}-3 & 7\\ 1 & -2\end{pmatrix}$$

$$=\frac{1}{2}\begin{pmatrix}1(-3)+1\times 1 & 1\times 7+1(-2)\\ 0(-3)+2(1) & 0\times 7+2(-2)\end{pmatrix}=\frac{1}{2}\begin{pmatrix}-2 & 5\\ 2 & -4\end{pmatrix}=\begin{pmatrix}-1 & 2.5\\ 1 & -2\end{pmatrix}$$

62. Solve a) $\begin{vmatrix}-2x & -3y\\ 4x & -y\end{vmatrix}=7y$ b) $\begin{vmatrix}2x-1 & x+1\\ x+2 & x-2\end{vmatrix}=-12$

Solution: a) $\begin{vmatrix}-2x & -3y\\ 4x & -y\end{vmatrix}=7y \Rightarrow 2xy+12xy=7y$

$$\Rightarrow 14xy=7y \Leftrightarrow 14x=7 \quad \therefore x=\frac{1}{2}$$

b) $\begin{vmatrix}2x-1 & x+1\\ x+2 & x-2\end{vmatrix}=-12 \Rightarrow x^2-8x=-12 \Rightarrow x^2-8x+12$

$$\Rightarrow (x-6)(x-2)=0$$

$$\therefore x=6 \text{ or } x=2$$

The inverse of the matrix $A=\begin{bmatrix}2 & 1 & 3\\ 0 & -1 & 2\\ 4 & 3 & 1\end{bmatrix}$ is the matrix

$A^{-1}=\frac{1}{6}\begin{bmatrix}-7 & 8 & m\\ 8 & -10 & -4\\ 4 & k & -2\end{bmatrix}$ where;

A. $m=3, k=2$ C. $m=1, k=-3$

B. $m=5, k=-2$ D. $m=2, k=-3$

Solution: $A^{-1}.A=I$

$$A^{-1}.A=\frac{1}{6}\begin{bmatrix}-7 & 8 & m\\ 8 & -10 & -4\\ 4 & k & -2\end{bmatrix}\begin{bmatrix}2 & 1 & 3\\ 0 & -1 & 2\\ 4 & 3 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

C_{11} = the product of row 1 of A^{-1} and column 1 of A.

$\Rightarrow C_{11}=\frac{1}{6}(-7\times2+8\times0+m\times4)=1\Rightarrow -14+4m=6$

$\Rightarrow 4m=20\Rightarrow m=5$

$C_{32}=\frac{1}{6}[4\times1+(-k)\times1+(-2)(3)]=0\Rightarrow -k-2=0$

$\therefore k=-2$ **Answer: B**

54. Which of the following are the values of y and z respectively if

$$\begin{pmatrix}-2 & x+1\\ y & 0\end{pmatrix}\begin{pmatrix}3 & z-1\\ 2 & 1\end{pmatrix}=\begin{pmatrix}0 & -7\\ -3 & -5\end{pmatrix}\ldots\ldots \text{UEE}$$

A. 2, -1 B. -1, 6 C. 2, 6 D. 6, -1

Solution:

$$\begin{pmatrix}-2 & x+1\\ y & 0\end{pmatrix}\begin{pmatrix}3 & z-1\\ 2 & 1\end{pmatrix}=\begin{pmatrix}-6+2x+2 & -2z+2+x+1\\ 3y & yz-y\end{pmatrix}$$

$$=\begin{pmatrix}0 & -7\\ -3 & -5\end{pmatrix}\Leftrightarrow 2x-4=0\Rightarrow x=2\Leftrightarrow 3y=-3\Rightarrow y=-1$$

$\Leftrightarrow yz-y=-5\Rightarrow -z+1=-5\Rightarrow z=6$ **Answer: B**

65. Let $A = \begin{bmatrix} 4 & 4 & -2 \\ 2 & 6 & 0 \\ 3x & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ 3 & 0 \\ -1 & 5 \end{bmatrix}$ are given

If $C = AB$ and $C_{32} = 11$, then what is the value of x.

A. $\frac{3}{5}$ B. $\frac{-4}{3}$ C. $\frac{-2}{3}$ D. -8

Solution: $C_{32} = \begin{pmatrix} 3x & 1 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} = 6x + 0 + 15 = 11$

$\Leftrightarrow 6x + 15 = 11 \Rightarrow 6x = -4 \Rightarrow x = \frac{-4}{6} = \frac{-2}{3}$ **Answer: C**

66. If $\begin{pmatrix} x & -3 \\ -1 & y \end{pmatrix}$ is the inverse of $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ then the value of x a

y respectively. (UEE)

A. 1, 3 B. 2, 2 C. 4, 1 D. $\frac{1}{2}, -\frac{3}{2}$

Solution: $\begin{pmatrix} x & -3 \\ -1 & y \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Leftrightarrow$

$\begin{pmatrix} 2x-3 & 3x-6 \\ -2+y & -3+2y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Leftrightarrow 2x-3 = 1 \Rightarrow 2x = 4 \Rightarrow x = 2$ and $-2 + y = 0 \Rightarrow y = 2$ **Answer**

67. What are the value of x and y so that

$\begin{pmatrix} x & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} y & y \\ 2 & 1 \end{pmatrix}$?

A. $x = 3, y = 2$ C. $x = 4, y = 5$

B. $x = 0, y = -1$ D. $x = -2, y = y$

Solution: $\begin{pmatrix} 2x-4 & x-1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} y & y \\ 2 & 1 \end{pmatrix} \Leftrightarrow 2x - 4 = y$

$x - 1 = y$

$\Leftrightarrow 2x-4=x-1 \Rightarrow x=3$

$\therefore y=3(2)-4=2$

Answer: A

68. If $A=\begin{pmatrix} x & -1 \\ 3x & 1 \end{pmatrix}$ then for what values of x is $\det(2A)=x^2$...(UEE)

A. 16 B. 8 C. 4 D. -8

Solution: $|A|=\begin{vmatrix} x & -1 \\ 3x & 1 \end{vmatrix}=x+3x=4x$

And $\det(2A)=2^2|A|=4|A|=4(4x)=x^2$

$\Rightarrow x^2=16x \Leftrightarrow x^2-16x=0$ $x=0$ or $x=16$ **Answer: A**

69. What should be the value of k so that the system $\begin{cases} 2kx-y=4 \\ -4x+ky=1 \end{cases}$ has a unique Solution?

A. $k\neq 2,-2,$ C. $k=\pm\sqrt{2}$

B. $k\neq-\sqrt{2}, k\neq\sqrt{2}$ D. $k\neq\pm 1$

Solution: $\begin{vmatrix} 2k & -1 \\ -4 & k \end{vmatrix}\neq 0\Rightarrow 2k^2-4\neq 0\Rightarrow k^2-2\neq 0,\Rightarrow$

$k\neq\pm\sqrt{2}$ **Answer: B**

70. Let $A=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ be 2 × 2 matrix. Then the value of A^{2k+1} for $k\in N$ is equal to:

A. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ C. $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$* A^{2k+1}=A^{2k}.A$, Now, $A^{2k}=(A^2)^k$

$$=\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right)^k=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^k=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}=I_2$$

$\Rightarrow A^{2k+1}=A^{2k}.A=I_2\cdot A=A=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, **Answer: D**

71. Solve for x and y: $\begin{bmatrix} 2 & 4 & -1 \\ -1 & 0 & 5 \end{bmatrix}\begin{bmatrix} 2x+1 \\ 4 \\ x+y \end{bmatrix} = \begin{bmatrix} 12 \\ 11 \end{bmatrix}$

Solution:
Each row of the first matrix is multiplied by each column of second matrix.
$2(2x + 1) + 4(4) + (-1)(x+y) = 12 \Rightarrow 3x - y = -6$ ..... (1)
$-1(2x + 1)+0+5(x+y) = 11 \Rightarrow 3x + 5y = 12$ ..... (2)
Subtracting equation (2) from equation (1)
$-6y = -18 \Rightarrow y = 3$ and $3x - 3 = -6 \Rightarrow x = -1$
Therefore, $x = -1$ and $y = 3$.

72. Determine the set of values x such that the matrix $A = \begin{pmatrix} x^2-1 & 5 \\ 3 & x^2+1 \end{pmatrix}$ is non singular (invertible)

Solution: A is non singular (Invertible) if $|A| \neq 0$

$$\Rightarrow \begin{vmatrix} x^2-1 & 5 \\ 3 & x^2+1 \end{vmatrix} \neq 0 \Leftrightarrow x^4 - 1 - 15 \neq 0 \Leftrightarrow x^4 - 16 \neq 0$$

$\Rightarrow x^2 - 4)(x^2 + 4) \neq 0 \Rightarrow (x^2 - 4) \neq 0$
$\therefore x \neq 2$ or $x \neq -2$
$\therefore$ S.S = R\{2, -2}

73. What should be the value of α and β respectively so that $\begin{pmatrix} \alpha & 1 \\ 1 & \beta \end{pmatrix}$ is the inverse of $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$

A. 0, 0 B. 1, 1 C. -1, 0 D. Not invertib

Solution: $\begin{pmatrix} \alpha & 1 \\ 1 & \beta \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 1 & \alpha+1 \\ \beta & \beta+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \alpha + 1 = 0 \Rightarrow \alpha = -1$ and $\beta = 0$, Answer: C

Given matrix $A=\begin{bmatrix}1 & 1 & 1\\ 1 & b & b\\ 1 & b & c\end{bmatrix}$, then the value of |A|

A. (1 – b)(c – b) C. (b – 1)(c – b)
B. (1 – c)(b – c) D. (c – 1)(b – c)

Solution: $|A|=\begin{vmatrix}1 & 1 & 1\\ 1 & b & b\\ 1 & b & c\end{vmatrix}=1\begin{vmatrix}b & b\\ b & c\end{vmatrix}-1\begin{vmatrix}1 & b\\ 1 & c\end{vmatrix}+1\begin{vmatrix}1 & b\\ 1 & b\end{vmatrix}$

$\Rightarrow bc - b^2 - 1(c- b) = b(c - b) - 1(c - b) = (b - 1)(c - b)$

Answer: C

Given the system of equation: find its Solution set

a) $\begin{cases}2x-3y=6\\ -4x+6y=-12\end{cases}$ b) $\begin{cases}3x-y=2\\ -6x+2y=-4\end{cases}$ c) $\begin{cases}5x-y=1\\ -5x+y=4\end{cases}$

Solution: a) Using row operation:

$$\begin{pmatrix}2 & -3 & 6\\ -4 & 6 & -12\end{pmatrix}\xrightarrow{R_2\to R_2+2R_1}\begin{pmatrix}2 & -3 & 6\\ 0 & 0 & 0\end{pmatrix}$$ ← it has infinite Solution.

$\therefore$ S.S = {(x, y): 2x - 3y = 6}

* Also we can express as, Let x = C, then $y=\frac{2c}{3}-2$

$$\therefore \text{S.S}=\left\{\left(c,\frac{2c}{3}-2\right)\right\}=\left\{(0,-2)+\left(1,\frac{2}{3}\right)t,\ t\in R\right\}$$

b) $$\begin{pmatrix}3 & -1 & 2\\ -6 & 2 & -4\end{pmatrix}\xrightarrow{R_2\to 2R_1+R_2}\begin{pmatrix}3 & -1 & 2\\ 0 & 0 & 0\end{pmatrix}$$ ← It has infinit many ***Solution*** $\Rightarrow 0 = 0$ (True)

$\therefore$ S.S = {(x,y): 3x – y = 2} or we can express as let x = c, then 3c – y = 2 $\Rightarrow$ y = 3c – 2

$\therefore$ S.S = {(c, 3c -2)}

$\therefore$ S.S = {(0, -2) + (1, 3)t, t $\in$ R} ← using vector equation of line.

c) $\begin{pmatrix} 5 & -1 & 1 \\ -5 & 1 & 4 \end{pmatrix} \xrightarrow{R_2 \to R_1 + R_2} \begin{pmatrix} 5 & -1 & 1 \\ 0 & 0 & 5 \end{pmatrix} \Rightarrow 0 = 5$ is a

$\therefore$ It has no Solution.

$\therefore$ S.S $= \varnothing$

76. Let $A = \begin{pmatrix} 0 & \alpha & \beta \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$, and $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

If det (A) = 4, then what is the solution set of the system $AX =$

Solution: Since det (A) $= 4 \neq 0$, so that we can use cramer's

$$\therefore D_x = \begin{vmatrix} 8 & \alpha & \beta \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} = 8\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + 0 + 0 = 24$$

$$\therefore D_y = \begin{vmatrix} 0 & 8 & \beta \\ 2 & 0 & 1 \\ 3 & 0 & 2 \end{vmatrix} = 0 - 8\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + 0 = -8$$

$$\therefore D_z = \begin{vmatrix} 0 & \alpha & 8 \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = 0 + 0 + 8\begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} = -32$$

$$\therefore \{(x, y, z)^T\} = \left\{\left(\frac{D_x}{D}, \frac{D_y}{D}, \frac{D_z}{D}\right)\right\} = \left\{\left(\frac{24}{4}, \frac{-8}{4}, \frac{-32}{4}\right)^T\right\}$$

$\therefore \{(x, y, z)^T\} = \{(6, -2, -8)^T\}$

77. Given the system of equations: $\begin{cases} x - 3y + z = 0 \\ 4x + z = 3 \\ -3x + y - z = -2 \end{cases}$ its solu

Set is

A. $\{(1, 0, -1) + (1, 1, 4)t, t \in \mathbb{R}\}$ C. $\{(1, 0, -1)\}$

B. $\{(0, 1, 3) + (1, -1, -4)t, t \in \mathbb{R}\}$ D. $\varnothing$

Solution: Here

$$D=\begin{vmatrix}1 & -3 & 1\\ 4 & 0 & 1\\ -3 & 1 & -1\end{vmatrix}=\begin{vmatrix}0 & 1\\ 1 & -1\end{vmatrix}+3\begin{vmatrix}4 & 1\\ -3 & -1\end{vmatrix}+\begin{vmatrix}4 & 0\\ -3 & 1\end{vmatrix}$$

$$= -1 + 3(-4 + 3) + 4 = 0$$

$$D_x=\begin{vmatrix}0 & -3 & 1\\ 3 & 0 & 1\\ -2 & 1 & -1\end{vmatrix}=3\begin{vmatrix}3 & 1\\ -2 & -1\end{vmatrix}+1\begin{vmatrix}3 & 0\\ -2 & 1\end{vmatrix}=-3+3=0$$

Since $D = 0, D_x = 0, D_y = 0, D_z = 0 \Rightarrow$ system has infinite solution

: from second equation: $4x + z = 3 \Rightarrow z = 3 - 4x$

: From 1st equation: $x - 3y + z = 0 \Rightarrow 3y = x + z.$

: Put $z = 3 - 4x$ in $3y = x + z \Rightarrow 3y = x + 3 - 4x = -3x + 3$

$\Rightarrow y = -x + 1 \Leftrightarrow y = 1 - x$

$\therefore$ S.S $= \{(x, y, z)\} = \{(x, 1 - x, 3 - 4x)\}$, Let $x = 0$

$$= \{(0,1,3) + (t,-t,-4t)\}$$

$$= \{(0,1,3) + (1,-1,-4)t \quad t \in R\}$$

Answer: B

78. Let $A=\begin{pmatrix}-2 & 0 & 5\\ x & -1 & 1\\ 2x+1 & 2 & -x\end{pmatrix}$ if the minor of an element 0 is -16, then the possible value of x is

A. 7, -3 B. 1, 3 C. 3, – 5 D. -3, 5

Solution: 0 is located in 1st row and 2nd column of A, so that we write a_{12}. Thus, minor of $a_{12} = \begin{vmatrix}x & 1\\ 2x+1 & -x\end{vmatrix} = -16 \Leftrightarrow -x^2 - 2x - 1 = -16$

$\Rightarrow x^2 + 2x - 15 = 0 \Leftrightarrow (x - 3)(x + 5) = 0$

$\therefore x = 3$ or $x = -5$ $\therefore$ Answer C

79. From Question # 78, if the cofactor of an element 5 is 25, then the value of x is

A. -1 B. 0 C. 6 D. No x-value

Solution: 5 is located 1st row and 3rd column of matrix A, so that cofactor A_{13} is given by

$$A_{13} = (-1)^{1+3} M_{13} = \begin{vmatrix}x & -1\\ 2x+1 & 2\end{vmatrix} = 25$$

$\Rightarrow 2x-(-2x-1)=25 \Leftrightarrow 4x+1=25 \Rightarrow 4x=24 \Rightarrow x=6$

$\therefore$ Answer: C

80. Solve each of the following.

a) $\begin{pmatrix}3&2&1\\2&4&0\\1&0&2\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-4\\-10\\5\end{pmatrix}$ b) $\begin{pmatrix}a-3&a+1\\a+1&a+3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}4\\2\end{pmatrix}$

Solution: We can solve by using Cramer's rule.

a) $D=\begin{vmatrix}3&2&1\\2&4&0\\1&0&2\end{vmatrix}=3\begin{vmatrix}4&0\\0&2\end{vmatrix}-2\begin{vmatrix}2&0\\1&2\end{vmatrix}+1\begin{vmatrix}2&4\\1&0\end{vmatrix}=24-8-4=12$

$D_x=\begin{vmatrix}-4&2&1\\-10&4&0\\5&0&2\end{vmatrix}=-4\begin{vmatrix}4&0\\0&2\end{vmatrix}-2\begin{vmatrix}-10&0\\5&2\end{vmatrix}+1\begin{vmatrix}-10&4\\5&0\end{vmatrix}$

$= -32+40-20=-12$

$D_y=-24, \quad D_z=36$

$\therefore x=\frac{D_x}{D}=\frac{-12}{12}=-1,\ y=\frac{D_y}{D}=\frac{-24}{12}=-2,$

$z=\frac{D_z}{D}=\frac{36}{12}=3$

b) $D=\begin{vmatrix}a-3&a+1\\a+1&a+3\end{vmatrix}=a^2-9-(a^2+2a+1)$

$=-2a-10,\ a\neq -5$

$D_x=\begin{vmatrix}4&a+1\\2&a+3\end{vmatrix}=4a+12-2a-2=2a+10,\ a\neq -5$

$D_y=\begin{vmatrix}a-3&4\\a+1&2\end{vmatrix}=2a-6-4a-4=-2a-10$

$\therefore\ x=\frac{D_x}{D}=\frac{2a+10}{-2a-10}=\frac{2a+10}{-(2a+10)}=-1,$

$$y=\frac{D_y}{D}=\frac{-2a-10}{-2a-10}=1$$

Let $A=(a_{ij})_{3\times3}$ given by $a_{ij}=\begin{cases}0 & if\ \ i>j\\ i+j & if\ \ i\le j\end{cases}$ then find a) $|A|$

b) A^{-1} c) $|A^{-1}|$ d) A.adjA

Solution:

a) $\begin{pmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{pmatrix}=\begin{pmatrix}1+1&1+2&1+3\\0&2+2&2+3\\0&0&3+3\end{pmatrix}=\begin{pmatrix}2&3&4\\0&4&5\\0&0&6\end{pmatrix}$

Since it is an upper triangular, $\therefore |A|=2\times4\times6=48$

b) To find adj.A, first determine, A^t then cofactor of (A^t) is adj.

$$A^t=\begin{pmatrix}2&0&0\\3&4&0\\4&5&6\end{pmatrix},\ adj.A=\begin{pmatrix}\begin{vmatrix}4&0\\5&6\end{vmatrix}&-\begin{vmatrix}3&0\\4&6\end{vmatrix}&\begin{vmatrix}3&4\\4&5\end{vmatrix}\\-\begin{vmatrix}0&0\\5&6\end{vmatrix}&\begin{vmatrix}2&0\\4&6\end{vmatrix}&-\begin{vmatrix}2&0\\4&5\end{vmatrix}\\\begin{vmatrix}0&0\\4&0\end{vmatrix}&-\begin{vmatrix}2&0\\3&0\end{vmatrix}&\begin{vmatrix}2&0\\3&4\end{vmatrix}\end{pmatrix}$$

$$=\begin{pmatrix}24&-18&-1\\0&12&-10\\0&0&8\end{pmatrix}$$

$$\therefore A^{-1}=\frac{adj.A}{|A|}=\frac{1}{48}\begin{pmatrix}24&-18&-1\\0&12&-10\\0&0&8\end{pmatrix}=\begin{pmatrix}\frac{1}{2}&\frac{-3}{8}&\frac{-1}{48}\\0&\frac{1}{4}&\frac{-5}{24}\\0&0&\frac{1}{6}\end{pmatrix}$$

c) $|A^{-1}|=\frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{6}=\frac{1}{48}$

Explanation: $|A^{-1}| = \frac{1}{|A|}$

d) $A.adj.A = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} \cdot \begin{pmatrix} 24 & -18 & -1 \\ 0 & 12 & -10 \\ 0 & 0 & 8 \end{pmatrix} = \begin{pmatrix} 48 & 0 & \\ 0 & 48 & \\ 0 & 0 & \end{pmatrix}$

explanation: (A)(adj.A) = |A|I

82. Find A^{-1} for each case

a) $A = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}$ b) $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ c) $A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$

Solution: a) $A^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$ b) $A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$

c) $A^{-1} = \begin{pmatrix} \frac{1}{a} & 0 & 0 \\ 0 & \frac{1}{b} & 0 \\ 0 & 0 & \frac{1}{c} \end{pmatrix}$ for abc ≠ 0

83. For what values of y is the matrix

$A = \begin{pmatrix} -1 & 1 & y \\ 0 & 6 & 1 \\ -y & 1 & 0 \end{pmatrix}$ non-singular: A $\frac{1}{12}$ B. $\frac{1}{2}, \frac{1}{3}$

C. $\mathbb{R}$ D. None

Solution: A is non singular (invertible) if |A| ≠ 0

$$\Rightarrow |A| = \begin{vmatrix} -1 & 1 & y \\ 0 & 6 & 1 \\ -y & 1 & 0 \end{vmatrix} = -1\begin{vmatrix} 6 & 1 \\ 1 & 0 \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ -y & 0 \end{vmatrix} + y\begin{vmatrix} 0 & 6 \\ -y & 1 \end{vmatrix} = 1 - y + 6y^2$$

For all real number y, $6y^2 - y + 1 \neq 0$, since $b^2 - 4ac < c$

$\Rightarrow 1 - 4(6)(1) = -23 < 0$ $\therefore$ Answer C

84. Let $A = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ *and* $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^T = 0$, then the value of y? . . . . UEE 2004/12

A. 0 B. $\frac{-13}{2}$ C. -8 D. any real number

Solution:

$$B^T = \begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} \Rightarrow 2B^T = \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix}$$

$$\Rightarrow A+2B^T = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow 2 - 2x + x = 0 \Rightarrow x = 2$ and $x+y+6 = 0 \Rightarrow 2 + y + 6 = 0$

$\Rightarrow y = -8$ **Answer: C**

85. Let A and B be 3× 3 matrices such that

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix} \text{ and } |B| = \frac{1}{10}, \textit{ what is } \left|2AB^T\right|?$$

A. 1 B. 4 C. 100 D. 400

Solution: $\left|2AB^T\right| = 2^3|A|\left|B^T\right| = 8(2 \times 5 \times \frac{1}{2})\left(\frac{1}{10}\right) = 4$

86. The Solution set of system of equation $\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases}$

A. $\{(0, 1, 0)\}$ B. $(-\infty, \infty)$

C. $\{(-3k,k+1,k)\}$ D. $\{(3k,k-1,k)|k\in\mathbb{R}\}$

Solution: $D=\begin{vmatrix}1&1&2\\1&2&1\\-2&-2&-4\end{vmatrix}=-2\begin{vmatrix}1&1&2\\1&2&1\\1&1&2\end{vmatrix}=0$

$D_x=\begin{vmatrix}1&1&2\\2&2&1\\-2&-2&-4\end{vmatrix}=-2\begin{vmatrix}1..&1&2\\2\,.\,2&1\\1&1&2\end{vmatrix}=0\,,D_y=0,\quad D_z=0$

$\Rightarrow$ The system has infinite many solution.

$$-\begin{cases}x+y+2z=1\\x+2y+z=2\end{cases}$$

$$0+-y+z=-1\Rightarrow y=z+1$$

Substitute $y=z+1$ in equation $x+y+2z=1$ and solve for x

$\Rightarrow x+z+1+2z=1\Rightarrow x=-3z$

$\Rightarrow S.S=\{(x,y,z)\}=\{(-3z,z+1,z)\}=\{(-3k,k+1,k)\}\ K\in\mathbb{R}\ \therefore$

Answer: C

87. If $\begin{pmatrix}\alpha&2&\beta\\2&1&3\\-1&0&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-5\\5\\0\end{pmatrix}$ and the determinant of coefficient matrix is -5, then the value of x is equal to:

A) 5 B) $\alpha+\beta$ C) -5α D) 3

Solution: We have $D=-5\neq 0$

$$\Rightarrow D_x=\begin{vmatrix}-5&2&\beta\\5&1&3\\0&0&1\end{vmatrix}=-5\begin{vmatrix}1&3\\0&1\end{vmatrix}-2\begin{vmatrix}5&3\\0&1\end{vmatrix}+\beta\begin{vmatrix}5&1\\0&0\end{vmatrix}=-15$$

$$\therefore x=\frac{D_x}{D}=\frac{-15}{-5}=3$$

Answer: D

88. What is the solution set of the system $\begin{cases}x+y+2z=1\\x+2y+z=1\\3x+4y+5z=3\end{cases}$

A. $\{(1,0,0)\}$ C. $\{(-3k+1,k,k|k\in\mathbb{R}\}$

B. $\{(k+1,-k,k)|k\in\mathbb{R}\}$ D. ϕ

Explanation:

$$D=\begin{vmatrix}1&1&2\\1&2&1\\3&4&5\end{vmatrix}=1\begin{vmatrix}2&1\\4&5\end{vmatrix}-1\begin{vmatrix}1&1\\3&5\end{vmatrix}+2\begin{vmatrix}1&2\\3&4\end{vmatrix}=6-2-4=0$$

$$D_y=\begin{vmatrix}1&1&2\\1&1&1\\3&3&5\end{vmatrix}=0,\ D_z=0 \text{ and } D_x=0$$

Since, $D=0, D_x=0, D_y=0, D_z=0$
therefore, the system has infinite solution.

$$\begin{cases}x+y+2z=1.....(1)\\x+2y+z=1.....(2)\\3x+4y+5z=3.....(3)\end{cases}$$

Eliminate x from equation (1) and (2)
by subtracting the two equation.

$$-\begin{cases}x+y+2z=1\\x+2y+z=1\end{cases}$$

$$0-y+z=0\Rightarrow y=z$$

Substitute $y=z$ in equation (1): $x+y+2z=1\Rightarrow x+z+2z=1$
$\Rightarrow x=-3z+1$. Let $z=k\in\mathbb{R}$.
$\therefore S.S=\{(-3k+1,k,k)|k\in\mathbb{R}\}$ **Answer: C**

89. If $A^{-1}=\begin{bmatrix}1&-1&3\\2&0&5\\-1&1&0\end{bmatrix}$. Find a matrix B such that $AB=\begin{bmatrix}1&-1&2\\0&1&1\\-1&0&0\end{bmatrix}$.

Solution: Here at
We know that $A^{-1}A=i$ (identity matrix).
Multiply both side of a matrix by A^{-1}, we get

$$\Rightarrow B = A^{-1}(AB) = A^{-1}\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow B = \begin{bmatrix} 1+0+3 & -1+1+0 & 2+(-1)+0 \\ 2+0+5 & -2+0+0 & 4+0+0 \\ -1+0+0 & 1+1+0 & -2+1+0 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 1 \\ 7 & -2 & 4 \\ -1 & 2 & -1 \end{bmatrix}$$

Review Exercise on unit six

1. **Conceptual exercise**

a) What is a matrix?

Explanation:

A matrix is a rectangular array of number.

b) What is the difference between row and column matrix?

Explanation:

A row runs horizontally and a column runs vertically.

c) What is the order of a matrix?

Explanation:

The number of rows and column.

d) What is the goal of row/column operation (Gauss – Jorda elimination).

Explanation:

The goal is to get 1 on the diagonal.

e) What is a determinant?

Explanation:

A real number associated with a square matrix.

f) Which systems can be solved using cramer's rule?

Explanation:

Cramer's rule works on systems that have exactly one solution.

g) What is a minor?

Explanation:

A minor for an element in a 3 × 3 matrix is the determinant of a 2 × 2 matrix.

Solve for x, y and z.

a) $\begin{bmatrix} 2x+3 & 8 \\ -5 & -1 \end{bmatrix} - \begin{bmatrix} 10 & 6 \\ 6y-1 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4z+8 \\ 8 & -8 \end{bmatrix}$

b) $\begin{bmatrix} 2x+3 \\ 5x-2 \\ y \end{bmatrix} + \begin{bmatrix} 3y \\ 2y+4 \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ -5 \end{bmatrix}$

c) $\begin{bmatrix} 2 & 3 \\ -1 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 10 \\ 0 \end{bmatrix}$

d) $\begin{bmatrix} 5x-6 & 7 \\ 2y+2 & z \end{bmatrix} - \begin{bmatrix} x & -2 \\ y & 3z+1 \end{bmatrix} = \begin{bmatrix} 6 & 9 \\ 5y & 3z \end{bmatrix}$

3. Find x and y if $\begin{pmatrix} x & 1 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} 5 & -4 \\ -2 & 7 \end{pmatrix} = \begin{pmatrix} y & y \\ 14 & 5 \end{pmatrix}$

4. Solve for x and y.

a) $\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2x \\ 8x & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 2y \\ -15 & 10 \end{bmatrix}$

b) $\begin{bmatrix} -2 & 1 & 2 \\ 3 & 2 & 4 \\ 0 & -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 6x & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -6 \\ 19 & 9 \\ 8 & y \end{bmatrix}$

5. Evaluate the determined of the matrix.

a) $\begin{pmatrix} -4 & 2 \\ -8 & 8 \end{pmatrix}$ b) $\begin{pmatrix} 7 & -4 \\ 3 & -1 \end{pmatrix}$ c) $\begin{pmatrix} 12 & 5 \\ 4 & 2 \end{pmatrix}$

6. Evaluate the determinant of the matrix.

a) $\begin{pmatrix} 5 & 0 & 1 \\ 1 & 4 & -1 \\ 3 & 2 & 0 \end{pmatrix}$ b) $\begin{pmatrix} -3 & 3 & -1 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{pmatrix}$ c) $\begin{pmatrix} 3 & 4 & 2 \\ -1 & 1 & 2 \\ 0 & 2 & 3 \end{pmatrix}$

7. Find the inverse of the matrix.

a) $\begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$ b) $\begin{pmatrix} -4 & 9 \\ 3 & -7 \end{pmatrix}$ c) $\begin{pmatrix} 9 & -2 \\ 5 & -1 \end{pmatrix}$

8. For what values of x and y the matrix $\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$ and $\begin{pmatrix} x & -2 \\ y & 1 \end{pmatrix}$ are inverse of each other.

9. Let $A = \begin{bmatrix} 11 & 2 & 4m \\ 4 & 1 & -3 \\ -8 & -1 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -4 & 2 \\ 0 & 2 & 1 \\ 4 & n & 3 \end{bmatrix}$. For what values of m and n the two matrices, A and B, to be inverse of each other.

10. Let $A = \begin{bmatrix} 11 & 5 \\ 2 & 1 \end{bmatrix}$, $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ 2 & 0 \end{bmatrix}$.
What is the solution set of the system $AX = B$.

11. If $A = \begin{bmatrix} -7 & 5 \\ 3 & -2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$, then $(AB)^{-1}$ is equal to _____.

12. Solve the matrix equation:

a) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$ b) $\begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$

13. Let $A = \begin{pmatrix} 0 & \alpha & \beta \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix}$, $b = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix}$, and $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$
If $\det(A) = 6$, then what is the solution set of the system $AX = b$.

14. If $A = \begin{pmatrix} 4 & -3 & 8 \\ 0 & 2 & 3 \\ 0 & 1 & 4 \end{pmatrix}$, then $\det(A^T A)$ is equal to _____

15. What elementary row operation does $-3R_1 + R_3$ denote? Which row can it replace? Explain

16. Use an augmented matrix to solve the linear system.

a) $\begin{cases} 2x+4y+5z=5 \\ x+3y+3z=2 \\ 2x+4y+4z=2 \end{cases}$ b) $\begin{cases} 3x+3y+z=1 \\ 2x+3y+z=-1 \\ 2x+4y+z=-2 \end{cases}$

17. What is the solution set of the system?

a) $\begin{cases} 2x+y+3z=-4 \\ 2x+3y+z=-4 \\ 6x+5y+7z=-12 \end{cases}$ b) $\begin{cases} x+y-2z=5 \\ x+2y+z=8 \\ 2x+3y-z=13 \end{cases}$

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Unit Seven

Vectors and Scalars

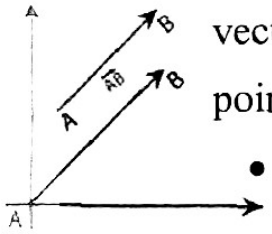
Quantities are categorized in to two.

i) **Scalar quantities** such as temperature area, volute, work, mass etc and have magnitude only.

ii) **Vector quantities**, such as velocity force, acceleration, magnetic field etc, have magnitude and direction.

Definition: A **vector** in the plane is a directed line segment. The directed line segment $\overrightarrow{AB}$ has **initial point A** and **terminal point B.**

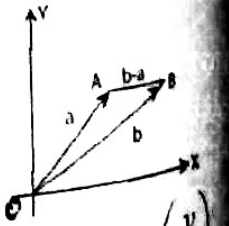
- The length (magnitude) of directed line segment is denoted by $\left|\overrightarrow{AB}\right|$.
- The component form by directed line segment of the vector $\overrightarrow{AB}$ with initial point A = (x_1, y_1) and terminal point B = (x_2, y_2) is $\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$



- The length of (magnitude) $\overrightarrow{AB}$ is $\left|\overrightarrow{AB}\right| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

- **Position vector** If the initial point of vector is fixed to origin (0,0) then it is called position vector, i.e. If the initial point is origin and the terminal point P is vector every point in plane is called **Position vector** $\overrightarrow{OP}$

- Any vector $\overrightarrow{AB}$ we can expressed interms of positioned vector
  - $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a}$



Note: $\overrightarrow{AB}$ = position vector of B relative to A

$\overrightarrow{OB}$ = position vector of B elative to O

- Direction of position vector $\overrightarrow{AB}$ is $\tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$

Illustrative Example

1. Find the position vector $\overrightarrow{AB}$, the direction and the magnitude of $\overrightarrow{AB}$.

a) A = (1, 6), B = (-3, -2)
b) A = (-2, 7), B = (1, -2)
c) A = (3,1); B = (5, 3)
d) A = (2,1); B = (7, -4)

***Solution*: Let O(0, 0) be origin**

a) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (-3,-2) - (1,6) = (-4, -8)$ ← position vector

Direction of $\overrightarrow{AB}$ is $\tan\theta = \frac{y}{x} = \frac{-8}{-4} = 2 \Rightarrow \theta = \tan^{-1}(2)$

$$\left|\overrightarrow{AB}\right| = \sqrt{(-4)^2 + (-8)^2} = \sqrt{16+64} = \sqrt{80} = 4\sqrt{5} \text{ unit.}$$

b) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (1, -2) - (-2,7) = (3,-9)$ ← position vector

Direction of $\overrightarrow{AB}$ is $\tan\theta = \frac{y}{x} = \frac{-9}{3} = -3 \Rightarrow \theta = \tan^{-1}(-3)$

$$\left|\overrightarrow{AB}\right| = \sqrt{(3)^2 + (-9)^2} = \sqrt{9+81} = \sqrt{90} = 3\sqrt{10}$$ ← magnitude

c) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (5, 3) - (3, 1) = (2, 2)$ ← position vector

Direction of $\overrightarrow{AB}$ is $\tan\theta = \frac{y}{x} = \frac{2}{2} = 1 \Rightarrow \theta = \tan^{-1}(1) = \frac{\pi}{4}$

$$\left|\overrightarrow{AB}\right| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2} \text{ unit.}$$

d) $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (7, -4) - (2, 1) = (5, -5)$

Direction of $\overrightarrow{AB}$ is $\tan\theta = \frac{y}{x} = \frac{-5}{5} = -1 \Rightarrow \theta = \tan^{-1}(-1) = \frac{7\pi}{4}$

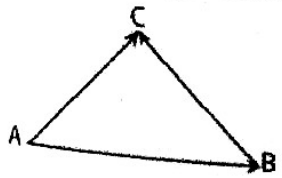
$$\left|\overrightarrow{AB}\right| = \sqrt{5^2 + (-5)^2} = \sqrt{50} = 5\sqrt{2} \text{ unit.}$$

Operation of Vector

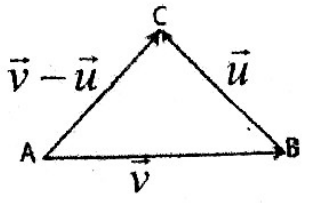
I. Vector Addition

Triangle method

a)



b)



- $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$
- $\overrightarrow{BC}+\overrightarrow{CA}=\overrightarrow{BA}$
- $\overrightarrow{AB}-\overrightarrow{CB}=\overrightarrow{AC}$
- $\overrightarrow{BC}-\overrightarrow{BA}=\overrightarrow{AC}$
- $-\overrightarrow{CB}+\overrightarrow{AB}=\overrightarrow{AC}$

Note: $\overrightarrow{AB}=-\overrightarrow{BA}$

2. ABCD is parallelogram, find a single vector of the following

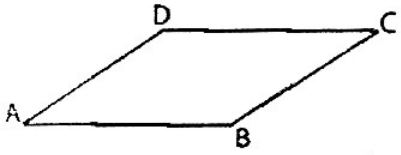
a) $\overrightarrow{AB}-\overrightarrow{CB}$ c) $\overrightarrow{DC}-\overrightarrow{DA}$

b) $\overrightarrow{AB}+\overrightarrow{BC}-\overrightarrow{DC}$ d) $\overrightarrow{DB}+\overrightarrow{BA}+\overrightarrow{DA}$

Solution: a) From triangle law of addition $\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$

$\Rightarrow \overrightarrow{AB}-\overrightarrow{CB}=\overrightarrow{AC}$ → Explanation $\overrightarrow{BC}=-\overrightarrow{CB}$

b) $\overrightarrow{AB}+\overrightarrow{BC}-\overrightarrow{DC}$



$\Rightarrow \overrightarrow{AC}-\overrightarrow{DC} \Rightarrow \overrightarrow{AC}+\overrightarrow{CD}=\overrightarrow{AD}$

$\therefore \overrightarrow{AB}+\overrightarrow{BC}-\overrightarrow{DC}=\overrightarrow{AD}$

c) $\overrightarrow{DC}-\overrightarrow{DA} \Rightarrow \overrightarrow{DC}+\overrightarrow{AD}=\overrightarrow{AD}+\overrightarrow{DC}=\overrightarrow{AC}$

$\therefore \overrightarrow{DC}-\overrightarrow{DA}=\overrightarrow{AC}=\overrightarrow{AD}-\overrightarrow{CD}$

d) $\overrightarrow{DB}+\overrightarrow{BA}+\overrightarrow{DA} \Rightarrow \overrightarrow{DA}+\overrightarrow{DA}=2\overrightarrow{DA}$

II. Multiplication of vectors by scalars

Definition: If $\vec{u}=(a, b)$, $\vec{v}=(c, d)$, k is scalar then

i) $k\vec{u}=k(a, b)=(ka, kb)$

ii) $k\vec{v}=k(c, d)=(kc, kd)$

iii) $|k\vec{u}|=|k||\vec{u}|=|k|\sqrt{a^2+b^2}$

iv) $|k\vec{v}|=|k||\vec{v}|=|k|\sqrt{c^2+d^2}$

4. Let $\vec{u}=(1, -2)$ and $\vec{v}=(-2, 4)$, then find

a) $2\vec{u}$ c) $\vec{u}+\vec{v}$ e) $3\vec{u}+\frac{1}{2}\vec{v}$

b) $\frac{-1}{2}\vec{v}$ d) $\left|\frac{-1}{2}\vec{v}\right|$ f) $\left|3\vec{u}+\frac{1}{2}\vec{v}\right|$

Solution:

a) $2\vec{u}=2(1, -2)=(2, -4)$

b) $\frac{-1}{2}\vec{v} = \frac{-1}{2}(-2, 4) = (1, -2)$

c) $\vec{u} + \vec{v} = (1, -2) + (-2, 4) = (-1, 2)$

d) $\left|\frac{-1}{2}\vec{v}\right| = \left|\frac{-1}{2}\right| |\vec{v}| = \frac{1}{2}\sqrt{(-2)^2 + (4)^2} = \frac{1}{2}\sqrt{4+16} = \sqrt{5}$

e) $3\vec{u} + \frac{1}{2}\vec{v} = 3(1,-2) + \frac{1}{2}(-2,4) = (3, -6) + (-1, 2) = (2, -4)$

f) $\left|3\vec{u} + \frac{1}{2}\vec{v}\right| = \sqrt{(2)^2 + (-4)^2} = \sqrt{4+16} = 2\sqrt{5}$

5. If the magnitude of $\vec{v}$ is 5 unit then find the magnitude of:

a) $4\vec{v}$ b) $-3\vec{v}$ c) $\frac{-1}{2}\vec{v}$ d) $\frac{2}{5}\vec{v}$

Solution: Given $|\vec{v}| = 5$ unit

a) $|4\vec{v}| = 4|\vec{v}| = 4(5) = 20$ unit

b) $|-3\vec{v}| = 3|\vec{v}| = 3(5) = 15$ unit

c) $\frac{-1}{2}\vec{v} = \frac{1}{2}|\vec{v}| = \frac{1}{2}(5) = \frac{5}{2} unit$

d) exercise

Parallel or collinear vector

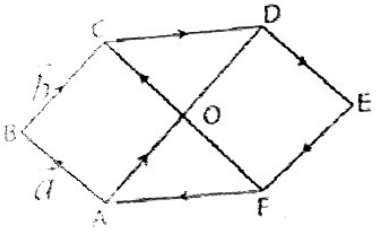
If two vectors are collinear or parallel then any one of them can be express as scalar multiple of the other.

i.e. $\vec{u} = k\vec{v} \Rightarrow \vec{u} \parallel \vec{v}$

4. ABCDEF are vertices of regular hexagon. In which $\overrightarrow{AB} = \vec{a}$ =(1,2) and $\overrightarrow{BC} = \vec{b}$ = (-2, 3). Express the other vector interms of the vector $\vec{a}$ *and* $\vec{b}$. (other vector are $\overrightarrow{CD}$, $\overrightarrow{DE}$, $\overrightarrow{EF}$ and $\overrightarrow{AF}$)

Solution:
- Let $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$
- since $\overrightarrow{AB} \parallel \overrightarrow{OC} \Rightarrow \overrightarrow{AB} = \vec{a}$ and $\overrightarrow{OC} = \vec{a}$
- $\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \vec{b} - \vec{a}$

$CD = (-2, 3) - (1, 2) = (-3, 1)$



- $\overrightarrow{DE} = \overrightarrow{OE} - \overrightarrow{OD}$ and $\overrightarrow{OE} \parallel \overrightarrow{CD}$

$= \vec{b} - \vec{a} - (\vec{b}) = -\vec{a}$

$\Rightarrow \overrightarrow{DE} = -(\vec{a}) = (1, 2) = (-1, -2).$

- $\overrightarrow{EF} = \overrightarrow{OF} - \overrightarrow{OE} = -\vec{a} - (\vec{b} - \vec{a}) = -\vec{b}$

$\Rightarrow \overrightarrow{EF} = -\vec{b} = (-2, 3) = (2, -3).$

- $\overrightarrow{FA} = \overrightarrow{OA} - \overrightarrow{OF} = -\vec{b} - (-\vec{a}) = \vec{a} - \vec{b}$

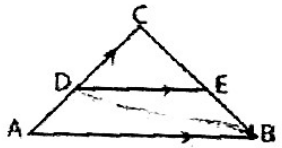
$\Rightarrow \overrightarrow{FA} = \vec{a} - \vec{b} = (1, 2) - (-1, 3) = (3, -1)$

$\therefore \overrightarrow{CD} = \vec{b} - \vec{a}$, $\overrightarrow{DE} = -\vec{a}$, $\overrightarrow{EF} = -\vec{b}$ and $\overrightarrow{FA} = \vec{a} - \vec{b}$

Mid – point theorem

5. Using vectors show that, the segment joining the mid-points of two side of a triangle is parallel to the third side and equal to half of it.

Solution:



$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

$\overrightarrow{DB} + \overrightarrow{BE} = \overrightarrow{DE}$

$\overrightarrow{DE} = \frac{1}{2}AB$, $\overrightarrow{BE} = \frac{1}{2}BC$

$\Rightarrow \frac{1}{2}\left(\overrightarrow{AB} + \overrightarrow{BC}\right) = \overrightarrow{DE} \Leftrightarrow \overrightarrow{DE} = \frac{1}{2}\overrightarrow{AC}$

6. If ABCD is parallelogram in Cartesian coordinate planes with A = (1, 1), B = (7, 5), and C = (10, 10). What are the coordinate of D(x, y)

A. (4, 6) B. (9, 9) C. (6, 3) D. (4, 5)

Solution: Since $\overrightarrow{AB} \parallel \overrightarrow{BC}$ and $\overrightarrow{AD} \parallel \overrightarrow{BC}$

$\Rightarrow$ AD = (x, y) – (1, 1) = (x – 1, y – 1)

$\Rightarrow \overrightarrow{BC} = (10, 10) - (7, 5) = (3, 5)$

$\Rightarrow$ (x – 1, y – 1) = (3, 5) $\Rightarrow$ x – 1 = 3 and y – 1 = 5

$\therefore$ x = 3 + 1 = 4 and y = 5 + 1 = 6

$\therefore$ D(x, y) = (4, 6)

Unit vector

- A vector whose magnitude is one is called a unit vector
- The standard unit vectors are i = (1, 0) and j = (0, 1)
- Any vector can express as linear combination of unit vector i and j

Let $\vec{v}$ = (a, b) expressed as

$\vec{v}$ = (a, b) = a(1,0) + b(0, 1) = ai + bj

$\vec{v}$ = (2, 3) = 2(1,0) + 3(0, 1) = 2i + 3j

Which of the following is unit vector

a) $\vec{v} = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ b) $\vec{v} = \left(\frac{3}{5}, \frac{-4}{5}\right)$ c) (-1, 1)

Solution:

a) $|v| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5} + \frac{1}{5}} = \frac{5}{5} = 1 \leftarrow$ unit vector

$\Rightarrow v = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ is unit vector

b) $\vec{v} = \left(\frac{3}{5}, \frac{-4}{5}\right) \Rightarrow |\vec{v}| = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1 \leftarrow$ unit vector

c) $\vec{v}(-1,1) \Rightarrow |\vec{v}| = \sqrt{(-1)^2 + (1)^2} = \sqrt{01+1} = \sqrt{2} \leftarrow$ not unit vector

Unit vector in the direction of $\vec{v}$

- If $\vec{v}$ is a non-zero vector in the plane, then the vector $\vec{u} = \frac{\vec{v}}{|\vec{v}|}$ has length 1 and the same direction of $\vec{v}$

8. Find unit vector:

i) in the direction of the vector $\vec{v}$

ii) in the direction opposite of $\vec{v}$ if given that

a) $\vec{v}$ = (-3, 4) b) $\vec{v}$ = (2, -4) c) $\vec{v}$ = (x, y)

Solution: i) Unit vector in the direction of $\vec{v}$ given by

a) i) $\frac{\vec{v}}{|\vec{v}|} = \frac{(-3,4)}{\sqrt{(-3)^2 + 4^2}} = \frac{(-3,4)}{\sqrt{25}} = \frac{1}{5}(-3,4) = \left(\frac{-3}{5}, \frac{4}{5}\right)$

ii) Unit vector in the direction opposite to v is

$$\frac{-v}{|v|} \Rightarrow \left(\frac{3}{5}, \frac{-4}{5}\right)$$

b) i) $\frac{v}{|v|} = \frac{1}{\sqrt{2^2+(-4)^2}}(2,-4) = \frac{1}{2\sqrt{5}}(2,-4) = \left(\frac{1}{\sqrt{5}}, \frac{-2}{\sqrt{5}}\right)$

unit vector in the same direction

ii) Unit vector in the direction opposite of $\vec{v} = (2, -4)$ is

$$\frac{-v}{|v|} = \left(\frac{-2}{2\sqrt{5}}, \frac{4}{2\sqrt{5}}\right) = \left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

c) i) $\left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}}\right)$ ii) $\left(\frac{-x}{\sqrt{x^2+y^2}}, \frac{-y}{\sqrt{x^2+y^2}}\right)$

9. If $\vec{v} = 2i + 3j$ and $\vec{u} = 3i - j$ then

Find a) $\vec{v} + \vec{u}$ b) $|\vec{v} + \vec{u}|$ c) $|2\vec{v} - 3\vec{u}|$

d) Unit vector in the direction of $\vec{u} + \vec{v}$

e) Unit vector in the direction opposite to vector $\vec{u} + \vec{v}$

Solution:

a) $\vec{v} + \vec{u} = 2i + 3j + 3i - j = (2 + 3)i + (3 - 1)j = 5i+2j$

b) $|\vec{v} + \vec{u}| = |5i + 2j| = \sqrt{5^2 + 2^2} = \sqrt{29}$

c) $|2\vec{v} - 3\vec{u}| = |-5i + 9j| = \sqrt{(-5)^2 + (9)^2} = \sqrt{106}$

d) Unit vector in the direction of $\vec{v} + \vec{u} = \frac{\vec{v}+\vec{u}}{|\vec{v}+\vec{u}|} = \frac{5i}{\sqrt{29}} + \frac{2j}{\sqrt{29}}$

e) Unit vector in the direction opposite to vector

$$\vec{v} + \vec{u} = \frac{-(\vec{v}+\vec{u})}{|\vec{v}+\vec{u}|} = \frac{-1}{\sqrt{29}}(5i + 2j)$$

10. If $\vec{a} = 4i + j$, $\vec{b} = 3i - 2j$ and $\vec{c} = -i - 2j$

- Find the unit vector parallel to $2\vec{a} - \vec{b} - \vec{c}$ but in opposite direction

Solution: $2\vec{a} - \vec{b} - \vec{c} = 2(4i + j) - (3i - 2j) - (-i - 2j) = 6i + 6j$

Unit vector in the direction opposite of $2\vec{a} - \vec{b} - \vec{c}$

$$\Rightarrow \frac{-(2\vec{a}-\vec{b}-\vec{c})}{|2\vec{a}-\vec{b}-\vec{c}|}=\frac{-1}{|6i+6j|}(6i+6j)$$

$$=\frac{-6i}{6\sqrt{2}}+\frac{-6j}{6\sqrt{2}}=\frac{-i}{\sqrt{2}}-\frac{j}{\sqrt{2}}$$

11. If A = (1, 2) and B = (3, 5) be two points, find the unit vector along a) $\overrightarrow{AB}$ b) $\overrightarrow{BA}$

Solution:

$\overrightarrow{AB}$ = (3, 5) – (1, 2) = (2, 3) ⟵ position vector of $\overrightarrow{AB}$

$\Rightarrow$ Unit vector along $\overrightarrow{AB}$ is $\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}=\frac{(2,3)}{\sqrt{4+9}}=\left(\frac{2}{\sqrt{13}},\frac{3}{\sqrt{13}}\right)$

b) $\overrightarrow{BA}$ = (1,2) –(3,5) = (-2, -3)

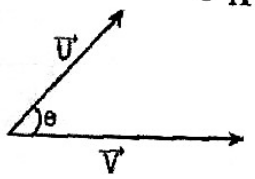
$\Rightarrow$ Unit vector along $\overrightarrow{BA}=\frac{\overrightarrow{BA}}{|\overrightarrow{BA}|}=\left(\frac{-2}{\sqrt{13}},\frac{-3}{\sqrt{13}}\right)$

Scalar (dot or Inner) product of vector

Definition: If $\vec{u}=(a_1, a_2)$, $\vec{v}=(b_1, b_2)$ and θ is the angle between $\vec{u}$ and $\vec{v}$ then the dot product of $\vec{u}$ and $\vec{v}$ is denoted by $\vec{u}\,.\,\vec{v}$ is defined by

i) $\vec{u}\,.\,\vec{v}=(a_1, a_2)\cdot(b_1, b_2)=a_1b_1+a_2b_2$

ii) $\vec{u}\,.\,\vec{v}=|\vec{u}|.|\vec{v}|\cos\theta$



- if θ is acute, then $\vec{u}\,.\,\vec{v}>0$
- if θ is obtuse, then $\vec{u}\,.\,\vec{v}<0$
- if θ is $\frac{\pi}{2}$, then $\vec{u}\,.\,\vec{v}=0$ $(\vec{u}\perp\vec{v})$
- if θ = 0, then $\vec{u}\parallel\vec{v}$
- if θ = π, then $\vec{u}\parallel\vec{v}$ but opposite in direction

Properties of dot product

- $\vec{u}\,.\,\vec{v}=\vec{v}\,.\,\vec{u}$
- $k(\vec{u}\,.\,\vec{v})=k\vec{u}\,.\,\vec{v}=\vec{u}\,.k\vec{v}$
- $\vec{u}\cdot(\vec{v}+\vec{w})=\vec{u}\cdot\vec{v}+\vec{u}\cdot\vec{w}$

- $|\vec{u}|^2 = \vec{u}\cdot\vec{u} = |\vec{u}||\vec{u}|$
- $|\vec{u}+\vec{v}|^2 = (\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v}) = |u|^2 + 2\vec{u}\cdot\vec{v} + |\vec{v}|^2$
- $|\vec{u}-\vec{v}|^2 = (\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v}) = |u|^2 - 2\vec{u}\cdot\vec{v} + |\vec{v}|^2$

$\Rightarrow |\vec{u}+\vec{v}| = \sqrt{(\vec{u}+\vec{v})\cdot(\vec{u}+\vec{v})} = \sqrt{|\vec{u}|^2 + 2\vec{u}\cdot\vec{v} + |\vec{v}|^2}$

$\Rightarrow |\vec{u}-\vec{v}| = \sqrt{(\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v})} = \sqrt{|\vec{u}|^2 - 2\vec{u}\cdot\vec{v} + |\vec{v}|^2}$

12. Find the dot product of the vector $\vec{u}$ and $\vec{v}$ and state, angle between acute or obtuse

a) $\vec{u} = (-2, 5)$ and $\vec{v} = (3, 4)$ c) $\vec{v} = 4i - j$ and $\vec{u} = (-3i + 2j)$

b) $\vec{u} = (4, -3)$ and $\vec{v} = (-2, 3)$ d) $\vec{v} = 8i - 3j$ and $\vec{u} = 2i + 7j$

Solution:

a) $\vec{u}\cdot\vec{v} = (-2,5)\cdot(3,4) = -6+20 = 14 \leftarrow$ Angle b/n $\vec{u}$ and $\vec{v}$ is acute

b) $\vec{u}\cdot\vec{v} = (4,-3)\cdot(-2,3) = -8-9 = -17 \leftarrow$ Angle b/n $\vec{u}$ and $\vec{v}$ is obtuse

c) $\vec{v}\cdot\vec{v} = (4i - j)\cdot(2i+7j) = 8 - 7 = 1 \longleftarrow$ Acute angle

d) $\vec{v}\cdot\vec{u} = (8i - 3j)\cdot(2i+7j) = 16 - 21 = -5 \longleftarrow$ Obtuse angle

13. Determine the value of k so that the angle between the vectors $\vec{u} = (3, -2)$ and $\vec{v} = (4, 3k)$ is

a) acute b) obtuse c) right angle (or thogional)

Solution:

a) Acute $\Rightarrow \vec{u}\cdot\vec{v} > 0, \Rightarrow (3, -2)\cdot(4, 3k) > 0$

$\Rightarrow 12 - 6k > 0 \Rightarrow 12 > 6k \Rightarrow k < 2$

b) obtuse $\Rightarrow \vec{u}\cdot\vec{v} < 0 \Rightarrow (3, -2)\cdot(4, 3k) < 0 \Rightarrow 12 - 6k < 0$

$\Rightarrow 12 < 6k \Rightarrow k > 2$

c) Right angle) $\Rightarrow \vec{u}\cdot\vec{v} = 0 \Rightarrow 12 - 6k = 0 \Rightarrow k = 2 \leftarrow \vec{u} \perp \vec{v}$

14. If $\vec{v} = (3, -1)$ and $\vec{u} = (1, 3)$. Find $(\vec{v}\cdot\vec{u})\vec{u}$

Solution:

$(\vec{v}\cdot\vec{u})\vec{u} \Rightarrow ((3, -1)\cdot(1, 3))(1, 3) = (3 + (-3))(1, 3) = (0, 0)$

15. Let $\vec{u} = (2, -2)$, $\vec{v} = (5, 8)$ and $\vec{w} = (-4, 3)$

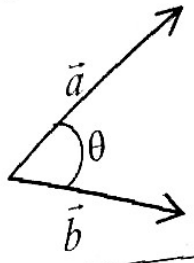
Find:

a) $\vec{u}\cdot\vec{v} = (2, -2)\cdot(5, 8) = 10 - 16 = -6$

b) $(\vec{u}\cdot\vec{v})\vec{w} = -6(-4, 3) = (24, -18).$

c) $|\vec{w}|^2 = \vec{w}\cdot\vec{w} = (-4, 3)\cdot(-4,3) = 16 + 9 = 25$

Angle between two vector



$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

$$\Rightarrow \cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} \Rightarrow \theta = \cos^{-1}\left(\frac{\vec{a}\cdot\vec{b}}{|a||b|}\right)$$

$$0 \le \theta \le \pi$$

16. Find the consin of the angle between $\vec{v}$ and $\vec{u}$ if

a) $\vec{v} = (-1, -1)$ and $\vec{u} = (3, 0)$ c) $\vec{v} = (-2, 1)$ and $\vec{u} = (0, 3)$

b) $\vec{v} = (1, 2)$ and $\vec{u} = (-1, -2)$ d) $\vec{v} = (4, 3)$ and $\vec{u} = (1, 7)$

Solution: Let θ be the angle between $\vec{v}$ and $\vec{u}$

$$\vec{v}\cdot\vec{u} = |v||\vec{u}|\cos\theta \Rightarrow \cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|}$$

a) $\cos\theta = \dfrac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|} \Rightarrow \dfrac{(-1,-1)\cdot(3,0)}{\sqrt{1^2+1^2}\sqrt{3^2+0^2}} = \dfrac{-3}{3\sqrt{2}} = \dfrac{-1}{\sqrt{2}} \Rightarrow \theta = \dfrac{3\pi}{4}$

b) $\cos\theta = \dfrac{(1,2)\cdot(-1,-2)}{(\sqrt{1^2+2^2})(\sqrt{1^2+2^2}} = \dfrac{-1-4}{\sqrt{5}\cdot\sqrt{5}} = \dfrac{-5}{5} = -1 \Rightarrow \theta = \cos^{-1}(-1) = \pi$

c) $\cos\theta = \dfrac{(-2,1)\cdot(0,3)}{\sqrt{4+1}\sqrt{3^2}} = \dfrac{3}{3\sqrt{5}} = \dfrac{1}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}\left(\dfrac{1}{\sqrt{5}}\right)$

d) $\cos\theta = \dfrac{(4,3)\cdot(1,7)}{(\sqrt{16+9})\sqrt{1+49}} = \dfrac{4+21}{5\sqrt{50}} = \dfrac{25}{25\sqrt{2}} = \dfrac{1}{\sqrt{2}} \Rightarrow \theta = \cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$

17. Let $\vec{v} = (1,7)$ and $\vec{u} = (4, 3)$. Find

a) the angle between $\vec{v}$ and $\vec{u} - \vec{v}$

b) the angle between $\vec{u}$ and $2\vec{u} - \vec{v}$

Solution: $\vec{v}\cdot(\vec{u} - \vec{v}) = |\vec{v}||\vec{u} - \vec{v}|\cos\theta$

Let θ be the angle between $\vec{v}$ and $\vec{u} - \vec{v}$

a) $\cos\theta = \dfrac{\vec{v}\cdot(\vec{u}-\vec{v})}{|\vec{v}||\vec{u}-\vec{v}|} \Rightarrow$

$$\frac{(1,7)\cdot(3,-4)}{(\sqrt{1^2+7^2})(\sqrt{3^2+4^2}} = \frac{3-28}{\sqrt{50}\sqrt{25}} = \frac{-25}{25\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

b) $\cos\theta = \dfrac{\vec{u}\cdot(2\vec{u}-\vec{v})}{|\vec{u}||2\vec{u}-\vec{v}|} \Rightarrow$

$$\frac{(4,3)\cdot(7,-1)}{\sqrt{4^2+3^2}\sqrt{7^2+1^2}} = \frac{28-3}{\sqrt{25}\sqrt{50}} = \frac{25}{25\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

Theorem Two vector $\vec{a}$ *and* $\vec{b}$ are

i) Orthogonal if and only if $\vec{a}\cdot\vec{b} = 0$

ii) Parallel if and only if $\dfrac{\vec{a}\cdot\vec{b}}{|a||b|} = \pm 1$

18. Determine **m** such that the two vectors are orthogonal.

a) $\vec{v} = 3i - 2j$, $\vec{u} = 4i + 5mj$ c) $\vec{v} = 3i + 4j$, $\vec{u} = mi + 2j$

b) $\vec{v} = 5mi + 3j$, $\vec{u} = 2i + 7j$ d) $\vec{v} = mi - 3j$, $\vec{u} = mi + 2mj$

Solution: a) $\vec{v}\cdot\vec{u} = 0 \Rightarrow (3i - 2j)\cdot(4i + 5mj) = 0$

$$\Rightarrow 12 - 10m = 0 \Rightarrow m = \frac{12}{10} = \frac{6}{5}$$

b) $\vec{v}\cdot\vec{u} = 0 \Rightarrow (5mi + 3j)\cdot(2i + 7j) = 0 \Rightarrow 10m + 21 = 0$

$\Rightarrow -10m = 21 \Rightarrow m = -2\cdot 1$

c) $\vec{v}\cdot\vec{u} = 0 \Rightarrow (3i + 4j)\cdot(mi + 2j) = 3m + 8 = 0 \Rightarrow m = \dfrac{-8}{3}$

d) $\vec{v}\cdot\vec{u} = 0 \Rightarrow (mi - 3j)\cdot(mi + 2mj) = 0 \Rightarrow m^2 - 6m = 0$

$\Rightarrow m(m - 6) = 0$

$\Rightarrow m = 0$ or $m = 6$

19. If $\vec{a} = i + 2j$, $\vec{b} = -i + 2j$ and $\vec{c} = 3i + j$. Find **t** such that $\vec{a} + t\vec{b}$ is perpendicular to C.

Solution: $(\vec{a} + t\vec{b})\cdot\vec{c} = 0 \Rightarrow (i + 2j + -ti + 2tj)\cdot(3i + j) = 0$

$\Rightarrow [(1 - t)i + (2 + 2t)j]\cdot(3i + j) = 0 \Rightarrow 3 - 3t + 2 + 2t = 0$

$\Rightarrow 5-t=0 \Rightarrow t=5$

20. Let $\vec{a}=2i-4j$ an $\vec{b}=\left(-\frac{1}{2}i+j\right)$ then show that $\vec{a}$ is parallel to $\vec{b}$

Solution:

$$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|a||b|}=\frac{(2i-4j)\cdot\left(\frac{-1}{2}i+j\right)}{\sqrt{4+16}\sqrt{\frac{1}{4}+1}}=\frac{-1-4}{2\sqrt{5}\cdot\frac{\sqrt{5}}{2}}=-1$$

$$\Rightarrow \theta=\cos^{-1}(-1)=\pi \Rightarrow \vec{a}\,\|\,\vec{b}$$

21. Vector $\vec{a}$ and $\vec{b}$ makes an angle of $\theta=\frac{2}{3}\pi$.

If $|\vec{a}|=3$ and $|\vec{b}|=4$, then find

a) $\vec{a}\cdot\vec{b}$ b) $(\vec{a}-\vec{b})\cdot(2\vec{a}+\vec{b})$ c) $|\vec{a}+\vec{b}|$

Solution:

a) $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta \Rightarrow (3)(4)\cos\frac{2\pi}{3}=12\left(\frac{-1}{2}\right)=-6$

b) $(\vec{a}-\vec{b})\cdot(2\vec{a}+\vec{b})=2\vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}-\vec{b}\cdot2\vec{a}-\vec{b}\cdot\vec{b}$

$=2|a|^2-\vec{a}\cdot\vec{b}-|b|^2$

$=2(3)^2-(-6)-(4)^2=8$

c) $|\vec{a}+\vec{b}|=\sqrt{(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})}=\sqrt{|a|^2+2\vec{a}\cdot\vec{b}+|b|^2}$

$=\sqrt{9-12+16}=\sqrt{13}$

22. Vectors $\vec{a}$ and $\vec{b}$ makes an angle $\theta=\frac{\pi}{4}$,

if $|\vec{a}|=\sqrt{2}$ and $|\vec{b}|=1$ then find

a) $|\vec{a}-\vec{b}|$ b) $|\vec{a}+2\vec{b}|$

c) angle between $\vec{a}-\vec{b}$ and $\vec{a}+2\vec{b}$

Solution: a) $|\vec{a}-\vec{b}|=\sqrt{(\vec{a}-\vec{b})\cdot(\vec{a}-\vec{b})}$

$$= \sqrt{|\vec{a}|^2 - 2|\vec{a}||\vec{b}|\cos\theta + |\vec{b}|^2}$$

$$\therefore |\vec{a}-\vec{b}| = \sqrt{\sqrt{2}^2 + 2(\sqrt{2})(1)\frac{\sqrt{2}}{2} + 1} = \sqrt{2-2+1} = 1$$

b) $|\vec{a}+\vec{b}| = \sqrt{(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})} = \sqrt{|a|^2 + 2|a||b|\cos\frac{\pi}{4} + |b|^2}$

$$\therefore |\vec{a}+\vec{b}| = \sqrt{\sqrt{2}^2 + 2(\sqrt{2})(1)\frac{\sqrt{2}}{2} + 1} = \sqrt{2+2+1} = 5$$

c) Let θ be angle between $\vec{a}-\vec{b}$ and $\vec{a}+2\vec{b}$ is given by

$$\cos\theta = \frac{(\vec{a}-\vec{b})\cdot(\vec{a}+2\vec{b})}{|\vec{a}-\vec{b}||\vec{a}+2\vec{b}|} \Rightarrow \frac{|a|^2 + |a||b|\cos\theta - 2|b|^2}{|\vec{a}-\vec{b}||\vec{a}+2\vec{b}|}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2}^2 + (\sqrt{2})(1) - 2(1)^2}{(1)(\sqrt{5})} = \frac{2+1-2}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

23. Let $|\vec{a}| = 7$, $|\vec{b}| = 3$, and $|\vec{a}+\vec{b}| = 5$

Find a) $\vec{a}\cdot\vec{b}$ b) $|\vec{a}-\vec{b}|$ c) $|2\vec{a}+3\vec{b}|$

Solution: To find $\vec{a}\cdot\vec{b}$ first interpret $|\vec{a}+\vec{b}|$

$$\Rightarrow |\vec{a}+\vec{b}| = \sqrt{(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})} = \sqrt{|a|^2 + 2\vec{a}\cdot\vec{b} + |b|^2}$$

$$\Rightarrow 5 = \sqrt{7^2 + 2\vec{a}\cdot\vec{b} + 3^2} = \sqrt{49+9+2a\cdot b}$$

$$\Rightarrow 5^2 = 49 + 9 + 2\vec{a}\cdot\vec{b} \Rightarrow 2\vec{a}\cdot\vec{b} = 25 - 58 = -33$$

$$\Rightarrow \vec{a}\cdot\vec{b} = \frac{-33}{2}$$

b) $|\vec{a}-\vec{b}| = \sqrt{(\vec{a}-\vec{b})\cdot(\vec{a}-\vec{b})} = \sqrt{|a|^2 - 2\vec{a}\cdot\vec{b} + |b|^2}$

$$\Rightarrow |\vec{a}-\vec{b}| = \sqrt{49 - (2)\left(\frac{-33}{2}\right) + 9} = \sqrt{91}$$

c) $|2\vec{a}+3\vec{b}| = \sqrt{4|\vec{a}|^2 + 12\vec{a}\cdot\vec{b}) + 9|b|^2}$

$$= \sqrt{4(49) + 12\left(\frac{-33}{2}\right) + 81}$$

$$\Rightarrow |2\vec{a} + 3\vec{b}| = \sqrt{196 - 198 + 81} = \sqrt{79}$$

24. Let $|\vec{a}| = 13$, $|\vec{b}| = 19$ and $|\vec{a} + \vec{b}| = 24$, then find

a) $\vec{a} \cdot \vec{b}$ b) Angle between $\vec{a}$ and $\vec{b}$

c) $|a - b|$ d) $|3\vec{a} - 4\vec{b}|$ e) $|3\vec{a} + 4\vec{b}|$

Solution: From $|\vec{a} + \vec{b}|$ we have

$$|\vec{a} + \vec{b}| = \sqrt{|\vec{a}|^2 + 2a \cdot b + b^2} = \sqrt{169 + 2a \cdot b + 361}$$

$$\Rightarrow (24) = \sqrt{2\vec{a} \cdot \vec{b} + 530} \Rightarrow 576 = 2\vec{a} \cdot \vec{b} + 530$$

$$\Rightarrow 2\vec{a} \cdot \vec{b} = 576 - 530 = 46 \Rightarrow \vec{a} \cdot \vec{b} = 23$$

b) Let θ be angle between $\vec{a}$ and $\vec{b}$, then

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{23}{|a||b|} = \frac{23}{(13)(19)} = \frac{23}{247} \Rightarrow \theta = \cos^{-1}\left(\frac{23}{247}\right)$$

c) $|\vec{a} - \vec{b}| = \sqrt{|a|^2 - 2\vec{a} \cdot \vec{b} + |b|^2}$

$$= \sqrt{169 - 2(23) + 361} = \sqrt{484} = 22$$

d) $|3\vec{a} - 4\vec{b}| = \sqrt{9|a|^2 - 24\vec{a} \cdot \vec{b} + 16|b|^2}$

$$= \sqrt{1521 - 552 + 5776} = \sqrt{6745}$$

e) $|3\vec{a} + 4\vec{b}| = \sqrt{9|a|^2 + 24\vec{a} \cdot \vec{b} + 16|b|^2}$

$$= \sqrt{1521 + 552 + 5776} = \sqrt{7849} \approx 88.6$$

23. Let $|\vec{a}| = 1$, $|\vec{b}| = 1$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, then find

a) $|\vec{a} + \vec{b}|$ b) angle between $\vec{a}$ and $\vec{a} + \vec{b}$

Solution: a) $|\vec{a} + \vec{b}| = \sqrt{|a|^2 + 2|\vec{a}||\vec{b}|\cos\frac{\pi}{3} + |b|^2}$

$$= \sqrt{1+2(1)(1)\frac{1}{2}+1} = \sqrt{3}$$

b) Let β be angle between $\vec{a}$ and $\vec{a}+\vec{b}$ is given by

$$\cos\beta = \frac{\vec{a}\cdot(\vec{a}+\vec{b})}{|a||\vec{a}+\vec{b}|} = \frac{|a|^2+|a||b|\cos\frac{\pi}{3}}{(1)(\sqrt{3})} = \frac{1+\frac{1}{2}}{\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \beta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

26. The cosine of the angle between the vector $\vec{u} = 2i + bj$ and $\vec{v} = bi + j$ is $\frac{3}{\sqrt{10}}$ find b

Solution:

$$\Rightarrow \frac{\vec{u}\cdot\vec{v}}{|u||v|} = \frac{(2i+bj)\cdot(bi+j)}{\sqrt{4+b^2}\sqrt{b^2+1}} = \frac{2b+b}{\sqrt{b^4+5b^2+4}}$$

$$\Rightarrow \frac{3}{\sqrt{10}} = \frac{3b}{\sqrt{b^4+5b^2+4}} \Rightarrow \frac{9}{10} = \frac{9b^2}{b^4+5b^2+4}$$

$\Rightarrow 10b^2 = b^4 + 5b^2 + 4$

$\Rightarrow b^4 - 5b^2 + 4 = 0 \Rightarrow (b^2-4)(b^2-1) = 0 \Rightarrow b = \pm 2, \pm 1$

27. If $\vec{u}$ and $\vec{v}$ are unit vector and angle between them is 30^0. Find

a) $|3u - v|$ b) $(\vec{u}+\vec{v})\cdot(2\vec{u}-\vec{v})$

Solution: since $\vec{u}$ and $\vec{v}$ are unit vector $\Rightarrow |u| = 1, |v| = 1$

a) $|3\vec{u}-\vec{v}| = \sqrt{9|\vec{u}|^2 - 6|\vec{u}||v|\cos 30^0 + |v|^2} = \sqrt{9 - \frac{6\sqrt{3}}{2} + 1}$

$$\Rightarrow |3\vec{u}-\vec{v}| = \sqrt{10-3\sqrt{3}}$$

b) $(\vec{u}+\vec{v})\cdot(2\vec{u}-\vec{v}) = 2|u|^2 + |u||v|\cos 30^0 - |v|^2$

$$= (2)(1) + (1)(1)\frac{\sqrt{3}}{2} - 1 = 1 + \frac{\sqrt{3}}{2}$$

28. If $\vec{a}+\vec{b}+\vec{c} = 0$, and $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$

Solution: $\vec{a}+\vec{b}+\vec{c}=0 \Rightarrow \vec{a}+\vec{b}=-\vec{c} \Rightarrow |\vec{a}+\vec{b}|^2=|-\vec{c}|^2$

$\Rightarrow |a|^2+2|a||b|\cos\theta+|b|^2=|-\vec{c}|^2 \leftarrow \theta$ is angle between $\vec{a}$ and $\vec{b}$

$\Rightarrow (3)^2+2(3)(5)\cos\theta+5^2=7^2 \Rightarrow \cos\theta=\dfrac{49-9-25}{30}=\dfrac{1}{2}$

$\Rightarrow \theta=\cos^{-1}\left(\dfrac{1}{2}\right)=\dfrac{\pi}{3}$

29. If $|2\vec{a}+\vec{b}|=|2\vec{a}-\vec{b}|$. Then find angle between $\vec{a}$ and $\vec{b}$.

Solution: $\Rightarrow |2a+b|^2=4|a|^2+4|\vec{a}||\vec{b}|\cos\theta+|\vec{b}|^2$

$\Rightarrow |2a-b|^2=4|\vec{a}|^2-4|\vec{a}||\vec{b}|\cos\theta+|\vec{b}|^2$

$\Rightarrow |2\vec{a}+\vec{b}|^2=|2\vec{a}-\vec{b}|^2$

$\Rightarrow 4|\vec{a}|^2+4|\vec{a}||\vec{b}|\cos\theta+|b|^2=4|\vec{a}|^2-4|\vec{a}||\vec{b}|\cos\theta+|b|^2$

$\Rightarrow 8|a||b|\cos\theta=0 \Rightarrow \cos\theta=0 \Rightarrow \theta=\dfrac{\pi}{2}$

30. If $\vec{u}=\vec{v}$ then find angle between $\vec{u}+\vec{v}$ and $\vec{u}-\vec{v}$

Solution: Let θ be the angle between $\vec{u}+\vec{v}$ and $\vec{u}-\vec{v}$

$$\Rightarrow \cos\theta=\frac{(\vec{u}+\vec{v})\cdot(\vec{u}-\vec{v})}{|\vec{u}+\vec{v}||u-\vec{v}|}=\frac{|\vec{u}|^2-|\vec{v}|^2}{|\vec{u}+\vec{v}||\vec{u}-\vec{v}|}$$

$$=\frac{0}{|\vec{u}+\vec{v}||\vec{u}-\vec{v}|}=0$$

$\Rightarrow \theta=\dfrac{\pi}{2}, \Rightarrow \vec{u}+\vec{v}$ and $\vec{u}-\vec{v}$ are perpendicular (orthogonal)

31. Let the angle between $\vec{v}=-2\vec{i}-\vec{j}+2\vec{k}$ and $\overrightarrow{AB}$ be 60°, where A and B are points in space. If $\vec{V}.\overrightarrow{AB}=2$ then what is the distance between A and B? (EUEE)

A. $\dfrac{3}{4}$ B. $\dfrac{4}{5}$ C. $\dfrac{4}{3}$ D. $\dfrac{5}{4}$

Solution: $\vec{V}.\overrightarrow{AB}=|V||AB|\cos 60°=2$

$$\Rightarrow |\overrightarrow{AB}|=\frac{2}{|V|\cos 60°}=\frac{2}{\sqrt{4+1+4}}(2)=\frac{4}{3}$$

Application of Vector

Definition: The **work w** done by a constant force F as its point of application moves along a vector $\vec{a}$ is $W = \vec{F} \cdot \vec{a} = |\vec{F}||\vec{a}|\cos\theta$.

32. Three forces $F_1 = 3i + 2j$, $F_2 = 3i - 2j$, $F_3 = 2i + j$ act on a particle which moves a point A to point B and then to point C if $A = i + 2j$, $B = (-2i - 5j)$ and $C = 3i + j$ then find the work done by
   a) each force in moving from B to C
   b) each force in moving from A to B.
   c) the combined force from A to C.
   d) the combined force from A to B.

Solution: $\overrightarrow{BC} = 3i + j - (2i - 5j) = i + 6j$

$\overrightarrow{AB} = 2i - 5j - (i + 2j) = i - 7j$

$\overrightarrow{AC} = 3i + j - (i + 2j) = 2i - j$

a) $\vec{F}_1 \cdot \overrightarrow{BC} = (3i + 2j)\cdot(i + 6j) = 3 + 12 = 15J$

$\vec{F}_2 \cdot \overrightarrow{BC} = (3i - 2j)\cdot(i + 6j) = 6 - 12 = -6J$

$\vec{F}_3 \cdot \overrightarrow{BC} = (2i + j)\cdot(i + 6j) = 2 + 6 = 8J$

b) $\vec{F}_1 \cdot \overrightarrow{AB} = (3i+2j)\cdot(i-7j) = 3 - 14 = -11J \leftarrow$ work done by F_1

$\Rightarrow \vec{F}_2 \cdot \overrightarrow{AB} = (3i-2j)\cdot(i-7j) = 3 + 14 = 17J \leftarrow$ work done by F_2

$\Rightarrow \vec{F}_3 \cdot \overrightarrow{AB} = (2i+j)\cdot(i-7j) = 2 - 7 = -52J \leftarrow$ work done by F_3

c) $(\vec{F}_1 + \vec{F}_2 + \vec{F}_3) \cdot (\overrightarrow{AC}) = (8i + j)\cdot(2i-j) = 16 - 1 = 15J$

d) $(F_1 + F_2 + F_3) \cdot \overrightarrow{AB} = (8i + j)\cdot(i-7j) = 8 - 7 = 1J \leftarrow$ work done by resultant

33. Two force $F_1 = 3i + 4j$ and $F_2 = -2i + j$ act on a particle which moves a point A(2, -1) to point B(-3, 1)
Find the work done by a) F_1 b) F_2 c) resultant

Solution: Determent $\overrightarrow{AB} = (-3, 1) - (2, -1) = (-5, 2) = -5i + 2j$

a) work done by $F_1 = \vec{F}_1 \cdot \overrightarrow{AB} = (3i+4j)\cdot(-5i + 2j) = -15 + 8 = -7J$

b) work done by $F_2 = \vec{F}_2 \cdot \overrightarrow{AB} = (-2i + j)\cdot(-5i+2j = 10 + 2 = 12J$

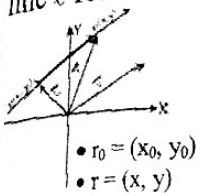
c) work done by resultant $(\vec{F}_1 + \vec{F}_2) = (3i+4j+-2i+j)\cdot(-5i + 2j)$
$= (i + 5j)\cdot(-5i+2j) = 5j$

Vectors and Lines

In plane line is determined by point on the line and given slope. On the other hand lines determined by a point and non zero vector. (WOW! It is surprising). The non zero vector $\vec{v}$ is the direction vector which is parallel to the line ℓ.

Vector equation of the line ℓ

Suppose that ℓ is a line in plane passing through a point $P_0(x_0, y_0)$ parallel to a non zero vector $\vec{v} = (a, b)$. Then P(x, y) is set of all point on line ℓ for which $\overrightarrow{P_0P}$ is parallel to $\vec{v}$.



$\Rightarrow \overrightarrow{P_0P} = t\vec{v}$, $-\infty < t < \infty$

$\Rightarrow (x, y) - (x_0, y_0) = t(a, b) \Rightarrow r = r_0 + t\vec{v}$

$\Rightarrow$ The set of all point (x, y) on ℓ is given by

$\Rightarrow (x, y) = (x_0, y_0) + t(a, b)$

If the line ℓ passing through $P_0(x_0, y_0)$ parallel to vector $\vec{v} = (a, b)$ then

i) $(x, y) = (x_0, y_0) + t(a, b)$ is vector equation of the line

ii) $\left.\begin{aligned} x &= x_0 + ta \\ y &= y_0 + tb \end{aligned}\right\}$ is called parametric equation of the line ℓ.

solving for t will get:

iii) $\dfrac{x - x_0}{a} = \dfrac{y - y_0}{b}$ is standard (rectangular) equation of line

35. Find vector, parametric, and standard (rectangular) equation of the line. If the line ℓ passing through the point

a) $P_0(1, 5)$ and $P(-2, -1)$

b) $P_0(-2, 4)$ and $P(4, -4)$

c) $P_0(-2, 4)$ and parallel to the vector $\vec{v} = (-2, 3)$

d) $P_0(1, -2)$ and parallel to the vector $\vec{v} = (2, 4)$

e) $P_0(3, -1)$ and perpendicular to the vector $\vec{v} = (4, 2)$

f) $P_0(0, 3)$ that is parallel to the line $x = -5 + t$, $y = 1 - 2t$

Solution: To find direction vector $\vec{v}$ for the line passing through P_0 and P is given by $\vec{v} = \overrightarrow{P_0P} \Rightarrow (x, y) = (x_0, y_0) + t\overrightarrow{P_0P}$

$\Rightarrow (x, y) = P_0 + t\overrightarrow{P_0P}$

a) $\vec{v} = \overrightarrow{P_0P} = (-2, -1) - (1, 5) = (-3, -6)$ ← parallel ve

$\Rightarrow (x, y) = (1, 5) + t(-3, -6)$ ← vector equation of the line

$\Rightarrow x = 1 - 3t,\ y = 5 - 6t$ ← parametric equation of the line

$\Rightarrow \dfrac{x-1}{-3} = \dfrac{y-5}{-6}$ ← standard (rectangular) equation of the line

$\Rightarrow 6x - 3y = -9$

b) $\vec{v} = \overrightarrow{P_0P} = (4, -4) - (-2, 4) = (6, -8)$ ← parallel vector

$\Rightarrow (x, y) = (-2, 4) + t(6, -8)$ ← vector equation of the line

$\Rightarrow x = -21 + 6t,\ y = 4 - 8t$ ← parametric equation of the line

$\Rightarrow \dfrac{x+2}{6} = \dfrac{y-4}{-8}$ ← standard (rectangular) equation of the line

$\Rightarrow 8x + 6y = 8$

c) direction (parallel) vector $\vec{v} = (-2, 3)$, and the line passing through $P_0 = (-2, 4)$

The set all (x, y) in ℓ is given by

$\Rightarrow (x, y) = (-2, 4) + t(-2, 3)$ ← vector equation of the line

$\Rightarrow x = -2 - 2t,\ y = 4 + 3t$ ← parametric equation of the line

$\Rightarrow$ solving for, t: $\dfrac{x+2}{-2} = \dfrac{y-4}{3}$ ← standard equation

$\Rightarrow 3x + 2y = 2$

d) $P_0 = (1, -2),\ \vec{v} = (2, 4)$

$\Rightarrow (x, y) = (1, -2) + t(2, 4)$ ← vector equation of the line

$\Rightarrow x = 1 + 2t,\ y = -2 + 4t$ ← parametric equation of the line

$\Rightarrow$ solving for t: $\dfrac{x-1}{2} = \dfrac{y+2}{4}$ ← standard equation

$\Rightarrow 4x - 2y = 8$

e) Let P(x, y) and $P_0(3, -1)$ be on the lien ℓ

Perpendicular vector $\vec{v} = (4, 2)$

$\Rightarrow \overrightarrow{P_0P} = (x,y) - (3, -1) = (x - 3, y + 1)$

Since $\overrightarrow{P_0P} \perp \vec{v} \Rightarrow \overrightarrow{P_0P} \cdot \vec{v} = 0$

$\Rightarrow (x-3, y+1)\cdot(4, 2) = 4x - 12 + 2y + 2 = 0$

$\Rightarrow 4x + 2y = 10 \Rightarrow 2x + y = 5$

To find the direction vector $\vec{v}$, set $2x + y = 0$

$\Rightarrow y = -2x$

$\Rightarrow$ The set of all parallel vector is given by

$\vec{v} = (x, y) = (x, -2x) = x(1, -2) = t(1, -2)$

We have $P_0 = (3, -1)$

$\Rightarrow (x, y_ = (3, -1) + t(1, -2) \longleftarrow$ vector equation of ℓ.

$\Rightarrow x = 3 + t, y = -1 - 2t \longleftarrow$ parametric equation of ℓ.

f) solving for $t = \dfrac{x+5}{1} = \dfrac{y-1}{-2} \Rightarrow \vec{v} = (1,-2)$

$\Rightarrow (x, y) = (0, 3) + t(1, -2)$

$\Rightarrow x = 0 + t, y = 3 - 2t.$

36. The line L passes through the point (1, -4) and parallel to the vector $\vec{v} = (-2, 5)$. Determine which point does not lie on L, and find t

A. (-1, 1) B. (3, -9) C. $\left(2, \dfrac{-13}{2}\right)$ D. $(-3,9)$

Solution: The set of all point, P(x, y) on ℓ is determined by vector equation of the line L:

$\Rightarrow (x, y) = (1, -4) + t(-2, 5) = (1 - 2t, -4 + 5t)$

$\Rightarrow$ Parametric equation of L: $x = 1 - 2t$ and $y = -4 + 5t$

Solving for t

$$\Rightarrow \frac{x-1}{-2} = \frac{y+4}{5} \Rightarrow 5x - 5 = -2y - 8 \Rightarrow 5x + 2y = -3$$

A. (-1, 1): $5(-1) + 2(1) = -3 \Rightarrow$ (-1, 1) on L; $t = 1$

B. (3, -9): $5(3) + 2(-9) = 15 - 18 = -3 \Rightarrow$ (3, -9) on L, $t = -1$

C. $\left(2, \dfrac{-13}{2}\right)$: $5(2)+2\left(\dfrac{-13}{2}\right) = 10-13 = -3 \Rightarrow \left(2, \dfrac{-13}{2}\right)$ on L, $t = \dfrac{-1}{2}$

D. $(-3,9)$: $5(-3) + 2(9) = -15 + 18 = 3 \Rightarrow$ (-3, 9) not on L. (has not the same t)

$\therefore$ (-1, 1), (3, -9), $\left(2, \frac{-13}{2}\right)$ are lie on ℓ and collinear. But (-1, 5)

is not on ℓ.

37. Are the following points collinear.

a) A(1, -1), B(-2, 5), C(3, -5)

b) A(2, 5) B(-1, -4), C(-2, -5)

Solution: The set of all point P(x, y) on L is determined by:

$P(x, y) = A + t\overrightarrow{AB} \Rightarrow (x, y) = (1, -1) + t(-3, 6)$

$\Rightarrow (x, y) = (1 - 3t, -1 + 6t)$

A. $(1, -1) = (1 - 3t, -1 + 6t) \Rightarrow 1 - 3t = 1$ and $-1 + 6t = -1$

$\Rightarrow -3t = 0$ and $6t = 0 \Rightarrow t = 0 \Rightarrow (1, -1)$ on ℓ,

B. $(-2, 5) = (1 - 3t, -1 + 6t) \Rightarrow 1 - 3t = -2$ and $-1 + 6t = 5$,

$\Rightarrow -3t = -3 \Rightarrow t = 1$ and $-1 + 6t = 5 \Rightarrow t = 1$

$\Rightarrow (-2, 5)$ lie on ℓ,

C. $(3, -5) = (1 - 3t, -1 + 6t) \Rightarrow 1 - 3t = 3$ and $-1 + 6t = -5$

$\Rightarrow -3t = 2 \Rightarrow t = \frac{-2}{3}$ and $6t = -5 + 1 = -4 \Rightarrow t = \frac{-4}{6} = \frac{-2}{3}$

$\Rightarrow (3, -5)$ lie on ℓ,

$\therefore$ (1, -1),(-2, 5) and (3, -5) are collinear.

d. $P(x, y) = A + t\overrightarrow{AB} \Rightarrow (x, y) = (2, 5) + t(-3, -9)$

$\Rightarrow (x, y) = (2 - 3t, 5 - 9t)$

$\Rightarrow A(2, 5) = (2 - 3t, 5 - 9t)$ when $t = 0$

$B(-1, -4) = (2 - 3t, 5 - 9t)$ when $t = 1$

$C(-2, -5) = (2 - 3t, 5 - 9t)$

$\Rightarrow 2 - 3t = -2 \Rightarrow t = \frac{4}{3}$ and $5 - 9t = -5 \Rightarrow t = \frac{10}{9}$

Since t is not the same

$\Rightarrow$ A, B and C are not collinear

38. Find Parametric equation of the line L

a) L: $3x + 4y = 10$ b) L: $\frac{x-1}{5} = \frac{y+6}{7}$

Solution: To find parallel (direction) vector, $\vec{v}$

a) set: $3x + 4y = 0 \Rightarrow y = \frac{-3}{4}x$ the set of all direction vector parallel to the Line

L: $3x + 4y = 10$ is $y = \frac{-3}{4}x$

$\Rightarrow \vec{v} = (x, y) = t(a, b) = t\left(x, \frac{-3}{4}x\right) = t\left(1, \frac{-3}{4}\right) = t(4, -3)$

We can choice any point on line $3x + 4y = 10$

Let $x = 2, \Rightarrow y = 1 \Rightarrow P_0 = (2, 1)$

$\Rightarrow (x, y) = (2, 1) + t(4, -3)$ ⟵ vector equation of ℓ,

$\Rightarrow x = 2 + 4t, y = 1 - 3t$ ⟵ parametric equation

b) L: $\frac{x-1}{5} = \frac{y+6}{7} \Rightarrow \vec{v} = (5,7)$, $P_0 = (1, -6)$

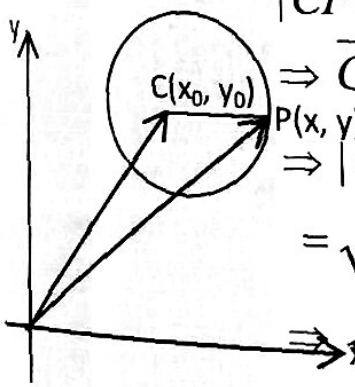
$\Rightarrow (x, y) = (1, -6) + t(5, 7)$ ⟵ vector equation of ℓ,

$\Rightarrow x = 1 + 5t$ and $y = -6 + 7t$ ⟵ parametric equation

Vector and circle

39. Using vector method find the general equation of the circle with center $C(x_0, y_0)$ and radius, R

Let P(x, y) be set of point in circle such that



$|\overrightarrow{CP}| = R$

$\Rightarrow \overrightarrow{CP} = (x, y) - (x_0, y_0) = (x - x_0, y - y_0)$

$\Rightarrow |\overrightarrow{CP}| = \sqrt{\overrightarrow{CP} \cdot \overrightarrow{CP}}$

$= \sqrt{(x - x_0, y - y_0) \cdot (x - x_0, y - y_0)}$

$\Rightarrow \sqrt{(x - x_0)^2 + (y - y_0)^2} = R$

Squaring both side

$\Rightarrow (x - x_0)^2 + (y - y_0)^2 = R^2$

40. Using vector method write the equation of the circle center (4,-3) radius 5

Solution: Let P(x, y) be point on the circle and C(4, -3) is center

$\Rightarrow |\overrightarrow{CP}| = R \Rightarrow |(x - 4, y + 3| = 5$

$\Rightarrow (x-4, y+3)\cdot(x-4, y+3) = 25$

$\Rightarrow (x-4)^2 + (y+3)^2 = 25$

41. Find the equation of circle with diameter with end points A(2, 3) and B(6, 9)

Solution: Center $C\left(\frac{2+6}{2}, \frac{3+9}{2}\right) = C(4,6)$

Radius $R = \sqrt{(6-4)^2 + (9-6)} = \sqrt{4+9} = \sqrt{13}$

Let P(x, y) be point on circle

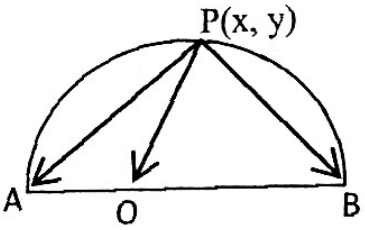
$\Rightarrow |\overrightarrow{CP}| = \sqrt{13} \Rightarrow |(x-4, y-6| = \sqrt{13}$

$\Rightarrow (x-4, y-6)\cdot(x-4, y-6) = 13$

$\Rightarrow (x-4)^2 + (y-6)^2 = 13$

42. Show that the angle in a semicircle is right angle

Solution:



- $|\overrightarrow{AO}| = |\overrightarrow{OB}| = |\overrightarrow{OP}|$ ⟵ radius
- $\overrightarrow{PA} = \overrightarrow{PO} + \overrightarrow{OA}$
- $\overrightarrow{PB} = \overrightarrow{PO} + \overrightarrow{OB} = \overrightarrow{PO} - \overrightarrow{OA}$

[$\overrightarrow{OB}$ *and* $\overrightarrow{OA}$ are equal magnitude but opposite direction

$\Rightarrow \overrightarrow{PA}\cdot\overrightarrow{PB} = (\overrightarrow{PO} + \overrightarrow{OA})\cdot(\overrightarrow{PO} - \overrightarrow{OA})$

$\Rightarrow |\overrightarrow{PO}|^2 - |\overrightarrow{OA}|^2 = 0$ since $|\overrightarrow{PO}| = |\overrightarrow{OA}|$

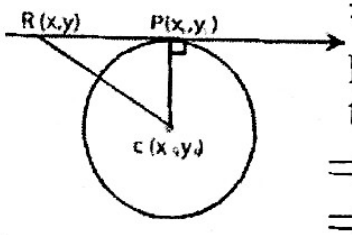
$\Rightarrow \overrightarrow{PA}$ is right angle to $\overrightarrow{PB}$

Equation of tangent line to Circle

Let R(x, y) = Point on tangent line

P = (x_1, y_1) = Point of tangency on circle

C(x_0, y_0) = center of circle.



$\Rightarrow \overrightarrow{RP}\cdot\overrightarrow{PC} = 0$. The radius of circle is perpendicular to the tangent line at point of tangency at $P(x_1, y_1)$

$\Rightarrow (x_1 - x, y_1 - y)\cdot(x_0 - x_1, y_0 - y_1) = 0$

$\Rightarrow \boxed{x_1x + y_1y = r^2}$ if center origin.

43. Find the equation of tangent line to the circle at given point P.

a) $x^2 + y^2 = 10$ at point P(3, 1)
b) $x^2 + y^2 = 5$ at point P(-1, 2)
c) $(x-3)^2 + (y+1)^2 = 8$ at point P(1, 1)
d) $x^2 + y^2 - 8x + 10y + 21 = 0$ at point P(2, -1)

Solution: Let R(x, y) point on tangent line.

a) $x^2 + y^2 = 10$, P(3, 1)and center C(0, 0)

$\Rightarrow \overrightarrow{RP} = (3\text{-}x, 1-y)$ and $\overrightarrow{PC} = (0\text{-}3, 0\text{-}1) = (-3,1)$

$\Rightarrow \overrightarrow{RP} \cdot \overrightarrow{PC} = 0 \Rightarrow (3-x, 1-y)\cdot(-3, -1) = 0$

$\Rightarrow -9 + 3x - 1 + y = 0 \Rightarrow 3x + y = 10$

$\therefore 3x + y = 10$ is equation of tangent line.

b) $x^2 + y^2 = 5$ at P(-1, 2), center (0, 0), Let R(x, y).

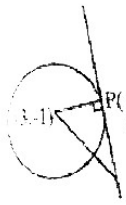
$\Rightarrow \overrightarrow{RP} = (-1\text{-}x, 2\text{–}y)$ and $\overrightarrow{PC} = (0,0) - (-1,2) = (1,-2)$

$\Rightarrow \overrightarrow{RP} \cdot \overrightarrow{PC} = 0 \Rightarrow (-1-x, 2-y)\cdot(1, -2) = 0$

$\Rightarrow -1\text{-}x\text{-}4+2y = 0 \Rightarrow 2y - x = 5$

$\therefore -x + 2y = 5$ is equation of tangent line.

c) $(x-3)^2 + (y+1)^2 = 8$, at P(1, 1), center, C(3, -1), R(x, y)



$\Rightarrow \overrightarrow{RP} = (1, 1) - (x, y) = (1-x, 1-y)$

$\Rightarrow \overrightarrow{PC} = (3,-1) - (1,1) = (2,-2)$

$\Rightarrow \overrightarrow{RP} \cdot \overrightarrow{PC} = 0 \Rightarrow (1-x, 1-y)\cdot(2, -2) = 0$

$\Rightarrow 2 - 2x + -2 + 2y = 0 \Rightarrow -2x + 2y = 0$

$\therefore -2x + 2y = 0$ is equation of tangent line.

d) $x^2 + y^2 - 8x + 10y + 21 = 0$ Let R= (x, y) on line

$\Rightarrow (x-4)^2 + (y+5)^2 = 20$, at P(2, -1), center (4, -5)

$\Rightarrow \overrightarrow{RP} = (2, -1) - (x, y) = (2-x, -1-y)$

$\Rightarrow \overrightarrow{PC} = (4,-5) - (2,-1) = (2,-4)$

$\Rightarrow \overrightarrow{RP} \cdot \overrightarrow{PC} = 0 \Rightarrow (2-x, -1-y)\cdot(2, -4) = 0$

$\Rightarrow 4 - 2x + 4 + 4y = 0 \Rightarrow -2x + 4y = -8$

$\Rightarrow -x + 2y = -4$

$\therefore -x + 2y = -4$ is equation of tangent line.

II. Vector in space (three dimensional vector)

Unit vector (i, j, k)

43. If A = (1, 2, 0) and B(3, 2, -1) be two points, find the unit vector in the direction of

a) $\overrightarrow{AB}$ b) $\overrightarrow{BA}$

Solution: $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3, 2, -1) - (1, 2, 0) = (2, 0, -1)$

$\Rightarrow$ Unit vector along

$$\overrightarrow{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \left(\frac{2}{\sqrt{5}}, 0, \frac{-1}{\sqrt{5}}\right) = \frac{2}{\sqrt{5}}i - \frac{k}{\sqrt{5}}$$

$\Rightarrow$ unit vector along $\overrightarrow{BA} = \frac{\overrightarrow{BA}}{|\overrightarrow{BA}|} = \left(\frac{-2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}\right) = \frac{-2}{\sqrt{5}}i + \frac{k}{\sqrt{5}}$

44. Determine the value of m, so that $\vec{a} = 2i + mj + k$ and $\vec{b} = 4i - 2j - 2k$ are perpendicular

Solution: $\vec{a} \cdot \vec{b} = 0 \Rightarrow (2i + mj + k) \cdot (4i - 2j - 2k) = 0$

$\Rightarrow 8 - 2m - 2 = 0 \Rightarrow -2m = -6 \Rightarrow m = 3$

45. Find the scalar (dot) product of each of the following pair of vector.

a) $\vec{v} = (2, -3, 1)$ and $\vec{u} = (3, 1, -2)$

b) $\vec{v} = (-1, 1, 2)$ and $\vec{u} = (2, -2, -1)$

Solution: a) $\vec{v} \cdot \vec{u} = (2, -3, 1) \cdot (3, 1, -2) = 6 + -3 - 2 = 1$

b) $\vec{v} \cdot \vec{u} = (-1, 1, 2) \cdot (2, -2, -1) = -2 - 2 - 2 = -6$

46. If $\vec{a} = 2i + j - k$, $\vec{b} = i + 2j + k$, $\vec{c} = 3i - 2j + 2k$.

Find a) unit vector parallel to $\vec{a} - 2b + c$

b) unit vector in direction opposite to $\vec{a} - 2b + c$

Solution: Firs, $\vec{a} - 2\vec{b} + \vec{c} = 2i + j - k - 2(i + 2j + k) + 3i - 2j + 2k \Rightarrow \vec{a} - 2\vec{b} + \vec{c} = 3i - 5j - k$

Unit vector along $\vec{a} - 2\vec{b} + \vec{c} = \frac{\vec{a} - 2\vec{b} + \vec{c}}{|\vec{a} - 2\vec{b} + \vec{c}|}$

$$\Rightarrow \frac{3i - 5j - k}{\sqrt{3^2 + 5^2 + 1}} = \frac{3i - 5j - k}{\sqrt{35}} = \left(\frac{3}{\sqrt{35}}, \frac{-5}{\sqrt{35}}, \frac{-1}{\sqrt{35}}\right)$$

$$= \frac{3i}{\sqrt{35}} - \frac{5j}{\sqrt{35}} - \frac{k}{\sqrt{35}}$$

b) Unit vector in direction opposite is $= \left(\frac{-3}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{1}{\sqrt{35}}\right)$

$$= \frac{-3i}{\sqrt{35}} + \frac{5j}{\sqrt{35}} + \frac{k}{\sqrt{35}}$$

47. Find the angle between the vector.

a) $\vec{v} = (-4, 0, 2)$ and $\vec{u} = (2, 0, -1)$

b) $\vec{v} = (2, 0, 2)$ and $\vec{u} = (2, 2, 0)$

c) $\vec{v} = 2i + 3j + k$ and $\vec{u} = -3i - j + 0k$

Solution: Let θ be the angle between vector.

a) $\cos\theta = \frac{\vec{v}\cdot\vec{u}}{|\vec{v}||\vec{u}|} = \frac{-8+0-2}{\sqrt{16+0+4}\sqrt{4+0+1}} = \frac{-1}{\sqrt{100}} = \frac{-10}{10} = -1$

$\Rightarrow \cos\theta = -1 \Rightarrow \theta = \cos^{-1}(-1) = \pi$

b) $\cos\theta = \frac{\vec{v}\cdot\vec{u}}{|\vec{v}||\vec{u}|} = \frac{4+0+0}{\sqrt{4+4}\sqrt{4+4}} = \frac{4}{\sqrt{64}} = \frac{4}{8} = \frac{1}{2}$

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

c) $\cos\theta = \frac{\vec{v}\cdot\vec{u}}{|\vec{v}||\vec{u}|} = \frac{(2,3,1)\cdot(-3,-1,2)}{\sqrt{4+9+1}\sqrt{9+1+4}} = \frac{-6-3+2}{\sqrt{14}\sqrt{14}} = \frac{-7}{14} = \frac{-1}{2}$

$\Rightarrow \cos\theta = -\frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{-1}{2}\right) = \pi - \cos^{-1}\left(\frac{-1}{2}\right) = \pi - \frac{\pi}{3}$

$$= \frac{2\pi}{3}$$

III. Transformation of the plane.

When a point P is given a new position, the point P is said to have a transformation.

There are two categories of transformation (motion)

i) Non-Rigid motion

ii) Rigid motion

i) Non-Rigid motion- A transformation that does not preserve distance.

$$\overline{AB} \neq \overline{A'B'}$$

ii) Rigid Motion- A transformation that preserves distance.

- segment moved into congruent segment

- angle in to congruent angle.
- Triangle into congruent triangle.
- $\overline{AB} = \overline{A'B'}$

iii) **Identity transformation:** A transformation the image of every point is itself.

Identity transformation is a rigid motion.

There are three types of **Rigid Motion** transformation)

1. **Translation** 2. **Reflection** 3. **Rotation**

1. Translation

Translation: A translation if every point in a line or a plane figure moves the same distance in the same direction.

Translation vector. If point A is translated to A', the vector $\overrightarrow{AA'}$ is called **translation vector.**

If $\vec{v}$ = (h, k) is translation vector then the image of

$$(x, y) \xrightarrow{\vec{v}=(h,k)} (x+h, y+k)$$

If the point P(4, 6) has been translated to p'(9, 8)

$$P(4, 6) \longrightarrow P'(9, 8)$$

Translating vector $\overrightarrow{PP'} = \vec{v} = (9-4, 8-6) = (5, 2)$

$\Rightarrow$ P(4, 6) moved 5 unit to right and 2 unit up.

T(x, y) = (x + 5, y + 2)

Illustrative Example

48. Which one of the following transformation does not describe a **Rigid motion?**

A. T((x, y)) = (2y, 2x) C. T((x, y)) = (x + 1, y + 1)

B. T((x, y)) = (-x, y) D. T((x, y)) = (-y, x)

Solution: Let A = (a, b), B = (c, d)

$\Rightarrow$ Distance $\overline{AB} = \sqrt{(a-c)^2 + (b-d)^2}$

A. A' = T((a, b)) = (2b, 2a), B' = T((c, d)) = (2d, 2c)

$\Rightarrow$ Distance $\overline{A'B'} = \sqrt{4(b-d)^2 + 4(a-c)^2} \Rightarrow \overline{AB} = 2\overline{A'B'}$

$\Rightarrow \overline{AB} \neq \overline{A'B'} \Rightarrow$ Not rigid.

B. A' = T((a, b) = (-a, b), B' = T((c, d)) = (-c, d)

$\Rightarrow$ A'B' = $\sqrt{(a-c)^2 + (b-d)} \Rightarrow \overline{AB} = A'B' \Rightarrow$ Rigid motion

C. A' = T((a, b)) = (a+1, b+1) and B' = T((c, d)) = (c +1, d

$\Rightarrow A'B' = \sqrt{((a+1-(c+1))^2 + [(b+1-d-1)]^2}$

$= \sqrt{(a-c)^2 + (b-d)^2}$

$\Rightarrow \overline{AB} = A'B' \Rightarrow$ rigid motion (Preserve distance)

d) A' = T((a, b)) = (-b, a) and B' = T((c, d)) = (-d, c)

$\Rightarrow A'B' = \sqrt{(d-b)^2 + (a-c)^2} \Rightarrow \overline{AB} = A'B' \Rightarrow$ Rigid motion

In choice A-distance $\overline{AB} \neq A'B$

Answer: A

49. Suppose T is transformation. Let A = (3, 1) and B = (2, 4) then which one is rigid and which is not rigid.

a) T((x, y)) = (x, y- 1) b) T((x, y)) = (3x, y + 2)

c) T((x, y)) = (x – 3, y – 2)

Solution: Distance $AB = \sqrt{(3-2)^2 + (1-4)^2} = \sqrt{10}$

a) A' = T((3, 1)) = (3, 1–1) = (3, 0) and B' = T((2, 4)) = (2, 3)

$\Rightarrow A'B' = \sqrt{(3-2)^2 + (0-3)^2} = \sqrt{1+9} = \sqrt{10} \Rightarrow \overline{AB} = \overline{A'B'}$

$\therefore$ T((x, y)) = (x, y – 1) is rigid motion

b) A' = T((3, 1)) = (9, 1 +2)=(9, 3) and B' = T((2, 4)) = (6, 6)

$\Rightarrow A'B' = \sqrt{(9-6)^2 + (3-6)^2} = \sqrt{18} \Rightarrow AB \neq A'B'$

$\therefore$ T((x, y)) = (3x, y + 2) is not rigid.

c) A' = T((3, 1)) = (3 – 3, 1 - 2) = (0, -1) and B' = T((2, 4))
= (2-3,4-2) = (-1, 2)

$\Rightarrow A'B' = \sqrt{(0+1)^2 + (-1-2)^2} = \sqrt{10} \Rightarrow AB = A'B'$

$\therefore$ T((x, y)) = (x - 3, y – 2) is rigid motion

50. If translation T takes the point (7, 2) to the point (5, 3) then find the image of

a) P(3, 4)

b) the triangle with vertices A(2, 3), B(3, -4) and C(2, -1)

c) the line ℓ: y = 2x – 3

d) the line ℓ: 3x – 4y = 7

e) the circle: x2 + y2 = 5

f) the circle: (x – 1)2 + y2 = 4

g) the circle: x2 + y2 – 4x + 6y + 7 = 0

h) the parabola: y = 2x2 + 4

i) the parabola: y2 = 3x

j) the ellipse: $3x2 + 4y2 - 12 = 0$

k) $y = \sin x$

Solution: Translation vector $\vec{v} = (5, 3) - (7, 2) = (-2, 1)$

$\Rightarrow$ image of$(x, y) = (x - 2, y + 1)$

$\Rightarrow$ the movement 2 unit to left, 1 unit up

a) $T((3, 4)) = (3 - 2, 4 + 1) = (1, 5)$

b) $T((2, 3)) = (2 - 2, 3 + 4) = (0, 4)$

$T((2, -1)) = (2 - 2, -1 + 1) = (0, 0)$

$\Rightarrow A'(0, 4), B'(1, -3)$ and $(0, 0)$

c) Under translation line maps in to parallel line

$\Rightarrow$ vector $\vec{v} = (-2, 1)$ the line moves 2 unit left and 1 unit up $\Rightarrow$ put $x + 2$ in x and $y - 1$ in y.

$\Rightarrow \ell'$: $y - 1 = 2(x + 2) - 3 \Rightarrow \ell'$: $y = 2x + 2$

d) ℓ': $3(x + 2) - 4(y - 1) = 7 \Rightarrow \ell'$: $3x - 4y = -3$

e) $(x + 2)^2 + (y - 1)^2 = 5$, putting $x + 2$ in x and $y - 1$ in y

f) the image of circle $(x-1)^2 + y^2 = 4$ is $(x+2-1)^2 + (y-1)^2 = 4$

$\Rightarrow (x + 1)^2 + (y - 1)^2 = 4$

g) the image of circle $x^2 + y^2 - 4x + 6y + 7 = 0$

$\Rightarrow (x + 2)^2 + (y - 1)^2 - 4(x+2) + 6(y-1) + 7 = 0$

$\Rightarrow x^2 + 4x + 4 + y^2 - 2y + 1 - 4x - 8 + 6y - 6 + 7 = 0$

$\Rightarrow x^2 + y^2 + 4y + 4 = 0$

h) Put $x + 2$ in x and $y - 1$ in y

the image of parabola $y = 2x^2 + 4$ will be

$\Rightarrow y - 1 = 2(x+2)^2 + 4 \Rightarrow y = 2(x^2 + 4x + 4) + 4 + 1$

$\Rightarrow y = 2x^2 + 8x + 13$

i) put $x + 2$ in x and $y - 1$ in y

the image of parabola $y^2 = 3x$ is:

$\Rightarrow (y - 1)^2 = 3(x + 2) \Rightarrow y^2 - 2y + 1 = 3x + 6$

$\Rightarrow y^2 - 2y - 3x - 5 = 0$

j) Put $x + 2$ in x and $y - 1$ in y

the image of ellipse: $3x^2 + 4y^2 - 12 = 0$ is

$\Rightarrow 3(x + 2)^2 + 4(y - 1)^2 - 12 = 0$

$\Rightarrow 3(x^2 + 4x + 4) + 4(y^2 - 2y + 1) - 12 = 0$

$\Rightarrow 3x^2 + 4y^2 + 12x + -8y + 4 = 0$

k) Put $x + 2$ in x and $y - 1$ in y

the image of $y = \sin x \Rightarrow y - 1 = \sin(x + 2)$

$\Rightarrow y = \sin(x + 2) + 1$

51. If the point A(3, 7) is transulated to the point A'(0, 2) then find the equation of the image of a) (0, 0) b) (2, 1) c) $y = 3x + 4$ d) $6x + 5y = 9$ e) $x^2 - y^2 = 8$

Solution: $\vec{v} = (h, k) = \overrightarrow{AA'} = (0, 2) - (3, 7) = (-3, -5)$

a) $T((0, 0)) = (0 + -3, 0 + -5) = (-3, -5)$

b) $T((2, 1)) = (2 + -3, -5 + 1) = (-1, -4)$

c) The line move 3 unit to left, 5 unit down so that substitute $x + 3$ in x and $y + 5$ in y for $y = 3x + 4$ image will be
$\Rightarrow y + 5 = 3(x + 3) + 4 \Rightarrow y = 3x + 8$

d) Image of $6x + 5y = 9$ will be $6(x + 3) + 5(y + 5) = 9$
$\Rightarrow 6x + 5y = -34$

e) $(x + 3)^2 - (y + 5)^2 = 8$

52. A Translation T takes the point (1, 2) to the point (5, -1) A second translation S takes the point (1, 2) to the point (-2, 3) then find a) T followed by S taken the point (1, 2)

b) S followed by T takes the point (1, 2)

c) What do you conclude?

Solution: $T((1, 2)) \xrightarrow{\vec{v}=(4,-3)} (5, -1)$

$S((1, 2)) \xrightarrow{\vec{u}=(-3,1)} (-2, 3)$

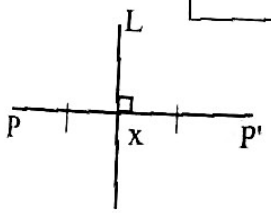
a) $S(T(1, 2)) = S(5, -1) = (5 + -3, -1 + 1) = (2, 0)$

b) $T(S(1, 2) = T(-2, 3) = (4 + -2, 3 + -3) = (2, 0)$

c) $S(T(x, y)) = T(S(x, y))$ and we observed
$S(T(1,. 2)) = T(S(1, 2)) = (2, 0)$

Reflections

If a point **P** is reflected in a mirror so that its image is P', the mirror (or line of reflection) L is the perpendicular bisector of $\overline{PP'}$



i) $\overline{PP'} \perp L$, L is line of reflection

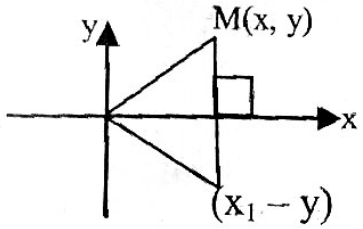
ii) $\overline{PX} = \overline{P'X}$

iii) if P is on L, its image P' in L is P itself

A. Reflection in the axes

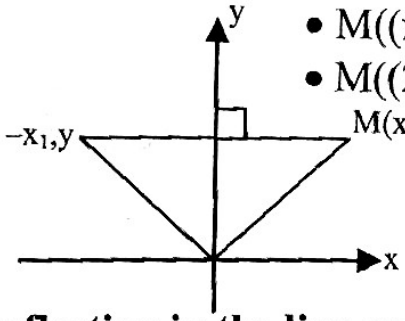
i) Reflection in the x-axis (y = 0)

M ((x, y)) = (x, -y)



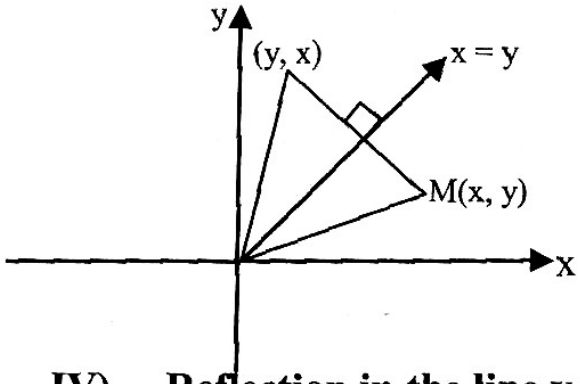
- M (3, 4) = (3, -4)
- M (-4, 8) = (-4, -8)

ii) Reflection in the y-axis (x = 0)



- M((x, y)) = (-x, y)
- M((2, 4)) = (-2, 4)
- M((x, a^x)) = (x, a^{-x})

iii) Reflection in the line y = x



- M((x, y)) = (y, x)
- M((2, 5)) = (5, 2)
- M((x, a^x)) = (x, $\log_a^x$)

IV) Reflection in the line y = -x

- M((x, y)) = (-y, -x)

Example:

- M((-2, 4)) = (-4, 2)
- M((3, 1)) = (-1, -3)

V) Reflection in the line x = C

- M((a, b)) $\underline{\underline{x = C}}$ (2c – a, b)

Example:

- M((3, 4)) $\underline{\underline{x = 5}}$ (2(5) – 3, 4) = (7, 4)

Illustrative Example

53. Find the image in the line of reflection x = 3

a) M(4, 1) b) y = 2x – 5 c) $(x-2)^2 + (y+1)$

Solution: M((a, b)) = (2(3) – a, b) = (6 – a, b)

a) M(4, 1) = (2(3) – 4, 1) = (2, 1)

b) take two point arbitrarly on $y = 2x - 5$

Let (1, -3) and (2, -1)

$\Rightarrow M((1, -3) \xrightarrow{x=3} (2(3) - 1, -3) = (5, -3)$

$M((2, -1)) \xrightarrow{x=3} (2(3) - 2, -1) = (4, -1)$

Equation of the line passing through M'(5,-3) and M'(4,-1) is given

$y_1 = m(x - x_1) \Rightarrow m = \dfrac{-3-(-1)}{5-4} = -2$

$\Rightarrow y - (-3) = -2(x - 5) \Rightarrow y + 3 = -2x + 10 \Rightarrow y = -2x + 7$

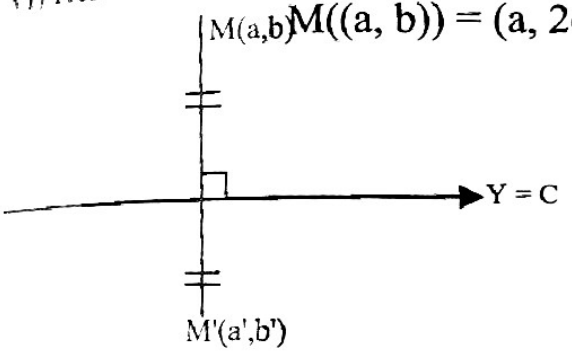
$\therefore$ image of $y = 2x - 5$ is $y = -2x + 7$

c) Take center (2, -1)

$\Rightarrow M((2, -1)) \xrightarrow{x=3} (2(3) - 2, -1) = (4, -1)$

$\therefore$ image of $(x - 2)^2 + (y + 1)^2 = 7$ is $(x - 4)^2 + (y + 1)^2 = 7$

VI) **Reflection on the line y = c**



54. Find the image in the line of reflection $y = 2$

a) $M((3, 4))$ b) $y = 3x + 1$ c) $x^2 + y^2 - 2y - 2 = 0$

Solution: $M((a, b)) \xrightarrow{y=c} (a, 2c - b)$

a) $M((3, 4)) \xrightarrow{y=2} (3, 2(2) - 4) = (3, 0)$

b) First, take two point arbitrarily on $y = 3x + 1$

Let (1, 4) and (3, 10)

$\Rightarrow M((1, 4)) \xrightarrow{y=2} (1, 2(2) - 4) = (1, 0)$

$\Rightarrow M((3, 10)) \xrightarrow{y=2} (3, 2(2) - 10) = (3, -6)$

The image of the given line is the equation of the line passing through (1, 0) and (3, -6) is given by $y - y_1 = m(x - x_1)$,

$m = \dfrac{0+6}{1-3} = -3$

$\Rightarrow y - 0 = -3(x - 1) \Rightarrow y = -3x + 3$

c) center of circle is $(0, 1) \Rightarrow M(0, 1) = (0, 2(2) - 1)) = (0, 3)$

Thus, the image will be $x^2 + y^2 - 6y + 6 = 0$

55. Find the image in the line y = 0(x – axis)

a) M((3, 5)) b) $y = 2^x$ c) $y = \log_2^x$ d) y = sinx

Solution: M((a, b)) = (a, -b)

a) M((3, 5)) = (3, -5),

b) $y = -2^x$

c) the image of $y = \log_2^x$ in the line of reflection x – axis (y = 0) is $y = -\log_2^x = \log_2^{\frac{1}{x}} = \log_{\frac{1}{2}}^x$

d) y = -sinx or y = sin(-x)

56. Find the image in the line, x = 0(y – axis)

a) y = x, b) $y = 3^x$ c) $y = \log_3^x$ d) y = 3x + 4

Solution: M((a, b)) = (-a, b)

a) y = -x b) $y = 3^{-x}$ c) $y = \log_3^{-x}$

d) First take two point on y = 3x + 4

Let (1, 7) and (-2, -2)

⇒ M((1, 7)) = (-1, 7) and M(-2, -2) = (2, -2)

Equation of the line passing through (-1, 7) and (2, -2) is given by $y - y_1 = m(x - x_1)$

$$\Rightarrow m = \frac{7-(-2)}{-1-2} = \frac{9}{-3} = -3 \Rightarrow y - 7 = -3(x+1)$$

⇒ the image of the line y = 3x + 4 is y = -3x + 4

57. Find the images after reflection in the line y = x

a) (3, -2) b) $y = 4^x$ c) y = lnx d) $y = \log_a^x$

Solution: M(a, b) = (b, a)

a) M(3, -2) = (-2, 3) b) $y = \log_4^x$ c) $y = e^x$ d) $y = a^x$

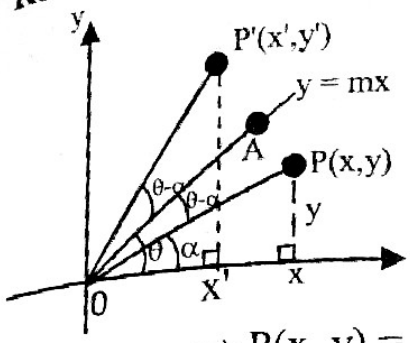
58. If a point P has the coordinates (5, 2), find its reflection in the

a) x – axis b) y-axis c) y = -x d) y = x

Solution: a) M((5, 2)) = (5, -2) b) M(5, 2) = (-5, 2)

c) $M((5, 2)) \xrightarrow{y=-x} (-2, -5)$ d) $M(5, 2) \xrightarrow{y=x} (2, 5)$

B. Reflection in the line y = mx, where m = tanθ



- Let P'(x', y') be the image of P(x, y)
- Let OP = r ⇒ OP' = r
- $\cos\alpha = \frac{x}{r} \Rightarrow x = r\cos\alpha$
- $\sin\alpha = \frac{y}{r} \Rightarrow y = r\sin\alpha$

$\Rightarrow P(x, y) = (r\cos\alpha, r\sin\alpha)$

$m(\angle P'OX') = \theta - \alpha + \theta = 2\theta - \alpha$

- $\cos(2\theta - \alpha) = \frac{x'}{r} \Rightarrow x' = r\cos(2\theta - \alpha)$

$\Rightarrow x' = r\cos(2\theta - \alpha) = r\cos2\theta\cos\alpha + r\sin2\theta\sin\alpha$

$\Rightarrow \boxed{x' = x\cos2\theta + y\sin2\theta}$

- $\sin(2\theta - \alpha) = \frac{y'}{r} \Rightarrow y' = r\sin(2\theta - \alpha)$

$\Rightarrow y' = r\sin2\theta\cos\alpha - r\cos2\theta\sin\alpha$

$\Rightarrow y' = x\sin2\theta - y\cos 2\theta$

The image of P(x, y), in the line of reflection y = mx where m = tanθ is
x' = xcos2θ + ysin2θ.
y' = xsin2θ - ycos2θ

Illustrative Example

59. Find the image of (4, 5) n the line of reflection

a) $y = \sqrt{3}x$ b) y = 3x

Solution: a) $y = \sqrt{3}x \Rightarrow m = \sqrt{3} \Rightarrow \tan\theta = \sqrt{3}$

$$\Rightarrow \theta = \tan^{-1}\left(\sqrt{3}\right) = \frac{\pi}{3}$$

$$x' = x\cos2\theta + y\sin2\theta \Rightarrow x' = 4\cos\frac{2\pi}{3} + 5\sin\frac{2\pi}{3}$$

$$\Rightarrow x' = 4\left(\frac{-1}{2}\right) + 5\frac{\sqrt{3}}{2} = -2 + \frac{5\sqrt{3}}{2}$$

$y' = x\sin 2\theta - y\cos 2\theta \Rightarrow y' = 4\sin\frac{2\pi}{3} - 5\cos\frac{2\pi}{3}$

$\Rightarrow y' = 2\sqrt{3} - 5\left(\frac{-1}{2}\right) = 2\sqrt{5} + \frac{5}{2}$

$\therefore P(4, 5) \xrightarrow{y=\sqrt{3}x} \left(-2 + \frac{5\sqrt{3}}{2}, 2\sqrt{3} + \frac{5}{2}\right)$

b) $y = 3x \Rightarrow m = 3 \Rightarrow \tan\theta = 3 \Rightarrow \sin\theta = \frac{3}{\sqrt{3^2+1}} = \frac{3}{\sqrt{10}}$

$\Rightarrow \cos\theta = \frac{1}{\sqrt{3^2+1}} = \frac{1}{\sqrt{10}}$

$\Rightarrow \cos 2\theta = \cos(\theta + \theta) = \cos^2\theta - \sin^2\theta = \left(\frac{1}{\sqrt{10}}\right)^2 - \left(\frac{3}{\sqrt{10}}\right)^2 = \frac{-4}{5}$

$\Rightarrow \sin 2\theta = 2\cos\theta\sin\theta = 2\left(\frac{1}{\sqrt{10}}\right)\left(\frac{3}{\sqrt{10}}\right) = \frac{3}{5}$

$\Rightarrow x' = x\cos 2\theta + y\sin 2\theta \Rightarrow x' = \frac{-4}{5}x + \frac{3}{5}y$

$\Rightarrow y' = x\sin 2\theta - y\cos 2\theta \Rightarrow y' = \frac{3}{5}x - \left(\frac{-4}{5}\right)y = \frac{3}{5}x + \frac{4}{5}y$

Then the image of (4, 5) will be

$\Rightarrow x' = \frac{-4}{5}x + \frac{3}{5}y \Rightarrow \frac{-4}{5}(4) + \frac{3}{5}(5) = \frac{-1}{5}$

$\Rightarrow y' = \frac{3}{5}x + \frac{4}{5}y \Rightarrow \frac{3}{5}(4) + \frac{4}{5}(5) = \frac{37}{5}$

$\Rightarrow M((x, y)) = \left(\frac{-4}{5}x + \frac{3}{5}y, \frac{3}{5}x + \frac{4}{5}y\right)$

$\Rightarrow M((4, 5)) = \left(\frac{-4}{5}(4) + \frac{3}{5}(5), \frac{3}{5}(4) + \frac{4}{5}(5)\right) = \left(\frac{-1}{5}, \frac{37}{5}\right)$

60. The image of the point (2, 2) reflected in line passing through the origin and making 30^0 with the x-axis …

A. (2, -2)
C. $(\sqrt{3}-1, 1+\sqrt{3})$
B. $(1+\sqrt{3}, \sqrt{3}-1)$
D. $(\sqrt{3}, -\sqrt{3})$

Solution: we use $x' = x\cos 2\theta + y\sin 2\theta$
$y' = x\sin 2\theta - y\cos 2\theta$
$\theta = 30^0$, $x = 2$, $y = 2$

$$\Rightarrow x' = 2\cos 60^0 + 2\sin 60^0 = 2\left(\frac{1}{2}\right) + 2\left(\frac{\sqrt{3}}{2}\right) = 1+\sqrt{3}$$

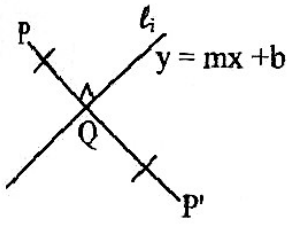
$$y' = 2\sin 60^0 - 2\cos 60^0 = 2\left(\frac{\sqrt{3}}{2}\right) - 2\left(\frac{1}{2}\right) = \sqrt{3}-1$$

$M((2, 2)) = (1 + \sqrt{3}, \sqrt{3} - 1)$

$\therefore$ Answer B

61. From question #60, find the image of (2, 2) if $\theta = 60^0, 150^0, \ldots$
Exercise left for you (Tip!)

C. Reflection in the line y = mx + b



To find the image of P(a, b) in the line of reflection ℓ: y – mx + b

1. Determine the slope of the line
   $\Rightarrow$ slope of $\ell = m$
2. Determine the slope of the line $\overline{PP'}$
   $\Rightarrow$ slope of $\overline{PP'}$ or $\ell' = \dfrac{-1}{m}$
   Since $\ell \perp PP'$
3. Determine the intersection point of the line ℓ and PP' or ℓ'
4. Q is the mid-point of PP', find coordinate of P'

Illustrative Example

62. Find the image of the point P(1, -2) after reflection in the line $y = 3x + 1$

Solution: Step 1: Slope of ℓ is $m = 3 \Rightarrow$ slope of PP' or ℓ' is $\frac{-1}{3}$

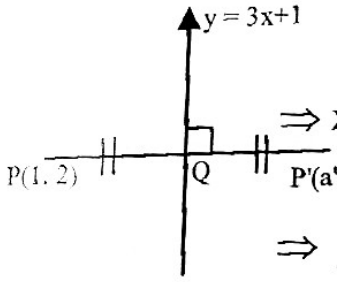
Step 2: Equation of ℓ' passing through P(1,-2) with slope $\frac{-1}{3}$

$$\Rightarrow y-(-2)=\frac{-1}{3}(x-1) \Rightarrow y+2=\frac{-1}{3}(x-1)$$

$$\Rightarrow 3y+6=-x+1 \Rightarrow x+3y=-5$$

Step 3: The intersection of ℓ and ℓ'

$$\Rightarrow \begin{cases} y=3x+1 \\ x+3y=-5 \end{cases}$$



$$\Rightarrow x+3(3x+1)=-5 \Rightarrow 10x=-8 \Rightarrow x=\frac{-4}{5}$$

$$\Rightarrow y=3\left(\frac{-4}{5}\right)+1=\frac{-7}{5}$$

$\therefore$ the intersection point $Q\left(\frac{-4}{5},\frac{-7}{5}\right)$

Step 4: $Q\left(\frac{-4}{5},\frac{-7}{5}\right)$ is the mid-point of PP', Let the image of P(1, -2) is P'(a', b')

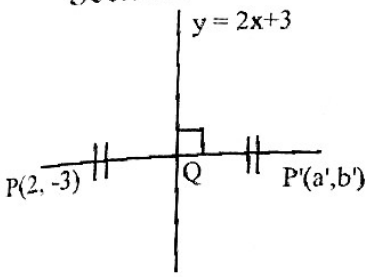
$$\Rightarrow \frac{1+a'}{2}=\frac{-4}{5} \text{ and } \frac{-2+b'}{2}=\frac{-7}{5}$$

$$\Rightarrow a'+1=\frac{-8}{5} \Rightarrow a'=\frac{-13}{5} \text{ and } b'=\frac{-14}{5}+2=\frac{-4}{5}$$

$\therefore$ The image of M((1, -2)) $= \left(\frac{-13}{5},\frac{-4}{5}\right)$

63. Find the image of the circle $(x-2)^2+(y+3)^2=4$ after reflection in the line $y=2x+3$.

Solution: The center of the given circle is (2, -3)



- slope of ℓ is $m=2$

$\Rightarrow$ slope of ℓ' is $m'=\dfrac{-1}{2}$

- Equation of the line ℓ' passing through (2,-3) with slope $\dfrac{-1}{2}$ is:

$$\Rightarrow y+3=\frac{-1}{2}(x-2)\Rightarrow 2y+6=-x+2$$

$$\Rightarrow x+2y=-4$$

The intersection of ℓ and ℓ' is at mid pt of PP': $\begin{cases} y=2x+3 \\ x+2y=-4 \end{cases}$

$\Rightarrow x+2(2x+3)=-4\Rightarrow x=-2$ and $y=-1$

Let the image of P(2, -3) is P' = (a', b')

- $Q(-2,-1)=\left(\dfrac{2+a'}{2},\dfrac{-3+b'}{2}\right)$

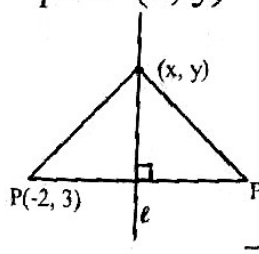
$$\Rightarrow \frac{a'+2}{2}=-2\Rightarrow a'=-6 \text{ and } \frac{-3+b'}{2}=-1\Rightarrow b'=1$$

$\therefore$ The image of M((2, -3)) = (-6, 1)

$\Rightarrow$ image of circle is $(x+6)^2+(y-1)^2=4$

64. In a reflection, the image of the point P(-2, 3) is P'(6, 7). Find the equation of line of reflection.

Solution; Since ℓ is the perpendicular bisector of PP', the set of all point (x, y) in ℓ are equidistant to P(-2, 3) and P'(6, 7)



$$\Rightarrow (x+2)^2+(y-3)^2=(x-6)^2+(y-7)^2$$

$$\Rightarrow x^2+4x+4+y^2-6y+9$$

$$=x^2-12x+36+y^2-14y+49$$

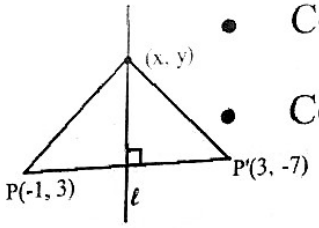
$$\Rightarrow 16x+8y=36+49-13=72$$

$\Rightarrow 2x+y=9 \leftarrow$ equation of line of reflection

65. The image of the circle $x^2 + y^2 + 2x - 6y = 0$ when it is reflected about the line ℓ is $x^2 + y^2 - 6x + 14y = 0$. Find the equation of line of reflection:

Solution: First we determine the center of each circle

- Center of $x^2 + y^2 + 2x - 6y = 0$ is $(-1, 3)$
- Center of $x^2 + y^2 - 6x + 14y = 0$ is $\left(\frac{-6}{-2}, \frac{-14}{2}\right)$ $= (3, -7)$



$\Rightarrow \ell$ is perpendicular bisector of PP', the set of all point (x, y) is ℓ are equidistant from $P(-1, 3)$ and $P'(3, -7)$

$\Rightarrow (x+1)^2 + (y-3)^2 = (x-3)^2 + (y+7)^2$

$\Rightarrow x^2 + 2x + 1 + y^2 - 6y + 9 = x^2 - 6x + 9 + y^2 + 14y + 49$

$\Rightarrow 8x - 20y = 48 \Rightarrow 2x - 5y = 12$

$\therefore$ The equation of the line of reflection is $4x - 10y = 25$

66. Find the image of the line L: $y = 2x + 1$ after it has been reflected along the line ℓ: $y = 2x + 6$

Solution: since $L \| \ell$

$\Rightarrow L'$: $y = 2x + b$

Take $(0, 1)$ on $y = 2x + 1$, the image of $(0, 1)$ in $y = 2x + 6$ is $(-4, 3)$ i.e $M((0, 1)) = (-4, 3)$ lies on L'

$\Rightarrow 3 = 2(-4) + b \Rightarrow b = 3 + 8 = 11$

$\Rightarrow L'$: $y = 2x + 11$ is the image of L: $y = 2x + 1$

67. Find the image of the line L: $y = 3x - 6$ after it has been reflected along the line ℓ: $y = x - 4$

Solution: Since L: $y = 3x - 6$ and ℓ: $y = x - 4$ are non-parallel

$$\text{The intersection of } \begin{cases} y = 3x - 6 \\ y = x - 4 \end{cases} \text{ is } (1, -3)$$

Then take point on L: $y = 3x - 6$ say $(2, 0)$

The image of $(2, 0)$ on line of reflection $y = x - 4$

$\Rightarrow M((2, 0)) = (4, -2)$

$\Rightarrow$ The equation of line L' passing through $(4, -2)$ and $(1, -3)$ is given by $y - y_1 = m(x, x_1)$

$$m = \frac{-2-(-3)}{4-1} = \frac{1}{3}$$

$\Rightarrow y-(-2)=\frac{1}{3}(x-4) \Rightarrow y+2=\frac{1}{3}x-\frac{4}{3} \Rightarrow y=\frac{1}{3}x-\frac{10}{3}$

$\therefore$ The image of $y = 3x - 6$ is $y=\frac{1}{3}x-\frac{10}{3}$

Double Transformation: A transformation. Translation followed by reflection or reflection followed by translation i.e. T(M(x, y)) or M(T(x,y)) or T(R(x, y)) or R(M(x, y)) ….etc.

- T(M(x, y)) ≡ Reflection of point followed by translation T.
- M(T(x, y)) ≡ Translation of point followed by reflection M.
- M(R(x, y)) ≡ Rotation R followed by reflection M.

Note: $T(M(x, y)) \neq M(T(x, y))$

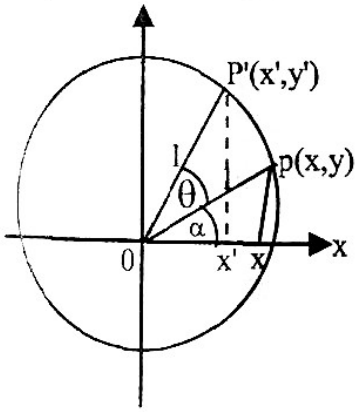
Rotation

A **rotation R** of θ about a point 0, is called center of rotation and θ is called an amount of rotation.

- If $\theta > 0$, then rotation is counter clock wise direction
- If $\theta < 0$, then rotation is clock wise direction

Rotation about the origin

Let R be a rotation through an angle θ about the origin then the image of P(x, y) obtained by formula



$x' = x\cos\theta - y\sin\theta$

$y' = x\sin\theta + y\cos\theta$

To proof: consider the following unit circle

$\Rightarrow OP = OP' = 1$ (radius of unit circle)

$\Rightarrow \cos\alpha=\frac{x}{OP}=\frac{x}{1}=x \Rightarrow x=\cos\alpha$

$\Rightarrow \sin\alpha=\frac{y}{OP}=\frac{y}{1}=1 \Rightarrow y=\sin\alpha$

In angle $P'OA'(\alpha+\theta)$

$\cos(\alpha+\theta)=\frac{x'}{OP'}=\frac{x'}{1}=x'$

$x' = \cos(\alpha+\theta) = \cos\alpha\cos\theta - \sin\alpha\sin\theta = x\cos\theta - y\sin\theta$

$\Rightarrow \boxed{x' = x\cos\theta - y\sin\theta}$

And $\sin(\alpha+\theta)=\frac{y'}{OP'}=\frac{y'}{1} \Rightarrow y'=\sin(\alpha+\theta)$

$\Rightarrow y' = \sin\alpha\cos\theta + \cos\alpha\sin\theta = y\cos\theta + x\sin\theta$

$\Rightarrow \boxed{y' = x\sin\theta + y\cos\theta}$

The six Basic rotation

i) **If R is Rotation through $\theta = \frac{\pi}{2}$ about (0, 0)**

$$R((x, y)) = (x\cos\frac{\pi}{2} - y\sin\frac{\pi}{2}, x\sin\frac{\pi}{2} + y\cos\frac{\pi}{2})$$

$$R_{\frac{\pi}{2}}(x, y) = (-y, x)$$

ii) **If R is rotation through $\theta = \frac{-\pi}{2}$ about(0, 0)**

$$\Rightarrow R((x, y)) = \left(x\cos\left(\frac{-\pi}{2}\right) - y\sin\left(\frac{-\pi}{2}\right), x\sin\left(\frac{-\pi}{2}\right) + y\cos\left(\frac{-\pi}{2}\right)\right)$$

$= (y, -x)$

iii) **If R is rotation through $\theta = \pi$ about (0, 0)**

$\Rightarrow R((x,y)) = (x\cos\pi - y\sin\pi, x\sin\pi + y\cos\pi)$

$= (-x, -y)$

iv) **If R is rotation through $\theta = \frac{3}{2}\pi$ about (0, 0)**

$$\Rightarrow R((x, y)) = \left(x\cos\frac{3}{2}\pi - y\sin\frac{3}{2}\pi, x\sin\frac{3}{2}\pi + y\cos\frac{3}{2}\pi\right)$$

$= (0 - y(-1), x(-1) + 0) = (y, -x)$

v) **If R is rotation through $\theta = \frac{-3}{2}\pi$ about (0, 0)**

$$\Rightarrow R((x,y)) = \left(x\cos\left(\frac{-3}{2}\pi\right) - y\sin\left(\frac{-3}{2}\pi\right), x\sin\left(\frac{-3}{2}\pi\right) + y\cos\left(\frac{-3}{2}\pi\right)\right)$$

$= (0 - y(1), x(1) + y(0)) = (-y, x)$

Note: i) $R_{90^0} = R_{-270^0}$

ii) $R_{-90^0} = R_{270^0}$

iii) $R_{120^0} = R_{-240^0}$

iv) $R_{180^0} = R_{-180^0}$

v) $R_{90^0} = R_{450^0}$

vi) **If R is rotation through $\theta = 2\pi k$, $k \in z$, about (0, 0)**
$R((x,y)) = (x, y)$ ..... which is called an identity transformation

71. Find the image of each part, in rotation through the indicated angle about the origin.

a) $(4, 2),\ \theta = \frac{\pi}{3}$

b) $(2, -3),\ \theta = \frac{-\pi}{6}$

c) $(-1, 4),\ \theta = \frac{\pi}{4}$ for $\frac{\pi}{2} \le \theta \le \pi$

d) $3x - 2y = 8,\ \theta = \frac{3}{2}\pi$

e) $3x + 4y = -5$, with $\tan\theta = \frac{-3}{4}$

f) $x^2 + y^2 - 4x + 2y = 0,\ \theta = \frac{-3\pi}{2}$

Solution: use $x' = x\cos\theta - y\sin\theta$, $y' = x\sin\theta + y\cos\theta$

a) $x = 4, y = 2,\ \theta = \frac{\pi}{3}$

$$\Rightarrow x' = 4\cos\frac{\pi}{3} - 2\sin\frac{\pi}{3} = 4\left(\frac{1}{2}\right) - 2\left(\frac{\sqrt{3}}{2}\right) = 2 - \sqrt{3}$$

$$\Rightarrow y' = 4\sin\frac{\pi}{3} + 2\cos\frac{\pi}{3} = 4\left(\frac{\sqrt{3}}{2}\right) + 2\left(\frac{1}{2}\right) = 2\sqrt{3} + 1$$

$$\therefore R((4, 2)) = (2 - \sqrt{3}, 2\sqrt{3} + 1)$$

b) $x = 2, y = -3,\ \theta = \frac{-\pi}{6}$

$$\Rightarrow x' = 2\cos\left(\frac{-\pi}{6}\right) - 3\sin\left(\frac{-\pi}{6}\right) = 2\left(\frac{\sqrt{3}}{2}\right) + \frac{3}{2} = \frac{2\sqrt{3}+3}{2}$$

$$\Rightarrow y' = 2\sin\left(\frac{-\pi}{6}\right) + 3\cos\left(\frac{-\pi}{6}\right) = 2\left(\frac{-1}{2}\right) + 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}-2}{2}$$

$$\therefore R((2,-3)) = \left(\frac{2\sqrt{3}+3}{2}, \frac{3\sqrt{3}-2}{2}\right)$$

c) $x = -1, y = 4, \theta = \frac{\pi}{4}$

$$\Rightarrow x' = -1\cos\frac{\pi}{4} - 4\sin\frac{\pi}{4} = \frac{-\sqrt{2}}{2} - \frac{4\sqrt{2}}{2} = \frac{-5\sqrt{2}}{2}$$

$$\Rightarrow y' = -1\sin\frac{\pi}{4} + 4\sin\frac{\pi}{4} = \frac{-\sqrt{2}}{2} + \frac{4\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}$$

$$\therefore R((-1, 4)) = \left(\frac{-5\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$$

d) Take two point arbitrary on $3x - 2y = 8$

let $x = 2, y = (-1) \Rightarrow (2, -1)$ and $(0, -4), \theta = \frac{3}{2}\pi$

To find image of (2, -1)

$$\Rightarrow x' = 2\cos\left(\frac{3\pi}{2}\right) - (-1)\sin\left(\frac{3\pi}{2}\right) = 2(0) - (1) = -1$$

$$\Rightarrow y' = 2\sin\left(\frac{3\pi}{2}\right) + (-1)\cos\left(\frac{3\pi}{2}\right) = 2(-1) + 0 = -2$$

$\therefore R((2, -1)) = (-1, -2)$ and

To find image of (0, -4)

$$\Rightarrow x' = 0\cos\frac{3\pi}{2} - (-4)\sin\frac{3\pi}{2} = 0 - 4 = -4$$

$$\Rightarrow y' = 0\sin\frac{3\pi}{2} + (-4)\cos\frac{3\pi}{2} = 0(-1) + (-4)(0) = 0$$

$\therefore ((R(0, -4)) = (-4, 0)$, Thus

Equation of the line passing through (-1, -2) and (-4, 0) is given by

$$y - y' = m(x - x_1) \Rightarrow y - (-2) = \frac{-2-0}{-1-(-4)}(x - (-1))$$

$$\Rightarrow y + 2 = \frac{-2}{3}(x+1) \Rightarrow 3y + 6 = -2x - 2 \Rightarrow 2x + 3y = -8$$

$\therefore$ The image of the line is $2x + 3y = -8$

e) First determine $\sin\theta$ and $\cos\theta$ from $\tan\theta = \frac{-3}{4}$

$$\Rightarrow \sin\theta = \frac{3}{5} \text{ and } \cos\theta = \frac{-4}{5}$$

- Take two point arbitrarly on $3x + 4y = -5$

let $x = 5, y = -5 \Rightarrow (5, -5)$ and

let $x = 1, y = -2 \Rightarrow (1, -2)$

To find image of the point (5, -5)

$$\Rightarrow x' = 5\left(\frac{-4}{5}\right) - (-5)\left(\frac{3}{5}\right) = -4 + 3 = 1$$

$$y' = 5\left(\frac{3}{5}\right) + (-5)\left(\frac{-4}{5}\right) = 3 + 4 = 7$$

$\therefore R((5, -5)) = (-1, 7)$

To find image of the point (1, -2)

$$\Rightarrow x' = 1\left(\frac{-4}{5}\right) - (-2)\left(\frac{3}{5}\right) = \frac{-4}{5} + \frac{6}{5} = \frac{2}{5}$$

$$\Rightarrow y' = 1\left(\frac{3}{5}\right) + (-2)\left(\frac{-4}{5}\right) = \frac{3}{5} + \frac{8}{5} = \frac{11}{5}$$

$$\therefore R((1, -2)) = \left(\frac{2}{5}, \frac{11}{5}\right)$$

The equation of the line passing through (-1, 7) and $\left(\frac{2}{5}, \frac{11}{5}\right)$ is:

$$m = \frac{\frac{11}{5} - 7}{\frac{2}{5} + 1} = \frac{-24}{7}$$

$$\Rightarrow y - 7 = \frac{-24}{7}(x + 1) \Rightarrow y = \frac{-24}{7}x + \frac{25}{7}$$

$\therefore 24x + 7y = 25$ is the image of $3x + 4y = -8$

f) center of $x^2 + y^2 - 4x + 2y = 0$ is $(2, -1)$, radius $= \sqrt{5}$

$$x = 2, y = -1, \theta = \frac{-3\pi}{2}$$

$$\Rightarrow x' = 2\cos\left(\frac{-3\pi}{2}\right) - (-1)\sin\left(\frac{-3\pi}{2}\right) = 0 - 1 = -1$$

$$\Rightarrow y' = 2\sin\left(\frac{-3\pi}{2}\right) + -1\cos\left(\frac{-3\pi}{2}\right) = -2 - 0 = -2$$

$\therefore R((2, -1)) = (-1, -2)$

$\therefore$ The image of circle is $(x+1)^2 + (y+2)^2 = 5$

$\Rightarrow x^2 + y^2 + 2x + 4y = 0$

Rotation about any point (a, b)

If P'(x', y') is the image of a point P(x, y) when rotated by an angle θ about (a, b), then

$$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$$
$$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$$

- To rotate P(x, y) in an angle θ about (a, b) we have the following step,

Step 1: Translate $(a, b) \longrightarrow (0, 0)$

- Translating vector $\vec{v} = (-a, -b)$

Step 2: Translate $(x, y) \xrightarrow{\vec{v}=(-a,-b)} (x - a, y - b)$

Step 3: Rotate, (x-a, y-b) by θ about (0,0) then translate by $(-v)=(a,b)$

72. Find the image of P(3,1) when it is rotated through $\theta = \frac{\pi}{2}$ about (-2, 3)

Method I

- Translate $(-2, 3) \longrightarrow (0, 0)$, $\vec{v} = (2, -3)$
- Translate $(3, 1) \xrightarrow{\vec{v}=(2,-3)} (5, -2)$
- Rotate (5, -2) by $\theta = \frac{\pi}{2}$ about (0, 0)

$$x = 5, y = -2, \theta = \frac{\pi}{2}$$

$$x' = 5\cos\frac{\pi}{2} - (-2)\sin\frac{\pi}{2} = 0 + 2 = 2$$

$y' = 5\sin\frac{\pi}{2} + (-2)\cos\frac{\pi}{2} = 5 + 0 = 5$

R((5, -2)) = (2, 5)

- Translate (2, 5) by $(-\vec{v}) = (-2, 3)$

T((2, 5)) = (2 + (-2), 5 + 3) = (0, 8)

$\therefore$ The image of the point (3, 1) about (-2, 3) with an angle $\theta = \frac{\pi}{2}$ is (0, 8)

i.e. R((3, 1)) = (0, 8) about (-2, 3) with $\theta = \frac{\pi}{2}$

Method II

$\Rightarrow x' = a+(x-a)\cos\theta - (y-b)\sin\theta = -2+(3-(-2))\cos\frac{\pi}{2} - (1-3)\sin\frac{\pi}{2} = 0$

$\Rightarrow y' = b+(x-a)\sin\theta + (y-b)\cos\theta = 3+(3+2)\sin\frac{\pi}{2} + (1-3)\cos\frac{\pi}{2} = 8$

$\therefore$ R((3, 1)) = (0, 8)

73. Find the equation of the line x + y = 0 after it has been rotated by $\theta = 270^0$ about (3, 2)

Solution:

- Choose any two arbitrary points, say (-1, 1) and (2, -2)
- Center of rotation, (a, b) = (3, 2) and $\theta = 270^0$

To find image of R((-1, 1)) about (a, b) = (3, 2)

$\Rightarrow x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$= 3 + (-1 - 3)\cos(270^0) - (1 - 2)\sin 270^0 = 3 + (-1) = 2$

$\Rightarrow y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$= 2 + (-1 - 3)\sin(270^0) + (1 - 2)\cos 270^0 = 6$

$\therefore$ R((-1, 1)) = (2, 6)

To find image of (2, -2) about (a, b) = (3, 2)

$x' = 3 + (2 - 3)\cos 270^0 - (-2 - 2)\sin 270^0 = 3 - 4 = -1$

$y' = 2 + (2 - 3)\sin 270^0 + (-2 - 2)\cos 270^0 = 2 + 1 = 3$

$\therefore$ R((2, -2)) = (-1, 3) thus,

Equation of line passing through (2, 6) and (-1, 3).

$\Rightarrow m = \frac{6-3}{2-(-1)} = \frac{3}{3} = 1$ use $y - y_1 = m(x - x_1)$

$\Rightarrow y - 6 = 1(x - 2) \Rightarrow y = x + 4 \Leftrightarrow -x + y = 4$

$\therefore$ Image of the line is $y - x = 4$

Note: In a rotation R, the image of A is A' and the image of B is B' then the intersection point of the perpendicular bisector of $\overline{AA'}$ and $\overline{BB'}$ is the center of rotation.

Illustrative Example

74. In rotation R, the image of A(3, 5) is A'(-2, 4) and the image of B(1, 4) is B'(-1, 2)

Find: a) center of rotation
b) the amount of rotation (θ)
c) the image of (-1, 0)

Solution: To determine center of rotation determine the intersection of the perpendicular bisector of $\overline{AA'}$ and $\overline{BB'}$

a)

- Mid-point of $\overline{AA'} = \left(\frac{3+(-2)}{2}, \frac{5+4}{2}\right) = \left(\frac{1}{2}, \frac{9}{2}\right)$
- Mid-point of $\overline{BB'} = \left(\frac{1+(-1)}{2}, \frac{4+2}{2}\right) = (0,3)$
- Slope of $AA' = \frac{4-5}{-2-3} = \frac{-1}{-5} = \frac{1}{5}$
- Slope of $\perp$ bisector of $\overline{AA'}$ is -5

$\Rightarrow$ Equation of $\perp$ bisector of $\overline{AA'}$ is $y - \frac{9}{2} = -5\left(x - \frac{1}{2}\right)$

$\Rightarrow y = -5x + 7 \Rightarrow y + 5x = 7$

- Slope of $\overline{BB'} = \frac{1-(-1)}{4-2} = \frac{2}{2} = 1$
- Slope of perpendicular bisector of BB' is -1

$\Rightarrow$ Equation of $\perp$ bisector of $\overline{BB'}$ passing through (0, 3)
$\Rightarrow y - 3 = -1(x - 0) \Rightarrow y = -x + 3$

Center of rotation is intersection of $\begin{cases} y + 5x = 7 \\ y = -x + 3 \end{cases}$

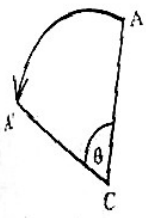
$\Rightarrow -x + 3 + 5x = 7 \Rightarrow 4x = 4 \Rightarrow x = 1$ and $y = 2$
$\therefore$ center of rotation (1, 2)

b) To determine amount of rotation θ

We use vector dot product between angle $\overrightarrow{AC}$ and $\overrightarrow{A'C}$ or $\overrightarrow{BC}$ and $\overrightarrow{B'C}$ where C is center of rotation.

$\Rightarrow \overrightarrow{AC} = (1, 2) - (3, 5) = (-2, -3)$

$\Rightarrow \overrightarrow{A'C} = (1, 2) - (-2, 4) = (3, -2)$



$$\cos\theta = \frac{\overrightarrow{AC}\cdot\overrightarrow{A'C}}{|\overrightarrow{AC}||\overrightarrow{A'C}|} = \frac{(-2,-3)\cdot(3,-2)}{\sqrt{4+9}\sqrt{9+4}} = \frac{-6+6}{13} = 0$$

$$\Rightarrow \theta = \cos^{-1}(0) = \frac{\pi}{2}$$

$\therefore$ The amount of rotation is $\theta = \frac{\pi}{2}$ about (1, 2)

c) The image of (-1, 0) when we rotate by about (1, 2) is obtained by:

$x' = a + (x - a)\cos\theta - (y - b)\sin\theta$

$$\Rightarrow x' = 1 + (-1 - 1)\cos\frac{\pi}{2} - (0 - 2)\sin\frac{\pi}{2} = 1 + 2(1) = 3$$

$y' = b + (x - a)\sin\theta + (y - b)\cos\theta$

$$\Rightarrow y' = 2 + (-1 - 1)\sin\frac{\pi}{2} + (0 - 2)\cos\frac{\pi}{2} = 2 + (-2) = 0$$

$\therefore R((-1, 0)) = (3, 0)$

75. If M is reflection in the line $y = 2x$ and R is rotation about the origin through $\theta = -270^0$. Find

a) R(M(3, 2)) b) M(R(3, 2))

c) Does R(M((x, y)) = M(R((x, y)))

Solution: $y = 2x, \Rightarrow \tan\theta = 2$

$$\Rightarrow \sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right) = \frac{4}{5}$$

$$\Rightarrow \cos 2\theta = \cos^2\theta - \sin^2\theta = \left(\frac{1}{\sqrt{5}}\right)^2 - \left(\frac{2}{\sqrt{5}}\right)^2 = \frac{1}{5} - \frac{4}{5} = \frac{-3}{5}$$

$\Rightarrow x' = x\cos 2\theta + y\sin 2\theta \Rightarrow 3\left(\frac{-3}{5}\right) + 2\left(\frac{4}{5}\right) = \frac{-1}{5}$

$\Rightarrow y' = x\sin 2\theta - y\cos 2\theta \Rightarrow 3\left(\frac{4}{5}\right) - 2\left(\frac{-3}{5}\right) = \frac{18}{5}$

$\therefore R(M((3, 2)) = R\left(\frac{-1}{5}, \frac{18}{5}\right) = \left(\frac{-18}{5}, \frac{-1}{5}\right)$

b) $M(R(3, 2) = M(-2, 3)$

$x' = -2\left(\frac{-3}{5}\right) + 3\left(\frac{4}{5}\right) = \frac{6}{5} + \frac{12}{5} = \frac{18}{5}$

$y' = -2\left(\frac{4}{5}\right) + (-3)\left(\frac{-3}{5}\right) = \frac{-8}{5} + \frac{9}{5} = \frac{1}{5}$

$\therefore M(R((3, 2)) = M(-2,3) = \left(\frac{18}{5}, \frac{1}{5}\right)$

c) $R(M(x, y)) \neq M(R)(x, y))$

Solved Problem

76. If $\vec{a} = 2i - 3j$, $\vec{b} = 3i - j$ and $\vec{c} = i - j$ then which of the following is equal to $\vec{a}.(2\vec{b} - \vec{c})$

A) -10 B. 7 C. 10 D. 13

Solution: $\vec{a}.(2\vec{b} - \vec{c}) = (2i - 3j)\cdot(5i - j) = 10 + 3 = 13$

77. If A = (2, 3), B = (3, 4) and c = (-1, -2) then the dot product of $\overrightarrow{AB}$ and $\overrightarrow{AC}$

A. -10 B. -2 C. 2 D. -8

Solution:

$\overrightarrow{AB} = (3, 4) - (2, 3) = (1,1)$ and $\overrightarrow{AC} = (-1,-2) - (3,4) = (-3,-5)$

$\Rightarrow \overrightarrow{AB} \cdot \overrightarrow{AC} = (1,1)\cdot(-3, -5) = -3 + -5 = -8$

78. Let $\vec{u} = (-1, 10)$ and $\vec{v} = (1, 3)$. A unit vector in the direction $\vec{u} - 2\vec{v}$ is ....(UEE)

A. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ B. $\left(\frac{-3}{5}, \frac{4}{5}\right)$ C. $\left(\frac{3}{5}, \frac{-4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{4}{5}\right)$

Solution: $\vec{u} - 2\vec{v} = (-1, 10) - (2, 6) = (-3, 4)$

Unit vector in the direction of $(-3, 4) = \frac{(-3,4)}{|(-3,4)|} = \frac{(-3,4)}{\sqrt{3^2+4^2}}$

$$= \left(\frac{-3}{5}, \frac{4}{5}\right)$$

79. If $(i + 5j) - \vec{x} = (3i + 2j) - 2(i - 3j)$ then the norm of the vector $\vec{x}$ is

A. 11 B. $\sqrt{10}$ C. 3 D. $\sqrt{3}$

Solution: $\vec{x} = i + 5j - 3i - 2j + 2i - 6j = -3j \Rightarrow |\vec{x}| = \sqrt{(-3)^2} = 3$

80. If the magnitude of vector $\vec{v}$ is 15 and its direction is 120^0 from the positive x-axis. What are the vector $\vec{v}$ ..... (UEE)

A. $\left(\frac{-15}{2}, \frac{15\sqrt{3}}{2}\right)$ C. $\left(\frac{-15}{2}, 15\sqrt{2}\right)$

B. $\left(10\sqrt{3}, \frac{15}{2}\right)$ D. $\left(12\sqrt{3}, \frac{-15}{\sqrt{2}}\right)$

Solution: $\cos 120^0 = \frac{a}{15} \Rightarrow a = 15\cos 120^0 = \frac{-15}{2}$

$\sin 120^0 = \frac{b}{15} \Rightarrow b = 15\sin 120^0 = \frac{15\sqrt{3}}{2}$

$\therefore \vec{v} = (a, b) = \left(\frac{-15}{2}, \frac{15\sqrt{3}}{2}\right)$ **Answer: A**

81. If $|\vec{a}| = 4$, $|\vec{b}| = \sqrt{3}$ and $\vec{a} \cdot \vec{b} = -3$ then the magnitude of $\vec{a} - \vec{b}$ .....EHEECE

A. 5 B. 7 C. $\sqrt{13}$ D. $\sqrt{22}$

Solution: $|\vec{a} - \vec{b}| = \sqrt{(\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})} = \sqrt{|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2}$

$= \sqrt{4^2 - 2(-3) + \sqrt{3}^2} = \sqrt{16 + 6 + 3} = \sqrt{25} = 5$

82. If $\vec{u} = -i + 2j$, $\vec{v} = -3i + j$ and $\vec{w} = -3j$ then the cosine of the angle between $\vec{v} + \vec{w}$ and $\vec{v}$

A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{2}$ C. $\frac{4}{\sqrt{20}}$ D. 0

Solution: $\cos\theta = \frac{(\vec{v}+\vec{w})\cdot(\vec{v})}{|\vec{v}+\vec{w}||\vec{v}|} = \frac{(-i-j)\cdot(-3+j)}{\sqrt{2}\sqrt{10}} = \frac{2}{\sqrt{20}} = \frac{1}{\sqrt{5}}$

83. Let $\vec{b}$ is unit vector, and $|\vec{a}| = 3$. The vector $\vec{a}$ and $\vec{b}$ makes an angle $\theta = \frac{\pi}{3}$. What is $|\vec{a} - 2\vec{b}|$

A. $\sqrt{5}$ B. $\sqrt{7}$ C. $\sqrt{8}$ D. $\sqrt{19}$

Solution: $|\vec{a} - 2\vec{b}| = \sqrt{|a|^2 - 4|\vec{a}||\vec{b}|\cos\theta + 4|b|^2}$, $|\vec{b}| = 1$

$$= \sqrt{9 - 4(3)(1)\left(\frac{1}{2}\right) + 4(1)} = \sqrt{13-6} = \sqrt{7}$$

84. Suppose that a triangle has vertices A = (1, 2), B = (3, 4) and C = (2, 5). What is the interior angle of triangle ABC at vertex B.

A. 30^0 B. 45^0 C. 60^0 D. 90^0

Solution: Position vector of $\overrightarrow{BA} = (1, 2) - (3, 4) = (-2, -2)$ and $\overrightarrow{BC} = (2, 5) - (3, 4) = (-1, 1)$

$$\Rightarrow \cos\beta = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{(-2,-2)\cdot(-1,1)}{\sqrt{8}\sqrt{2}} = \frac{2-2}{\sqrt{16}} = \frac{0}{4} = 0$$

$\Rightarrow B = \cos^{-1}(0) = 90^0$, **Answer: D**

85. What is vector equation of the line passing through (-1, 4) and parallel to the line $y - 3x - 10 = 0$...UEE

A. (x,y) = (0, -7) + t(1,3) C) (x,y) = (0,7) + t(1, -3)

B. (x,y) = (1, 10) + t(1, 3) D) (x, y) = (-1, 4) + t(1, 3)

Solution: $y - 3x - 10 = 0 \Rightarrow y = 3x + 10$

Direction vector $\vec{v}$ is obtained by setting y = 3x

$\Rightarrow \vec{v} = (a, b) = (a, 3a) = a(1, 3) = t(1, 3)$

$\Rightarrow (x, y) = (-1, 4) + t(1, 3)$

Answer: D

86. Suppose that the equation of a line ℓ is given by ℓ: $2x-3y-7=0$. What is vector equation of ℓ.

A. $(x, y) = (0, -7) + t(2, -3)$ C. $(x,y) = (-3, -4) + t(1, -6)$

B. $(x, y) = (2, -1) + t(3, 2)$ D. $(x, y) = (-4, 2) + t(1, 5)$

Solution: $2x-3y+7=0 \Rightarrow 3y = 2x-7 \Rightarrow y = \frac{2}{3}x - \frac{7}{3}$

$\Rightarrow$ Direction vector $\vec{v}$ parallel to ℓ is $y = \frac{2}{3}x$

$$\Rightarrow \vec{v} = (a, b) = \left(a, \frac{2}{3}a\right) = a\left(1, \frac{2}{3}\right) = t(3,2)$$

$\therefore$ Take any point on: $y = \frac{2}{3}x - \frac{7}{3}$, say $(2, -1)$

$\Rightarrow (x, y) = (2, -1) + t(3, 2)$ **Answer: B**

87. Let $(x, y) = (2, 5) + t(1, -2)$ is vector equation of ℓ. What is the standard equation of ℓ

A. $y + 2x = 9$ C. $y = 3x - 5$

B. $3y - 4x = 3$ D. $-2y + x = 3$

Solution: $(x, y) = (2, 5) + t(1, -2) \Rightarrow x = 2 + t,\ y = 5 - 2t$

Solving for, $t = \frac{x-2}{1} = \frac{y-5}{-2} \Rightarrow y + 2x = 9,$ **Answer: A**

89. If $\vec{u} = 2i - j$, $|\vec{v}| = 5$ and $\vec{u}$ and $\vec{v}$ are perpendicular vectors then $|\vec{u} + 3v|$ ....

A. $\sqrt{55}$ B. $9\sqrt{5}$ C. $\sqrt{230}$ D. 20

Solution: $|\vec{u} + 3\vec{v}| = \sqrt{(\vec{u}+3\vec{v})\cdot(\vec{u}+3\vec{v})}$

$$= \sqrt{|\vec{u}|^2 + 6|\vec{u}||\vec{v}|\cos\theta + 9|\vec{v}|^2}$$

$$= \sqrt{\sqrt{5}^2 + 6(\sqrt{5})(5)\cos 90^0 + 9(5)^2} = \sqrt{230}$$

90. If $A\left(-\sqrt{3}, 3\right)$, $B(0, 0)$ and $C\left(-2\sqrt{3}, 0\right)$ are vertices of triangle, then what is the interior angle at C

A. 30^0 B. 45^0 C. 60^0 D. 135^0

Solution: $\overrightarrow{CA} = \left(-\sqrt{3}, 3\right) - \left(-2\sqrt{3}, 0\right) = \left(\sqrt{3}, 3\right)$

$$\overrightarrow{CB} = (0,0) - \left(-2\sqrt{3}, 0\right) = \left(2\sqrt{3}, 0\right)$$

$$\Rightarrow \cos(c) = \frac{\overrightarrow{CA}\cdot\overrightarrow{CB}}{|\overrightarrow{CA}||\overrightarrow{CB}|} = \frac{(\sqrt{3},3)\cdot(2\sqrt{3},0)}{\sqrt{3+9}\sqrt{12+0}} = \frac{6}{\sqrt{144}} = \frac{6}{12} = \frac{1}{2}$$

$$\Rightarrow C = \cos^{-1}\left(\frac{1}{2}\right) = 60^0,$$

Answer: C

91. Let ℓ be a line given by $x - 2y = 0$ parametric equation of the line passes through (5, 0) and perpendicular to ℓ is ..... UEE

A. $(x, y) = (5, 0) + t(1, 2)$ C. $(x, y) = (5 + t, -2t), t\in(-\infty,\infty)$

B. $(x,y) = (5,0)+t(1,-2), t\geq 0$ D. $(x,y) = (5 - t, 2t), [0, \infty)$

Solution: $x - 2y = 0 \Rightarrow x = 2y \Rightarrow y = \frac{1}{2}x \Rightarrow \left(x, \frac{1}{2}x\right) = t(2,1)$

Let $\vec{v} = (a, b) \Rightarrow (a, b)\cdot(2, 1) = 0$

$\Rightarrow 2a + b = 0 \Rightarrow b = -2a \Rightarrow (a, b) = (a, -2a)$

$\Rightarrow \vec{v} = t(1, -2)$

$\therefore$ Vector equation $(x, y) = (5, 0) + t(1, -2)$

$\Rightarrow$ Parametric equation $(x, y) = (5+ t, -2t)$

Answer: C

92. A translation takes $x^2 + (y + 1)^2 = 5$ to $(x - 2)^2 + y^2 = 5$ what is the image of (1, 3) under this transformation (UEE)

A. (3, 4) B. (3, 2) C. (-1, 2) D. (-2, 3)

Solution:

Center of $x^2+(y+1)^2 = 5$ is (0, -1) center of $(x - 2)^2 + y^2 = 5$ is (2, 0)

$T(0, -1) = (2, 0)$, $\vec{v} = (2, 1)$

$T(1, 3) = (2 + 1, 3 + (1)) = (3, 4)$

Answer: A

93. A circle that passes through A = (2, 3) and center at C = (1, 1). The equation of the line tangent to the circle at A is ....(UEE)

A. $3x + y = 9$ C. $3x - 2y = 0$

B. $x + 2y = 8$ D. $x+y = 5$

Solution: Let R(x, y) be point on ℓ

$$\Rightarrow \overrightarrow{CA}\cdot\overrightarrow{AR} = 0 \Rightarrow ((2,3) - (1,1)\cdot(x-2, y-3)$$

$\Rightarrow (1, 2)\cdot(x - 2, y - 3) = x - 2 + 2y - 6 = x + 2y = 8$

Answer:

94. A translation T takes point P(-1, 0) to point Q(1, 3). What is the image of the ellipse $9(x - 1)^2 + 4(y + 2)^2 = 36$

A. $9(x - 3)^2 + 4(y + 2)^2 = 36$ C. $9(x + 1)^2 + 4y^2 = 36$

B. $9(x + 3)^2 + 4(y - 1)^2 = 36$ D. $9(x - 2)^2 + 4(y - 1)^2 = 36$

Solution: Translating vector $\vec{v} = \overrightarrow{PQ} = (1, 3) - (-1, 0) = (2, 3)$
move 2 unit to right and 3 unit up $\Rightarrow (x - 2, y - 3)$
$\Rightarrow 9(x - 2 - 1)^2 + 4(y - 3 + 2)^2 = 36 \Rightarrow 9(x - 3)^2 + 4(y - 1)^2 = 36$

Answer: A

95. In a translation T the image of P(1, -1) is P'(3, 2). If a line ℓ, its parametric equation is given by $(x, y) = (2, -1) + t(1, 1)$ for are $t \in R$ is translated by T, what is image of ℓ.....(UEE)

A. $y = x + 3$ C. $x - y = 2$
B. $2x + y = 1$ D. $3y - x = 3$

Solution: Translating vector $\vec{v} = \overrightarrow{PP'} = (3 - 1, 2 - (-1)) = (2, 3)$
$\Rightarrow (x, y) = (2 + t, -1 + t) \Rightarrow x - 2 = y + 1 \Rightarrow x - y = 3$
Image, move 2 unit to left and 3 unit up
i.e. $(x - 2, y - 3)$
$\Rightarrow x - 2 - (y - 3) = 3 \Rightarrow x - y + 1 = 3 \Rightarrow x - y = 2$

Answer: C

96. If T is translation that sends (0, 0) to (3, -2) and M is reflection that maps (0, 0) to (2, 4) then what is T(M(1, 3))?

A. (0, -1) C. $\left(\frac{3}{2}, 4\right)$
B. (4, 1) D. (3, 2)

Solution: Slope of $MM' = \frac{4-0}{2-0} = 2 \Rightarrow$ slop of $\ell \Rightarrow \frac{-1}{2}$

Mid-point of MM' = (1, 2)

$\Rightarrow$ Equation of ℓ $\frac{y-2}{x-1} = \frac{-1}{2} \Rightarrow y - 2 = -\frac{1}{2}x + \frac{1}{2}$

$\Rightarrow y = -\frac{1}{2}x + \frac{5}{2} \longrightarrow$ line of reflection

$M(1, 3) \xrightarrow{y=-\frac{1}{2}x+\frac{5}{2}} (-3, 1)$

$T(M(1, 3)) = (-3, 1) = \left(3 + (-3), 1 + (-2)\right) = (0, -1)$

Answer: A

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97. If R is rotation through $\theta = 90^0$ and is followed by reflection in the x-axis what is the image of ellipse given by $\frac{1}{4}(x-1)^2 + (y+2)^2 = 1$ under the two transformation

A. $4(x-1)^2 + (y+1)^2 = 4$ C. $(x-1)^2 + 4(y+2)^2 = 4$

B. $4(x-2)^2 + (y-1)^2 = 4$ D. $(x+1)^2 + 4(y-2)^2 = 4$

Solution:

Center (1, -2) $\xrightarrow{R_{90^0}}$ (2, 1) $\longrightarrow$ Rotation with $\theta = 90^0$

M(2, 1) = (2, -1) $\longrightarrow$ reflection on x-axis

x-ellipse become y-ellipse.

$\Rightarrow$ Image will be $4(x-2)^2 + (y+1)^2 = 4$.

Answer: A

98. Determine the value of m for which the vectors $\vec{a} = -2i + 2j + mj$ and $\vec{b} = 3i + 6j$ are i) Perpendicular ii) parallel

Solution: a) $\vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$

$\Rightarrow (-2i + (2 + m)j) \cdot (3i + 6j) = -6 + 12 + 6m = 0$

$\Rightarrow 6m = -6 \Rightarrow m = -1$

b) $\vec{a} \parallel \vec{b} \Rightarrow \cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|a||b|} = \pm 1$

$$\Rightarrow \frac{(-2i + (2+m)j) \cdot (3i + 6j)}{\sqrt{4 + (2+m)^2}\sqrt{3^2 + 6^2}} = \pm 1$$

$$\Rightarrow \frac{6m + 6}{3\sqrt{5m^2 + 20m + 40}} = \pm 1$$

$$\Rightarrow \frac{2m + 2}{\sqrt{5m^2 + 20m + 40}} = \pm 1$$

$\Rightarrow (2m + 2)^2 = 5m^2 + 20m + 40 \longleftarrow$ squaring and simplifying

$\Rightarrow m^2 + 12m + 36 = 0 \Leftrightarrow (m+6)(m+6) = 0$

$\Rightarrow m = -6$

99. Find unit vector parallel to the line y = 3x + 2

Solution: Vector $\vec{v}$ = (a, b) parallel to the line is set y = 3x

$\Rightarrow \vec{v} = (\vec{a}, \vec{b}) = (a, 3a) = a(1, 3) = (1, 3)$

Unit vector in the direction of $\vec{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{(1,3)}{\sqrt{10}} = \left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

100. Which of the following is a vector that lies on the line through (0, 0) and (2, 4).....UEE2004/12

A. $\vec{u} = (2,1)$ C. $\vec{u} = \left(\frac{1}{2}, 2\right)$

B. $\vec{u} = (-1,-2)$ D. $\vec{u} = (-2,-6)$

Solution: Let A = (0, 0) and B = (2, 4)

$\Rightarrow \overrightarrow{AB} = (2, 4) - (0, 0) = (2, 4)$

Vector equation of ℓ: (x, y) = (0, 0) + t(2, 4) = (2t, 4t)

$\Rightarrow$ (x, y) = (2t, 4t) if t = $\frac{-1}{2}, \vec{u} = (-1,-2)$

Answer: B

101. Let i and j be standard unit vector, A = (-1, 0) and B = (2, 2). If $\vec{v} = 3\overrightarrow{AB} - 3\vec{i} + 2\vec{j}$ then unit vector in the direction of $\vec{v}$,...(UEE)

A. $\left(\frac{3}{5}, \frac{4}{5}\right)$ C. $\left(\frac{-3}{5}, \frac{4}{5}\right)$

B. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

Solution: $\overrightarrow{AB}$ = (2, 2) – (-1, 0) = (3, 2) = 3i + 2j

$\vec{v}$ = 3(3i + 2j) – 3i + 2j = 6i + 8j = (6, 8)

Unit vector in direction of $\vec{v}$ is $\frac{\vec{v}}{|\vec{v}|} = \frac{(6,8)}{\sqrt{6^2+8^2}} = \left(\frac{3}{5}, \frac{4}{5}\right)$

Answer: A

102. Which of the following is a vector equation of the line tangent to the circle $x^2 + y^2 + 2x - 7 = 0$ at (1, 2) UEE2004/12

A. (x, y) = (0, 3) + λ(-1, 2) C. (x, y) = (0, 3) + λ(1, -1)

B. (x, y) = (1, 2) + λ(2, -1) D. (x, y) = (1, 2) + λ(-1, 2)

Solution: $x^2 + y^2 + 2x - 7 = 0 \Rightarrow (x + 1)^2 + (y - 0)^2 = 8$

$\Rightarrow$ center, C(-1, 0), Let P(1, 2) and R(x, y) point on tangent line.

$\overrightarrow{CP} = (1, 2) - (-1, 0) = (2, 2)$

$\Rightarrow \overrightarrow{CP} \cdot \overrightarrow{PR} = 0 \Rightarrow (2, 2)\cdot(x - 1, y - 2) = 0$

$\Rightarrow 2x - 2 + 2y - 4 = 0 \Rightarrow x + y = 3$

- Direction vector $\vec{v} = (a, b)$ parallel to the line $x + y = 3$ is $x + y = 0 \Rightarrow y = -x$, $\vec{v} = (a, b) = (a, -a) = \lambda(1, -1)$
- take any arbitrary point on $x + y = 3$, say, $(0, 3)$

$\therefore$ vector equation of ℓ: $(x, y) = (0, 3) + \lambda(1, -1)$, Answer(

103. Let ℓ be line given by $2x - y = 10$. what is the equation of the image of ℓ after a reflection in the line $y = 2x - 5$ followed by rotation through the angle of $\theta = 90^0$ about the origin ...UE 2004/12

A. $x + 2y = 0$ C. $x + 2y = 5$

B. $2x + y = 0$ D. $x - 2y = 5$

Solution: ℓ: $y = 2x - 10$ is parallel to the reflecting axis.

Hence, the image ℓ': $y = 2x + b$ we need to find b.

Let (a, b) be any point on ℓ. say (0, -10)

So that its lies on ℓ': $y = 2x + b$

$$\Rightarrow M((0, -10)) = (a', b') \Rightarrow \frac{b'+10}{a'-0} = \frac{-1}{2} \Rightarrow a' = -2b' - 20$$

Also, mid-point of (0, -10) and (a', b')

$$\Rightarrow \left(\frac{0+a'}{2}, \frac{b'-10}{2}\right) \text{ lies on reflecting axis}$$

$$\Rightarrow \frac{b'-10}{2} = 2\left(\frac{a'}{2}\right) - 5 \Rightarrow a' = \frac{b'}{2}$$

$$\text{But } a' = -2b' - 20 \Rightarrow -2b' - 20 = \frac{b'}{2} \Rightarrow b' = -8$$

$a' = -4 \Rightarrow (-4, -8)$ lies on $y = 2x + b$

$\Rightarrow -8 = 2(-4) + b \Rightarrow b = 0 \Rightarrow \ell'$: $y = 2x$

$\Rightarrow$ Image of ℓ': $y = 2x$ when rotated by $\theta = 90^0$ about origin

$R(x, y) = (-y, x)$

$R(x, 2x) = (-2x, x)$

$\Rightarrow x = -2y \Rightarrow x + 2y = 0$

Answer: A

104. Let ℓ_1 and ℓ_2 be two lines in space intersecting at the origin (0,0,0). If ℓ_1 and ℓ_2 pass through point, A(1, 1, 0) and B(0, 1, 1) respectively, then the angle between ℓ_1 and ℓ_2 is equal to … UEE 2004/12.

A. 30° B. 45° C. 60° D. 90°

Solution: Let origin be 0 = (0, 0, 0)

$\Rightarrow \overrightarrow{OA} = (1, 1, 0)$ and $\overrightarrow{OB} = (0, 1, 1)$

Let θ be angle between ℓ_1 and ℓ_2

$$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} = 60°$$

Answer: C

105. Let $\vec{a} = -i + 3k$ and $\vec{b} = -i + j$ be vector in the spaced. What is the cosine of the angle between $\vec{a}$ and $\vec{a} - \vec{b}$ …UEE 2004/12.

A. $\frac{9}{10}$ B. $\frac{3}{5}$ C. $\frac{3}{\sqrt{10}}$ D. $\frac{-9}{10}$

Solution: Let θ be angle between $\vec{a}$ and $\vec{a} - \vec{b}$

$$\Rightarrow \cos\theta = \frac{\vec{a}\cdot(\vec{a}-\vec{b})}{|\vec{a}||\vec{a}-\vec{b}|} = \frac{(-i+0j+3k)\cdot(0i-j+3k)}{\sqrt{1+9}\sqrt{1+9}} = \frac{9}{10}$$

Answer: A

106. Let $\vec{v} = 3i - 4k$, i, j and k be standard unit vector in x, y, z axis and A = (0, 1, 2)

If $\overrightarrow{AB}$ is parallel to $\vec{v}$ and $\|\overrightarrow{AB}\| = 10$, then point B is at … UEE 2004/12

A. (-6, -1, 10) C. (6, 1, -6)

B. (-6, -1, -10) D. (-6,-1, 6)

Solution: Let B = (x, y, z)

Since $\overrightarrow{AB} \| \vec{v} \Rightarrow \overrightarrow{AB} = t\vec{v} \Rightarrow |\overrightarrow{AB}| = |t\vec{v}|$

$\Rightarrow t\vec{v} = t(3, 0, -4) \Rightarrow (3t, 0, -4t)s$

$$\Rightarrow |t\vec{v}| = 10 \Rightarrow \sqrt{9t^2 + 0^2 + 16t^2} = \sqrt{25t^2} = |t|5$$

$\Rightarrow 5|t| = 10 \Rightarrow t = 2$

$\overrightarrow{AB} = B - A = (x, y, z) - (0, 1, 2) = (x, y-1, z-2) = (2)(3, 0, -4)$

$\Rightarrow x = 6, y - 1 = 0 \Rightarrow y = 1, z - 2 = -8 \Rightarrow z = -6$

$\therefore B = (x, y, z) = (6, 1, -6)$

Answer: C

Supplementary exercise

1. The position vectors of four points A, B, C D and $\vec{a}, \vec{b}, 2\vec{a}+3\vec{b}$ and $\vec{a}-2\vec{b}$ respectively. Find $\overrightarrow{AC}$, $\overrightarrow{BC}$, $\overrightarrow{DA}$ and $\overrightarrow{DB}$ interms of $\vec{a}$ and $\vec{b}$
2. Verify whether the three points $\vec{a}-2\vec{b}+3\vec{c}$, $2\vec{a}+3\vec{b}-4\vec{c}$, $7\vec{b}+10\vec{c}$ are collinear
3. Show that the following vectors are coplanar $5\vec{a}+6\vec{b}+7\vec{c}$, $7\vec{a}-8\vec{b}+9\vec{c}$, $3\vec{a}+20\vec{b}+5\vec{c}$
4. For the vectors $\vec{a}$ and $\vec{b}$ where $\vec{a}=2\mathrm{i}+3\mathrm{j}+\vec{k}$, $\vec{b}=3\mathrm{i}-\mathrm{j}-$ k, find the value of $\left|2\vec{a}+3\vec{b}\right|$
5. ABCD is parallelogram. Find single vector of the following.
   a) $\overrightarrow{CB}-\overrightarrow{CA}$
   b) $\overrightarrow{DC}-\overrightarrow{DA}$
   c) $\overrightarrow{AC}+\overrightarrow{CB}$
   d) $\overrightarrow{AD}+\overrightarrow{DC}$
   e) $\overrightarrow{DB}+\overrightarrow{BA}+\overrightarrow{DA}$
   f) $\overrightarrow{CD}-\overrightarrow{CB}$
6. If the vectors $\vec{c}$ and $\vec{d}$ are two adjacent sides of regular hexagon, determine the other sides interms of $\vec{c}$ and $\vec{d}$
7. Find the magnitude and unit vector in the direction of the sum of the vectors $\vec{a}=i+4j+2\vec{k}$, $\vec{b}=3i-3j-2k$, and $\vec{c}=-2i+2j+6k$
8. find the unit vector in the direction of the vector $\vec{r}_1-\vec{r}_2$ where $\vec{r}_1=i+2j-k$ and $\vec{r}_2=3j+j-5k$
9. Find the angle between the vector
   a) $\vec{a}=-3i+2j$, $\vec{b}=2i+3j$
   b) $\vec{a}=i+j$, $\vec{b}=3i$
   c) $\vec{a}=-2i$, $\vec{b}=\sqrt{3}i-$
10. Find the value of t for which the vectors 3i + j +tk and 2i -2j+ are perpendicular to each other.
11. If $\vec{a}=2i-3j+k$, $\vec{b}=i+2j+5k$, $\vec{c}=-3i+j-2k$
    Find a) $\vec{a}\cdot\vec{b}$ b) $\vec{b}\cdot\vec{c}$

12. Given the vector $\vec{a} = i$, $\vec{b} = i + j$, find scalar k such that
    a) the vector $\vec{a} + k\vec{b}$ is perpendicular to $\vec{a}$
    b) the vector $\vec{a} + k\vec{b}$ is perpendicular to $\vec{b}$

13. Vectors $\vec{a}$ and $\vec{b}$ make an angle $\theta = \frac{\pi}{3}$. If $|\vec{a}| = 3$, $|\vec{b}| = 2$ then find a) $|2\vec{a} + \vec{b}|$ b) Angle between $\vec{b}$ and $(\vec{a} + \vec{b})$

14. Show that A(1, 2), B(3, 4) and C(5, 2) are vertices of a right triangle by considering the sides of the triangle as vectors.

15. Find the angle between the vectors
    a) $\vec{a} = (1, 2)$ and $\vec{b} = \left(1, -\frac{1}{2}\right)$
    b) $\vec{a} = \mathrm{i} - \sqrt{3}\,\mathrm{j}$ and $\vec{b} = \mathrm{i} + \sqrt{3}\,\mathrm{j}$

16. If $\vec{a} = 2\mathrm{i} - 2\mathrm{j} + \mathrm{k}$, $\vec{b} = 2\mathrm{i} + 3\mathrm{j} + 6\mathrm{k}$, $\vec{c} = -\mathrm{i} + 2\mathrm{k}$
    Find the direction cosine of $\vec{a} - \vec{b} + 2\mathrm{c}$

17. Find a vector equation and rectangular form of equation for the line that contains the given point and is parallel to a vector.
    a) $\vec{r}_0 = (-3, 4)$; $\vec{a} = (2\mathrm{i} + \mathrm{j})$ b) $r_0 = (5, 1)$; $\vec{a} = 3\mathrm{i} - 2\mathrm{j}$

18. Let $\vec{u} = \mathrm{i} - 3\mathrm{j} + 2\mathrm{k}$, $\vec{v} = \mathrm{i} + \mathrm{j}$, $\vec{w} = 2\mathrm{i} + 2\mathrm{j} - 4\mathrm{k}$
    Find
    a) $|\mathrm{u} + \mathrm{v}|$ c) $|3\mathrm{u} - 5\mathrm{v} + \mathrm{w}|$
    b) $|\mathrm{u}| + |\mathrm{v}|$ d) $\frac{1}{|w|}^{w}$ f) $\left|\frac{1}{|w|}^{w}\right|$

19. Find the component form of the vector $\vec{v}$ in 2-space that has the stated length and makes the stated angle θ with the positive x – axis
    a) $|\mathrm{v}| = 3$, $\theta = \frac{\pi}{4}$ c) $|\mathrm{v}| = 5$, $\theta = 120^0$
    b) $|\mathrm{v}| = 2$, $\theta = 90^0$ d) $|\mathrm{v}| = 1$ $\theta = \pi$

20. Let $\vec{u} = (1, 3)$, $\vec{v} = (2, 1)$, $\vec{w} = (3, 4)$, find the vector $\vec{x}$ that satisfies $2\vec{u} - \vec{v} + \vec{x} = 7\vec{x} + \vec{w}$

21. Find parametric equations for the line through P_1 and P_2
    a) $P_1(3, -2)$, $P_2(5, 1)$ b) $P_1(5, -2, 1)$, $P_2(2, 4, 2)$

22. Find thee equation of the line that passes through (3, 2) and
    a) Parallel to vector $2i - 4j$
    b) Perpendicular to the vector $-i + j$
23. Using vector methods, find the equation for circles whose cen and radius are given below
    a) center (3, 4), r = 2 b) center (-1, 1), $r = \sqrt{3}$
24. Find the parametric equation of the line that is tangent to t circle $x^2 + y^2 = 25$ at the point (3, -4)
25. A translation T(-2, 3) mapped to the point (4, -1) then find t image of
    a) (0, 1) b) 1, 1)
    c) the circle with equation of $(x + 10^2 + (y - 2)^2 = 5$
26. Find the image of the following under the line of reflection x =
    a) $y = 2x + 1$, b) (1, 3) c) $y = 2$ d) x = 1
27. Find the image of the following under the line of reflectio
    ℓ: $x - y = 3$
    a) (1, 1) b) $(x - 1)^2 + (y - 1)^2 = 6$ c) x =
28. Find the coordinate of the image of the following after a rotation about the origin through each of i) 45^0 ii) 150^0 iii) 300^0
    a) (6, -2) b) (2, 1) c) $y = 1$ d) $y = 3x + 1$
    e) $x^2 + y^2 = 2$
29. Find the image of the following point after clockwise rotation through 90^0
    a) (4, -3) about i) (0, 0), ii) (1, 2)
    b) $y = 2x - 1$, about i) (0, 0) ii) (1, 1)
    c) $(x - 4)^2 + (y + 3)^2 = 4$ about i) (0, 0) ii) (1, 2)

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Unit Eight

The set of complex number

1. The concept of "complex Number".

Now consider the equation $x^2 + 1 = 0$. If we attempt to solve this equation, we have

$$x^2 + 1 = 0 \Leftrightarrow x^2 = -1 \Rightarrow x = \pm\sqrt{-1} = \pm i$$

A number whose square is -1 (Negative) is called imaginary number denoted by **i**, defined as $i = \sqrt{-1},\ i^2 = -1$

A complex number system (C) which contains both R (real number) and number whose square is Negative denoted by $\mathbb{C}$ and given by

$\mathbb{C} = \{Z/Z = x + yi\}$ where x and y are real number and

$$i = \sqrt{-1}$$

Illustrative Example

1. Solve each of the following

a) $x^2 + 4 = 0$ b) $x^2 + 9 = 0$ c) $x^2 + 64 = 0$

d) $x^2 + 2x + 10 = 0$

Solution:

a) $x^2 + 4 = 0 \Leftrightarrow x^2 = -4 \Rightarrow x = \pm\sqrt{-4} = \pm 2i$ $\therefore$ x=2i or x=-2i

b) $x^2 + 9 = 0 \Leftrightarrow x^2 = -9 \Rightarrow x = \pm\sqrt{-9} = \pm 3i$ $\therefore$ x=3i or x=-3i

c) $x^2 + 64 = 0 \Leftrightarrow x^2 = -64 \Rightarrow x = \pm\sqrt{-64} = \pm 8i$ $\therefore$ x=8i or x=-8i

d) $x^2 + 2x + 10 = 0 \Leftrightarrow x^2 + 2x + 1 + 10 - 1 = 0 \Leftrightarrow (x+1)^2 + 9 = 0$

$\Rightarrow (x+1)^2 = -9 \Leftrightarrow x + 1 = \pm\sqrt{-9} = \pm 3i$

$\Rightarrow x + 1 = \pm 3i$ $\therefore$ x = {-1 + 3i, -1 -3i}

2. Write the expression in the form: z = x + yi

a) $z = 3 - \sqrt{-4}$

b) $z = 1 + \sqrt{-25}$

c) $z = \sqrt[3]{-8} + \sqrt{-8}$

g) $z = \dfrac{\sqrt{-27}}{\sqrt{3}}$

h) $z = \sqrt{3}\sqrt{-4}$

i) $z = \sqrt{(-4)(-3)}$

d) $z=\sqrt{-12}+\sqrt{-3}$ j) $z=\left(2-\sqrt{-125}\right)-\left(3-\sqrt{-20}\right)$

e) $z=\sqrt{-3}\sqrt{-4}$ k) $z=\left(-2+\sqrt{-4}\right)-\left(\sqrt{12}+\sqrt{-3}\right)$

f) $z=\sqrt{-18}$

Solution: Here

a) $z=3-\sqrt{-4}\Leftrightarrow z=3-2i$

b) $z=1+\sqrt{-25}\Leftrightarrow z=1+5i$

c) $z=\dfrac{\sqrt[3]{-8}+\sqrt{-8}}{2}\Leftrightarrow z=-1+\sqrt{2}i$

d) $z=\sqrt{-12}+\sqrt{-3}\Leftrightarrow 2\sqrt{3}i+\sqrt{3}i \quad \therefore z=0+3\sqrt{3}i$

e) $z=\sqrt{-3}\cdot\sqrt{-4}\Leftrightarrow(\sqrt{3}i)\cdot(2i)\Rightarrow z=-2\sqrt{3}$

f) $z=\dfrac{\sqrt{-18}}{\sqrt{-2}}\Leftrightarrow z=\dfrac{3\sqrt{2}i}{\sqrt{2}i}=3+0i$

g) $z=\dfrac{\sqrt{-27}}{\sqrt{3}}=\dfrac{3\sqrt{3}i}{\sqrt{3}}=3i \quad \therefore z=0+3i$

h) $z=\sqrt{3}\sqrt{-4}=2\sqrt{3}i \quad \therefore z=0+2\sqrt{3}i$

i) $z=\sqrt{(-4)(-3)}=2\sqrt{3} \quad \therefore z=2\sqrt{3}+0i$

j) $z=2-\sqrt{-125}-3+\sqrt{-20}=-1-5\sqrt{5}i+2\sqrt{5}i$

$\therefore z=-1-3\sqrt{5}i$

k) $z=-2+\sqrt{-4}-(\sqrt{12}+\sqrt{-3})$

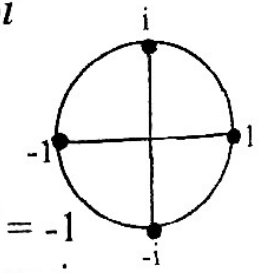
$\therefore z=-2+2i-2\sqrt{3}-\sqrt{3}i$

$\therefore z=(-2-2\sqrt{3})+(2-\sqrt{3})i$

Power of i

*A few power of **i** are listed here*

- $i^1 = i$
- $i^2 = -1$
- $i^3 = -i$
- $i^4 = 1$
- $i^5 = i$
- $i^6 = -1$
- $i^7 = -i$
- $i^8 = 1$
- $i^9 = i$
- $i^{10} = -1$
- $i^{11} = -i$
- $i^{12} = 1$



The power of i rotate through the four numbers i, -1, -i, and 1.

Note: $i^n = i^r$, where $\Rightarrow n = 4m + r$
- n is positive integer
- r is the remainder when n is divided by 4
- $0 \le r \le 3$

Illustrative Example

3. Simplify and write as $z = x + yi$

a) i^7 f) i^{543} k) i^{205}
b) i^{13} g) i^{176} l) i^{131}
c) i^{15} h) $i^{298} + i^{723} + i^{10}$
d) i^{74} i) $i + i^2 + i^3 + i^4 + \ldots + i^{28} + i^{29} + i^{30}$
e) i^{102} j) $i^{4n+3} + (i^{2n+2})^2$

Solution:

a) Divide 7 by 4, remainder, $r = 3$, $\therefore i^7 = i^3 = -i$

b) Divide 13 by 4, remainder, $r = 1$
$\therefore i^{13} = i \longleftarrow 13 = 4(3) + 1, r = 1$

c) Divide 15 by 4, Remainder, $r = 3$
$\therefore i^{15} = i^3 = -i$

d) $74 = 4(18) + 2$, $\therefore i^{74} = i^2 = -1 \longleftarrow$ Dividing 74 by 4 remainder 2

e) $i^{102} = i^{4(25)+2} \Rightarrow i^{102} = i^2 = -1 \leftarrow$ Dividing 102 by 4 remainder 2

f) Dividing 543 by 4, remainder, $r = 3$
$\therefore i^{543} = i^3 \longleftarrow 543 = 4(135) + 3$
$\therefore i^{543} = i^3 = -i$

g) $i^{176} = i^{4(44)+0} = i^0 = 1 \longleftarrow 176 = 4(44) + 0$, Dividing 176 by 4 remainder, $r = 0$

h) $i^{298} = i^2$, $\longleftarrow$ Dividing 298 by 4, remainder $r = 2$
$i^{723} = i^3$, $\longleftarrow$ Dividing 723 by 4, remainder $r = 3$
$i^{10} = i^2$, $\longleftarrow$ Dividing 10 by 4, remainder $r = 2$
$\therefore i^{298} + i^{723} + i^{10} = i^2 + i^3 + i^2 = -1 - i - i = -1 - 2i$

i) $i + i^2 + i^3 + i^4 + \ldots + i^{28} + i^{29} + i^{30}$
$= (i + -1 + -i + 1 + \ldots + i) + i - 1 = i = 1$

j) $i^{4n+3} = i^3$ and $(i^{2n+2})^2 = i^{4n+4} = i^4 = 1$
$\therefore i^{4n+3} + i^{4n+4} = i^3 + i^4 = -i + 1 = 1 - i$

k) $i^{205} = i^{4(51)+1} = i \longleftarrow$ Dividing 205 by 4, remainder $r = 1$

l) $i^{131} = i^{4(32)+3} = i^3 \longleftarrow$ Dividing 131 by 4, remainder $r = 3$
$\therefore i^{131} = i^3 = -i$

4. Simplify a) i^{2n} b) i^{2n+1}

Solution: a) $i^{2n} = \begin{cases} -1 \text{ if n is odd, } n = 1,3,5,7..... \\ 1 \text{ if n is even, } n = 2,4,6,8,..... \end{cases}$

b) $i^{2n+1} = i^{2n} \cdot i = \begin{cases} -i \text{ if n is odd,} \\ i \text{ if n is even} \end{cases}$

Equality of complex Number

Let $z_1 = x + yi$ and $z^2 = a + bi$ then

$z_1 = z_2$ if and only if $x = a$ and $y = b$

Example: a) $x + yi = 3 + 4i$ iff $x = 3$ and $y = 4$

b) $(2x - 4) + 9i = 8 + 3yi$, iff $2x - 4 = 8$ and $3y = 9$

$\Rightarrow 2x = 12$ and $3y = 9$

$\therefore x = 6$ and $y = 3$

Operation on complex Number

Let $z_1 = x + yi$ and $z_2 = a + bi$ be complex number

i) Addition and subtraction:

$z_1 + z_2 = (x + yi) + (a + bi) = (x + a) + (y + b)i$

$z_1 - z_2 = (x + yi) - (a + bi) = (x - a) + (y - b)i$

ii) Multiplication of complex Number

$z_1 \cdot z_2 = (x + y_i)(a + bi) = x(a + bi) + yi(a + bi)$

$= ax + xbi + ayi - yb$

$= (ax - yb) + (xb + ay)i$

If $z_1 \cdot z_2 = 1 + 0i$ then

- z_1 and z_2 are multiplicative inverse of each other
- $1 + 0i$ is the identity element for multiplication of complex number

iii) Division of complex Number

- Let $z_1 = 1 + 0i = 1$ and $z_2 = a + bi$, then

$$\frac{z_1}{z_2} = \frac{1+0i}{a+bi} = \frac{1}{(a+bi)}\frac{a-bi}{(a-bi)} = \frac{a-bi}{a^2+b^2}$$

$$\therefore \frac{1}{z_2} = \frac{1}{a+bi} = \frac{a}{a^2+b^2} - \frac{bi}{a^2+b^2}$$

Example: a) $\frac{1}{2+3i} = \frac{1}{2+3i} = \frac{2}{2^2+3^2} - \frac{3i}{2^2+3^2} = \frac{2}{13} - \frac{3i}{13}$

b) $\frac{1}{3-4i}=\frac{1}{3-4i}\cdot\frac{3+4i}{3+4i}=\frac{3}{3^2+4^2}+\frac{4i}{3^2+4^2}=\frac{3}{25}+\frac{4i}{25}$

Let $z_1 = a + bi$ and $z_2 = c + di$ then

$$\frac{z_1}{z_2}=\frac{a+bi}{c+di}=(a+bi)\left(\frac{1}{c+di}\right)$$

$$=(a+bi)\left(\frac{c}{c^2+d^2}-\frac{di}{c^2+d^2}\right)$$

$$=\frac{ac+bci-adi+bd}{c^2+d^2}$$

$$\therefore \frac{a+bi}{c+di}=\frac{(ac+bd)}{c^2+d^2}+\frac{(bc-ad)i}{c^2+d^2}$$

Example C: $\frac{2+3i}{1+5i}=\frac{2+15}{1^2+5^2}+\frac{(3-10)i}{1^2+5^2}=\frac{17}{26}-\frac{7i}{26}$

Example d: $\frac{1-2i}{2+4i}=\frac{(2-8)}{2^2+4^2}-\frac{(4+4)i}{2^2+4^2}=\frac{-6}{20}-\frac{8i}{20}$

Multiplicative Inverse

For every $z \neq 0$ in $\mathbb{C}$ there is multiplicative inverse $\frac{1}{z}$ such that

$$z\cdot\frac{1}{z}=1=\frac{1}{z}\cdot z$$

Example: Find the multiplicative inverse of each of the following

a) a + bi b) x + yi c) 2 + 3i d) $1-\sqrt{3}i$

Solution: a) multiplicative inverse of a + bi is $\frac{1}{a+bi}$

Thus, $\frac{1}{a+bi}=\frac{a}{a^2+b^2}-\frac{bi}{a^2+b^2}$ ← multiplicative inverse of a +bi

b) Multiplicative inverse of x + yi is $\frac{1}{x+yi}$

$$\therefore \frac{1}{x+yi}=\frac{x}{x^2+y^2}-\frac{yi}{x^2+y^2} \leftarrow \text{multiplicative inverse of } x+yi$$

c) Multiplicative inverse of $2+3i$ is $\frac{1}{2+3i}$

$$\therefore \frac{1}{2+3i}=\frac{2}{2^2+3^2}-\frac{3i}{2^2+3^2}=\frac{2}{13}-\frac{3i}{13} \leftarrow \text{multiplicative inverse of } 2+3i$$

d) Multiplicative inverse of $1-\sqrt{3}i$ is $\frac{1}{1-\sqrt{3}i}$

$$\therefore \frac{1}{1-\sqrt{3}i}=\frac{1}{1-\sqrt{3}i}\cdot\frac{1+\sqrt{3}i}{1+\sqrt{3}i}=\frac{1}{1^2+\sqrt{3}^2}=\frac{1}{4}+\frac{\sqrt{3}i}{4}$$

5. Find the multiplicative inverse and write as $z = x + yi$

a) $z=\frac{1+i}{2+i}$ b) $z=\frac{3-4i}{2+3i}$ c) $z=\frac{2-5i}{i}$

Solution: Multiplicative inverse of z is $\frac{1}{z}$

a) $z=\frac{1+i}{2+i} \Leftrightarrow \frac{1}{z}=\frac{2+i}{1+i}=(2+i)\left(\frac{1}{1+i}\right)=(2+i)\left(\frac{1}{2}-\frac{i}{2}\right)$

$$\Rightarrow \frac{1}{z}=\frac{2+i-i(2+i)}{2}=\frac{2+i-2i+1}{2}=\frac{3}{2}-\frac{i}{2}$$

$$\therefore \frac{1}{z}=\frac{3}{2}-\frac{i}{2} \leftarrow \text{Multiplicative inverse of } z=\frac{1+i}{2+i}$$

b) $\frac{1}{z}=\frac{2+3i}{3-4i}=(2+3i)\left(\frac{1}{3-4i}\right)=(2+3i)\left(\frac{3}{3^2+4^2}+\frac{4i}{3^2+4^2}\right)$

$$\frac{1}{z}=\frac{3(2+3i)+4i(2+3i)}{25}=\frac{-6}{25}+\frac{17i}{25} \leftarrow \text{multiplicative inverse of } \frac{3-4i}{2+3i}$$

c) $\frac{1}{z}=\frac{i}{2-5i}=(i)\left(\frac{1}{2-5i}\right)=i\left(\frac{2}{2^2+5^2}+\frac{5i}{2^2+5^2}\right)=\frac{-5}{29}+\frac{2i}{29}$

Illustrative Example

6. In each part solve for x and y

a) $(x + yi)(1 + 2i) = 4 + 7i$

b) $(x - yi)(4 + i) = 14 - 5i$

c) $\dfrac{x+yi}{3-i} = \dfrac{1}{5} + \dfrac{2}{5}i$

d) $(x + yi)i + 2 - 3x - y + 4i = 0$

e) $2x + yi = (3+5i)(2-4i)$

f) $\dfrac{3+i}{x+yi} = \dfrac{1}{2} + \dfrac{1}{2}i$

g) $\dfrac{i+2}{x-4yi} = 3i^{23}$

h) $\dfrac{2+3i}{3x-4yi} = i+1$

i) $(x + yi)^2 = i^5$

Solution: Here

a) $(x+yi)(1+2i)=4+7i \Leftrightarrow x+yi$

$= \dfrac{4+7i}{1+2i}$ ← dividing both side by $(1 + 2i)$

$$\therefore x + yi = \frac{4+7i}{1+2i} = (4+7i)\left(\frac{1}{1+2i}\right) = (4+7i)\left(\frac{1}{1^2+2^2} - \frac{2i}{1^2+2^2}\right)$$

$$\Rightarrow x + yi = \frac{4+7i-2i(4+7i)}{5} = \frac{4+14}{5} - \frac{i}{5}$$

$$\therefore x + yi = \frac{18}{5} - \frac{i}{5} \qquad \therefore x = \frac{18}{5}, \quad y = -\frac{1}{5}$$

b) $(x - yi)(4 + i) = 14 - 5i \Leftrightarrow x - yi = \dfrac{14-5i}{4+i}$

$$\Rightarrow x - yi = (14-5i)\left(\frac{1}{4+i}\right) = (14+5i)\left(\frac{4}{4^2+1^2} - \frac{i}{4^2+1}\right)$$

$$\Rightarrow x - yi = \frac{4(14-5i) - i(14-5i)}{17} = \frac{51}{17} - \frac{34i}{17}$$

$$\therefore x = 3, \quad y = \frac{34}{17} = 2$$

c) $\dfrac{x+yi}{3-i} = \dfrac{1}{5} + \dfrac{2}{5}i \Leftrightarrow x + yi = (3-i)\left(\dfrac{1}{5} + \dfrac{2}{5}i\right)$

$$\Rightarrow x+yi=(3-i)\left(\frac{1+2}{5}i\right)=\frac{3(1+2i)-i(1+2i)}{5}$$

$$\Rightarrow x+yi=\frac{3+6i-i+2}{5}=\frac{5}{5}+\frac{5i}{5}=1+i$$

$\therefore$ x = 1 and y = 1

d) $(x + yi)i + 2 - 3x - y + 4i = 0$
$\Leftrightarrow (x + yi)i = -2 + 3x + y - 4i$
$\Leftrightarrow ix - y = -2 + 3x + y - 4i$
$\Rightarrow ix = -4i$ and $-y = -2 + 3x + y$
$\Rightarrow x = -4$ and $-2y = -2 + 3x$, put $x = -4$
$\Rightarrow -2y = -2 + 3(-4) = -14$
$\Rightarrow x = -4$ and $-2y = -14, \Rightarrow y = 7$
$\therefore x = -4$ and $y = 7$

e) $2x + yi = (3 + 5i)(2 - 4i) = 6 - 12i + 10i + 20$
$\Rightarrow 2x + yi = 26 - 2i \Rightarrow 2x = 26$ and $y = -2$
$\therefore x = 13$ and $y = -2$

f) $$\frac{3+i}{x+yi}=\frac{1}{2}+\frac{1}{2}i \Leftrightarrow 3+i=(x+yi)\left(\frac{1+i}{2}\right)$$

$$\Leftrightarrow x+yi=(3+i)\cdot\frac{2}{1+i}=(6+2i)\left(\frac{1}{1+i}\right)$$

$$\Leftrightarrow x+yi=(6+2i)\cdot\left(\frac{1}{1^2+1^2}-\frac{i}{1^2+1^2}\right)$$

$$\Rightarrow x+yi=\frac{6+2i}{2}-\frac{i(6+2i)}{2}=3+i-3i+1$$

$\Rightarrow x + yi = 4 - 2i \quad \therefore x = 4, y = -2$

g) $$\frac{i+2}{x-4yi}=3i^{23} \Leftrightarrow 2 + i = 3i^{23}(x - 4yi)$$

$$\Rightarrow x-4yi=\frac{2+i}{3i^{23}}$$ Note, $i^{23} = i^3 = -i$

$$\Rightarrow x-4yi=\frac{2+i}{-3i}=\left(\frac{2+i}{-3i}\right)\left(\frac{i}{i}\right)=\frac{2i-1}{3}$$

$$\Rightarrow x-4yi=-\frac{1}{3}+\frac{2i}{3}$$

$$\Rightarrow x=\frac{-1}{3}\ and\ -4y=\frac{2}{3}\Rightarrow y=-\frac{1}{6}$$

$$\therefore x=\frac{-1}{3}\ and\ y=-\frac{1}{6}$$

h) $$\frac{2+3i}{3x-4yi}=i+1\Leftrightarrow 2+3i=(i+1)(3x-4yi)$$

$$\Rightarrow 3x-4yi=\frac{2+3i}{1+i}=(2+3i)\left(\frac{1}{1+i}\right)$$

$$=2+3i\left(\frac{1}{1^2+1^2}-\frac{i}{1^2+1^2}\right)$$

$$\Rightarrow 3x-4yi=(2+3i)\left(\frac{1}{2}-\frac{i}{2}\right)=\frac{2+3i-2i+3}{2}$$

$$\Rightarrow 3x-4yi=\frac{5}{2}+\frac{i}{2}$$

$$\Rightarrow 3x=\frac{5}{2}\ and\ -4y=\frac{1}{2}$$

$$\therefore x=\frac{5}{6}\ \ and\ \ y=-\frac{1}{8}$$

i) $(x+yi)^2=i^5\Leftrightarrow x^2+2xyi-y^2=i$

$\Rightarrow x^2-y^2=0$ and $2xy=1$

$\Rightarrow x=y$ or $x=-y$ and $2xy=1$

Substitute $y=x$ and $2xy=1$

$\Rightarrow 2x\cdot x=1\Rightarrow 2x^2=1$

$$\Rightarrow x^2=½\Rightarrow x=\pm\frac{1}{\sqrt{2}}=\pm\frac{\sqrt{2}}{2}$$

Since $x=y$, therefore, $y=\pm\frac{\sqrt{2}}{2}$, $\therefore x=\pm\frac{\sqrt{2}}{2}$ and $y=\pm\frac{\sqrt{2}}{2}$

Complex Conjugate

Definition: The complex conjugate of a complex number $z = x + yi$ denoted by $\bar{z}$ is given by $\bar{z} = x - yi$

Properties of complex conjugate

i) If $z = x + yi$, then

a) $\bar{z} = x - yi$

b) $z + \bar{z} = x + yi + x - yi = 2x$

c) $z - \bar{z} = x + yi - (x - yi) = 2yi$

d) $z \cdot \bar{z} = (x + yi)(x - yi) = x^2 + y^2$

ii) If z_1 and z_2 be complex number then

a) $\bar{\bar{z}}_1 = z_1$

b) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

c) $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$

d) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, z_2 \neq 0$

e) $\overline{(z^n)} = (\bar{z})^n$

Definition: Modulus (the absolute value of complex number $z = x + yi$, denoted by $|z|$ is defined to be $|z| = \sqrt{x^2 + y^2}$

Note: If $z_1 = x + yi$ and $z_2 = a + bi$, then

$$|z_1 - z_2| = |(x - a) + (y - b)i| = \sqrt{(x-a)^2 + (y-b)^2}$$

Properties of Modulus

- $z \cdot \bar{z} = |z|^2$
- $|z^n| = |z|^n,\ n \in R$
- $|z| = |\bar{z}|$
- $|z_1 + z_2| \leq |z_1| + |z_2|$
- $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
- $|z_1 - z_2| \geq |z_1| - |z_2|$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

Illustrative Example

8. Find the conjugate of the expression

a) $z = -3 + 4i$

b) $z = 1 - 2i$

d) $z = \frac{2+3i}{4-2i}$

e) $z = \frac{1}{1+i}$

c) $z = x + yi$

f) $z = \dfrac{2i}{2+i} + \dfrac{3}{2-i}$

Solution: Conjugate of $z = x + yi$ is given by $\bar{z} = x - yi$, thus

a) $z = -3 + 4i$ then $\bar{z} = -3 - 4i$

b) $z = 1 - 2i$, then $\bar{z} = 1 + 2i$

c) $z = x + yi$, then $\bar{z} = x - yi$

d) we use $\left(\dfrac{\bar{z}}{z_2}\right) = \dfrac{\bar{z}_1}{\bar{z}_2}$, thus

$$\bar{z} = \overline{\left(\frac{2+3i}{4-2i}\right)} = \frac{\overline{2+3i}}{4-2i} = \frac{2-3i}{4+2i}$$

e) $$\bar{z} = \left(\frac{\bar{1}}{1+i}\right) = \frac{\bar{1}}{\overline{1+i}} = \frac{1}{1-i} = \frac{1}{2} + \frac{i}{2}$$

f) Exercise left for you (Answer $\bar{z} = \dfrac{9}{5} - \dfrac{6i}{5}$)

8. Find the modulus of the expression

a) $z = 2 + 3i$

b) $z = 2 - 6i$

c) $z = 3i^{497}$

d) $z = \dfrac{(1-i)^{20}}{(8+8\sqrt{3}i)(1-\sqrt{3}i)^6}$

e) $z = \sqrt{2+3i}$

f) $z = \sqrt[3]{8+8\sqrt{3}i}$

Solution: we know, if $z = x + yi$ then $|z| = \sqrt{x^2 + y^2}$, hence

a) $$|z| = |2+3i| = \sqrt{2^2+3^2} = \sqrt{4+9} = \sqrt{13}$$

b) $$|z| = |2-6i| = \sqrt{2^2+6^2} = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$$

c) $$|z| = |3i^{497}| = |3i| = \sqrt{0^2+3^2} = 3$$

d) $$|z| = \frac{|1-i|}{|8+8\sqrt{3}i||1-\sqrt{3}i|} = \frac{\left(\sqrt{1^2+(1)^2}\right)^{20}}{\sqrt{8^2+(8\sqrt{3})^2}\cdot\left(\sqrt{(1)^2+\left(\sqrt{-3}\right)^2}\right)^6}$$

$$= \frac{\left(\sqrt{2}\right)^{20}}{16 \cdot 2^6} = \frac{2^{10}}{2^4 \cdot 2^6} = 1$$

$$\therefore |z| = 1$$

e) $|z| = |\sqrt{2+3i}| = |2+3i|^{\frac{1}{2}} = \left(\sqrt{2^2+3^2}\right)^{\frac{1}{2}} = \left(\sqrt{13}\right)^{\frac{1}{2}} = \sqrt[4]{13}$

f) $|z| = \left|\sqrt[3]{8+8\sqrt{3}i}\right| = \left|8+8\sqrt{3}i\right|^{\frac{1}{3}} = \left(\sqrt{8^2+(8\sqrt{3})^2}\right)^{\frac{1}{3}}$

$$= \sqrt[3]{16} = 2\sqrt[3]{2}$$

9. Simplifying the indicated operation and write the result in the form x + yi

a) $\frac{2}{1+i} - \frac{3}{1-i}$

b) $\frac{(2-3i)(5i)}{2+3i}$

c) $\frac{2i}{2+i} + \frac{3}{2-i}$

d) $-6i^3 + i^2$

e) $(1-2i)^2 - (1+2i)^2$

f) $i^{44} + i^{150} - i^{74} + i^{109} + i^{61}$

g) $(3+\sqrt{-5})(7-\sqrt{-10})$

h) $-i(\sqrt{-4}-1)$

i) $4i^2 - 2i^3$

j) $\left(\sqrt{-2}\right)^6 + \left(\sqrt{-5}\right)^3$

Solution:

a) $\frac{2}{1+i} - \frac{3}{1-i} = \frac{2(1-i)-3(1+i)}{(1+i)(1-i)} = \frac{2-2i-3-3i}{1^2+1^2} = \frac{-1}{2} - \frac{5i}{2}$

b) $\frac{(2-3i)(5i)}{2+3i} = \frac{10i+16}{2+3i} = \frac{16+10i}{2+3i} \cdot \left(\frac{2-3i}{2-3i}\right)$

$$= \frac{16(2-3i)+10i(2-3i)}{2^2+3^2}$$

$$= \frac{32-48i+20i-30}{13} = \frac{2}{13} - \frac{28i}{13}$$

c) $\frac{2i}{2+i} + \frac{5}{2-i} = \frac{2i(2-i)+5(2+i)}{(2+i)(2-i)} = \frac{4i+2+10+5i}{2^2+1^2} = \frac{12}{5} + \frac{9i}{5}$

d) $-6i^3 + i^2 = -6(-i) - 1 = -1 + 6i$

e) $(1-2i)^2 - (1+2i) = 1^2 - 4i + (2i)^2 - (1^2 + 4i + (2i)^2)$
$= 1 - 4i - 4 - 4i + 4 = 0 - 8i$

f) $i^{44} + i^{50} - i^{74} + i^{109} + i^{61} = i^4 + i^2 - i^2 + i + i = 1 + 2i$

g) $(3+\sqrt{-5})(7-\sqrt{-10}) = (3+\sqrt{5}i)(7-\sqrt{10}i)$
$= 3(7-\sqrt{10}i) + \sqrt{5}i(7-\sqrt{10}i)$
$= 21 - 3\sqrt{10}i + 7\sqrt{5}i + 5\sqrt{2}$
$= 21 + 3\sqrt{10}i + 7\sqrt{5}i + 5\sqrt{2}$

h) $-i(\sqrt{-4}-1) = -i(2i-1) = 2+i$

i) $4i^2 - 2i^3 = 4(-1) - 2(-i) = -4 + 2i$

j) $\left(\sqrt{-2}\right)^6 + \left(\sqrt{-5}\right)^3 = \left(\sqrt{2}i\right)^6 + \left(\sqrt{5}i\right)^3 = -8 - 5\sqrt{5}i$

10. Solve

a) $(3 + 4i)^2 - 2(x - iy) = x + iy$

b) $\left(\frac{1+i}{1-i}\right)^2 + \frac{1}{x+yi} = 1+i$

c) $(3 - 2i)(x + yi) = 2(x - 2iy) + 2i - 1$

Solution:

a) $(3 + 4i)^2 - 2(x - iy) = x + iy$
$\Leftrightarrow 9 + 24i - 16 - 2x + 2iy = x + iy$
$\Rightarrow 24i + 2iy - iy = x + 2x - 9 + 16$
$\Rightarrow 24i + iy = 3x + 7 + 0i$
$\Leftrightarrow (y + 24)i + 0 = (3x + 7) + 0i$
$\Leftrightarrow y + 24 = 0$ and $3x + 7 = 0$
$\therefore y = -24$ and $x = \frac{-7}{3}$

b) $\left(\frac{1+i}{1-i}\right)^2 + \frac{1}{x+yi} = 1+i \Leftrightarrow \frac{1}{x+yi} = 1+i-\left(\frac{1+i}{1-i}\right)^2$

$\Leftrightarrow \frac{1}{x+yi} = 1+i-\left(\frac{1+i}{1-i}\cdot\frac{1+i}{1+i}\right)^2$

$$\Leftrightarrow \frac{1}{x+yi} = 1+i-\left(\frac{2i}{2}\right)^2 \Rightarrow \frac{1}{x+yi} = 1+i+1$$

$$\Rightarrow \frac{1}{x+yi} = 2+i \Leftrightarrow x+yi = \frac{1}{2+i} = \frac{1}{2+i}\cdot\frac{2-i}{2-i}1 = \frac{2}{5} - \frac{i}{5}$$

$$\therefore x = \frac{2}{5} \text{ and } y = -\frac{1}{5}$$

c) $(3-2i)(x+yi) = (2x-4iy) + 2i - 1$

$\Rightarrow (3-2i)(x+yi) = (2x-1) + (2i-4iy)$

$\Rightarrow 3(x+yi) - 2i(x+yi) = (2x-1) + (2-4y)i$

$\Rightarrow 3x + 3iy - 2ix + 2y = (2x-1) + (2-4y)i$

$\Rightarrow (3x+2y) + (3y-2x)i = (2x-1) + (2-4y)i$

$\Rightarrow 3x + 2y = 2x - 1$ and $3y - 2x = 2 - 4y$

$\Rightarrow x + 2y = -1$ and $7y - 2x = 2$

Solving together $\begin{cases} -2x+7y=2 \\ x+2y=-1 \end{cases} \Rightarrow \begin{cases} -2x+7y=2 \\ 2x+4y=-2 \end{cases}$

$11y = 0 \Rightarrow y=0,\ x=-1$

11. Find a) $\left|4i^2 - 2i^3\right|$ b) $|-6i^3 + i^2|$ c) $\left|\frac{-\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right|^{100}$

Solution:

a) $|4i^2 - 2i^3| = |-4 + 2i| = \sqrt{(-4)^2 + (2)^2} = \sqrt{20} = 2\sqrt{5}$

b) $|-6i^3 + i^2| = |6i - 1| = \sqrt{6^2 + (-1)^2} = \sqrt{36+1} = \sqrt{37}$

c) $\left|\frac{-\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right| = \left(\sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2}\right)^{100} = \left(\sqrt{\frac{2}{4}+\frac{2}{4}}\right)^{100} = 1$

12. Solve each of the following: let $z = x + yi$

a) $x^2 + 4x + 20 = 0$

b) $z^2 - i = 0$

c) $z^2 = -5 + 12i$

d) $z = \sqrt{3-4i}$

e) $z = \sqrt{2i}$

Solution: Here

a) $x^2+4x+20=0 \Leftrightarrow x^2+4x+4+20-4=0$

$\Rightarrow (x+2)^2=-16$

$\Rightarrow x+2=\pm\sqrt{-16} \Rightarrow x+2=\pm 4i \Rightarrow x=-2\pm 4i$

$\therefore$ S.S = {-2 -4i, -2 + 4i}

b) $z^2-i=0 \Leftrightarrow z^2=i \Leftrightarrow z=\sqrt{i}$

$\Rightarrow (x+yi)^2=i \Rightarrow x^2+2ixy-y^2=i+0$

$\Rightarrow x^2-y^2=0$ and $2ix=i$

$\Rightarrow x=y$ and $2xy=1$ put x in y

$$\Rightarrow 2x^2=1 \Rightarrow x^2=\frac{1}{2} \Rightarrow x=\pm\frac{1}{\sqrt{2}}=\pm\frac{\sqrt{2}}{2}$$

Since $x=y$, $\therefore x=\pm\frac{\sqrt{2}}{2}$ and $y=\pm\frac{\sqrt{2}}{2}$

$$\therefore x+iy=\pm\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)$$

$$\therefore \text{S.S}=\left\{\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i,-\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i\right\}$$

c) $z^2=-5+12i \Leftrightarrow (x+yi)^2=-5+12i$

$\Rightarrow x^2+2xyi-y^2=-5+12i$

$$\Rightarrow x^2-y^2=-5 \text{ and } 2xy=12 \Rightarrow y=\frac{12}{2x}=\frac{6}{x}$$

$$\Rightarrow x^2-\left(\frac{6}{x}\right)^2=-5 \Rightarrow x^2-\frac{36}{x^2}=-5 \Rightarrow \frac{x^4-36}{x^2}=-5$$

$\Rightarrow x^4+5x^2-36=0$

$$\Rightarrow x^2=\frac{-5\pm\sqrt{25-4(36)}}{2}=\frac{-5\pm\sqrt{169}}{2}=\frac{-5\pm 13}{2}$$

$$\Rightarrow x^2=\frac{-5+13}{2}=\frac{8}{2}=4$$

$$\Rightarrow x=\pm\sqrt{4}=\pm 2 \text{ and } y=\frac{6}{\pm 2}=\pm 3$$

$\therefore$ the ***Solution*** set are z = {2 + 3i, -2-3i}

d) $z=\sqrt{3-4i} \Leftrightarrow z^2=3-4i \Leftrightarrow (x+yi)^2=3-4i$

$\Rightarrow x^2+2xyi-y^2=3-4i$

$\Rightarrow x^2-y^2=3$ and $2xy=-4 \Rightarrow y=\frac{-2}{x}$

$\Rightarrow x^2\left(\frac{-2}{x}\right)^2=3 \Rightarrow x^2-\frac{4}{x^2}=3 \Rightarrow x^4-3x^2-4=0$

$\Rightarrow x^2=\frac{3\pm\sqrt{9-4(-4)}}{2}=\frac{3\pm5}{2}=\frac{3+5}{2}=4$

$\Rightarrow x^2=4 \Rightarrow x=2$ or -2 and $y=\frac{-2}{x}=\frac{-2}{2}=-1$

$y=\frac{-2}{-2}=1$

$\therefore z=2-i$ or $z=-2+i$

$\therefore$ S.S $=\{2-i, -2+i\} \longrightarrow \sqrt{3-4i}=2-i$ or $-2+i$

e) $z=\sqrt{2i} \Leftrightarrow z^2=2i \Leftrightarrow (x+yi)^2=2i$

$\Rightarrow x^2+2xyi-y^2=2i$

$\Rightarrow x^2-y^2=0$ and $2xy=2 \Rightarrow y=\frac{1}{x}$

$\Rightarrow x^2-\left(\frac{1}{x}\right)^2=0 \Rightarrow x^2-\frac{1}{x^2}=0$

$\Rightarrow x^4-1=0 \Rightarrow (x^2-1)(x^2+1)=0$

$\Rightarrow x^2-1=0$

$\Rightarrow x=1$ and $y=1$ and $x=-1, y=-1$

$\therefore z=1+i$ or $z=-1-i$

$\therefore$ S.S $=\{1+i, -1-i\}$

13. Find the cube root of each of the following.

a) 1 b) -1 c) -8 d) 27 e) 64

Solution: a) $z=\sqrt[3]{1} \Leftrightarrow z^3=1$

$\Rightarrow z^3-1=0 \Leftrightarrow (z-1)(z^2+z+1)=0$

$\Rightarrow z-1=0$ or $z^2+z+1=0$

$\Rightarrow z = 1$ or $z = \dfrac{-1 \pm \sqrt{1-4(1)}}{2} = \dfrac{-1 \pm \sqrt{3}i}{2}$

$\therefore$ S.S $= \left\{1, \dfrac{-1-\sqrt{3}i}{2}, \dfrac{-1+\sqrt{3}i}{2}\right\}$ ← cube root of 1

b) $z^3 + 1 = 0 \Leftrightarrow (z+1)(z^2 - z + 1) = 0$

$\Rightarrow z = -1$ or $z^2 - z + 1 = 0$

$\Rightarrow z = -1$ or $z = \dfrac{1 \pm \sqrt{1-4(1)}}{2} = \dfrac{1 \pm \sqrt{3}i}{2}$

$\therefore$ S.S $= \left\{-1, \dfrac{1-\sqrt{3}i}{2}, \dfrac{1+\sqrt{3}i}{2}\right\}$ ← cube root of -1

c) $z^3 + 8 = 0 \Leftrightarrow (z+2)(z^2 - 2z + 4) = 0$

$\Rightarrow z + 2 = 0$ and $z^2 - 2z + 4 = 0$

$\Rightarrow z = -2$ and $z = \dfrac{2 \pm \sqrt{4-4(4)}}{2} = \dfrac{2 \pm 2\sqrt{3}i}{2}$

$\therefore$ S.S $= \left\{-2, 1+\sqrt{3}i, 1-\sqrt{3}i\right\}$ ← cube root of -8

d) 27, 64 ← exercise left for you

14. Find the fourth root of

a) 1 b) 16 c) 81 d) 4 e) 3

Solution: a) $z = \sqrt[4]{1} \Leftrightarrow z^4 = 1 \Leftrightarrow z^4 - 1 = 0$

$\Rightarrow z^4 - 1 = 0 \Leftrightarrow (z^2 - 1)(z^2 + 1) = 0$

$\Rightarrow (z-1)(z+1)(z+i)(z-i) = 0$

$\Rightarrow z = 1, -1, -i, i$

$\therefore$ S.S $= \{1, -1, -i, i\}$ ← cube root of 1

b) $z = \sqrt[4]{16} \Leftrightarrow z^4 = 16 \Leftrightarrow z^4 - 16 = 0$

$\Rightarrow (z^2 - 4)(z^2 + 4) = 0 \Rightarrow z^2 - 4 = 0$ and $z^2 + 4 = 0$

$\Rightarrow (z-2)(z+2)(z+2i)(z-2i) = 0$

$\Rightarrow z = 2, -2, -2i, 2i$

$\therefore$ S.S $= \{2, -2, -2i, 2i\}$ ← 4^{th} root of 16

c) $z = \sqrt[4]{81} \Leftrightarrow z^4 = 81 \Leftrightarrow z^4 - 81 = 0$

$\Rightarrow (z^2 - 9)(z^2 + 9) = 0 \Rightarrow (z - 3(z+3)(z+3i)(z-3i)$

$\Rightarrow z = 3, -3, -3i, 3i$

$\therefore$ S.S = $\{3,-3,-3i,3i\}$

d) $z=\sqrt[4]{4} \Leftrightarrow z^4=4 \Leftrightarrow z^4-4=0 \Leftrightarrow (z^2-2)(z^2+2)=0$

$\Rightarrow (z-\sqrt{2})(z+\sqrt{2})(z+\sqrt{2}i)(z-\sqrt{2}i)=0$

$\therefore$ S.S = $\{\sqrt{2},-\sqrt{2},-\sqrt{2}i,\sqrt{2}i\}$

e) $z=\sqrt[4]{3} \Leftrightarrow z^4=3 \Leftrightarrow z^4-3=0$

$\Rightarrow (z^2-\sqrt{3})(z^2+\sqrt{3})=0 \Rightarrow (z-\sqrt{3})(z+\sqrt{3})(z+\sqrt{3}i)(z-\sqrt{3}i)=0$

$\therefore$ S.S = $\{\sqrt{3},-\sqrt{3},-\sqrt{3}i,\sqrt{3}i\}$ ⟵ fourth root of 3

15. Find the sixth root of:

a) 1 b) 64

Solution: Here

a) $z=\sqrt[6]{1} \Leftrightarrow z^6=1 \Leftrightarrow z^6-1=0$

$\Rightarrow (z^3)^2-1^2=0 \Rightarrow (z^3-1)(z^3+1)=0$

$\Rightarrow (z-1)(z^2+z+1)(z+1)(z^2-z+1)=0$

$\Rightarrow z=1$ and $z=\dfrac{-1\pm\sqrt{1-4}}{2}, z=\dfrac{1\pm\sqrt{1-4}}{2}$

$\Rightarrow z=1,-1,\dfrac{-1-\sqrt{3}i}{2},\dfrac{-1+\sqrt{3}i}{2},\dfrac{1+\sqrt{3}i}{2},\dfrac{1-\sqrt{3}i}{2}$

$\therefore$ S.S = $\left\{1,-1,\dfrac{-1-\sqrt{3}i}{2},\dfrac{-1+\sqrt{3}i}{2},\dfrac{1+\sqrt{3}i}{2},\dfrac{1-\sqrt{3}i}{2}\right\}$

b) $z=\sqrt[6]{64} \Leftrightarrow z^6=64 \Leftrightarrow z^6-64=0$

$\Rightarrow (z^3)^2-8^2=0 \Rightarrow (z^3-8)(z^3+8)=0$

$\Rightarrow (z-2)(z+2)(z^2+2z+4)(z^2-2z+4)=0$

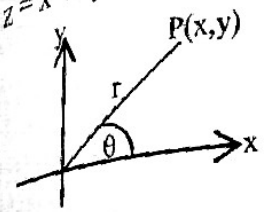
$\therefore$ z = 2, -2 and $z=\dfrac{-2\pm\sqrt{4-16}}{2}$ *and* $z=\dfrac{2\pm\sqrt{4-16}}{2}$

$\therefore$ S.S = $\{2,-2,-1-\sqrt{3}i,-1+\sqrt{3}i,1-\sqrt{3}i,1+\sqrt{3}i\}$ ⟵ sixth root of 64

Argand diagram and polar representation of complex number

Argand Diagrams

There are two geometrical representation of the complex number $z = x + iy$



i) as the point p(x, y) in the xy plane
ii) as the vector from the origin to P(polar representation)

In each representation, the x-axis is called real axis and the y-axis is called the imaginary axis.

- A coordinate plane with a complex number assigned to each point is referred to as a complex (or Argand) plane instead of an xy-plane.
- Geometrical interpretation of a complex number z = x+yi in complex plane:

The length of the line segment from the origin to the point P(x, y) in the complex plane is $|z| = \sqrt{x^2 + y^2}$

The distance between the points to z_1 and z_2 in the complex plane is $|z_1 - z_2|$

Example: The point corresponding to the complex numbers, z with |z| = 1 are on the unit circle.

16. Describe each of the following geometrically (graph, z = x + yi).

a) $|z-2| = 3$ c) $|z+2-i| = 1$

b) $|z+i| < 2$ d) $|z+2-i| > 1$

Solution: a) $|z-2| = 3 \Rightarrow |x+yi - 2| = 3 \Rightarrow |(x+2) + yi| = 3$

$$\Rightarrow \sqrt{(x-2)^2 + y^2} = 3 \Rightarrow (x-2)^2 + y^2 = 9$$

Point on the circle $(x-2)^2 + y^2 = 9$ with center (2, 0), r = 3

b) $|z+i| < 2 \Rightarrow |x + yi + i| < 2 \Rightarrow |x+(y+1)i| < 2$

$$\Rightarrow \sqrt{x^2 + (y+1)^2} < 2$$

$$\Rightarrow (x^2 + (y+1)^2) < 4$$

$\therefore$ Point inside the circle $x^2 + (y+1)^2 = 4$

c) $|z+2-i| = 1 \Rightarrow |x + yi + 2 - i| = 1 \Rightarrow |(x+2)+(y-1)i| = 1$

$\Rightarrow (x+2)^2 + (y-1)^2 \; ? \; 1$

$\therefore$ Point outside the circle $(x+2)^2 + (y-1)^2 = 1$

17. Describe each of the following geometrically

a) $|z-1| = |z+2|$ c) $|z+2i| = |z-3|$

b) $|z+i| = |z-1|$ d) $|z+1| \geq |z|$

Solution: a) $|z-1| = |z+2| \Rightarrow |(x-1)+yi| = |(x+2)+yi|$

$\Rightarrow (x-1)^2 + y^2 = (x+2)^2 + y^2$

$\Rightarrow x^2 - 2x + 1 + y^2 = x^2 + 4x + 4 + y^2$

$\Rightarrow -6x = 3 \Rightarrow x = \frac{-3}{6} = \frac{-1}{2}$

$\therefore$ Points on the line $x = \frac{-1}{2}$

b) $|z+i| = |z-1| \Rightarrow |(x+(y+1)i| = |(x-1)+yi|$

$\Rightarrow x^2 + (y+1)^2 = (x-1)^2 + y^2 \Rightarrow 2y+1 = -2x+1$

$\Rightarrow y = -x \longrightarrow$ point on the line $y = -x$

c) $|z+2i| = |z-3| \Rightarrow |x+(y+2)i| = |(x-3)+yi|$

$\Rightarrow x^2 + (y+2)^2 = (x-3)^2 + y^2 \Rightarrow 4y+4 = -6x+9$

$\therefore$ point on the line $6x + 4y = 5$

18. How may following complex numbers, be obtained from $z = x + yi$ geometrically

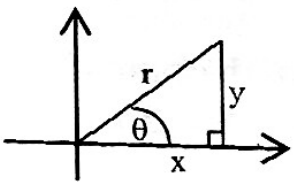
a) $\bar{z}$ b) $\overline{(-z)}$

Solution: $z = x + yi \Rightarrow \bar{z} = x - yi$

$\therefore$ a) $\bar{z} = x - yi$, by reflecting z on real x – axis

b) $\overline{(-z)} = -x + yi$ by reflecting z on imaginary (y) axis.

ii. Polar representation of complex Number (Trigonometric form of a complex number)



- A complex number $z = x + yi$ can be write interms of trigonometric

$\Rightarrow x = r\cos\theta$, $|z| = r = \sqrt{x^2 + y^2}$

$\Rightarrow y = r\sin\theta$

$\therefore z = x + yi = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta)$

This is called polar representation of z

- The polar angle θ is called the argument of z and is written

$$\theta = \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

- r is the modulus of a complex number z.

- $r=\sqrt{x^2+y^2}$
- $r(\cos\theta+i\sin\theta)=r(\cos(\theta+2k\pi)+i\sin(\theta+2k\pi)$
  Where K is integer

Principal argument of z = x + yi

The principal argument is polar angle θ which is denoted by arg(z) is an angle between $-\pi < \arg(z) \le \pi$

- The principal argument of z is **calculated by:**

$$Arg(z)=\theta=\begin{cases}\tan^{-1}\left(\frac{y}{x}\right) & \text{if } z=x+yi \text{ lies on } 1^{st} \text{ quadrant}\\ \pi+\tan^{-1}\left(\frac{y}{x}\right) & \text{if } z=x+yi \text{ lies } 2^{nd} \text{ quadrant } (x<0,y>0)\\ \tan^{-1}\left(\frac{y}{x}\right)-\pi & \text{if } z=x+yi \text{ lies on } 3^{rd} \text{ quadrant } (x<0,y<0)\\ \tan^{-1}\left(\frac{y}{x}\right) & \text{if } z=x+yi \text{ lies on } 4^{th} \text{ quadrant}\end{cases}$$

Note: $\tan^{-1}(-x)=-\tan^{-1}(x)$

Note: If $z_1=r_1(\cos\theta_1+i\sin\theta_1)$, $z_2=r_2(\cos\theta_2+i\sin\theta_2)$
then $z_1=z_2 \Leftrightarrow r_1=r_2$ and $\theta_1=\theta_2+2\pi k$, $k\in z$.

Example:

a) $2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)=2\left(\cos\frac{13\pi}{6}+i\sin\frac{13\pi}{6}\right)=2\left(\cos\frac{11\pi}{6}-i\sin\frac{11\pi}{6}\right)$

b) $3\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)=3\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)=3\left(\cos\frac{7\pi}{4}-i\sin\frac{7\pi}{4}\right)$

c) $6\left(\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}\right)=6\left(\cos\frac{11\pi}{5}+i\sin\frac{11\pi}{5}\right)=6\left(\cos\frac{9\pi}{5}-i\sin\frac{9\pi}{5}\right)$

d) $4\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)=4\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)=z=-2-2\sqrt{3}i$

19. Find the principal argument, and polar form of each of the following.

a) $z=1+i$

b) $z=1+\sqrt{3}i$

c) $z=3\sqrt{3}+3i$

d) $z=4+3i$

Solution: since all of them are in first quadrant thus, the principal argument is given by.

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$, therefore

a) $\text{Arg}(1+i) = \theta = \tan^{-1}\left(\frac{1}{1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$ ← principal argument

Modulus of z, $r = \sqrt{1^2+1^2} = \sqrt{2}$

$\therefore \left(\sqrt{2}, \frac{\pi}{4}\right) = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

c) $\text{Arg}(1+\sqrt{3}i) = \theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \tan^{-1}\left(\sqrt{3}\right) = \frac{\pi}{3}$ ← principal argument

$r = \sqrt{1^2+\sqrt{3}^2} = \sqrt{1+3} = 2$

$\therefore (r,\theta) = \left(2, \frac{\pi}{3}\right) = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

c) $\text{Arg}\left(3\sqrt{3}+3i\right) = \theta = \tan^{-1}\left(\frac{3}{3\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$ ← principal argument

$r = \sqrt{\left(3\sqrt{3}\right)^2+3^2} = 6$

$\therefore (r,\theta) = \left(6, \frac{\pi}{6}\right) = 6\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

d) $\text{Arg}(4+3i) = \theta = \tan^{-1}\left(\frac{3}{4}\right)$

$r = \sqrt{3^2+4^2} = 5$

$\therefore (r,\theta) = \left(5, \tan^{-1}\left(\frac{3}{4}\right)\right) = 5\left(\cos\left(\tan^{-1}\left(\frac{3}{4}\right)\right) + i\sin\left(\tan^{-1}\left(\frac{3}{4}\right)\right)\right)$

20. Find the principal argument and express in polar form

a) $z = -2 + 2i$ b) $z = -1 + \sqrt{3}i$ c) $-2\sqrt{3}+2i$

Solution: $z = -2 + 2i$, $z = -1 + \sqrt{3}i$ and $z = -2\sqrt{3} + 2i$ are found in **quadrant two.**
Therefore, the principal argument is given by

$\text{Arg}(z) = \pi + \tan^{-1}\left(\frac{y}{x}\right)$, hence

a) $\text{Arg}(-2 + 2i) = \theta = \pi + (\tan^{-1}\left(\frac{2}{-2}\right) = \pi + -\tan^{-1}(1)$

$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ ⟵ principal argument

$r = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$

$\therefore (r, \theta) = \left(2\sqrt{2}, \frac{3\pi}{4}\right) = 2\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

b) $\text{Arg}\,(1+\sqrt{3}i) = \theta = \pi + \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right) = \pi - \tan^{-1}\left(\sqrt{3}\right) = \pi - \frac{\pi}{3} = \frac{2}{3}\pi$

$r = \sqrt{(-1)^2 + \sqrt{3}^2} = \sqrt{1+3} = 2$

$\therefore (r, \theta) = \left(2, \frac{2}{3}\pi\right) = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

$\text{Arg}(-2\sqrt{3}+2i) = \theta = \pi + \tan^{-1}\left(\frac{2}{-2\sqrt{3}}\right) = \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5}{6}\pi$

$r = \sqrt{\left(-2\sqrt{3}\right)^2 + 2^2} = \sqrt{12+4} = \sqrt{16} = 4$

$\therefore (r, \theta) = \left(4, \frac{5}{6}\pi\right) = 4\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{6}\right)$

Find the principal argument and express in polar form:

a) $z = -2 - 2i$ b) $z = -2\sqrt{3} - 2i$ c) $z = -4 - 4\sqrt{3}i$

Solution: $z = -2 - 2i$, $z = -2\sqrt{3} - 2i$) *and* $z = -4 - 4\sqrt{3}i$ are all in 3rd quadrant, therefore the principal argument θ is given by

$\text{Arg}(z) = \theta = \tan^{-1}\left(\frac{y}{x}\right) - \pi$, Hence

a) $\text{Arg}(-2-2i) = \theta = \tan^{-1}\left(\frac{-2}{-2}\right) - \pi = \tan^{-1}(1) - \pi = \frac{\pi}{4} - \pi = \frac{-3\pi}{4}$

$r = \sqrt{(-2)^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$

$\therefore (r, \theta) = \left(2\sqrt{2}, \frac{-3\pi}{4}\right) = 2\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) - i\sin\frac{3\pi}{4}\right)$

It is also possible, as: $\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$

But $\frac{5\pi}{4}$ is not principal argument.

b) $\text{Arg}(-2\sqrt{3} - 2i) = \theta = \tan^{-1}\left(\frac{-2}{-2\sqrt{3}}\right) - \pi = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - \pi$

$\Rightarrow \frac{\pi}{6} - \pi = \frac{5\pi}{6} \leftarrow$ principal argument

$r = \sqrt{\left(-2\sqrt{3}\right)^2 + (-2)^2} = \sqrt{16} = 4$

$\therefore (r, \theta) = \left(4, \frac{-5\pi}{6}\right) = 4\left(\cos\left(\frac{5\pi}{6}\right) - i\sin\left(\frac{5\pi}{6}\right)\right)$

c) $\text{Arg}(-4 - 4\sqrt{3}i) = \theta = \tan^{-1}\left(\frac{-4\sqrt{3}}{-4}\right) - \pi = \tan^{-1}\left(\sqrt{3}\right) - \pi$

$\Rightarrow \frac{\pi}{3} - \pi = \frac{-2}{3}\pi \longleftarrow$ principle argument

$r = \sqrt{(-4)^2 + (-4\sqrt{3})^2} = \sqrt{64} = 8$

$\therefore (r, \theta) = \left(8, \frac{-2}{3}\pi\right) = 8\left(\cos\frac{2}{3}\pi - i\sin\frac{2\pi}{3}\right)$

22. Find the principal argument and express in polar form.

a) $z = 4 - 4i$ b) $z = 1 - \sqrt{3}i$ c) $z = 2\sqrt{3} - 2i$

Solution: $z = 4 - 4i$, $z = 1 - \sqrt{3}i$ and $z = 2\sqrt{3} - 2i$ are all in 4th quadrant, therefore, the principal argument θ is calculated by $\arg(z) = \theta = \tan^{-1}\left(\frac{y}{x}\right)$. Hence

a) $\text{Arg}(4-4i) = \theta = \tan^{-1}\left(\frac{-4}{4}\right) = -\tan^{-1}(1) = -\left(\frac{\pi}{4}\right)$ ← principal argument

$r = \sqrt{4^2 + (-4)^2} = 8$

$\therefore (r, \theta) = \left(8, \frac{-\pi}{4}\right) = 4\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)$

It is also possible as: $8\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right)$

b) $\text{Arg}(1-\sqrt{3}i) = \theta = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right) = \tan^{-1}\left(-\sqrt{3}\right)$

$= -\tan^{-1}\left(\sqrt{3}\right) = \frac{-\pi}{3}$

$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$

$\therefore (r, \theta) = \left(2, \frac{-\pi}{3}\right) = 2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

Possible like to express:

$\frac{-\pi}{3} + 2\pi = \frac{5\pi}{3} \Rightarrow 2\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)$

c) $\text{Arg}(2\sqrt{3} - 2i) = \theta = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right)$

$= \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{-\pi}{6}$

$r = \sqrt{\left(2\sqrt{3}\right)^2 + (-2)^2} = \sqrt{12+4} = 4$

$$\therefore (r, \theta) = \left(4, \frac{-\pi}{6}\right) = 4\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$$

Possible like to express: $\frac{-\pi}{6} + 2\pi = \frac{11\pi}{6}$

$$\Rightarrow 4\left(\cos\left(\frac{11\pi}{6}\right) + i\sin\left(\frac{11\pi}{6}\right)\right)$$

23. Find the principal argument θ

a) $z = \frac{\sqrt{3}}{2} - \frac{3}{2}i$ b) $z = -4i$ c) $z = -5$

d) $z = 2i$ e) $z = -2 + 2\sqrt{3}i$ f) $z = -3 - 3i$

Solution: a) $z = \frac{\sqrt{3}}{2} - \frac{3}{2}i$ it is on 4th quadrant.

$$\therefore \text{principal, } \arg(z) = \theta = \tan^{-1}\left(\frac{\frac{3}{-2}}{\frac{\sqrt{3}}{2}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\therefore \arg(z) = -\frac{\pi}{6}$$

b) since $z = -4i$ concide with negative y axis

$$\therefore \text{principal, } \arg(-4i) = \theta = \tan^{-1}\left(\frac{-4}{0}\right) = -\frac{\pi}{2}$$

c) since $z = -5 + 0i$ concide with negative x-axis

$\therefore$ principal, $\arg(-5) = \theta = \pi$

d) since $z = 2i$, concide wit positive y-axis

$$\therefore \arg(z) = \arg(2i) = \frac{\pi}{2}$$

e) since $z = -2 + 2\sqrt{3}i$ lies on 2nd quadrant

$$\therefore \arg(-2 + 2\sqrt{3}i) = \theta = \pi + \left(\tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right) = \pi - \tan^{-1}\left(\sqrt{3}\right)\right)$$

$$= \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

f) since $z = -3 - 3i$ lies on 3rd quadrant

$$\therefore \arg(-3-3i) = \theta = \tan^{-1}\left(\frac{-3}{-3}\right) - \pi = \tan^{-1}(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

24. Find the corresponding complex number in rectangular form

a) $\left(4, \frac{2\pi}{3}\right)$ b) $\left(3, \frac{4\pi}{3}\right)$ c) $\left(2, \frac{-\pi}{3}\right)$ d) $(4, \pi)$

e) $\left(7, \frac{3}{2}\pi\right)$ f) $\left(5, \frac{-\pi}{4}\right)$ g) $\left(1\frac{5\pi}{6}\right)$

Solution:

a) $\left(4, \frac{2\pi}{3}\right) = 4\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right) = 4\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -2 + 2\sqrt{3}i$

b) $\left(3, \frac{4\pi}{3}\right) = 3\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) = 3\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = \frac{-3}{2} - \frac{\sqrt{3}}{2}i$

c) $\left(2, \frac{-\pi}{3}\right) = 2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right) = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 1 - \sqrt{3}i$

d) $(4, \pi) = 4(\cos\pi + i\sin\pi) = (-1 + 0i) = -4$

e) $\left(7, \frac{3\pi}{2}\right) = 7\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 7(0 - i) = -7i$

f) $\left(5, \frac{-\pi}{4}\right) = 5\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right) = 5\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = \frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

g) $\left(1, \frac{5\pi}{6}\right) = 1\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

Product and quotient of complex number

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$

$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ be complex number, then

i) $z_1 \cdot z_2 = r_1 \cdot r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$

$$= \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

f) since $z = -3 - 3i$ lies on 3rd quadrant

$$\therefore \arg(-3-3i) = \theta = \tan^{-1}\left(\frac{-3}{-3}\right) - \pi = \tan^{-1}(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

24. Find the corresponding complex number in rectangular form

a) $\left(4, \frac{2\pi}{3}\right)$ b) $\left(3, \frac{4\pi}{3}\right)$ c) $\left(2, \frac{-\pi}{3}\right)$ d) $(4, \pi)$

e) $\left(7, \frac{3}{2}\pi\right)$ f) $\left(5, \frac{-\pi}{4}\right)$ g) $\left(1\frac{5\pi}{6}\right)$

Solution:

a) $\left(4, \frac{2\pi}{3}\right) = 4\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right) = 4\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -2 + 2\sqrt{3}i$

b) $\left(3, \frac{4\pi}{3}\right) = 3\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) = 3\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = \frac{-3}{2} - \frac{\sqrt{3}}{2}i$

c) $\left(2, \frac{-\pi}{3}\right) = 2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right) = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 1 - \sqrt{3}i$

d) $(4, \pi) = 4(\cos\pi + i\sin\pi) = (-1 + 0i) = -4$

e) $\left(7, \frac{3\pi}{2}\right) = 7\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 7(0 - i) = -7i$

f) $\left(5, \frac{-\pi}{4}\right) = 5\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right) = 5\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = \frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

g) $\left(1, \frac{5\pi}{6}\right) = 1\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

Product and quotient of complex number

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$

$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ be complex number, then

i) $z_1 \cdot z_2 = r_1 \cdot r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$

ii) $\frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)$

- **Demoivies theorem**

iii) $z^n = r^n(\cos(n\theta) + i\sin\theta)$ for any integer

iv) $z^{-1} = r^{-1}(\cos\theta - i\sin\theta)$. If $n = -1$

$$\frac{1}{z} = \frac{1}{r}(\cos\theta - i\sin\theta)$$

Note: • $\text{Arg}(z_1 \cdot z_2) = \text{Arg}z_1 + \text{Arg}z_2 + 2\pi k$

• $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2) + 2\pi k$

• $\text{Arg}(z^n) = n\text{Agr}(z) + 2\pi k$, where k, is integer

25. Find the indicated operation and express the results in rectangular form.

i) Let $z_1 = 4(\cos 15 + i\sin 15)$

$z_2 = 3(\cos 45 + i\sin 45)$ then find

a) $z_1 \cdot z_2$ b) $\frac{z_2}{z_1}$ c) $\frac{z_1}{z_2}$ d) $\frac{1}{z_2}$

e) $\text{Arg}(z_1 \cdot z_2)$ f) $\text{Arg}\left(\frac{z_1}{z_2}\right)$ g) $\text{Arg}\left(\frac{1}{z_2}\right)$

Solution: Here

a) $z_1 \cdot z_2 = (4 \cdot 3)(\cos(15^0 + 45^0) + i\sin(15^0 + 45^0))$

$= 12(\cos 60^0 + i\sin 60^0) = 12\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 6 + 6\sqrt{3}i$

b) $\frac{z_2}{z_1} = \frac{3}{4}(\cos(45^0 - 15^0) + i\sin(45^0 - 15^0)$

$= \frac{3}{4}(\cos 30^0 - i\sin 30^0) = \frac{3}{4}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = \frac{3\sqrt{3}}{8} + \frac{3}{8}i$

c) $\frac{z_1}{z_2} = \frac{4}{3}(\cos(15^0 - 45^0) + i\sin(15^0 - 45^0)$

$$= \frac{4}{3}(\cos 30^0 - i\sin 30^0) = \frac{4}{3}\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) = \frac{2\sqrt{3}}{3} - \frac{2}{3}i$$

d) $\frac{1}{z_2} = \frac{1}{3}(\cos 45^0 - i\sin 45^0) = \frac{1}{3}\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = \frac{\sqrt{2}}{6} - \frac{\sqrt{2}}{6}i$

e) $\text{Arg}(z_1 \cdot z_2) = \text{Arg}z_1 + \text{Arg}z_2 = 15^0 + 45^0 = 60^0 + 2\pi k$, k is integer

f) $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2) = 15^0 - 45^0 = -30^0 + 2\pi k$, k is integer

g) $\text{Arg}\left(\frac{1}{z_2}\right) = -1\ \arg(z_2) = -(45^0) = -45^0$

26. Let $z_1 = \sqrt{2}(\cos 84^0 + i\sin 84^0)$
$z_2 = \sqrt{3}(\cos 126^0 + i\sin 126^0)$ then find rectangular from of $z_1 \cdot z_2$:

Solution: $z_1 \cdot z_2 = \sqrt{2} \cdot \sqrt{3}(\cos(84^0+126^0) + i\sin(84^0+126^0))$

$$= \sqrt{6}(\cos 210^0 + i\sin 210^0)$$

$$= \sqrt{6}\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \frac{-3\sqrt{2}}{2} - \frac{\sqrt{6}i}{2}$$

27. Let $z = 1(\cos 20^0 + i\sin 20^0)$ then find rectangular form of $z = x + yi$ and principal argument of each.

a) z^{30} b) z^{-15} c) z^9 d) z^{-12}

Solution: a) $z^{30} = 1^{30}(\cos 20^0 + i\sin 20^0)^{30}$

$= 1(\cos 600^0 + i\sin 600^0)$

$= \cos 240^0 + i\sin 240^0) \longrightarrow 600^0 - 360^0 = 240^0$

$= \cos 120^0 - i\sin 120^0) \longrightarrow 240^0 - 360^0 = -120^0$

$$= -\frac{1}{2} - \frac{\sqrt{3}}{2} \longrightarrow \sin 120^0 = \frac{\sqrt{3}}{2}$$

$$\cos 120^0 = -\frac{1}{2}$$

$\text{Arg}(z^{30}) = 30(\text{Arg}z) = 30 \times 20^0 = 600^0$

$\therefore$ Principal argument is $600^0 - 720^0 = -120^0$

b) $z^{-15} = (\cos 20^0 + i\sin 20^0)^{-15} = \cos 300^0 - i\sin 300^0$
$= \cos 60^0 + i\sin 60^0 \longrightarrow 360^0 - 300^0 = 60^0$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$\therefore$ Principal $\arg(z^{-15}) = -15 \arg(z) + 2\pi$
$= -15(20^0) + 2\pi = -300^0 + 360^0 = 60^0$

c) $z^9 = 1^9(\cos 20^0 + i\sin 20^0)^9 = 1(\cos 180^0 + i\sin 180^0)$
$= -1 + 0 = -1$

$\therefore$ Principal $\arg(z^9) = 9 \arg(z) = (9)(20^0) = 180^0$

d) $z^{-12} = 1(\cos 240^0 - i\sin 240^0) = 1(\cos 120^0 + i\sin 120^0)$

$$= \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

$\therefore \text{Arg}(z^{-12}) = -12\text{Arg}(z) = -12 \times 20^0 = -240$
$\therefore$ Principal $\arg(z^{-12}) = -12 \arg(z) + 360^0 = -240^0 + 360^0$
$= 120^0$

28. Let $z_1 = 2\sqrt{3} - 2i$ and $z_2 = -1 + \sqrt{3}i$, then find

a) polar form of $(z_1 \cdot z_2)$ c) $\text{Arg}(z_1 \cdot z_2)$

b) polar form of $\left(\frac{z_1}{z_2}\right)$ d) $\text{Arg}\left(\frac{z}{z_2}\right)$

Solution: • $|z_1| = r_1 = \sqrt{(2\sqrt{3})^2 + (-2)^2} = \sqrt{12+4} = 4 \leftarrow r_1$

• $Arg(z_1) = \theta_1 = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$

• $|z_2| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2 \leftarrow r_2$

• since $z_2 = -1 + \sqrt{\sqrt{3}^2}\,i$ lies on 2nd quadrant principal

$$\text{Arg}(z_2) = \pi + \tan^{-1}\left(\frac{y}{x}\right) = \pi + \tan^{-1}(-\sqrt{3}) = \pi - \tan^{-1}(\sqrt{3})$$

$$= \pi - \frac{\pi}{3} = \frac{2}{3}\pi$$

Thus, a) $z_1 \cdot z_2 = r_1 r_2((\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$

$$= 2.4\left(\cos\left(\frac{-\pi}{6}+\frac{2\pi}{3}\right)+i\sin\left(\frac{-\pi}{6}+\frac{2}{3}\pi\right)\right)$$

$$= 8\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right) \leftarrow \textit{polar form of } z_1 \cdot z_2$$

$= 8(0 + i) - 8i \longleftarrow$ rectangular form

b) $\frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)$

$$= \frac{4}{2}\left(\cos\left(-\frac{\pi}{6}-\frac{2\pi}{3}\right)+i\sin\left(\frac{-\pi}{6}-\frac{-2\pi}{3}\right)\right)$$

$$= 2\left(\cos\frac{5\pi}{3}-i\sin\frac{5\pi}{3}\right) \longleftarrow \text{polar form of } \frac{z_1}{z_2}$$

$$= 2\left[-\frac{\sqrt{3}}{2}+i\left(\frac{-1}{2}\right)\right] = -\sqrt{3}-i \leftarrow \text{rectangular form of } \frac{z_1}{z_2}$$

c) $\text{Arg}(z_1 \cdot z_2) = \text{Arg}(z_1) + \text{Arg}(z_2) = \frac{-\pi}{6}+\frac{2}{3}\pi = \frac{\pi}{2}$

d) $\text{Arg}\left(\frac{z}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2) = \frac{-\pi}{6}-\left(\frac{2}{3}\pi\right) = \frac{-5\pi}{6}$

29. Find, polar form, principal argument modulus of z^{12}

a) $z = \frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i$ b) $z = \frac{-\sqrt{3}}{2}-\frac{1}{2}i$ c) $z = \sqrt{2}-\sqrt{6}i$

Solution: If $z = r(\cos\theta + i\sin\theta)$, then $z^n = r^n(\cos n\theta + i\sin n\theta)$ and substitute $n = 12$

a) $$Arg\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right) = \tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$|z| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2+\left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{\frac{2}{4}+\frac{2}{4}} = 1$$

$$z=\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\Rightarrow z^{12}=1^{12}\left(\cos\left(12\left(\frac{\pi}{4}\right)\right)+i\sin\left(12\left(\frac{\pi}{4}\right)\right)\right)$$

$$=\cos3\pi+i\sin3\pi \longleftarrow \text{polar form} = -1+0=-1$$

$$\therefore\ z^{12}=\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)^{12}=\cos3\pi+i\sin3\pi=-1$$

$$|z^{12}|=|-1|^{12}=1 \longleftarrow \text{modulus of } \left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i\right)^{12}$$

b) $$z=\left(\frac{-\sqrt{3}}{2}-\frac{1}{2}i\right)^{12}$$

$$Arg(z)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)-\pi=\frac{\pi}{6}-\pi=\frac{-5\pi}{6}$$

$$z^{12}=\left(1\left(\cos\frac{5\pi}{6}-i\sin\frac{5\pi}{6}\right)\right)^{12}=\cos10\pi-i\sin10\pi)$$

$$=\cos0^0-i\sin0^0 \longleftarrow \text{polar form}$$

$$\therefore \text{Principal } \arg(z^{12})=0^0$$

$$\therefore\ \left(\frac{-\sqrt{3}}{2}-\frac{1}{2}i\right)^{12}=1+0i$$

c) $$z=\sqrt{2}-\sqrt{6}i,\Rightarrow z^{12}=\left(\sqrt{2}-\sqrt{6}i\right)^{12}$$

$$\text{Arg}(z)=\tan^{-1}\left(\frac{-\sqrt{6}}{\sqrt{2}}\right)=-\tan^{-1}\left(\sqrt{3}\right)=-\frac{\pi}{3}$$

$$|z|=\sqrt{(\sqrt{2})^2+(-\sqrt{6})^2}=\sqrt{2+6}=\sqrt{8}$$

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$$z=\sqrt{8}\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)\Rightarrow z^{12}=\sqrt{8}^{12}\left(\cos\left(12\cdot\frac{\pi}{3}\right)-i\sin\left(12\cdot\frac{\pi}{3}\right)\right)$$

$$\Rightarrow z^{12}=\left(2^{\frac{3}{2}}\right)^{12}(\cos 4\pi - i\sin 4\pi)=2^{18}(1-0i)=2^{18}$$

$$\therefore z^{12}=\left(\sqrt{2}-\sqrt{6}i\right)^{12}=2^{18}(\cos 0^0-i\sin 0^0)\longleftarrow \text{polar form}$$

- Modulus: $|z^{12}|=2^{18}$
- Principal $\arg(z^{12})=0^0$

30. Find Modulus, principal argument and polar form:

a) $z=\dfrac{\sqrt{3}+\sqrt{3}i}{\left(1-\sqrt{3}i\right)\left(2-2i\right)}$ b) $z=\left(\dfrac{2\sqrt{3}+2i}{2+2i}\right)$

Solution: a) $z=\left|\dfrac{\left(\sqrt{3}+\sqrt{3}i\right)}{\left(1-\sqrt{3}i\right)\left(2-2i\right)}\right|=\dfrac{\left|\sqrt{3}+\sqrt{3}i\right|}{\left|1-\sqrt{3}i\right|\left|2-2i\right|}$

$$=\frac{\sqrt{\sqrt{3}^2+\sqrt{3}^2}}{\sqrt{1^2+\left(\sqrt{3}\right)^2}\cdot\sqrt{2^2+(-2)^2}}=\frac{\sqrt{6}}{2\sqrt{8}}=\frac{\sqrt{3}}{4}$$

- $Arg\left(\dfrac{\sqrt{3}+\sqrt{3}i}{\left(1-\sqrt{3}i\right)\left(2-2i\right)}\right)=Arg\left(\sqrt{3}+\sqrt{3}i\right)-[(Arg1-\sqrt{3})+Arg(2-2i)]$

$$=\tan^{-1}\left(\frac{\sqrt{3}}{\sqrt{3}}\right)-\left(\tan^{-1}\left(\frac{-\sqrt{3}}{1}\right)+\tan^{-1}\left(\frac{-2}{2}\right)\right)$$

$$=\tan^{-1}(1)-\left(-\tan^{-1}\left(\sqrt{3}\right)-\tan^{-1}(1)\right)$$

$$=\frac{\pi}{4}-\left(\frac{-\pi}{3}-\frac{\pi}{4}\right)=\frac{10\pi}{12}=\frac{5}{6}\pi$$

$\therefore$ Principal $\arg(z)=\ =\dfrac{5\pi}{6}$

$$\therefore z=\frac{\sqrt{3}+\sqrt{3}i}{\left(1-\sqrt{3}i\right)\left(2-2i\right)}=\frac{\sqrt{3}}{4}\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)$$

$$= \frac{\sqrt{3}}{4}\left(-\frac{\sqrt{3}}{2}+\frac{1}{2}i\right) = \frac{-3}{8}+\frac{\sqrt{3}}{8}i$$

b) $$z = \left|\frac{2\sqrt{3}+2i}{2+2i}\right| = \frac{\left|2\sqrt{3}+2i\right|^{16}}{\left|2+2i\right|^{16}}$$

$$= \frac{\left(\sqrt{\left(2\sqrt{3}\right)^2+2^2}\right)^{16}}{\left(\sqrt{2^2+2^2}\right)^{16}} = \frac{4^{16}}{\sqrt{8}^{16}} = \left(\sqrt{\frac{16}{8}}\right)^{16} = 2^8 = 256$$

- Arg(z) = 16(Arg(z) = 16((Arg $\left(2\sqrt{3}+2i\right)$ - Arg(2+2i))

$$= 16\left(\frac{\pi}{6}-\frac{\pi}{4}\right) = \frac{-16\pi}{12} \longleftarrow \text{It is not principal argument of (z}$$

$$\therefore \text{Principal arg(z)} = \frac{-16\pi}{12}+2\pi = \frac{8\pi}{12} = \frac{2}{3}\pi$$

$$\therefore z = \left(\frac{2\sqrt{3}+2i}{2+2i}\right)^{16} = 256\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right) \longleftarrow \text{polar form}$$

$$= 256\left(\frac{-1}{2}+\frac{\sqrt{3}}{2}i\right) = -128+128\sqrt{3}i \longleftarrow \text{rectangular form}$$

Solved Problems

31. The modulus of $\frac{1}{i^3}+\frac{-5+3i}{12-5i}$ is equal to .... UEE 2007

A. $\frac{\sqrt{15}}{13}$ B. $\frac{15}{13}$ C. $\frac{1+\sqrt{34}}{13}$ D. $\frac{20}{13}$

Solution:

$$\left|\frac{1}{i^3}+\frac{-5+3i}{12-5i}\right| = \left|\frac{12-5i+5i+3}{-i(12-5i)}\right| = \frac{|15|}{|-12i-5|} = \frac{15}{\sqrt{144+25}} = \frac{15}{13}$$

Answer: B

32. Simplified form of $i\left(1+\sqrt{3}i\right)^6 - 65i$ is equal to

A. 1 B. i C. –i D. i + 1

Solution:

$$i\left(1+\sqrt{3}i\right)^6 - 2^6\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)^6 = 2^6(\cos 2\pi + i\sin 2\pi) = 64$$

$$\therefore i\left(1+\sqrt{3}i\right)^6 - 65i = i(64) - 65i = 64i - 65i = -i$$

Answer: C

33. If the complex number a + bi corresponds to the point (a, b) to what point does 2i - πi correspond ------2008.

A. $(2, -\pi)$ B. $(2-\pi, 0)$ C. $(0, 2-\pi)$ D. $(2, \pi)$

Solution: $2i - \pi i = 0 + (2-\pi)i = (0, 2-\pi)$, ∴ Answer C

34. What are the value of real number x and y that satisfies $\frac{x+3i}{4-2i} = \frac{2+yi}{20}$ ---------- UEE 2009

A. x = 2, y = 3 C. x = 2, y = 16

B. x = 6, y = 10 D. x = 4, y = 6

Solution: $\frac{x+3i}{4-2i} = \frac{2+yi}{20} \Rightarrow 20x + 60i = (4-2i)(2+yi)$

$= 8 + 4yi - 4i + 2y$

$\Rightarrow 20x + 60i = 8 + (4y-4)i + 2y$

$\Rightarrow 4y - 4 = 60 \Rightarrow 4y = 64 \Rightarrow y = 16$

$\Rightarrow 20x = 8 + 2y \Leftrightarrow 20x = 8+2(16) = 40$

$\Rightarrow 20x = 40 \Leftrightarrow x = 2$ **Answer: C**

35. For z = x + yi, which of the following represent the circle $(x-3)^2 + (y+1)^2 = 4$ ..... (UEE 2001)

A. $|z+3-i| = 4$ C. $|z+3-i| = 2$

B. $|z-3+i| = 4$ D. $|z-3+i| = 2$

Solution: $|z-3-i| = 2 \Rightarrow |x+yi-3+i| = 2$

$\Rightarrow |(x-3| + (y+1)i| = 2$

$= (x-3)^2 + (y+1)^2 = 4$

Answer: D

36. Let z be a complex number such that $\left|\frac{3z}{-4+3i}\right| = (2-i)\overline{(2-i)}$

Then the modulus of is equal to: ---------- UEE

A. $\frac{25}{3}$ B. $\frac{10}{3}$ C. $\frac{14}{3}$ D.

Solution:

$$\left|\frac{3z}{-4+3i}\right| = (2-i)\overline{(2-i)} = \frac{3|z|}{\sqrt{4^2+3^2}} = (2-i)(2+i) = 5$$

$$\Rightarrow \frac{3|z|}{5} = 5 \Rightarrow 3|z| = 25 \Rightarrow |z| = \frac{25}{3}$$

37. In the set of complex number, the ***Solution*** set of $z^3 - 8 = 0$ is UEE 2010

A. {2}

B. $\left\{2, 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)\right\}$

C. {2, 2 +8i, 2-8i}

D. $\left\{2, 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) 2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)\right\}$

Solution: $z^2 - 8 = 0 \Rightarrow z^3 - 2^3 = 0 \Rightarrow (z-2)(z^2+2z+4) = 0$

$\Rightarrow z - 2 = 0$ and $z^2 + 2z + 4 = 0$

$\therefore z = 2$ and $\frac{-2 \pm \sqrt{4-4(4)}}{2} = \frac{-2 \pm 2\sqrt{3}i}{2} = -1 \pm \sqrt{3}i$

$\therefore z = 2$ and $z = -1 + \sqrt{3}i$ or $z = -1 - \sqrt{3}i$

In polar form, $2, 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right), 2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$

$\therefore$ Answer D

38. Polar form of $\left[\frac{2}{1+\sqrt{3}i}\right]^2$ equal -----

A. $\cos 120^0 + i\sin 120^0$ C. $\cos 60^0 + i\sin 60^0$

B. $\cos 240^0 + i\sin 240^0$ D. $\cos 120^0 + i\sin 120^0$

Solution: $\left[\frac{2}{1+\sqrt{3}i}\right]^2 = \frac{2^2(\cos 0^0 + i\sin 0^0)}{2^2(\cos 120^0 + i\sin 120^0)}$

$= \cos 120^0 - i\sin 120^0$

$\Rightarrow \cos 120^0 - i\sin 120^0 = \cos(240^0) + i\sin 240^0 \longrightarrow -120^0 + 360^0$

$= 240^0$

Answer: B

39. Solve each of the following expression over complex number

a. $x^3 + 2x^2 + 5x + 4 = 0$

b. $x^2 + 2x + 6 = 0$

c. $x^4 - x^2 - 20 = 0$

d. $x^3 + 8x = 0$

Solution: a) using rational root test $(-1)^3 + 2(-1)^2 + 5(-1) + 4 = 0$

$\therefore x^3 + 2x^2 + 5x + 4 = 0 \Rightarrow (x + 1)(x^2 + x + 4) = 0$

$\Rightarrow x + 1 = 0$ or $x^2 + x + 4 = 0$

$\Rightarrow x = -1$ or $x = \frac{-1 \pm \sqrt{1-4(4)}}{2} = \frac{-1 \pm \sqrt{15}i}{2}$

$\therefore S.S = \left\{-1, \frac{-1+\sqrt{15}i}{2}, \frac{-1-\sqrt{15}i}{2}\right\}$

b) $x^2 + 2x + 6 = 0 \Rightarrow x^2 + 2x + 1 + 6 - 1 = 0 \Rightarrow (x + 1)^2 + 5 = 0$

$\Rightarrow (x + 1)^2 = -5 \Rightarrow x + 1 = \pm\sqrt{-5} = \pm\sqrt{5}i$

$\Rightarrow x = -1 \pm \sqrt{-5}i \Rightarrow x = -1 + \sqrt{5}i$ or $x = -1 - \sqrt{5}i$

$\therefore S.S = \{-1 + \sqrt{5}i\ ,\ -1 - \sqrt{5}i\}$

c) $x^4 - x^2 - 20 = 0 \Rightarrow x = \frac{-1 \pm \sqrt{1-4(-20)}}{2} = \frac{1 \pm 9}{2} = \frac{10}{2}$ or $-\frac{8}{2}$

$\Rightarrow x^2 = 5$ or $x^2 = -4$

$\Rightarrow x = \pm\sqrt{5}$, or $x = \pm\sqrt{-4} = \pm 2i$

$\therefore S.S = \{-\sqrt{5}, \sqrt{5}, 2i, -2i\}$

d) $x^3 + 8x = 0 \Rightarrow x(x^2 + 8) = 0$

$\Rightarrow x(x - \sqrt{8}i)(x + \sqrt{8}i) = 0$

$\Rightarrow x = 0, x = \sqrt{8}i = 2\sqrt{2}i$ or $x = -\sqrt{8}i = -2\sqrt{2}i$

$\therefore S.S = \{0, 2\sqrt{2}i, -2\sqrt{2}i\}$

40. The simplified form of $z = \dfrac{(1-3i)(5+i)}{1+i}$ is ..... UEE 2005

A. -3 – 11i B. $3\sqrt{2}$ C. $\dfrac{2}{i} - \dfrac{5}{4}i$ D. 2i + 3

Solution:

$$\frac{(1-3i)(5+i)}{1+i} = \frac{(5+i-15i+3)}{1+i} = \frac{8-14i}{1+i} = 8-14i\left(\frac{1}{2} - \frac{i}{2}\right)$$

$$= \frac{8-14i-8i-14}{2}$$

$$\Rightarrow \frac{-6}{2} - \frac{22i}{2} = -3-11i$$

Answer: A

41. If z = -1 – i, then what is the value of z^{-6}

A. $1+\dfrac{1}{i}$ B) 8 = i C. $\dfrac{i}{8}$ D. $\dfrac{1}{360}$

Solution:

$$\frac{1}{z^6} = \frac{1}{(-1-i)^6} = \frac{1}{\left((-1-i)^2\right)^3} = \frac{1}{(1+2i-1)^3} = \frac{1}{(2i)^3} = \frac{1}{-8i} =$$

Answer: C

42. A polar form of $z = \dfrac{1+\sqrt{3}i}{1+i}$ is ..... UEE

A. $\sqrt{2}\,(\cos 60^0 + i\sin 60^0)$ C. $\dfrac{\sqrt{2}}{2}(\cos 105^0 + i\sin 105^0)$

B. $2(\cos 30^0 + i\sin 30^0)$ D. $\sqrt{2}\,(\cos 15^0 + i\sin 15^0)$

Solution: $|z| = \left|\dfrac{1+\sqrt{3}i}{1+i}\right| = \dfrac{|1+\sqrt{3}i|}{|1+i|} = \dfrac{\sqrt{1^2+\sqrt{3}^2}}{\sqrt{1^2+1^2}} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$

$$\text{Arg}(z) = Arg\left(\frac{1+\sqrt{3}i}{1+i}\right) = Arg(1+\sqrt{3}i) - Arg(1+i)$$

$= 60^0 - 45^0 = 15^0$

$\therefore$ polar form of $z = \sqrt{2}\,(\cos 15^0 + i\sin 15^0)$

Answer: D

43. A polar form of $z = \dfrac{1-i}{(\sqrt{3}+i)^2}$ with $a = \dfrac{\sqrt{2}}{4}$ is ..... UEE

A. $a[\cos 15^0 + i\sin 15^0]$ C. $a[\cos 255^0 + i\sin 255^0]$

B. $a[\cos 165^0 + i\sin 165^0]$ D. $a[\cos 275^0 + i\sin 275^0]$

Solution: $|z| = \dfrac{\sqrt{2}}{4}$

$$\text{Arg}(z) = \arg\left(\frac{1-i}{(\sqrt{3}+i)^2}\right) = \arg(1-i) - 2\arg(\sqrt{3}+i)$$

$$= \tan^{-1}\left(-\frac{1}{1}\right) - 2\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= -45^0 - 2(30^0) = -105^0 + 360^0 = 255^0$$

$\therefore$ Polar form $= a(\cos 105^0 - i\sin 105^0) = a(\cos 255 + i\sin 255^0)$

Answer: C

44. If $z = x + yi$, then when the equation $z(1+3i) - 4i = z$ is solved what are the value of x and y, respectively .....UEE 2006.

a) 3, 1 b) 0, 4 c) $\dfrac{4}{3}, 0$ d) 4, 3

Solution: $z(1 + 3i) - 4i = z \Rightarrow z + 3zi - 4i = z$

$\Rightarrow x + yi + 3(x + yi)I - 4i = z \Rightarrow x + yi + 3ix - 3y - 4i = x + yi$

$\Rightarrow (3x - 4)i - 3y = 0 \Rightarrow 3y = 0$ and $3x - 4 = 0$

$\therefore y = 0$ and $x = \dfrac{4}{3}$,

Answer: C

45. Find x + yi form of

a) $\left(-\sqrt{3}+i\right)^9$ b) $\left(2-2\sqrt{3}i\right)^5$ c) $\left(1+\sqrt{3}i\right)^8$

Solution: $\left|-\sqrt{3}+i\right| = \sqrt{3+1} = 2$

$$\text{Arg}\left(-\sqrt{3}+i\right)^9 = 9(\text{Arg}\left(-\sqrt{3}+i\right) = 9\left(\pi + \left(-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)\right)$$

$$= 9\left(\pi - \frac{\pi}{6}\right) = \frac{45\pi}{6} = \frac{15\pi}{2}$$

$$\therefore \left(-\sqrt{3}+i\right)^9 = 2^9\left(\cos\frac{15\pi}{2}+i\sin\frac{15\pi}{2}\right) = 2^9\left(\cos\frac{3}{2}\pi+i\sin\frac{3}{2}\pi\right)$$

$$= 2^9(-i) = -512i$$

b) $$\left(2-2\sqrt{3}i\right)^5 = \left(4\left(\cos\left(\frac{\pi}{3}\right)-i\sin\frac{\pi}{3}\right)\right)^5$$

$$= 2^{10}\left(\cos\frac{5\pi}{3}-i\sin\frac{5\pi}{3}\right)$$

$$= 2^{10}\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right) \rightarrow \frac{-5\pi}{3}+2\pi = \frac{\pi}{3}$$

$$= 2^{10}\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)$$

$$= 512 + 512\sqrt{3}i$$

c) $$|1+\sqrt{3}i| = \sqrt{1^2+\sqrt{3}^2} = \sqrt{1+3} = 2$$

$$Arg(1+\sqrt{3}i) = \tan^{-1}\left(\sqrt{3}\right) = \frac{\pi}{3}$$

$$\therefore \left[1+\sqrt{3}i\right]^8 = \left[2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)\right]^8 = 2^8\left(\cos\frac{8\pi}{3}+i\sin\frac{8\pi}{3}\right)$$

$$= 2^8\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right) = -128+128\sqrt{3}i$$

46. If $i^n = -i$ then which of the following is not possible values of n

a) 75 b) 307 c) 371 d) 397

Solution: A) If n = 75, then divide 75 by 4 $\Rightarrow$ r = 3

$\therefore i^{75} = i^3 = -i \Leftrightarrow 75 = 4(18) + 3 \longleftarrow r = 3$

B) If n = 307 $\Rightarrow$ 307 = 4(76) + 3 $\leftarrow$ divide 307 by 4, r = 3

C) If n = 371 $\Rightarrow$ 371 = 4(92) + 3 $\leftarrow$ divide 371 by 4, r = 3

D) If n = 397 $\Rightarrow$ 397 = 4(99) + 1 $\leftarrow$ divide 397 by 4, r = 1

$\Rightarrow i^{75} = i^3 = -i$, $i^{307} = i^3 = -i$, $i^{371} = i^3 = -i$

But $i^{397} = i^1 = i$

$\therefore$ possible values of n = 75, 307, and 371

Answer: D

47. For given cartesion coordinate, find polar coordinate.

a) (-1, 1) b) (0, 3) c) (0, -2) d) $\left(-\sqrt{3},-1\right)$

Solution: a) $r=\sqrt{(-1)^2+(1)^2}=\sqrt{2}$, and P(-1, 1) = -1 + i lies on 2nd quadrant.

$\text{Arg}(-1+i)=\pi+\tan^{-1}\left(\frac{-1}{1}\right)=\pi-\left(\frac{\pi}{4}\right)=\frac{3\pi}{4}$

$\therefore (r,\theta)=\left(\sqrt{2},\frac{3\pi}{4}\right)$

b) $r=\sqrt{0^2+3^2}=3$, $\text{Arg}(0+3i)=\frac{\pi}{2}$

$\therefore (0,3)=\left(3,\frac{\pi}{2}\right)$

c) $r=\sqrt{0^2+(-2)^2}=2$, $\text{Arg}(0-2i)=\frac{-\pi}{2}$

$\therefore (0,-2)=\left(2,\frac{-\pi}{2}\right)$

d) $r=\sqrt{\left(\sqrt{3}\right)^2+(-1)^2}=2$,

$\text{Arg}\left(-\sqrt{3}-i\right)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)-\pi \Rightarrow \frac{\pi}{6}-\pi=\frac{-5\pi}{6}$

$\therefore P(x,y)=P\left(-\sqrt{3},-1\right)=(r,\theta)=\left(2,\frac{-5\pi}{6}\right)$

Let z be a complex number. which of the following is the solution set of $z^3 - iz = 0$? .....UEE

A. $\left\{\pm\frac{1}{\sqrt{2}}(1+i)\right\}$ C. $\left\{\pm\sqrt{2}(1+i)\right\}$

B. $\left\{0,\pm\frac{1}{\sqrt{2}}(1-i)\right\}$ D. $\left\{0,\pm\frac{1}{\sqrt{2}}(1+i)\right\}$

Solution: $z^3 - iz = 0 \Leftrightarrow z(z^2 - i) = 0 = \Rightarrow z = 0$ or $z^2 - i = 0$

$\Rightarrow z^2 = i \Rightarrow (x + yi)^2 = i \Leftrightarrow x^2 - y^2 + 2xyi = 0 + i$

$\Rightarrow x^2 - y^2 = 0$ and $2xyi = i$

$\Rightarrow x = \pm y$ and $2xy = 1$

So that, $2x\cdot(x) = 1 \Rightarrow 2x^2 = 1$

$$\Rightarrow x = \pm\frac{1}{\sqrt{2}} \text{ and } y = \pm\frac{1}{\sqrt{2}}$$

$$\Rightarrow z = x + yi = \pm\frac{1}{\sqrt{2}}(1+i)$$

$$\therefore \text{S.S} = \left\{0, \pm\frac{1}{\sqrt{2}}(1+i)\right\}$$

Supplementary problem

1. Simplify write the answer in the form a+bi

   a) $\sqrt{-16} + \sqrt{-4}$ c) $\dfrac{\sqrt[3]{-27} + \sqrt{-27}}{3}$

   b) $\sqrt{-12} + \sqrt{-8}$ d) $\sqrt{32} + \sqrt{-32}$

2. Represent each of the following complex numbers as ordered pairs of number

   a) 2-3i b) $\left(\sqrt{2}+1\right) + \left(\sqrt{3}-4\right)i$ c) 4i d) 5

3. In each of the following find the product in the form x+yi

   a) (2+3i)(3-8i) c) $(5-2i)^2(3+i)$

   b) $\left(\sqrt{2}+i\right) + \left(\sqrt{3}-i\right)$ d) 2i(3-5i)

4. In each of the following solve for x and y

   a) (x+yi)(x-2i)=4+5i c) (x+yi)(2+3i)(2-i)=5+3i

   b) (x-yi)(8+i)=6-I d) (2+3i)(x+yi)=7-3i

5. Compute $\overline{Z_1 + Z_2}, \overline{Z_1} + \overline{Z_2}, \overline{Z_1 Z_2}$ and $\overline{Z_1} \cdot \overline{Z_2}$ in each of the following

   a) $Z_1 = 2, -i; Z_2 = 3 + i$ c) $Z_1 = 2, Z_2 = \sqrt{-3}$

   b) $Z_1 = -1 - 2i,\ Z_2 = i$ d) $Z_1 = \sqrt{-5}, Z_2 = 3 - \sqrt{-4}$

6. Simplify each of the following

   a) $\dfrac{1}{2-3i} + \dfrac{3+5i}{2-i}$ b) $\dfrac{2+3i}{3-2i} + \dfrac{3+4i}{2+4i}$ c) i^{47}

7. Compute each of the following

a) $|(4-2i)(4+3i)|$ b) $\left|\sqrt{\frac{1+i}{1-1}}\right|$

8. Express each of the following complex numbers in polar form

a) $\frac{1}{2}-\frac{\sqrt{3}}{2}i$ b) 1-4i c) 1- i d) $\frac{1+2i}{1-3i}$

9. Find a) $(1-i)^{10}$ b) $(2+2i)^7$ c) $\left(\frac{\sqrt{2}}{2}+\frac{i\sqrt{2}}{2}\right)^6$

10. Find all roots of the equation

a) $x^3+8=0$ b) $x^3+\left(1+i\sqrt{3}\right)=0$

11. Find $\sqrt{-16+16\sqrt{3}i}$

12. Find real values of x and y for which the following equations are satisfied

a) (x+yi)(2-3i)=4+i b) $\frac{x-y}{3+i}+\frac{y-i}{3-i}=i$

c) $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y-i}{3-i}=i$

13. Find the center and radius

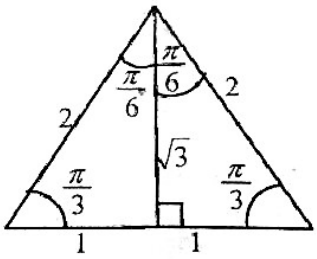
a) $z\bar{z}+(1+i)z+(1-i)\bar{z}=0$

b) $z\bar{z}-(2+3i)z-(2-3i)\bar{z}+9=0$

14. If $y+\frac{1}{y}=2\cos\theta$, *Find y*

Further on Trigonometric Function

- The function y = sec x, y = csc x and y = cot x
- $\sec x = \frac{1}{\cos x}$,
- $\csc x = \frac{1}{\sin x}$
- $\cot x = \frac{1}{\tan x}$



- Table of Trigonometric function of special angle

X	Sinx	Cosx	Tanx	Cscx	Secx	Cotx
30 $\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45 $\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60 $\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$

- The graph of y = secx, y = cscx and y = cot x
  To sketch the graph, each of the following elements are very important. These are:
- Domain, Range, period, asymptote.
- Determine for what x, the function is positive or negative
- Describe the behavior as x increase from:

$$0 \to \frac{\pi}{2}, \frac{\pi}{2} \to \pi, \pi \to \frac{3\pi}{2}, \frac{3\pi}{2} \to 2\pi$$

1. The graph of y = csc x

$$f(x) = \csc x = \frac{1}{\sin x}, \sin x \neq 0 \Rightarrow x \neq k\pi$$

- Domain = $\{x: x \neq k\pi, k \text{ is integer}\}$
- Range = $R\backslash\{(-1,1)\} = (-\infty,-1] \cup [1, \infty)$
- Period = 2π i.e $f(x + 2\pi) = f(x)$ for all x.
- The asymptote = $\begin{cases} k\pi, \text{ where k is integer and} \\ \text{at which } \sin x = 0 \text{ and } \sin x \\ \text{crosses the x - axis} \end{cases}$
- For what x, csc x is positive or negative

i) $\csc x > 0 \Rightarrow \dfrac{1}{\sin x} > 0 \Rightarrow \sin x > 0$

$\Rightarrow \csc x > 0$ when $-2\pi < x < -\pi$ or $0 < x < \pi$

$\therefore$ the graph lies above the x – axis

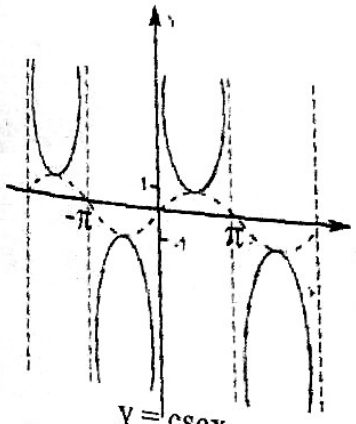
ii) $\csc x < 0 \Rightarrow \dfrac{1}{\sin x} < 0 \Rightarrow \sin x < 0$

$\therefore \csc x < 0$ when $-\pi < x < 0$ or $\pi < x < 2\pi$

That is the graph of y = csc x, lies below the x – axis on this-interval

- Describing the behavior as x increase from

$$0 \to \frac{\pi}{2}, \frac{\pi}{2} \to \pi, \pi \to \frac{3\pi}{2}, \frac{3\pi}{2} \to 2\pi$$



y = cscx

- Domain: $x \neq k\pi$
- Range: $(-\infty, -1] \cup [1, \infty)$

i) **As x increase from 0** $\to \dfrac{\pi}{2}$

- sin x increase from $0 \to 1$

$\Rightarrow$ csc x decease from $\infty \to 1$

ii) **As x increase from** $\dfrac{\pi}{2} \to \pi$

- sin x decrease from $1 \to 0$
- csc x increase from $1 \to \infty$

iii) **As x increase from** $\pi \to \dfrac{3}{2}\pi$

- sin x decrease from $0 \to -1$
- csc x increase from $-\infty \to -1$

iv) **As x increase from** $\frac{3}{2}\pi \to 2\pi$

- sin x increase from -1 $\to$ 0
- csc x decrease from -1 $\to$ -∞

- **Symmetric property**

Csc (-x) = -csc x

$\Rightarrow$ it is an odd function

$\therefore$ The graph is symmetric with respect to the **origin.**

2. The graph of y = sec x

We have g(x) = sec x = $\frac{1}{\cos x}, \cos x \neq 0 \Rightarrow x \neq \left(\frac{2k+1}{2}\right)\pi$

- Domain = $\left\{x \neq \frac{(2k+1)\pi}{2}, k \in Z\right\}$
- Range = (-∞ - 1] ∪[1 ∞)
- Period = 2π
- Asymptote = $\begin{cases} \frac{(2k+1)\pi}{2}, k \in Z \text{ and at which } \cos x = 0 \\ \text{and cos x crosses the x - axis} \end{cases}$

Explanation: $-1 \leq \cos x \leq 1$

$\Rightarrow |\cos x| \leq 1 \Rightarrow \frac{1}{|\cos x|} \geq 1$

$\Rightarrow |\sec x| \geq 1$

$\therefore$ Rang: $(-\infty, -1] \cup [1, \infty)$

- For what interval x, secx is positive or negative

i) sec x > 0 $\Rightarrow \frac{1}{\cos x} > 0 \Rightarrow \cos x > 0$

$\therefore$ sec x > 0 when x $\in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and x$\in$ (-2π $\frac{-3\pi}{2}$)

- That is the graph of g(x) = sec x lies above the x-axis (g(x)≥ 1)

ii) sec x < 0 $\Rightarrow \frac{1}{\cos x} < 0 \Rightarrow \cos x < 0$

$\therefore$ sec x < 0 when x $\in \left(\frac{-3\pi}{2}, \frac{-\pi}{2}\right)$ and

$x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

- When $x \in \left(\frac{-3\pi}{2}, \frac{-\pi}{2}\right)$ and $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$g(x) = \sec x$ has maximum value -1, at $x = -\pi$ and $x = \pi$
but has no finite minimum value

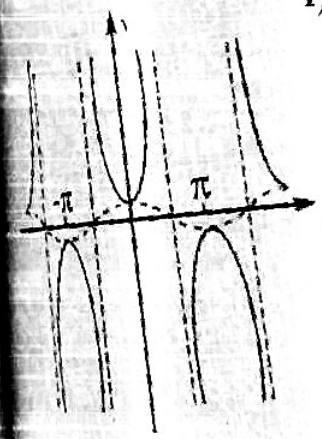
- Describing the behavior as x increase from (0 0)

$$\rightarrow \frac{\pi}{2}, \frac{\pi}{2} \rightarrow \pi, \pi \rightarrow \frac{3\pi}{2}, \frac{3\pi}{2} \rightarrow 2\pi$$

i) **As x increase from** $0 \rightarrow \frac{\pi}{2}$

- cos x decrease from $1 \rightarrow 0$
- sec x increase from $1 \rightarrow \infty$

(The graph goes up from $1 \rightarrow \infty$)



y = secx

- Domain: $x \neq \frac{\pi k}{2}$
- Range: $(-\infty, -1] \cup [1, \infty)$

ii) **As x increase from** $\frac{\pi}{2} \rightarrow \pi$

- cos x decrease from $0 \rightarrow -1$
- sec x increase from $-\infty \rightarrow -1$

(The graph goes up from $-\infty \rightarrow -1$)

iii) **As x increase from** $\pi \rightarrow \frac{3\pi}{2}$

- cos x increase from $-1 \rightarrow 0$
- sec x decrease from $0\ -1 \rightarrow -\infty$

(The graph goes down from $-1 \rightarrow -\infty$)

iv) **As x increase from** $\frac{3}{2}\pi \rightarrow 2\pi$

- cos x increase from $0 \rightarrow 1$
- sec x decrease from $\infty \rightarrow 1$
- The graph goes down from $\infty \rightarrow 1$)

Symmetry

Since $\cos(-x) = \cos x \Rightarrow \sec(-x) = \sec x$

$\Rightarrow$ [illegible] function

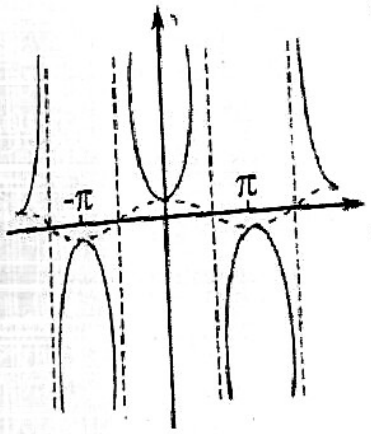
$x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

- When $x \in \left(\frac{-3\pi}{2}, \frac{-\pi}{2}\right)$ and $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$g(x) = \sec x$ has maximum value -1, at $x = -\pi$ and $x = \pi$ but has no finite minimum value

- Describing the behavior as x increase from (0 0)

$$\rightarrow \frac{\pi}{2}, \frac{\pi}{2} \rightarrow \pi, \pi \rightarrow \frac{3\pi}{2}, \frac{3\pi}{2} \rightarrow 2\pi$$



$y = \sec x$

- Domain: $x \neq \frac{\pi k}{2}$
- Range: $(-\infty, -1] \cup [1, \infty)$

i) As x increase from $0 \rightarrow \frac{\pi}{2}$

- cos x decrease from $1 \rightarrow 0$
- sec x increase from $1 \rightarrow \infty$

(The graph goes up from $1 \rightarrow \infty$)

ii) As x increase from $\frac{\pi}{2} \rightarrow \pi$

- cos x decrease from $0 \rightarrow -1$
- sec x increase from $-\infty \rightarrow -1$

(The graph goes up from $-\infty \rightarrow -1$)

iii) As x increase from $\pi \rightarrow \frac{3\pi}{2}$

- cos x increase from $-1 \rightarrow 0$
- sec x decrease from $0\ -1 \rightarrow -\infty$

(The graph goes down from $-1 \rightarrow -\infty$)

iv) As x increase from $\frac{3}{2}\pi \rightarrow 2\pi$

- cos x increase from $0 \rightarrow 1$
- sec x decrease from $\infty \rightarrow 1$
- The graph goes down from $\infty \rightarrow 1$)

Symmetry

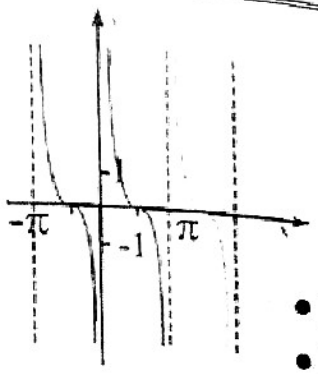
Since $\cos(-x) = \cos x \Rightarrow \sec(-x) = \sec x$

$\Rightarrow g(x) = \sec x$ is an even function

$\therefore g(x) = \sec x$ is symmetric w.r.t y-axis

3. The graph of y = cotx

We have $k(x) = \cot x = \dfrac{\cos x}{\sin x}$



- Range = $\mathbb{R}$
- Period = π
- Asymptote, = $\{x = k\pi.\ x \neq \in Z\}$
- x-intercept = $\left\{x = \left(\dfrac{2k+1}{2}\right)\pi\right.$
- for what interval x, $\cot x \geq 0$ $\cot x \leq 0$

i) $\cot x \geq 0$ when $x \in \left(-2\pi, \dfrac{-3\pi}{2}\right] \cup \left(-\pi, \dfrac{-\pi}{2}\right]$

$$\cup \left(0, \frac{\pi}{2}\right] \cup \left(\pi, \frac{3\pi}{2}\right]$$

That is the graph decrease from $\infty \rightarrow 0$

ii) $\cot x \leq 0$ when

$$x \in \left[\frac{-3\pi}{2}, -\pi\right) \cup \left[\frac{-\pi}{2}, 0\right) \cup \left[\frac{\pi}{2}, \pi\right) \cup \left[\frac{3\pi}{2}, 2\pi\right]$$

the graph lies below the x-axis

The graph lies below the x-axis

- the graph k(x) = cot x is decreasing in its domain.

Illustrative example

1. Find the interval between $-2\pi \leq x \leq 2\pi$ on which:
   a) g(x) = secx is i) increasing ii) Decreasing
   b) f(x) = csc x is i) increasing ii) Decrease
   c) k(x) = cot x is i) increasing ii) decreasing.
   d) Both cosecant and secant function are
      i) increasing at the same time.
      ii) decreasing at the same time
   e) i) $\sec x \geq 1$ ii) $\sec x \leq -1$
   f) i) $\csc x \geq 1$ ii) $\csc x \leq -1$

Solution: Here by observing the graph.

a) i) g(x) = sec x is increasing on: $[-2\pi, -\pi] \cup [0, \pi]$.

$x \neq \frac{-3}{2}\pi$ and $x \neq \frac{\pi}{2}$

ii) $g(x) = \sec x$ is decreasing on: $[-\pi\ 0] \cup [\pi, 2\pi]$,

$x \neq \frac{-\pi}{2}$, and $x \neq \frac{3}{2}, \pi$

b) i) $f(x) = \csc x$ is increasing: $\left[\frac{-3\pi}{2}, \frac{-\pi}{2}\right] \cup \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

for $x \neq -\pi$ and $x \neq \pi$.

ii) $f(x) = \csc x$ is decreasing on:

$\left(-2\pi, \frac{-3\pi}{2}\right] \cup \left[\frac{-\pi}{2}, \pi\right) \cup \left[\frac{3\pi}{2}, 2\pi\right)$ except

at $x = 0$,

c) i) $k(x) = \cot x$ has no interval on which $\cot x$ is increasing.

$\therefore$ s.s $= \phi$

ii) $k(x) = \cot x$ is decreasing on interval $(-2\pi < x < 2\pi)$

except at $x = -\pi$, $x = 0$ $x = \pi$ and $x = 2\pi$.

d) i) $f(x) = \csc x$ and $g(x) = \sec x$ both are increasing at the same time on $\left(\frac{-3\pi}{2}, -\pi\right) \cup \left(\frac{\pi}{2}, \pi\right)$

ii) $f(x) = \csc x$ and $g(x) = \sec x$ both are decreasing at the same time on interval $\left(\frac{-\pi}{2}, 0\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$

e) i) $\sec x \geq 1$ on $\left[-2\pi, \frac{-3\pi}{2}\right) \cup \left(\frac{-\pi}{2}, \frac{\pi}{2}\right] \cup \left(\frac{3}{2}\pi\ 2\pi\right]$

on this interval $\sec x = 1$ is the minimum value and No finite maximum value

ii) $\sec x \leq -1$ on $\left(\frac{-3}{2}\pi, \frac{-\pi}{2}\right) \cup \left(\frac{\pi}{2}\ \frac{3\pi}{2}\right)$

- on this interval $\sec x = -1$ is the maximum value at $x = -\pi$ and $x = \pi$

f) i) $\csc x \geq 1$ on $(-2\pi, -\pi) \cup (0\ \pi)$

- on this interval csc x = 1 is the minimum value at $x = \frac{-3}{2}\pi \text{ and } x = \frac{\pi}{2}$ but has no finite maximum value

ii) csc x ≤ - 1 on $(-\pi \quad 0) \cup (\pi \quad 2\pi)$

- on this interval csc x = -1 is the maximum value at $x = \frac{-\pi}{1} \text{ and } x = \frac{3}{2}\pi$
- But has no finite minimum value

2. Find the exact value

a) $\csc\left(\frac{2\pi}{3}\right)$

b) $\sec\left(\frac{2\pi}{3}\right)$

c) $\sec\left(\frac{-\pi}{6}\right)$

d) $\cot\left(\frac{33\pi}{4}\right)$

e) $\sec\left(\frac{23\pi}{3}\right)$

f) $\csc\left(\frac{-23\pi}{3}\right)$

g) $\csc\left(\frac{-5\pi}{4}\right)$

Solution:

a) $\csc\left(\frac{2\pi}{3}\right) = \frac{1}{\sin\left(\frac{2}{3}\pi\right)} = \frac{1}{\sin\left(\frac{\pi}{3}\right)} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

b) $\sec\left(\frac{2\pi}{3}\right) = \frac{1}{\cos\left(\frac{2}{3}\pi\right)} = \frac{1}{-\cos\left(\frac{\pi}{3}\right)} = \frac{1}{\frac{-1}{2}} = -2$

c) $\sec\left(\frac{-\pi}{6}\right) = \frac{1}{\cos\left(\frac{\pi}{6}\right)} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

d) $\cot\left(\frac{33\pi}{4}\right) = \frac{\cos\left(\frac{33\pi}{4}\right)}{\sin\left(\frac{33\pi}{3}\right)} = \frac{\cos\left(\frac{33\pi}{4} - 8\pi\right)}{\cos\left(\frac{33\pi}{4} - 8\pi\right)} = \frac{\cos\left(\frac{\pi}{4}\right)}{\sin\frac{\pi}{4}} = 1$

e) $\sec\left(\frac{23\pi}{3}\right) =$

$$\frac{1}{\cos\left(\frac{23}{3}\pi\right)} = \frac{1}{\cos\left(\frac{23}{3} - 8\pi\right)} = \frac{1}{\cos\left(\frac{-\pi}{3}\right)} = \frac{1}{\frac{1}{2}} = 2$$

f) $\csc\left(\frac{-23\pi}{3}\right) =$

$$\frac{1}{\sin\left(\frac{-23}{3}\pi + 8\pi\right)} = \frac{1}{\sin\left(\frac{\pi}{3}\right)} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$$

- g) $\csc\left(\frac{-5\pi}{4}\right) = \frac{1}{\sin\left(\frac{-5\pi}{4} + 2\pi\right)} = \frac{1}{\sin\left(\frac{3\pi}{4}\right)} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$

Point about Inverse function Relationships between f^{-1} and f

- $y = f^{-1}(x)$ iff $x = f(y)$, x is in the domain of f^{-1} } and y is in the domain of f.
- Domain of f^{-1} = Range of f
- Range of f^{-1} = Domain of f.
- $f(f^{-1}(x)) = x$, for all x in the domain of f^{-1}
- $f^{-1}(f(y)) = y$, for all y in the domain of f.
- If $(a,b) \in f$ then $(b, a) \in f^{-1}$
- The graph of f^{-1} and f are reflection of each other through the line $y = x$ (i.e. M(a,b) $\underline{y = x}$ (b,a)

Inverse trigonometric function

All the six basic trigonometric function are not one – to – one because they all repeated periodically and hence the horizontal line test crosses

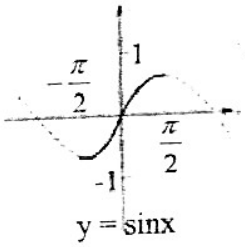
many time. Thus, to define inverse trigonometric function we must first restrict the domains of trigonometric function to make them **one – to – one**

Example: $\sin\left(\frac{5\pi}{6}\right)=\sin\left(\frac{-7\pi}{6}\right)=\sin\left(\frac{\pi}{6}\right)=\frac{1}{2}$

$\therefore$ sine function is not one to one

A. Inverse sine function

The graph of y = sin x is one to one function on restricted domain:



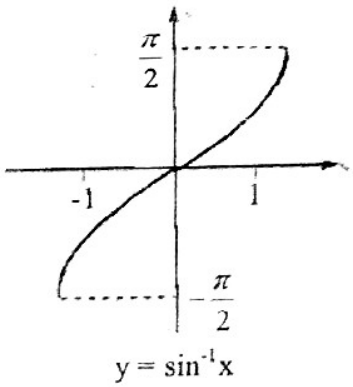
$-\frac{\pi}{2}\le x\le\frac{\pi}{2}$.

Thus, y = sin x is invertible on $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

Definition: the inverse sine function (arc sine function) denoted by $\sin^{-1}$ or arc sin is defined by $y=\sin^{-1}x$ or y = arc sin x iff x = sin y for

$$\frac{-\pi}{2}\le y\le\frac{\pi}{2}\ and\ -1\le x\le 1.$$

In function notation: Define $f^{-1}:[-1,1]\to\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$ is



$f^{-1}(x)=\sin^{-1}x$.

- Domain of $\sin^{-1}x$ is [-1.1]
- Range of $\sin^{-1}x$ is $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

- The graph of y = sin x on $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$ is increasing implies graph of $y=\sin^{-1}x$ is also increasing on [-1, 1].

Note a) • $\sin^{-1}(-x)=-\sin^{-1}x$.

- $\sin\frac{\pi}{2}=1\Rightarrow\sin^{-1}(1)=\frac{\pi}{2}$

- $\sin\frac{\pi}{3}=\frac{\sqrt{3}}{2}\Rightarrow \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$
- $\sin\frac{\pi}{6}=\frac{1}{2}\Rightarrow \sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

Illustrative example

3. If possible find the exact value

a) $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$

b) $\sin^{-1}\left(\frac{-\sqrt{2}}{2}\right)$

c) $\sin^{-1}\left(-\frac{1}{2}\right)$

d) $\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)$

e) $\sin^{-1}(1)$

f) $\sin^{-1}(0)$

Solution: Let $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=y\Rightarrow \sin y=\frac{\sqrt{2}}{2}\Rightarrow y=\pi/4$

a) $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=\frac{\pi}{4}$

b) $\sin^{-1}\left(\frac{-\sqrt{2}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=-\frac{\pi}{4}$

c) $\sin^{-1}\left(-\frac{1}{2}\right)=-\sin^{-1}\left(\frac{1}{2}\right)=-\frac{\pi}{6}$

d) $\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=-\frac{\pi}{3}$

e) $\sin^{-1}(-1)=-\sin^{-1}(1)=\frac{-\pi}{2}$

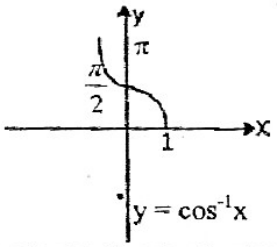
f) $\sin^{-1}(0)=0$

B. Inverse cosine function

- The graph of y = cos x is one to one (decreasing) function on restricted domain: $0 \le x \le \pi$.
- Thus y = cos x is invertible on $o \le x \le \pi$ and $-1 \le y \le 1$

Definition: The inverse cosine function (arc cosine denoted by $\cos^{-1}$ or arc cos is defined by $y = \cos^{-1}x$ or y = arc cos iff $x = \cos y$ for $-1 \le x \le 1$ and $o \le y \le \pi$

Function notation: $f^{-1} : [-1,1] \longrightarrow [0, \pi]$ is $f^{-1}(x) = \cos^{-1}x$



- Domain of $\cos^{-1}$ x is [-1, 1]
- Range of (x) is $[0, \pi]$

The graph of y = cos x is decreasing on $[0,\pi]$, thus the graph of $y = \cos^{-1}$ x is also decreasing on [-1, 1].

Note. 1) $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$

Example: $\cos^{-1}\left(\frac{-1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2}{3}\pi$

$$\therefore \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2}{3}\pi$$

4) If possible find the exact value of y

a) $y = \cos^{-1}\left(\frac{1}{2}\right)$

b) $y = \cos^{-1}\left(-\frac{1}{2}\right)$

c) $y = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

d) $y = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$

e) $y = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

f) $y = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$

Solution: Here, the value of y in the range $0 \le y \le \pi$ of $\cos^{-1}$

- $\text{Cos}^{-1}(-x) = \pi - \cos^{-1}(x)$

a) $y = \cos^{-1}\left(\frac{1}{2}\right) \Leftrightarrow \cos y = \frac{1}{2} \Rightarrow y = \frac{\pi}{3}$

$\therefore \cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$

b) $y=\cos^{-1}\left(-\frac{1}{2}\right)\Rightarrow y=\pi-\cos^{-1}\left(\frac{1}{2}\right)\Rightarrow y=\pi-\frac{\pi}{3}=\frac{2}{3}\pi$

$\therefore y=\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2}{3}\pi$

c) $y=\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)\Leftrightarrow \cos y=\frac{1}{\sqrt{2}}\Rightarrow y=\frac{\pi}{4}$

$\therefore \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)=\frac{\pi}{4}$

d) $y=\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)\Rightarrow y=\pi-\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)=\pi-\frac{\pi}{4}=\frac{3\pi}{4}$

$\therefore y=\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)=\frac{3\pi}{4}$

e) $y=\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)\Leftrightarrow \cos y=\frac{\sqrt{3}}{2}\Rightarrow y=\frac{\pi}{6}$

f) $y=\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)\Rightarrow y=\pi-\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)=\pi-\frac{\pi}{6}=\frac{5\pi}{6}$

$\therefore y=\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)=\frac{5\pi}{6}$

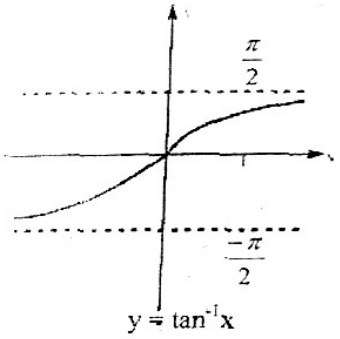
The inverse tangent function

- The inverse tangent function (arc tangent function is denoted $\tan^{-1}$ (arc tan) is defined by
  $y=\tan^{-1} x$ = arc tan if and only if

$x = \tan y$ for $-\infty < x < \infty$ and for $\frac{-\pi}{2} < y < \frac{\pi}{2}$

- Function notation

Define: $f^{-1} : \mathbb{R} \to \left(\frac{-\pi}{2}\ \frac{\pi}{2}\right)$ is $f^{-1}(x)=\tan^{-1}x$



- Horizontal asymptote of are Tangent: $y = \frac{-\pi}{2}$ *and* $y = \frac{\pi}{2}$
- Domain of $\tan^{-1} x$ is IR
- Range of $\tan^{-1}x$ is $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$
- $\tan^{-1}(-x) = -\tan^{-1} x$

5) If possible find the exact value of y

a) $y = \tan^{-1}(1)$

b) $y = \tan^{-1}(-1)$

c) $y = \tan^{-1}(\sqrt{3})$

d) $y = \tan^{-1}(-\sqrt{3})$

e) $y = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

f) $y = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

Solution: Here the value of y in the range $\frac{-\pi}{2} < y < \frac{\pi}{2}$

a) $y = \tan^{-1}(1) \Leftrightarrow \tan y = 1 \Rightarrow y = \frac{\pi}{4}$, $\therefore \tan^{-1}(1) = \frac{\pi}{4}$

b) $y = \tan^{-1}(-1) = -\tan^{-1}(1) = \frac{-\pi}{4}$ $\therefore \tan^{-1}(-\sqrt{3}) = \frac{\pi}{4}$

c) $y = \tan^{-1}(\sqrt{3}) \Rightarrow \tan y = \sqrt{3} \Rightarrow y = \frac{\pi}{3}$,

$\therefore \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

d) $y = \tan^{-1}(-\sqrt{3}) \Rightarrow y = -\tan^{-1}(\sqrt{3}) = \frac{-\pi}{3}$,

$\therefore \tan^{-1}(-\sqrt{3}) = \frac{-\pi}{3}$

e) $y = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \Rightarrow \tan y = \frac{1}{\sqrt{3}} \Rightarrow y = \frac{\pi}{6}$,

$\therefore \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

f) $y = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \Rightarrow y = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{-\pi}{6}$,

$\therefore \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \frac{-\pi}{6}$

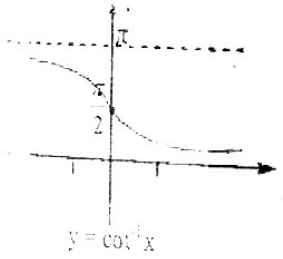
D. Inverse cotangent function

Definition: Inverse cotangent function is denoted by $\cot^{-1}$ is defined by
$y = \cot^{-1} x$ if and only if $x = \cot y$
real number x and $0 < y < \pi$

Function notation

$f^{-1}: IR \longrightarrow (0, \pi)$ is $f^{-1}(x) = \cot^{-1}(x)$

- Domain of $\cot^{-1} x$ is IR
- Range of $\cot^{-1} x$ is $(0,\pi)$



$$\text{Cot}^{-1}x = \frac{\pi}{2} - \tan^{-1} x.$$

nd the exact value of y.

a) $y = \cot^{-1}(\sqrt{3})$

b) $y = \cot^{-1}(-\sqrt{3})$

c) $y = \cot^{-1}(-1)$

d) $y = \cot^{-1}(1)$

e) $y = \cot^{-1}\left(\frac{\sqrt{3}}{3}\right)$

f) $y = \cot^{-1}\left(\frac{-\sqrt{3}}{3}\right)$

Solution: use $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$

a) $y = \cot^{-1}(\sqrt{3}) = \frac{\pi}{2} - \tan^{-1}(\sqrt{3}) = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$

b) $y = \cot^{-1}(-\sqrt{3}) = \frac{\pi}{2} - \tan^{-1}(\sqrt{3}) = \frac{\pi}{2} + \tan^{-1}(\sqrt{3})$

$= \frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

c) $y = \cot^{-1}(-1) = \frac{\pi}{2} - \tan^{-1}(-1) = \frac{\pi}{2} - (-\tan^{-1}(1))$

$= \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$

d) $y = \cot^{-1}(1) = \frac{\pi}{2} - \tan^{-1}(1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

e) $y = \cot^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$

f) $y = \cot^{-1}\left(\frac{-\sqrt{3}}{3}\right) = \frac{\pi}{2} - (-\tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2}{3}$

E) The inverse cosecant function $\csc^{-1} x$ or arc csc x is defined by $y = \csc^{-1} x$ if and only if $x = \csc y$ for $\frac{-\pi}{2} \leq y \leq \frac{\pi}{2}$, $y \neq 0$, $|x|$

Note: If $|x| \geq 1$ then

- $\csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right)$, let $y = \csc^{-1} x$

$$\Rightarrow \csc y = x \Rightarrow \frac{1}{\sin y} = x \Rightarrow \sin y = \frac{1}{x} \Rightarrow y = \sin^{-1}\left(\frac{1}{x}\right)$$

$$\therefore \csc^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$$

7) Find the exact value of y:

a) $y = \csc^{-1}(2)$ c) $y = \csc^{-1}\left(\frac{-\sqrt{2}}{3}\right)$

b) $y = \csc^{-1}(-\sqrt{2})$ d) $y = \csc^{-1}\left(\frac{1}{3}\right)$

Solution: we have $\csc^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$

a) $y = \csc^{-1}(2) \Rightarrow y = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$, $\therefore \csc^{-1}(2) = \frac{\pi}{6}$

b) $y = \csc^{-1}(-\sqrt{2}) \Rightarrow y = \sin^{-1}\left(\frac{-1}{\sqrt{2}}\right) = -\left(\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\right) = -\frac{\pi}{4}$

c) $y = \csc^{-1}\left(\frac{-2}{\sqrt{3}}\right) \Rightarrow y = \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) = -\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{-\pi}{3}$

d) $y = \csc^{-1}\left(\frac{1}{3}\right) \Rightarrow y = \sin^{-1}(3)$ not defined:

F) The inverse secant function

The inverse secant function is denoted by $\sec^{-1}$ (arc sec) is defined $y = \sec^{-1}x$, if and only if $x = \sec y$ where $0 \le y \le \pi$, $y \neq \pi$, $|x| \ge 1$

Note: i) $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right)$ for $|x| \ge 1$

Let $y = \sec^{-1}x \Rightarrow \sec y = x$

$$\Rightarrow \frac{1}{\cos y} = x \Rightarrow \cos y = \frac{1}{x} \Rightarrow y = \cos^{-1}\left(\frac{1}{x}\right)$$

$\therefore \sec^{-1}x = \cos^{-1}\left(\frac{1}{x}\right)$ and

- $\sec^{-1}(-x) = \pi - \cos^{-1}\left(\frac{1}{x}\right) = \pi - \sec^{-1}x$

8) Find the exact value of y:

a) $y = \sec^{-1}(\sqrt{2})$

b) $y = \sec^{-1}(-\sqrt{2})$

c) $y = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

d) $y = \sec^{-1}\left(\frac{-2}{\sqrt{3}}\right)$

e) $y = \sec^{-1}(-1)$

f) $y = \sec^{-1}\left(\frac{-1}{2}\right)$

Solution:

a) $y = \sec^{-1}(\sqrt{2}) \Rightarrow y = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$, $\therefore \sec^{-1}(\sqrt{2}) = \frac{\pi}{4}$

b) $y = \sec^{-1}(\sqrt{2}) \Rightarrow y = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

c) $y = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) \Rightarrow y = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$ $\therefore -\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$

d) $y = \sec^{-1}\left(\frac{-2}{\sqrt{3}}\right) \Rightarrow y = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) = \pi - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$$= \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

e) $y = \sec\text{-}1(-1) \Rightarrow y = \cos\text{-}1(-1) = \pi - \cos\text{-}1(1) = \pi - \pi = 0$

$\therefore -\sec^{-1}(-1) = 0$

f) $y = \sec^{-1}\left(\frac{-1}{2}\right) \Rightarrow y = \cos^{-1}(2) \longleftarrow$ not defined

Composition of trigonometric function, and their inverse

- **If $-1 \le x \le 1$, then**

i) $\sin(\sin^{-1}x) = x = \sin^{-1}(\sin x)$, if $\frac{-\pi}{2} \le x \le \frac{\pi}{2}$

ii) $\cos(\cos^{-1}x) = x = \cos^{-1}(\cos x)$, if $0 \le x \le \pi$

iii) $\tan(\tan^{-1}x) = x = \tan^{-1}(\tan x)$, if $\frac{-\pi}{2} < x < \frac{\pi}{2}$

Note: $\sin^2 x + \cos^2 x = 1 \Rightarrow \sin x = \pm\sqrt{1-\cos^2 x}$

$\cos^2 x = 1 - \sin^2 x \Rightarrow \cos x = \pm\sqrt{1-\sin^2 x}$

9. Find the exact value:

a) $\sin\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)$

b) $\cos\left(\sin^{-1}\left(\frac{12}{13}\right)\right)$

c) $\sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

d) $\cos\left(\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)\right)$

e) $\sin\left(2\cos^{-1}\left(\frac{3}{4}\right)\right)$

f) $\sin\left(2\sin^{-1}\left(\frac{4}{5}\right)\right)$

g) $\sin\left(\tan^{-1}\left(\frac{3}{4}\right)\right)$

h) $\sin\left(2\tan^{-1}\left(\sqrt{3}\right)\right)$

i) $\cos\left(\sec^{-1}(-2)\right)$

j) $\sec\left(\tan^{-1}\left(\frac{2}{3}\right)\right)$

k) $\sin\left(2\cos^{-1}\left(\frac{-1}{2}\right)\right)$

Solution: Use $\sin\theta = \pm\sqrt{1-\cos^2\theta}$, $\cos\theta = \pm\sqrt{1-\sin^2\theta}$

a) $\sin\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

b) $\cos\left(\sin^{-1}\left(\frac{12}{13}\right)\right)$, Let $\sin^{-1}\left(\frac{12}{13}\right) = \theta \Rightarrow \sin\theta = \frac{12}{13}$,

$$\cos\theta = \sqrt{1-\frac{144}{169}} = \frac{5}{13} \Rightarrow \cos\left(\sin^{-1}\left(\frac{12}{13}\right)\right) = \cos\theta = \frac{5}{13}$$

c) $$\sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) = \sin\left(\pi - \frac{\pi}{6}\right) = \sin\frac{5\pi}{6} = \frac{1}{2}$$

d) $$\cos\left(\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)\right) = \cos\left(\frac{-\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

e) $\sin\left(2\cos^{-1}\left(\frac{3}{4}\right)\right)$, Let $\cos^{-1}\left(\frac{3}{4}\right) = \theta \Rightarrow \cos\theta = \frac{3}{4}$

$$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\left(\frac{3}{4}\right)^2} = \frac{\sqrt{7}}{4}$$

$$\Rightarrow \sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{\sqrt{7}}{4}\right)\left(\frac{3}{4}\right) = \frac{6\sqrt{7}}{16}$$

f) $\sin\left(2\sin^{-1}\left(\frac{4}{5}\right)\right)$, Let $\sin^{-1}\left(\frac{4}{5}\right) = \theta \Rightarrow \sin\theta = \frac{4}{5}$

$$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\left(\frac{16}{25}\right)} = \sqrt{\frac{25-16}{25}} = \frac{3}{5}$$

$$\Rightarrow \sin\left(2\sin^{-1}\left(\frac{4}{5}\right)\right) = \sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}$$

g) $\sin\left(\tan^{-1}\left(\frac{3}{4}\right)\right)$, Let $\tan^{-1}\left(\frac{3}{4}\right) = \theta \Rightarrow \tan\theta = \frac{3}{4}$

$$\Rightarrow \sin\theta = \frac{3}{5}, \Rightarrow \sin\left(\tan^{-1}\left(\frac{3}{4}\right)\right) = \sin\theta = \frac{3}{5}$$

h) $$\sin\left(2\tan^{-1}\left(\sqrt{3}\right)\right) = \sin\left(2\left(\frac{-\pi}{3}\right)\right) = \sin\left(\frac{-2\pi}{3}\right) = -\sin\frac{2\pi}{3} = \frac{-\sqrt{3}}{2}$$

i) $\cos\left(\sec^{-1}(-2)\right) \Rightarrow \cos\left(\cos^{-1}\left(\frac{-1}{2}\right)\right) = \cos\left(\frac{2\pi}{3}\right) = \frac{-1}{2}$

j) $\sec\left(\tan^{-1}\left(\frac{2}{3}\right)\right)$. Let $\tan^{-1}\left(\frac{2}{3}\right) = \theta \Rightarrow \tan\theta = \frac{2}{3}$

hypotenuse: $\sqrt{3^2+2^2} = \sqrt{13}$. $\sin\theta = \frac{2}{\sqrt{13}}, \cos\theta = \frac{3}{\sqrt{13}}$

$\Rightarrow \sec\left(\tan^{-1}\left(\frac{2}{3}\right)\right) = \sec(\theta) = \frac{1}{\cos\theta} = \frac{\sqrt{13}}{3}$

k) $\sin\left(2\cos^{-1}\left(\frac{-1}{2}\right)\right)$

$\Rightarrow \cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$\Rightarrow \sin\left(2\cos^{-1}\left(-\frac{1}{2}\right)\right) = \sin\left(2\cdot\frac{2\pi}{3}\right) = \sin\frac{4\pi}{3} = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}$

10. Find the exact value

a) $\sin^{-1}\left(\sin\frac{5\pi}{6}\right)$

b) $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$

c) $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)$

d) $\cos^{-1}\left(\sin\frac{11\pi}{6}\right)$

e) $\tan^{-1}\left(\tan\frac{4\pi}{3}\right)$

f) $\sin^{-1}\left(\sin\frac{5\pi}{3}\right)$

g) $\sin^{-1}\left(\cos\frac{5\pi}{6}\right)$

Solution:

a) $\sin^{-1}\left(\sin\frac{5\pi}{6}\right)=\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

b) $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)=\cos^{-1}\left(-\frac{1}{2}\right)=\pi-\cos^{-1}\left(\frac{1}{2}\right)=\pi-\frac{\pi}{3}$

c) $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)=\sin^{-1}\left(\frac{-\sqrt{2}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=-\frac{\pi}{4}$

d) $\cos^{-1}\left(\sin\frac{11\pi}{6}\right)=\cos^{-1}\left(-\frac{1}{2}\right)=\pi-\cos^{-1}\left(\frac{1}{2}\right)=\pi-\frac{\pi}{3}$

e) $\tan^{-1}\left(\tan\frac{4\pi}{3}\right)=\tan^{-1}\left(\sqrt{3}\right)=\frac{\pi}{3}$

f) $\sin^{-1}\left(\sin\frac{5\pi}{3}\right)=\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=-\frac{\pi}{3}$

g) $\sin^{-1}\left(\cos\frac{5\pi}{6}\right)=\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=-\frac{\pi}{3}$

11. Simplify each of the expression interms of x.

a) sin(arc cos(3x))
b) cos(arc sin(2x))
c) $\sin(2\cos^{-1}x)$
d) $\cos(\sin^{-1}x)$
e) $\sin(\tan^{-1}x)$
f) $\cos(2\tan^{-1}x)$

Solution: use $\sin x=\sqrt{1-\cos^2 x}$, $\cos x=\pm\sqrt{1-\sin^2 x}$

a) $\sin(\cos^{-1}(3x))$ · Let $\cos^{-1}(3x)=\theta\Rightarrow\cos\theta=3x$

$\Rightarrow \sin\theta=\sqrt{1-\cos^2\theta}=\sqrt{1-(3x)^2}=\sqrt{1-9x^2}$ *if* $-\frac{1}{3}\le x\le\frac{1}{3}$

$\Rightarrow \sin(\cos^{-1}(3x))=\sin\theta=\sqrt{1-9x^2}$ · *if* $-\frac{1}{3}\le x\le\frac{1}{3}$

b) $\cos(\sin^{-1}(2x))$·Let $\sin^{-1}(2x)=\theta\Rightarrow\sin\theta=2x$. $\cos\theta=\sqrt{1-4x^2}$

$\Rightarrow \cos\theta=\cos(\sin^{-1}(2x))=\sqrt{1-4x^2}$ · *if* $-\frac{1}{2}\le x\le\frac{1}{2}$

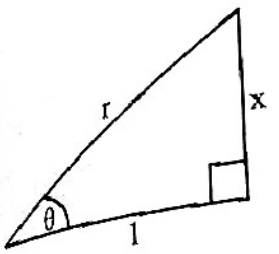
c) $\sin(2\cos^{-1}x)$ · Let $\cos^{-1}x = \theta \Rightarrow \cos\theta = x$, $\sin\theta = \sqrt{1-x^2}$

$\Rightarrow \sin(2\cos^{-1}x) = \sin(2\theta) = 2\sin\theta\cos\theta = 2x\sqrt{1-x^2}$

d) $\cos(\sin^{-1}x)$ · Let $\sin^{-1}x = \theta \Rightarrow \sin\theta = x$, $\cos\theta = \sqrt{1-x^2}$

$\Rightarrow \cos(\sin^{-1}x) = \cos\theta = \sqrt{1-x^2}$

e) $\sin(\tan^{-1}x)$ · Let $\tan^{-1}x = \theta \Rightarrow \tan\theta = x$



$r^2 = x^2 + 1 \Rightarrow r = \sqrt{x^2+1}$ · $\sin\theta = \dfrac{x}{r} = \dfrac{x}{\sqrt{x^2+1}}$

$\Rightarrow \sin(\tan^{-1}x) = \sin\theta = \dfrac{x}{\sqrt{x^2+1}}$

Example: $\sin(\tan^{-1}(2) = \dfrac{2}{\sqrt{4+1}} = \dfrac{2}{\sqrt{5}}$

f) $\cos(2\tan^{-1}x)$ · Let $\tan^{-1}x = \theta \Rightarrow \tan\theta = x$

$\Rightarrow \sin\theta = \dfrac{x}{\sqrt{x^2+1}}$, $\cos\theta = \dfrac{1}{r} = \dfrac{1}{\sqrt{x^2+1}}$

$\Rightarrow \cos(2\tan^{-1}x) = \cos2\theta = \cos^2\theta - \sin^2\theta = \dfrac{1}{x^2+1} - \dfrac{x^2}{x^2+1} = \dfrac{1-x^2}{x^2+1}$

Example: $\cos(2\tan^{-1}(3)) = \dfrac{1-(3)^2}{3^2+1} = \dfrac{-8}{10} = \dfrac{-4}{5}$

12. Express x interms of y and determine the range of values x and y.

a) $y = \dfrac{1}{2}\sin^{-1}(x-3)$ c) $y = 2 + 3\cos^{-1}(5x-1)$

b) $y = 2 + 3$ arc sin $(5x-1)$

Solution: Here

a) $y = \dfrac{1}{2}\sin^{-1}(x-3) \Leftrightarrow 2y = \sin^{-1}(x-3)$

$\Rightarrow \dfrac{-\pi}{2} \le 2y \le \dfrac{\pi}{2}$ *and* $-1 \le x-3 \le 1$

$\Rightarrow x-3 = \sin 2y, \dfrac{-\pi}{4} \le y \le \dfrac{\pi}{4}$ *and* $2 \le x \le 4$

$$\therefore\ x = 3 + \sin 2y, \frac{-\pi}{4} \le y \le \frac{\pi}{4} \text{ and } 2 \le x \le 4$$

b) $\dfrac{y-2}{3} = \sin^{-1}(5x-1)$

$$\Rightarrow \frac{-\pi}{2} \le \frac{y-2}{3} \le \frac{\pi}{2} \text{ and } -1 \le 5x-1 \le 1$$

$$\Rightarrow \sin\left(\frac{y-2}{3}\right) = 5x-1 \text{ if } 0 \le x \le \frac{2}{5}$$

$$\therefore\ x = \frac{1}{5} + \frac{1}{5}\sin\left(\frac{y-2}{3}\right) \text{ if } 0 \le x \le \frac{2}{5} \text{ and } 2-3\pi \le y \le 2+3\pi$$

c) $\dfrac{y-2}{3} = \cos^{-1}(5x-1)$ if $0 \le \dfrac{y-2}{3} \le \pi$ and $-1 \le 5x-1 \le 1$

$$\Rightarrow \cos\left(\frac{y-2}{3}\right) = 5x-1 \text{ if } 0 \le y-2 \le 3\pi \text{ and } 0 \le 5x \le 2$$

$$\Rightarrow 5x = 1 + \cos\left(\frac{y-2}{3}\right) \text{ if } 2 \le y \le 2+3\pi \text{ and } 0 \le x \le \frac{2}{5}$$

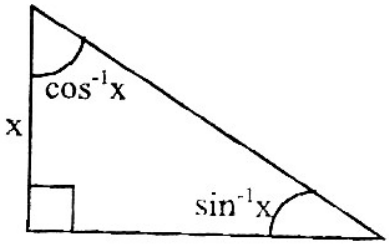
$$\therefore\ x = \frac{1}{5} + \frac{1}{5}\cos\left(\frac{y-2}{3}\right) \text{ if } 2 \le y \le 2+3\pi \text{ and } 0 \le x \le \frac{2}{5}$$

13. Simplify each of the following

a) $\sin^{-1}x + \cos^{-1}x$
b) $\cos^{-1}(-x) + \cos^{-1}(x)$
c) $\sin(\sin^{-1}x + \cos^{-1}x)$
d) $\cos(\cos^{-1}(-x) + \cos^{-1}x)$
e) $\cos(\sin^{-1}x + \sin^{-1}(-x))$

Solution:

a) $\sin^{-1}x + \cos^{-1}x$, let $\sin^{-1}x = \alpha \Rightarrow \sin\alpha = x$



let $\cos^{-1}x = \beta \Rightarrow \cos\beta = x$

since it is right angle $\therefore \alpha + \beta = \dfrac{\pi}{2}$

$$\therefore \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}.$$

b) $\cos^{-1}(-x) = \pi - \cos^{-1}x \Rightarrow \pi - \cos^{-1}x + \cos^{-1}x = \pi$

$\therefore \cos^{-1}(-x) + \cos^{-1}x = \pi$

$$\Rightarrow \sin^{-1}\left(\frac{3}{5}\right) = \pi - \cos^{-1} x, \text{ Recall.} \cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$\Rightarrow \sin^{-1}\left(\frac{3}{5}\right) = \cos^{-1}(-x)$$

$$\Rightarrow -x = \cos\left(\sin^{-1}\left(\frac{3}{5}\right)\right), \text{ Let } \sin^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$$

$$\Rightarrow -x = \frac{4}{5}, \text{ therefore } x = \frac{-4}{5}$$

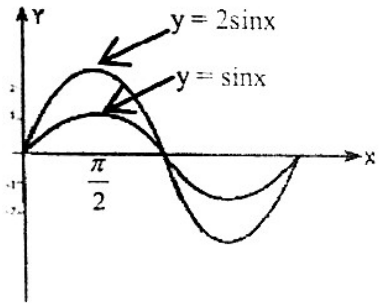
Graph of some trigonometric function

Let y = sinx and y = 2sinx. Then state:

a) Period
b) x-intercept
c) the amplitude
d) range
e) the maximum and minimum value and the value of x at which the maximum and minimum occur for y = 2sinx
f) sketch the graph on the same axes.

Solution:

a) The period: 2π
b) x-intercept: $x = \pi k$, k is an integer
c) Amplitude for, y = sinx is |1| = 1
   Amplitude for y = 2sinx is |2| = 2
d) Range for y = sinx is [-1, 1]
   Range for y = 2 sinx is [-2, 2]
e) The maximum is 2 which occurs



at $x = \frac{\pi}{2} + 2\pi k$, k is integer

Minimum is -2, which occurs at $x = \frac{3\pi}{2} + 2\pi k$

Multiplying periodic function by a constant affects: the amplitude

ition: Amplitude |a|

c) $\sin(\sin^{-1}x + \cos^{-1}x) = \sin\left(\frac{\pi}{2}\right) = 1$

d) $\cos(\cos^{-1}(-x) + \cos^{-1}x = \cos\pi = -1$

e) $\cos(\sin^{-1}x + \sin^{-1}(-x)) = \cos(\sin^{-1}x + -\sin^{-1}x) = \cos 0 = 1$

14. Find the exact value of the expression

a) $\cos\left(\sin^{-1}\left(\frac{3}{4}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right)$

b) $\sin\left(\sin^{-1}\left(\frac{2}{3}\right) + \cos^{-1}\left(\frac{1}{2}\right)\right)$

Solution:

a) Let $\sin^{-1}\left(\frac{3}{4}\right) = \alpha \Rightarrow \sin\alpha = \frac{3}{4} \Rightarrow \cos\alpha = \sqrt{1-\left(\frac{3}{4}\right)^2} = \frac{\sqrt{7}}{4}$

Let $\cos^{-1}\left(\frac{5}{13}\right) = \beta \Rightarrow \cos\beta = \frac{5}{13}$

$\Rightarrow \sin\beta = \sqrt{1-\left(\frac{5}{13}\right)^2} = \frac{12}{13}$

$\Rightarrow \cos\left(\sin^{-1}\left(\frac{3}{4}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right) = \cos(\alpha+\beta)$

$\Rightarrow \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$= \frac{\sqrt{7}}{4}\cdot\frac{5}{13} - \frac{3}{4}\cdot\frac{12}{13} = \frac{5\sqrt{7}-36}{52}$

$\therefore \cos\left(\sin^{-1}\left(\frac{3}{4}\right) + \cos^{-1}\left(\frac{5}{13}\right)\right) = \frac{5\sqrt{7}-36}{52}$

b) Exercise left for you

15. Solve each of the following

a) $\sin^{-1}x = \cos^{-1}\left(\frac{5}{13}\right)$ c) $\cos^{-1}\left(x-\frac{1}{2}\right) = \frac{\pi}{3}$

b) $\sin^{-1}(x-1) = \frac{\pi}{2}$ d) $\tan^{-1}(x) = \sin^{-1}\left(\frac{3}{5}\right)$

e) $\sin^{-1}\left(\frac{3}{5}\right)+=\frac{\pi}{3}$

Solution:

a) $\sin^{-1}x=\cos^{-1}\left(\frac{5}{13}\right)\Leftrightarrow x=\sin\left(\cos^{-1}\left(\frac{5}{13}\right)\right)$

$$Let\ \cos^{-1}\left(\frac{5}{13}\right)=\theta\Rightarrow\cos\theta=\frac{5}{13}$$

$$\Rightarrow\sin\theta=\sqrt{1-\left(\frac{5}{13}\right)^2}=\sqrt{1-\frac{25}{169}}=\frac{12}{13}$$

$$\therefore x=\sin\left(\cos^{-1}\left(\frac{5}{13}\right)\right)=\sin\theta=\frac{12}{13}$$

b) $\sin^{-1}(x-1)=\frac{\pi}{2}\Rightarrow x-1=\sin\frac{\pi}{2}=1\Rightarrow x=1+1=2$

c) $\cos^{-1}\left(x-\frac{1}{2}\right)=\frac{\pi}{3}\Rightarrow x-\frac{1}{2}=\cos\frac{\pi}{3}=\frac{1}{2}$

$$\Rightarrow x-\frac{1}{2}=\frac{1}{2}$$

$$\therefore x=\frac{1}{2}+\frac{1}{2}=1$$

d) $\tan^{-1}x=\sin^{-1}\left(\frac{3}{5}\right)\Leftrightarrow x=\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$

$$\Rightarrow \text{Let } \sin^{-1}\left(\frac{3}{5}\right)=\theta\Rightarrow\sin\theta=\frac{3}{5}\Rightarrow\cos\theta=\frac{4}{5}$$

$$\therefore \tan\theta=\frac{\sin\theta}{\cos\theta}=\frac{3}{4},\ \therefore x=\frac{3}{4}$$

e) $\sin^{-1}\left(\frac{3}{5}\right)+\cos^{-1}x=\pi$

$$\Rightarrow \sin^{-1}\left(\frac{3}{5}\right) = \pi - \cos^{-1} x, \text{ Recall.} \cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$\Rightarrow \sin^{-1}\left(\frac{3}{5}\right) = \cos^{-1}(-x)$$

$$\Rightarrow -x = \cos\left(\sin^{-1}\left(\frac{3}{5}\right)\right), \; Let \; \sin^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$$

$$\Rightarrow -x = \frac{4}{5}, \; therefore \quad x = \frac{-4}{5}$$

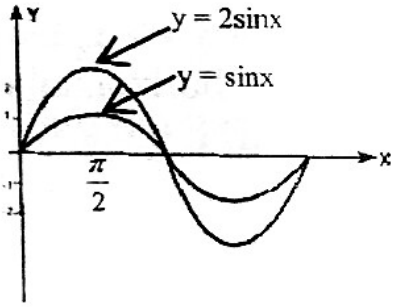
Graph of some trigonometric function

16. Let y = sinx and y = 2sinx. Then state:

a) Period
b) x-intercept
c) the amplitude
d) range
e) the maximum and minimum value and the value of x at which the maximum and minimum occur for y = 2sinx
f) sketch the graph on the same axes.

Solution:

a) The period: 2π
b) x-intercept: $x = \pi k$, k is an integer
c) Amplitude for, y = sinx is |1| = 1
   Amplitude for y = 2sinx is |2| = 2
d) Range for y = sinx is [-1, 1]
   Range for y = 2 sinx is [-2, 2]
e) The maximum is 2 which occurs



at $x = \frac{\pi}{2} + 2\pi k$, k is integer

Minimum is -2, which occurs at $x = \frac{3\pi}{2} + 2\pi k$

Note: Multiplying periodic function by a constant affects: the amplitude

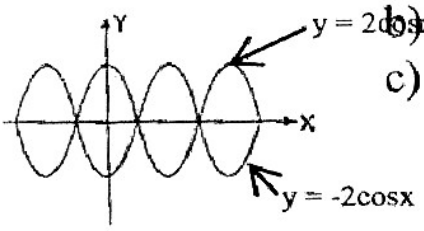
Definition: Amplitude |a|

The amplitude of periodic function with maximum M and minimum m is

$$|a| = \frac{M - m}{2}$$

17. Let y = -2cosx, then state
    a) Amplitude
    b) the range
    c) period
    d) how the graph differ from the graph of y = 2cosx

Solution: a) Amplitude is |-2| = 2



b) Range is [-2, 2]

c) period is 2π

d) It is the reflection of the graph of y = 2cos about the x-axis.

Graph of y = asin(kx) and y = acos kx

- Amplitude = |a|
- Period = $\frac{2\pi}{|k|}$
- Effect of k to stretch or compress the cycle of the graph by factor of $\frac{1}{|k|}$

  i) if k > 1, the resulting graph is compressed (contract) horizontally (not vertically)

  ii) If 0 < k < 1, thee resulting graph is stretched horizontally

18. Find, thee amplitude, period, x-intercept, the value of x wh give minimum or maximum value sketch the graph

a) f(x) = 3sin(2x)

b) f(x) = 4cos(πx)

c) $f(x) = \cos\left(\frac{1}{4}x\right)$

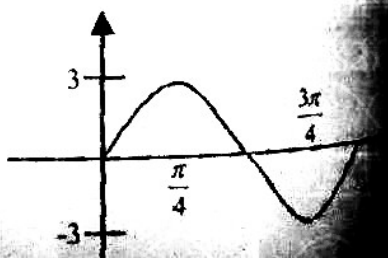
d) f(x) = 3cos2x

e) $f(x) = 4\cos\left(\frac{-3x}{2}\right)$

Solution:

a) f(x) = 3sin(2x)

- Amplitude = |3| = 3,



- period, $P = \frac{2\pi}{2} = \pi$
- x-intercept (the graph crosses the x-axis:

  $3\sin 2x = 0 \Rightarrow 2x = k\pi$, k is integer $\Rightarrow x = \frac{\pi}{2}k$

  $\therefore$ the graph cross the x-axis at

  $$(-\pi,0),\left(\frac{-\pi}{2},0\right),(0,0),\left(\frac{\pi}{2},0\right),(\pi,0)$$

- maximum value is 3 which occurs:

  $\Rightarrow 3\sin 2x = \Rightarrow \sin 2x = 1$

  $\Rightarrow 2x = \frac{\pi}{2} + 2\pi k \Rightarrow x = \frac{\pi}{4} + \pi k, k \in z$

  $\therefore$ maximum attain, at $x = \frac{\pi}{4} + \pi k$

- $\sin 2x = -1 \Rightarrow 2x = \frac{3\pi}{2} + 2\pi k$, k is integer

  $\Rightarrow x = \frac{3\pi}{2} + \pi k$

  $\therefore$ The function attains its minimum value at

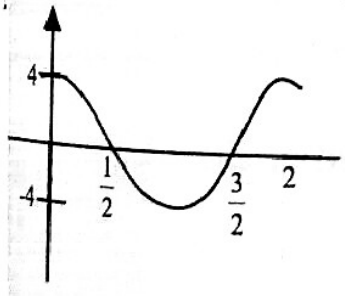
  $x = \frac{3\pi}{4} + \pi k$

- One cycle of each graph is completed on the interval:

  $0 \le 2x \le 2\pi \Rightarrow 0 \le x \le \pi$

b) $f(x) = 4\cos \pi x$

- Amplitude = 4,
- Period = $\frac{2\pi}{\pi} = \pi$
- The graph crosses the x-axis: $\cos \pi x = 0$



$\Rightarrow \pi x = \frac{\pi}{2}, \frac{3}{2}\pi, \frac{5\pi}{2}, \ldots\ldots$

$\Rightarrow x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots\ldots$

$\therefore$ x-intercept $\left(\frac{1}{2},0\right),\left(\frac{3}{2},0\right),\left(\frac{5}{2},0\right)\ldots$

- $3\cos\pi x = 3 \Rightarrow \cos\pi x = 1$

  $\cos\pi x = 1 \Rightarrow \pi x = 0 + 2\pi k \Rightarrow x = 2k$, k is integer

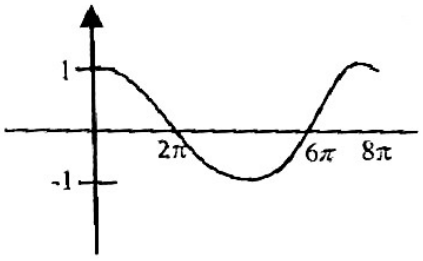
  $\therefore$ the function attain maximum at $x = 2k$
- $3\cos\pi x = -3 \Rightarrow \cos\pi x = -1$

  $\Rightarrow \pi x = \pi + 2\pi k \Rightarrow x = 1 + 2k,\ k \in z$

  $\therefore$ the function attain minimum value at $x = 1+2k$
- One cycle is completed on the interval:

  $0 \le \pi x \le 2\pi \Rightarrow 0 \le x \le 2$

c) $f(x) = \cos\left(\frac{1}{4}x\right)$



- Amplitude = |1| = 1, • Period, $P = \frac{2\pi}{\frac{1}{4}} = 8\pi$
- The graph crosses the x-axis: $\cos\left(\frac{1}{4}x\right) = 0$

$$\Rightarrow \frac{1}{4}x = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow x = 2\pi,\ 6\pi$$

$\therefore$ x-intercept $(-6\pi, 0), (-2\pi, 0), (2\pi, 0), (6\pi, 0)$

- $\cos\left(\frac{1}{4}x\right) = 1 \Rightarrow \frac{1}{4}x = 0 + 2\pi k \Rightarrow x = 8\pi k$

  $\therefore$ the function attain maximum value at $x = 8\pi k$, k is integer
- $\cos\left(\frac{1}{4}x\right) = -1 \Rightarrow \frac{1}{4}x = \pi + 2\pi k \Rightarrow x = 4\pi + 8\pi k$

  $\therefore$ the function attain minimum value at $x = (1+2k)4\pi$, k is integer
- One cycle of each graph is completed on the interval:

$$0 \le \frac{1}{4}x \le 2\pi \Rightarrow 0 \le x \le 8\pi$$

d, e. exercise left for you

18b) Find the amplitude and period fo the following graph.

a) $f(x) = 6\sin\left(\frac{2x}{5}\right)\cos\left(\frac{2x}{5}\right)$

b) $f(x) = 4\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right)$

Solution: Use $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

a) $6\sin\left(\frac{2x}{5}\right)\cos\left(\frac{2x}{5}\right) = (6)\frac{1}{2}\left[\sin\left(\frac{2x}{5}+\frac{2x}{5}\right) + \sin\left[\frac{2x}{5} - \frac{2x}{5}\right]\right]$

$\therefore f(x) = 6\sin\left(\frac{2x}{5}\right)\cos\left(\frac{2x}{5}\right) = 3\sin\left(\frac{4x}{5}\right) + 0$

Then comparing $y = 3\sin\left(\frac{4x}{5}\right)$ to $y = A\sin(wx)$

Amplitude $= |A| = 3$

Period: $P = \frac{2\pi}{w} = \frac{2\pi}{\frac{4}{5}} = \frac{10\pi}{4} = \frac{5\pi}{2}$

b) Exercise left for you. (Ans. 2, 3π)

19. Find an equation for a sine function with the given information.

a) Amplitude a = 2, period, $P = 3\pi$

b) Amplitude a = 2, period, $P = \frac{2\pi}{3}$

c) Amplitude a = 4, period, P = 2

Solution: We have, $y = a\sin(kx)$

a) $|a| = 2$, period, $P = \frac{2\pi}{k} = 3\pi \Rightarrow 3\pi k = 2\pi$

$\Rightarrow k = \frac{2}{3}$ $\qquad \therefore y = \pm 2\sin\left(\frac{2x}{3}\right)$

b) $|a| = 2$, Period, $P = \frac{2\pi}{k} = \frac{2\pi}{3} \Rightarrow k = 3,$

$\therefore y = \pm 2\sin(3x)$

c) $|a| = 2$, Period, $P = \frac{2\pi}{k} = 2 \Rightarrow k = \pi$

$\therefore y = \pm 2\sin \pi x$

20. Find a function of the form $f(x) = a\cos kx$ satisfying the given properties

a) Amplitude, $a = \frac{2}{3}$, and $f(4) = 0$

b) Amplitude, $a = 3$, and $f(2) = 3$

c) Amplitude, $a = 2$, and the graph pass through $\left(\frac{\pi}{4}, 0\right)$

d) `Peak at $\left(\frac{\pi}{5}, 3\right)$

Solution: $f(x) = a \cos kx$

a) $|a| = \frac{2}{3}$, $f(4) = 0 \Rightarrow \cos(4k) = 0$

$\Rightarrow 4k = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow k = \frac{\pi}{8}$ *or* $k = \frac{3\pi}{8}$

$\therefore f(x) = \pm\frac{2}{3}\cos\left(\frac{\pi}{8}x\right)$ or $f(x) = \pm\frac{2}{3}\cos\left(\frac{3\pi}{8}x\right)$

b) $|a| = 3 \Rightarrow a = \pm 3$ and $f(2) = a\cos(2k) = 3$

$\Rightarrow 3\cos(2k) = 3 \Rightarrow \cos(2k) = 1$

$\Rightarrow 2k = 0, 2\pi, 4\pi \ldots \Rightarrow k = 0, \pi$

$\therefore f(x) = 3\cos(\pi x)$

c) $f\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}k\right) = 0 \Rightarrow \frac{\pi k}{4} = \frac{\pi}{2} \Rightarrow k = 2$

$\therefore f(x) = \pm 2\cos(2x)$

d) Peak at $x = \frac{\pi}{5} \Rightarrow f\left(\frac{\pi}{5}\right) = \cos\left(\frac{\pi}{5}k\right) = 3 \Rightarrow a = 3$

$\Rightarrow \cos\left(\frac{\pi}{5}k\right) = 1 \Rightarrow \frac{\pi}{5}k = 2\pi \Rightarrow k = 10$

$\therefore f(x) = 3\cos(10x)$

Graph of $f(x) = a\sin(kx + b) + c$ and $f(x) = a\cos(kx + b) + c$

- Amplitude = |a|
- Period, $P = \frac{2\pi}{k}, k > 0$
- Range = [c - |a|, c + |a|]
- Phase angle = -b
- Phase shift = $\frac{-b}{k}$ ⟵ the horizontal shift of the graph from y = a sinkx and y = acoskx along x –axis.
- One cycle of each graph is completed on the interval: $0 \le kx + b \le 2\pi \Rightarrow 0 - b \le kx \le 2\pi - b$

$$\Rightarrow \frac{-b}{k} \le x \le \frac{-b}{k} + \frac{2\pi}{k}$$

i) $x = \frac{-b}{k}$ ⟵ left end point of graph

ii) $x = \frac{-b}{k} + \frac{2\pi}{k}$ ⟵ Right end point of graph

- Vertical shift C – unit up in the positive y-direction if c > 0 and C unit down if C < 0 in negative y – direction

21. Find the amplitude, period, phase shift, phase angle, interval for one cycle, the range.

a) $f(x) = -\frac{1}{2}\sin(4x - 2)$

b) $f(x) = \frac{3}{2}\sin\left(2x + \frac{\pi}{4}\right)$

c) $f(x) = 3\cos\left(3x + \frac{3\pi}{4}\right)$

d) $f(x) = 4\sin\left(\frac{2}{3}x + \frac{\pi}{6}\right)$

e) $f(x) = \frac{5}{4}\cos(3x + 2)$

f) $f(x) = 3\sin\left(\frac{1}{2}x + 3\right) - 2$

g) $f(x) = 2\cos\left(\frac{3}{2}x + \frac{\pi}{4}\right) + 4$

h) $f(x) = -3\sin(\pi x + 1) - 1$

Solution:

a) $f(x) = -\frac{1}{2}\sin(4x - 2)$

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c) $|a| = 2$, Period, $P = \frac{2\pi}{k} = 2 \Rightarrow k = \pi$

$\therefore y = \pm 2\sin \pi x$

20. Find a function of the form $f(x) = a\cos kx$ satisfying the given properties

a) Amplitude, $a = \frac{2}{3}$, and $f(4) = 0$

b) Amplitude, $a = 3$, and $f(2) = 3$

c) Amplitude, $a = 2$, and the graph pass through $\left(\frac{\pi}{4}, 0\right)$

d) `Peak at $\left(\frac{\pi}{5}, 3\right)$

Solution: $f(x) = a \cos kx$

a) $|a| = \frac{2}{3}$, $f(4) = 0 \Rightarrow \cos(4k) = 0$

$\Rightarrow 4k = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow k = \frac{\pi}{8}$ *or* $k = \frac{3\pi}{8}$

$\therefore f(x) = \pm\frac{2}{3}\cos\left(\frac{\pi}{8}x\right)$ or $f(x) = \pm\frac{2}{3}\cos\left(\frac{3\pi}{8}x\right)$

b) $|a| = 3 \Rightarrow a = \pm 3$ and $f(2) = a\cos(2k) = 3$

$\Rightarrow 3\cos(2k) = 3 \Rightarrow \cos(2k) = 1$

$\Rightarrow 2k = 0, 2\pi, 4\pi \ldots \Rightarrow k = 0, \pi$

$\therefore f(x) = 3\cos(\pi x)$

c) $f\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}k\right) = 0 \Rightarrow \frac{\pi k}{4} = \frac{\pi}{2} \Rightarrow k = 2$

$\therefore f(x) = \pm 2\cos(2x)$

d) Peak at $x = \frac{\pi}{5} \Rightarrow f\left(\frac{\pi}{5}\right) = \cos\left(\frac{\pi}{5}k\right) = 3 \Rightarrow a = 3$

$\Rightarrow \cos\left(\frac{\pi}{5}k\right) = 1 \Rightarrow \frac{\pi}{5}k = 2\pi \Rightarrow k = 10$

$\therefore f(x) = 3\cos(10x)$

Graph of $f(x) = a\sin(kx + b) + c$ and $f(x) = a\cos(kx + b) + c$

- Amplitude, $|a| = \left|\frac{-1}{2}\right| = \frac{1}{2}$, Period, $P = \frac{2\pi}{k} = \frac{2\pi}{4} = \frac{\pi}{2}$
- Phase shift, $4x - 2 = 0 \Rightarrow x = \frac{2}{4} = \frac{1}{2} \leftarrow$ Horizontal shift
- One cycle, $0 \le 4x - 2 \le 2\pi \Rightarrow \frac{1}{2} \le x \le \frac{1}{2} + \frac{\pi}{2}$
- Range $\left[\frac{-1}{2}, \frac{1}{2}\right]$

b) $f(x) = \frac{3}{2}\sin\left(2x + \frac{\pi}{4}\right)$, Amplitude, $a = \left|\frac{-3}{2}\right| = \frac{3}{2}$,

- Period, $P = \frac{2\pi}{k} = \frac{2\pi}{2} = \pi$
- Phase shift, $2x + \frac{\pi}{4} = 0 \Rightarrow 2x = \frac{-\pi}{4} \Rightarrow x = \frac{-\pi}{8}$
- Phase angle $\frac{\pi}{4}$
- One cycle, covers, $0 \le 2x + \frac{\pi}{4} \le 2\pi$

  $\Rightarrow \frac{-\pi}{8} \le x \le \frac{-\pi}{8} + \frac{2\pi}{2} \Rightarrow \frac{-\pi}{8} \le x \le \frac{7\pi}{8}$
- Range $\left[\frac{-3}{2}, \frac{3}{2}\right]$
- The graph has maximum at: $2x + \frac{\pi}{4} = \frac{\pi}{2}$

  $\Rightarrow x = \frac{\pi}{4} - \frac{\pi}{8} = \frac{\pi}{8}$
- The graph has minimum, $\sin\left(2x + \frac{\pi}{4}\right) = -1$

$$\Rightarrow 2x+\frac{\pi}{4}=\frac{3\pi}{2}\Rightarrow x=\frac{5\pi}{8}$$

c) $f(x)=3\cos\left(3x+\frac{3\pi}{4}\right)$, Amplitude, $|a|=|3|\Rightarrow a=3$

- Period, $P=\frac{2\pi}{k}\Rightarrow P=\frac{2\pi}{3}$
- Phase shift, $\frac{-b}{k}=\frac{-3\pi}{(3)(4)}=\frac{-\pi}{4}$ *or* $3x+\frac{3\pi}{4}=0\Rightarrow x=\frac{-\pi}{4}$
- Phase angle $\frac{3\pi}{4}$
- One cycle, covers, $0\le 3x+\frac{3\pi}{4}\le 2\pi$

$$\Rightarrow \frac{-\pi}{4}\le x\le \frac{-\pi}{4}+\frac{2\pi}{3}$$

$$\Rightarrow \frac{-\pi}{4}\le x\le \frac{5\pi}{12}$$

- The graph has maximum when $\cos\left(3x+\frac{3\pi}{4}\right)=1$

$$\Rightarrow 3x+\frac{3\pi}{4}=2\pi\Rightarrow 3x=\frac{5\pi}{4}\Rightarrow x=\frac{5\pi}{12}$$

and at $x=\frac{-\pi}{4}$

- The graph has minimum when, $\cos\left(3x+\frac{3\pi}{4}\right)=-1$

$$\Rightarrow 3x+\frac{3\pi}{4}=\pi\Rightarrow 3x=\frac{\pi}{4}\Rightarrow x=\frac{\pi}{12}$$

d and e are exercise left for you

f) $f(x)=3\sin\left(\frac{1}{2}x+3\right)-2$, Amplitude, $a=3$

- Period, P = $\frac{2\pi}{k} = \frac{2\pi}{\frac{1}{2}} \Rightarrow P = 4\pi$
- Phase shift, $\frac{-b}{k} = \frac{-3}{\frac{1}{2}} = -6$
- Phase angle = –3
- One cycle, completed, $0 \le \frac{1}{2}x + 3 \le 2\pi$

  $\Rightarrow -6 \le x \le -6 + 4\pi$
- The graph has maximum when $\sin\left(\frac{1}{2}x + 3\right) = 1$

  $\Rightarrow \frac{1}{2}x + 3 = \frac{\pi}{2} \Rightarrow \frac{1}{2}x = \frac{\pi}{2} - 3 \Rightarrow x = \pi - 6$
- The graph has minimum when, $\sin\left(\frac{1}{2}x + 3\right) = -1$

  $\Rightarrow \frac{1}{2} + 3 = \frac{3\pi}{2} \Rightarrow \frac{1}{2}x = \frac{3\pi}{2} - 3 \Rightarrow x = 3\pi - 6$
- Range

  $[c - |a|,\ c + |a|] = [-2 - 3, -2 + |3|] = [-5, 1]$

g) f(x) = $2\cos\left(\frac{3}{2}x + \frac{\pi}{4}\right) + 4$, Amplitude, a = |2| = 2

- Period, P = $\frac{2\pi}{k} = \frac{2\pi}{\frac{3}{2}} = \frac{4\pi}{3}$
- Phase shift, $\frac{-b}{k} = \frac{\frac{-\pi}{4}}{\frac{1}{2}} = \frac{-\pi}{4} \cdot \frac{2}{3} = \frac{-\pi}{6}$
- Phase angle $\frac{\pi}{4}$

- One cycle of each graph is completed on interval $0 \le \frac{3}{2}x + \frac{\pi}{4} \le 2\pi \Rightarrow \frac{-\pi}{6} \le x \le \frac{-\pi}{6} + \frac{4\pi}{3}$
- Range [-2+4, 2+4] = [2, 6]
- The graph has maximum when $\cos\left(\frac{3}{2}x + \frac{\pi}{4}\right) =$

$$\Rightarrow \frac{3}{2}x + \frac{\pi}{4} = 0 \ \ and \ \ \frac{3}{2}x + \frac{\pi}{4} = 2\pi$$

$$\Rightarrow x = \frac{-\pi}{6} \ \ and \ \ x = \frac{7\pi}{6}$$

h) exercise left for you

22. Find a function of the form f(x) = a sin(kx + b) + c satisfying the given properties.

a) Amplitude, a = 2, period, P = π, and phase shift $\frac{\pi}{3}$.

b) Amplitude, a = 3, period, P = 6, and phase shift $\frac{\pi}{4}$.

Solution: a) $|a| = 2 \Rightarrow a = \pm 2$, Period, $P = \frac{2\pi}{k} = \pi \Rightarrow k =$

- Phase shift, $\frac{-b}{k} = \frac{\pi}{3} \Rightarrow \frac{-b}{2} = \frac{\pi}{3} \Rightarrow b =$

$$\therefore f(x) = \pm 2\sin\left(2x - \frac{2\pi}{3}\right)$$

b) exercise left for you

Trigonometric equation

Particular and General solution

Summary of features of trigonometric equation

Equation	General solution of equation
• $\sin x = 0$,	$\Rightarrow x = \pi k$, k is integer,
• $\sin x = 1$,	$\Rightarrow x = \frac{\pi}{2} + 2\pi k$, k is integer,

- $\sin x = -1, \Rightarrow x = \frac{3\pi}{2} + 2\pi k$, k is integer,
- $\cos x = 0, \Rightarrow x = \frac{\pi}{2} + 2\pi k$, k is integer,
- $\cos x = 1, \Rightarrow x = 2\pi k$, k is integer,
- $\cos x = -1, \Rightarrow x = \pi + 2\pi k \Rightarrow x = (1 + 2k)\pi$, k is integer
- $\tan x = 0, \Rightarrow x = \pi k$, k is integer,
- $\tan x = 1, \Rightarrow x = \frac{\pi}{4} + \pi k$, since the period of tangent is π,
- $\tan x = -1, \Rightarrow x = \frac{3}{4}\pi + \pi k$, k is integer

Solving trigonometric Equation

There are two type of solution: These are

- **Particular Solution** is the Solution in its period.
- **General Solution** is particular Solution plus integer multiple of period of given trig. due to periodic nature of trigonometric function.

23. Find
    i) Particular Solution
    ii) General Solution for each of the following trigonometric equation.

a) $\sin x = \frac{1}{2}$

b) $\sin x = \frac{-1}{2}$

c) $\sin(3x) = \frac{1}{2}$

d) $\sin(3x) = \frac{-1}{2}$

e) $\sin\left(2x - \frac{\pi}{2}\right) = \frac{1}{2}$

f) $\sin(4x) = \frac{-\sqrt{3}}{2}$

g) $\sin x = \frac{-\sqrt{2}}{2}$

h) $\sin 6x = \frac{-\sqrt{2}}{2}$

i) $\cos x = \frac{\sqrt{3}}{2}$

j) $\cos(3x) = \frac{-\sqrt{3}}{2}$

k) $\tan x = \sqrt{3}$

Solution: a) $\sin x = \frac{1}{2}$

i) Particular Solution, $\left\{x : x = \frac{\pi}{6} \text{ and } x = \pi - \frac{\pi}{6}\right\} = \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$

in $0 \le x \le 2\pi$

ii) General Solution $= \left\{x : x = \frac{\pi}{6} + 2\pi k, \frac{5\pi}{6} + 2\pi k\right\}$

b) $\sin x = \frac{-1}{2}$

i) Particular Solution lies in 3rd and 4th quadrant

$\therefore S.S = \left\{\frac{\pi}{6} + \pi, \pi + \frac{5\pi}{6}\right\} = \left\{\frac{7\pi}{6}, \frac{11\pi}{6}\right\}$

ii) General Solution $= \left\{\frac{7\pi}{6} + 2\pi k, \frac{11\pi}{6} + 2\pi k\right\}$

c) $\sin(3x) = \frac{1}{2} \Rightarrow 3x = \frac{\pi}{6} \text{ and } 3x = \frac{5\pi}{6}$

$\Rightarrow x = \frac{\pi}{18} \text{ and } x = \frac{5\pi}{18}$

Period, $P = \frac{2\pi}{3}$

i) Particular Solution $= \left\{x : x = \frac{\pi}{18}, \frac{5\pi}{18}\right\}$

ii) General Solution $= \left\{x : x = \frac{\pi}{18} + \frac{2\pi k}{3}, \frac{5\pi}{18} + \frac{2\pi k}{3}\right\}$

d) $\sin(3x) = \frac{-1}{2} \Rightarrow -3x = \frac{\pi}{6} \text{ and } -3x = \frac{5\pi}{6}$

$\Rightarrow x = \frac{-\pi}{18} \text{ and } x = \frac{-5\pi}{18}$

$\Rightarrow x = \frac{-\pi}{18} + \frac{2\pi}{3} \text{ and } x = \frac{-5\pi}{18} + \frac{2\pi}{3}$

$$\Rightarrow x=\frac{11\pi}{18} \text{ and } x=\frac{7\pi}{18}$$

i) Particular Solution in $\left[0,\frac{2\pi}{3}\right]$ are $\left\{x: x=\frac{11\pi}{18}, \frac{7\pi}{18}\right\}$

ii) General Solution $=\left\{x: x=\frac{11\pi}{18}+\frac{2}{3}\pi k,\ \frac{7\pi}{18}+\frac{2\pi k}{3}\right\}$

e) $\sin\left(2x-\frac{\pi}{2}\right)=\frac{1}{2}\Rightarrow 2x-\frac{\pi}{2}=\frac{\pi}{6}$ and $2x-\frac{\pi}{2}=\frac{5\pi}{6}$

$$\Rightarrow 2x=\frac{4\pi}{6}\Rightarrow x=\frac{\pi}{3} \text{ and } x=\frac{2\pi}{3}$$

$$\text{Period, P}=\frac{2\pi}{2}=\pi$$

i) Particular Solution interval $[0,\pi]$ are $\left\{\frac{\pi}{3}, \frac{5\pi}{6}\right\}$

ii) General Solution $=\left\{\frac{\pi}{3}+\pi k,\ \frac{5\pi}{6}+\pi k\right\}$

f) $\sin(4x)=\frac{-\sqrt{3}}{2}\Rightarrow -4x=\frac{\pi}{3}$ and $-4x=\pi-\frac{\pi}{3}=\frac{2\pi}{3}$

$\Rightarrow x=\frac{-\pi}{12}$ and $x=-\frac{\pi}{6}$ Period $=\frac{2\pi}{4}=\frac{\pi}{2}$

i) Particular Solution interval $\left[0,\frac{\pi}{2}\right]$ are $\left\{\frac{-\pi}{12}+\frac{\pi}{2}, \frac{-\pi}{6}+\frac{\pi}{2}\right\}$

$$=\left\{\frac{5\pi}{12}, \frac{\pi}{3}\right\}$$

ii) General Solution $=\left\{\frac{\pi}{3}+\frac{\pi}{2}k,\ \frac{5\pi}{12}+\frac{\pi}{2}k\right\}$

g) $\sin x=\frac{-\sqrt{2}}{2}$ ⟵ Exercise left for you

h) $\sin 6x = \frac{-\sqrt{2}}{2} \Rightarrow -6x = \frac{\pi}{4}$ *and* $-6x = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$\Rightarrow x = \frac{-\pi}{24}$ *and* $x = \frac{-\pi}{8}$

Period $= \frac{2\pi}{6} = \frac{\pi}{3}$

i) Particular solution on interval $\left[0, \frac{\pi}{3}\right]$ *are* $\left\{\frac{-\pi}{24} + \frac{\pi}{3}; \frac{-\pi}{8} + \frac{\pi}{3}\right\}$

$= \left\{\frac{7\pi}{24}, \frac{5\pi}{24}\right\}$

ii) General solution S.S $= \left\{\frac{5\pi}{24} + \frac{\pi k}{3}, \frac{7\pi}{24} + \frac{\pi k}{3}\right\}$.

i) $\cos x = \frac{\sqrt{3}}{2} \Rightarrow x = \frac{\pi}{6}$ *and* $x = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$

i) Particular solution $= \left\{\frac{\pi}{6}, \frac{11\pi}{6}\right\}$

ii) General solution $= \left\{\frac{\pi}{6} + 2\pi k, \frac{11\pi}{6} + 2\pi k\right\}$

j) $\cos(3x) = \frac{-\sqrt{3}}{2} \Rightarrow 3x = \pi - \frac{\pi}{3}$ *and* $3x = \pi + \frac{\pi}{3}$

$\Rightarrow x = \frac{2\pi}{9}$ *and* $x = \frac{4\pi}{9}$

Period $= \frac{2\pi}{3}$

i) Particular solution on interval $\left[0, \frac{2\pi}{3}\right]$ *are* $\left\{\frac{2\pi}{9}, \frac{4\pi}{9}\right\}$

ii) General solution S.S $= \left\{\frac{2\pi}{9} + \frac{2\pi k}{3}, \frac{4\pi}{9} + \frac{2\pi k}{3}\right\}$

k) $\tan x = \sqrt{3} \Rightarrow x = \frac{\pi}{3}$, Period, P = π

i) Particular solution = $\left\{\frac{\pi}{3}\right\}$

ii) General solution = $\left\{\frac{\pi}{3} + \pi k\right\}$

24. Find the general solution set for each of the followi trigonometric equation.

a) $2\sin^2 x + 3\cos x - 3 = 0$

b) $2\cos x + \sin 2x = 0$

c) $\sec^2(2x) - 2 = 0$

Solution: Here

a) $2\sin^2 x + 3\cos x - 3 = 0$

$\Rightarrow 2(1 - \cos^2 x) + 3\cos x - 3 = 0$

$\Rightarrow 2 - 2\cos^2 x + 3\cos x - 3 = 0$

$\Rightarrow -2\cos^2 x + 3\cos x - 1 = 0$

$\Rightarrow 2\cos^2 x - 3\cos x + 1 = 0$

Let $\cos x = u$

$\Rightarrow 2u^2 - 2u + 1 = 0$

$\Rightarrow u = \frac{3 \pm \sqrt{9 - 4(2)}}{4} = \frac{3 \pm 1}{4}$

$\Rightarrow u = \frac{3+1}{4} = 1 \text{ and } u = \frac{3-1}{4} = \frac{2}{4} = \frac{1}{2}$

But $u = \cos x \Rightarrow \cos x = 1$ and $\cos x = \frac{1}{2}$

Equation	**Solution**
$\cos x = 1$	$\Rightarrow x = 2\pi k$
$\cos x = \frac{1}{2}$	$\Rightarrow x = \frac{\pi}{3} + 2\pi k, \frac{5\pi}{3} + 2\pi k$

$\therefore$ S.S $= \left\{\frac{\pi}{3} + 2\pi k, \frac{5\pi}{3} + 2\pi k, 2\pi k\right\}$

b) $2\cos x + \sin 2x = 0$

$\Rightarrow 2\cos x + 2\sin x \cos x = 0$

131

$\Rightarrow 2\cos x(1+\sin x)=0$

$\Rightarrow 2\cos x=0$ or $\sin x+1=0 \Rightarrow \cos x=0$ or $\sin x=-1$

Equation	Solution
$\cos x=0$	$\Rightarrow x=\frac{\pi}{2}+2\pi k$
$\sin x=-1$	$\Rightarrow x=\frac{3}{2}\pi+2\pi k$

$$\therefore S.S=\left\{\frac{\pi}{2}+2\pi k,\frac{3}{2}\pi+2\pi k\right\}$$

c) $\sec^2(2x)-2=0 \Rightarrow (\sec(2x)-\sqrt{2})(\sec(2x)+\sqrt{2})=0$

$\Rightarrow \sec(2x)=\sqrt{2}$ or $\sec(2x)=-\sqrt{2}$

$\Rightarrow \cos 2x=\frac{1}{\sqrt{2}}$ or $\cos(2x)=\frac{-1}{\sqrt{2}}$

Equation	Solution for 2x	Solution for x
$\cos(2x)=\frac{1}{\sqrt{2}}$	$\Rightarrow 2x=\frac{\pi}{4}+2\pi k$	$\Rightarrow x=\frac{\pi}{8}+\pi k$
	$\Rightarrow 2x=\frac{7\pi}{4}+2\pi k$	$\Rightarrow x=\frac{7\pi}{8}+\pi k$
$\cos(2x)=-\frac{1}{\sqrt{2}}$	$\Rightarrow 2x=\frac{3\pi}{4}+2\pi k$	$\Rightarrow x=\frac{3\pi}{8}+\pi k$
	$\Rightarrow 2x=\frac{5\pi}{4}+2\pi k$	$\Rightarrow x=\frac{5\pi}{8}+\pi k$

$$\therefore S.S=\left\{\frac{\pi}{8}+\pi k,\frac{7\pi}{8}+\pi k,\frac{3\pi}{8}+\pi k,\frac{5\pi}{8}+\pi k\right\}=\left\{\frac{\pi}{8}+\frac{\pi k}{4}\right\}$$

25. Solve each of the following trigonometric equation

a) $\sin 2x=2\sin x$

b) $2\cos x+\sin 2x=0$

c) $\cos\left(\frac{\pi}{3}x-2\right)=\frac{1}{2}$

d) $\sec\left(\frac{1}{3}x+\frac{\pi}{3}\right)=2$ *and* $\tan x<0$

e) $\tan\left(\frac{x}{2}\right)-2\sin x=0$

Solution: a) $\sin 2x = 2\sin x \Rightarrow \frac{\sin 2x}{\sin x}=2 \Rightarrow \frac{2\sin x\cos x}{\sin x}=2$

$\Rightarrow 2\cos x = 2 \Rightarrow \cos x = 1$

$\therefore x = 2\pi k$

b) $2\cos x + \sin 2x = 0 \Rightarrow 2\cos x = -\sin 2x$

$\Rightarrow \frac{\sin 2x}{\cos x}=-2 \Rightarrow \frac{2\sin x\cos x}{\cos x}=-2$

$\Rightarrow \sin x=-1 \Rightarrow x=\frac{3}{2}\pi \therefore x=\frac{3}{2}\pi+2\pi k$

c) $\cos\left(\frac{\pi}{3}x-2\right)=\frac{1}{2} \Rightarrow \frac{\pi}{3}x-2=\frac{\pi}{3} \Rightarrow \frac{\pi}{3}x=\frac{\pi}{3}+2$

$\Rightarrow x=1+\frac{6}{\pi}$, since period, $P=\frac{2\pi}{\frac{\pi}{3}}=6$

$\therefore$ General Solution $=\left\{1+\frac{6}{\pi}+6k\right\}$, k is integer

d) $\sec\left(\frac{1}{3}x+\frac{\pi}{3}\right)=2$ *and* $\tan x<0$. Period $=\frac{2\pi}{\frac{1}{3}}=6\pi$

$\Rightarrow \cos\left\{\frac{1}{3}x+\frac{\pi}{3}\right\}=\frac{1}{2}$, since tan x < 0 and cosine value is positive, so lies on 4^{th} quadrant

$\Rightarrow \frac{1}{3}x+\frac{\pi}{3}=\frac{5\pi}{3} \Rightarrow \frac{1}{3}x=\frac{5\pi}{3}-\frac{\pi}{3}=\frac{4\pi}{3} \Rightarrow x=4\pi$

$\therefore$ General Solution $=\{4\pi + 6\pi k\}$

e) $\tan\left(\frac{x}{2}\right)-2\sin x=0 \Rightarrow \tan\left(\frac{x}{2}\right)-2\left(\sin\left(\frac{x}{2}+\frac{x}{2}\right)\right)=0$

$$\Rightarrow \frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)} - 2\left(2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)\right) = 0$$

$$\Rightarrow \sin\left(\frac{x}{2}\right)\left(\frac{1}{\cos\left(\frac{x}{2}\right)} - 4\cos\left(\frac{x}{2}\right)\right) = 0 \Rightarrow \sin\left(\frac{x}{2}\right)$$

$$= 0,\ 1 - 4\cos^2\frac{x}{2} = 0$$

$$\sin\left(\frac{x}{2}\right) = 0 \Rightarrow \frac{x}{2} = 0, \pi \ldots., \text{x} = 2\pi + 4\pi\text{k}, \text{ and } \cos\left(\frac{x}{2}\right) = \pm\frac{1}{2}$$

$$\Rightarrow \frac{x}{2} = \frac{\pi}{3}, \frac{2\pi}{3}, \ldots x = \frac{2\pi}{3} + 4\pi k, \frac{4}{3}\pi + 4\pi k \ldots.$$

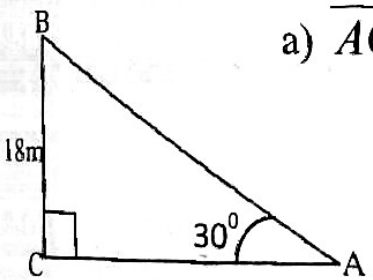
$$\therefore S.S = \left\{2\pi k, \frac{2}{3}\pi + 4\pi k, \frac{4}{3}\pi + 4\pi k\right\}$$

Solving triangles

I. Solving right triangle

26. For the right angled triangle given below. If $\overline{BC} = 18cm$ and $m(\angle A) = 30^0$ then how long is

a) $\overline{AC}$ b) $\overline{AB}$



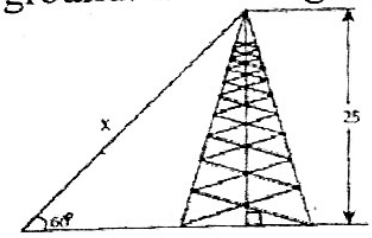
Solution: To find $\overline{AC}$: $\tan 30^0 = \frac{\overline{BC}}{AC} = \frac{18}{\overline{AC}}$

$$\Rightarrow \overline{AC} = \frac{18cm}{\tan 30^0} = \frac{18}{\frac{1}{\sqrt{3}}} = 18\sqrt{3}cm$$

To find $\overline{AB}$: $\sin 30^0 = \frac{BC}{AB} = \frac{18}{AB}$

$$\Rightarrow \frac{1}{2} = \frac{18}{AB} \Rightarrow \overline{AB} = 36$$

27. A support cable for 25 meter tower is to make 60^0 angle with the ground. How long must the cable be.

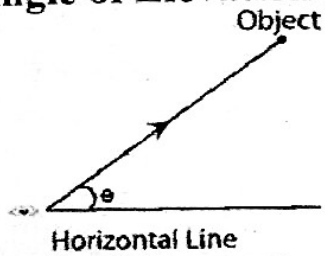


Solution:

$$\sin 60^0 = \frac{25}{x} \Rightarrow x = \frac{25}{\sin 60^0} = \frac{50}{\sqrt{3}} m.$$

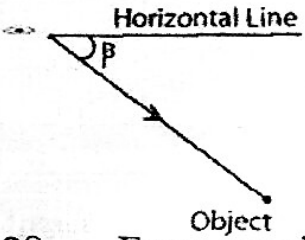
Angle of Elevation and Angle of Depression

Angle of Elevation: Looking up from a point.



The angle formed between the horizontal line and line of sight when we looking up is called **angle of elevation.**

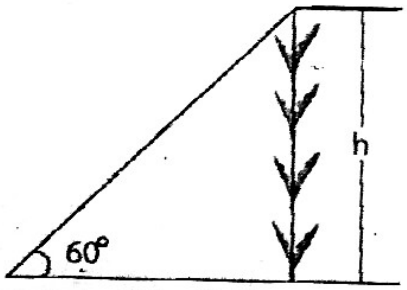
Angle of depression: Looking down from appoint.



The angle formed between the horizontal line and line of sight when we looking down is called angle of depression.

Using angle of elevation

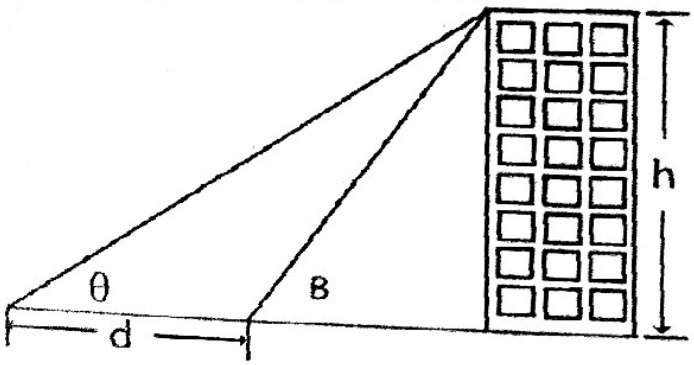
28. From a point 40meter from the base of kosso zaf tree, the angle of elevation to the top of tree is 60^0. Find the height of the tree.



Solution: $\tan 60^0 = \frac{h}{40m}$

$$\Rightarrow h = (40m)\tan 60^0 = (40)\sqrt{3} = 40\sqrt{3}m$$

Height of building When building top is looking from the point A, the angle of elevation is θ.



From a point B, which is closer to building, the angle of elevation increase to β and the distance between A to B is d, then the height of the

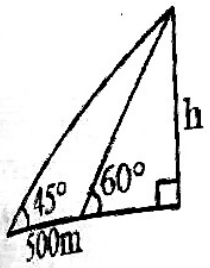
building h given by

$$h = \frac{d}{\cot\theta - \cot\beta},$$

• h is height • d is distance between

29. The angle of elevation to an air plane from two point A and B on level ground are 45^0 and 60^0 respectively. The distance between A and B is 500 meter and the air plane is to the east of A to B in the same vertical plane. Find the altitude of the plane

Solution: $h = \frac{d}{\cot\theta - \cot\beta}$, $\theta = 45^0$, $\beta = 60^0$



$$\Rightarrow h = \frac{500m}{\cot 45^0 - \cot 60^0} = \frac{500m}{1 - \frac{1}{\sqrt{3}}} = \frac{500\sqrt{3}}{\sqrt{3}-1} m.$$

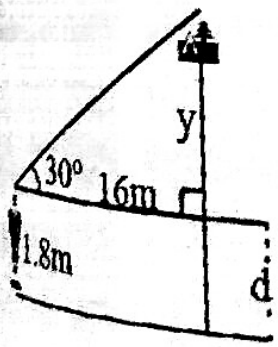
30. The angle of elevation from point A to the top of mountain is 30^0. From point B, 800m far there away, the angle of elevation is 45^0. Find the height of the mountain.

Solution: $h = \frac{d}{\cot\theta - \cot\beta}$, $\theta = 30^0$, $\beta = 45^0$

$$\Rightarrow h = \frac{800m}{\cot 30^0 - \cot 45^0} = \frac{800m}{\sqrt{3}-1} = \frac{800m}{\sqrt{3}-1} = 400(\sqrt{3}+1)m$$

31. A man 1.8m tall observes the angle of elevation of a tree to be 30^0. If he is standing 16m from the tree, find the height of the tree.

Solution: • $\tan 30^0 = \frac{y}{16} \Rightarrow y = 16\tan 30^0 = \frac{16}{\sqrt{3}}$



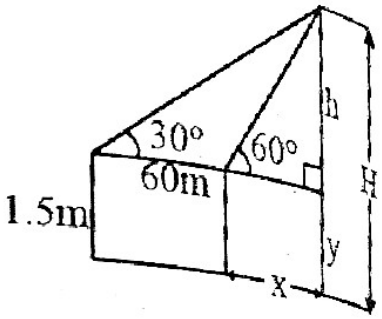
• height of tree, $h = y + d = \left(\frac{16}{\sqrt{3}} + 1.8\right)m$

d = 1.8 is height of man

32. Find the height of tower if the survey or measures the angle of elevation of its top changes from 30^0 to 60^0 at the survey or advances 60m toward its base, and that the instrument is 1.5m above ground level.

Solution:

$$\Rightarrow h = \frac{60m}{\cot 30^0 - \cot 60^0} = \frac{60m}{\sqrt{3} - \frac{1}{\sqrt{3}}}$$



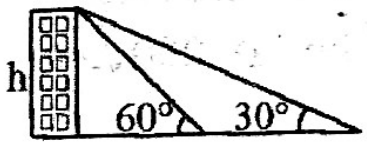
$$= \frac{60\sqrt{3}m}{2} = 30\sqrt{3}$$

$H = h + y = (30\sqrt{3} + 1.5)m \leftarrow$ Height of tower

41. Find the height of a building if the angle of elevation to the top of the building changes from 30^0 to 60^0 as the observer moves a distance of 100 meter toward the building.

Solution:

$$\Rightarrow h = \frac{100m}{\cot 30^0 - \cot 60^0} = \frac{100m}{\sqrt{3} - \frac{1}{\sqrt{3}}} = \frac{100\sqrt{3}m}{3-1}$$

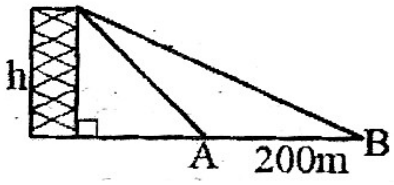


$\therefore h = 50\sqrt{3}m$

42. From the top of a tower T, the angle of depression to point B is 30^0 and the angle of depression to point A is 45^0. Point A and B are 200 meter a part.

By alternate interior angle

$\Rightarrow \angle B = 30^0$, and $\angle A = 45^0$



$$\Rightarrow h = \frac{200m}{\cot 30^0 - \cot 45^0}$$

43. The angle of depression of the top of the flag pole from the top of a building that is 100 meters away from the flag pole is 30°. If the flag pole is 20 meters tall, then what is the height of the building ..... (UEE)

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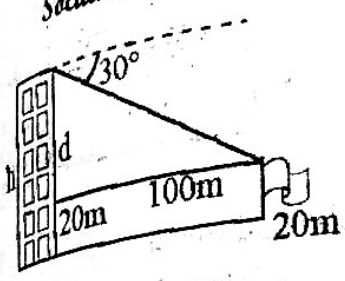
YouTube Digital[0,1] Ethiopia

A. $20\left(\frac{3+5\sqrt{3}}{3}\right)meter$ C. $20(5\sqrt{3}-1)meter$

B. $20(1+5\sqrt{3})meter$ D. $100\sqrt{3}meter$

Solution: $\Rightarrow \tan 30^0 = \frac{d}{100} \Rightarrow d = 100(\tan 30^0)$



$$\Rightarrow d = (100)\left(\frac{1}{\sqrt{3}}\right) = \frac{100\sqrt{3}}{3}m$$

$h = d + 20m$

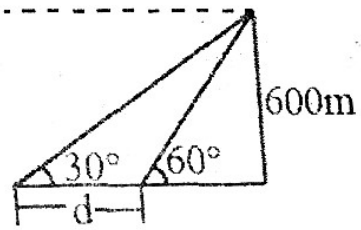
$$\Rightarrow h = \frac{100\sqrt{3}}{3} + 20m = 20\left(\frac{3+5\sqrt{3}}{3}\right)$$

Answer: A

44. If from the top of hill 600m above the sea level the angle of depressions of two boats in line with the observer were found to be 30^0 and 60^0 then the distance between the two boats was...

A. $400\sqrt{3}m$ B. 400m C. $300\sqrt{3}m$ D. $200\sqrt{3}m$

Solution: Use $h = \frac{d}{\cot\theta - \cot\beta}$



$h = 600m, \theta = 30^0, \beta = 60^0, d = ?$

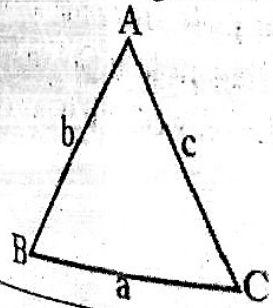
$$\Rightarrow 600m = \frac{d}{\cot 30^0 - \cot 60^0}$$

$\Rightarrow d = 600m\,(\cot 30^0 - \cot 60^0)$

$$\Rightarrow d = 600\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = 600\left(\frac{3-1}{\sqrt{3}}\right) = \frac{1200}{\sqrt{3}} = \frac{1200\sqrt{3}}{\sqrt{3}\cdot\sqrt{3}} = 400\sqrt{3}$$

Law of sines

* For any triangle ABC



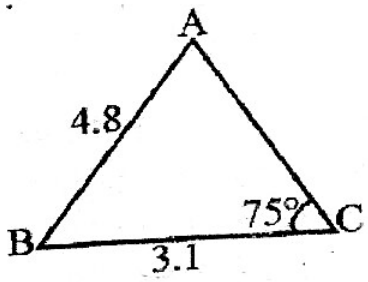
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

- The length of the sides of a triangle are proportional to the sines of the measure of the opposite angle.

Note: The law of sine we apply when the following two cases satisfy
I. **SSA** ← two side and an angle opposite one of them.
II. **AAS or ASA** ← two angle and any sides

45. In ΔABC, m(c) = 75^0, $\overline{BC} = 3.1 unit$ and $\overline{AB} = 4.8 units$. If $\cos 15^0 = 0.96$ then what is the value of sin A? ... UEE.
A. 0.62 B. 0.48 C. 0.31 D. 0.71

Solution:

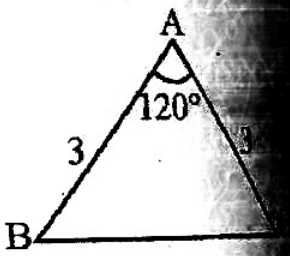


$$\frac{4.8}{\sin 75^0} = \frac{3.1}{\sin A} \Rightarrow \sin A = \left(\frac{\sin 75^0}{4.8}\right)(3.1)$$

$\Rightarrow \sin 75^0 = \cos 15^0$ ← complementary relation

$$\Rightarrow \sin A = \left(\frac{\sin 15^0}{4.8}\right)(3.1) = \left(\frac{0.96}{4.8}\right)(3.1) = 0.62$$

46. In ΔABC, If $\overline{AB} = \overline{AC} = 3$ units and $m(\angle A) = 120^0$ then how long is $\overline{BC}$?
A. 3 unit C. $3\sqrt{3}\, units$
B. $3\sqrt{2}\, units$ D. $3\sqrt{3}\, units$



Solution:

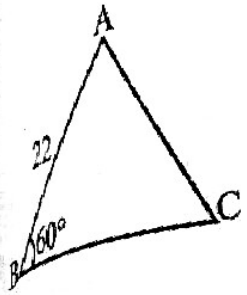
$$\frac{BC}{\sin 12^0} = \frac{3}{\sin 30^0} \Rightarrow BC = \frac{3}{\sin 30^0} \cdot \sin 120^0$$

$$\Rightarrow BC = \frac{3}{\frac{1}{2}} \cdot \frac{\sqrt{3}}{2} = \frac{6\sqrt{3}}{2} = 3\sqrt{3}$$ **Answer:**

47. In ΔABC, $m(\angle A) = 75^0$, $m(\angle B) = 60^0$, and AB = 22 units wh... the length of $\overline{AC}$

A. $11\sqrt{3}\, units$ C. $\frac{11}{2}\sqrt{3}\, units$

B. $11\sqrt{6}$ *units* D. $11\sqrt{2}$ *units*



Solution: To find $(\angle C)$

$\Rightarrow 180^0 = (75^0 + 60^0 - m(C))$

$\Rightarrow m(C) = 180^0 - (75^0 + 60^0) = 45^0$

Thus,

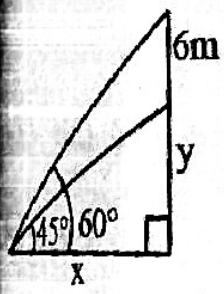
$$\frac{AC}{\sin 60^0} = \frac{22}{\sin 45^0} \Rightarrow \overline{AC} = \left(\frac{22}{\frac{1}{\sqrt{2}}}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$\therefore \overline{AC} = 11\sqrt{6}\ units$$

Answer: B

48. From the top of building, the angle of depression to the base of tree on the ground 60^0. If the angle of depression from a window 6 meter below the top of building is 45^0. Find the height of the building.

Solution: $\tan 60^0 = \frac{6+y}{x} \Rightarrow \sqrt{3} = \frac{6+y}{x} \Rightarrow \sqrt{3}x = 6 + y$



$$\tan 45^0 = \frac{y}{x} \Rightarrow y = x\tan 45^0 \Rightarrow y = x$$

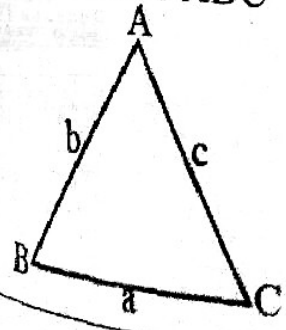
Solving together, $\begin{cases} \sqrt{3}x = 6 + y \\ y = x \end{cases}$

$$\Rightarrow \sqrt{3}x = 6 + x \Rightarrow \sqrt{3}x - x = 6 \Rightarrow y = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1)$$

$\Rightarrow$ Height, of the building, is $h = 6 + y = 6 + 3(\sqrt{3}+1) = 9 + 3\sqrt{3}$

The law of cosines

For any triangle ABC



- $a^2 = b^2 + c^2 - 2bc\cos A$
- $b^2 = a^2 + c^2 - 2ac\cos B$
- $c^2 = a^2 + b^2 - 2ab\cos C$

The law of cosines can be apply to solve triangle for which these data are given

I. SSS ⟵ length of three side

II. SAS ⟵ two side and the included angle

48. Use the law of cosines to solve ΔABC if

a) $a = 5,\ b = 4\ \ m(\angle C) = 30^0$

b) $a = 2,\ c = 3,\ m(\angle B) = 60^0$

c) $b = 6,\ c = 8,\ m(\angle A) = 120^0$

Solution: a) $c^2 = a^2 + b^2 - 2ab(\cos c)$

$$\Rightarrow c^2 = 5^2 + 4^2 - 2(5)(4)\cos 30^0$$

$$\Rightarrow c = \sqrt{25+16-10\left(\frac{\sqrt{3}}{2}\right)} = \sqrt{41-5\sqrt{3}}$$

b) $b^2 = a^2 + c^2 - 2ac(\cos B)$

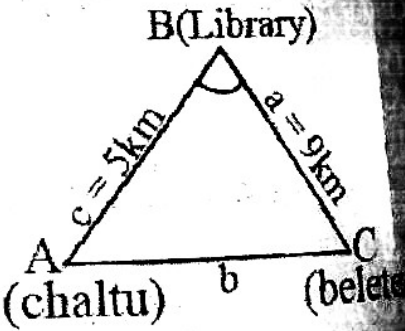
$$\Rightarrow b = \sqrt{4+9-2(2)(3)\left(\frac{1}{2}\right)} = \sqrt{7}$$

c) $a = \sqrt{6^2+8^2-2(8)(6)\left(\frac{-1}{2}\right)} = 2\sqrt{37}$

49. In the town where Chaltu and Beletech live, the public library is 5km away from Chaltu's home and 9km away from Beletech's home as shown in the figure. What is the distance between Chaltu's home and Beletech home?

A. $\sqrt{61}km$ C. $106+45\sqrt{3}km$

B. $\sqrt{151}km$ D. $106-45\sqrt{3}km$



Solution:

$\Rightarrow b^2 = a^2 + c^2 - 2ac\cos B$

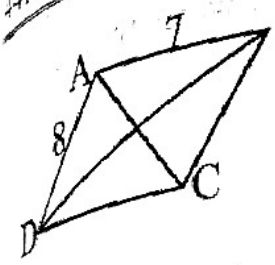
$= 9^2 + 5^2 - 2(5)(9)\cos(120^0) = 81 + 25 + 45 = 151$

$\therefore b = \sqrt{151}$ **Answer:**

50. ABCD is parallelogram with $\overline{AB}$ = 7cm and $\overline{AD} = 8cm$, $m(\angle DAB) = 60^0$ what is the length of the diagonal $\overline{AC}$

A. 13cm B. $\sqrt{113}cm$ C. 15cm D. $\sqrt{57}cm$

Solution: $m(\angle A) = 60^0 \Rightarrow m(\angle D) = 120^0$



$\Rightarrow \overline{AC}^2 = (AD)^2 + (DC)^2 - 2(AD)(DC)\cos D$
$= 8^2 + 7^2 - 2(8)(7)\cos(120^0)$

$\Rightarrow \overline{AC}^2 = 64 + 49 + 59 = 169$

$\therefore AC = \sqrt{169} = 13$ **Answer: A**

51. A rhombus has side of length 6cm and the angle at one vertex is 60^0. Find the length of the diagonal.

Solution: $D^2 = 6^2 + 6^2 - 2(6)(6)(\cos 60^0) = 72 - 36 = 36$

$\Rightarrow D = \sqrt{36} = 6$

52. In ΔDEF, $d = 4$, $e = 3$, $\cos F = \frac{5}{6}$. Find f

Solution: $f^2 = e^2 + d^2 - 2ed\cos F$

$\Rightarrow f^2 = 3^2 + 4^2 - 2(3)(4)\left(\frac{5}{6}\right) = 5$

53. In ΔPQR, $P = 5$, $q = 10$, $\cos R = 0$, Find r

Solution: $r^2 = p^2 + q^2 - 2pq\cos R \Rightarrow r^2 = 5^2 + 10^2 - 2(5)(10)(0)$

$\Rightarrow r^2 = 125 \Rightarrow r = \sqrt{125} = 5\sqrt{5}$

54. Find the cosine of each angle in the given triangle.

A. In ΔABC, $a = 2$, $b = 3$, $c = 4$

B. In ΔDEF, $a = 12$, $b = 5$, $c = 13$

Solution: a) $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{3^2 + 4^2 - 2^2}{2(3)(4)} = \frac{21}{24}$

$* \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{2^2 + 4^2 - 3^2}{2(2)(4)} = \frac{11}{16}$

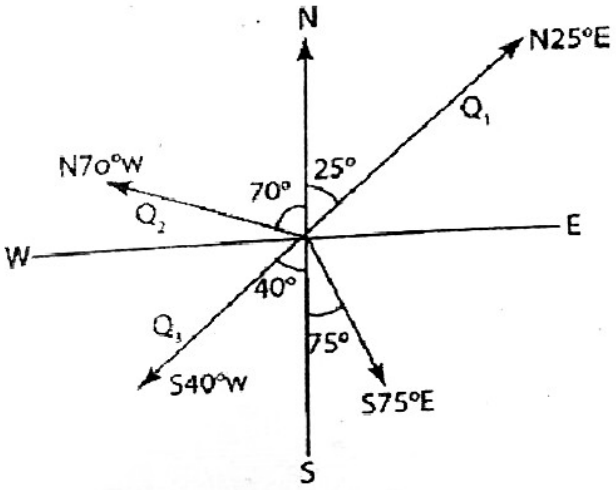
$* \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{2^2 + 3^2 - 4^2}{2(2)(3)} = \frac{-3}{12} = \frac{-1}{4}$

$\therefore \Delta ABC$ is an obtuse angle triangle

c) Exercise left for you.

Navigation

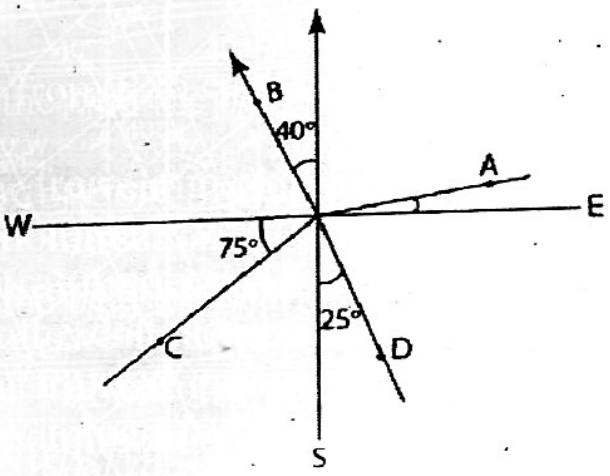
In certain navigation, surveying problems, **the direction**, or **bearing.**



* The angular direction used to locate one object in relation to another object is called bearing.
* **Bearing** is expressed interms of the acute angle formed by a **north-south line** and the line of direction.

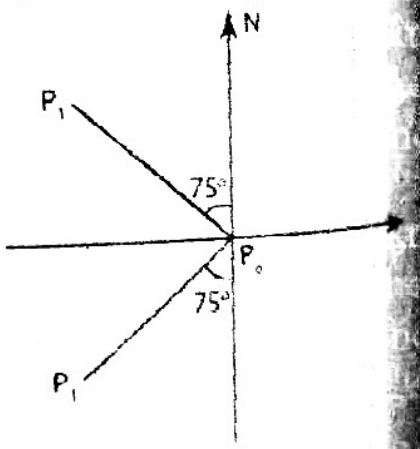
55. Find the bearing from P to each of the points A, B, C and D

Solution:



A. $N80^0E$
B. $N40^0W$
C. $S15^0W$
D. $S25^0E$

56. A car traveling from a parking lot P_0 travels 20kms in the direction N 75^0W to parking 10t P_1. Another car starting from P_0 travels 30kms in a direction S 75^0W to another lot P_2. which of the following is the distance between P_1 and P_2.



A. $10\sqrt{13-6\sqrt{3}}km$
B. $10\sqrt{10}km$
C. $10\sqrt{7}km$
D. $10\sqrt{19}km$

Solution:

The angle $(\angle P_1P_0P_2) = 180^0 - (75^0 + 75^0) = 30^0$

$\Delta P_1P_0P_2$, $\overline{P_1P_0} = 20km$, $\overline{P_0P_2} = 30km$

From cosine law,

$$(P_1P_1)^2 = (P_0P_1)^2 + (P_0P_2)^2 - 2(P_0P_1)(P_0P_2)\cos 30^0$$

$$= 20^2 + 30^2 - 2(20)(30)\left(\frac{\sqrt{3}}{2}\right)$$

$$\therefore \overline{P_1P_2} = \sqrt{1300 - 600\sqrt{3}} = 10\sqrt{13 - 6\sqrt{3}}$$ **Answer: A**

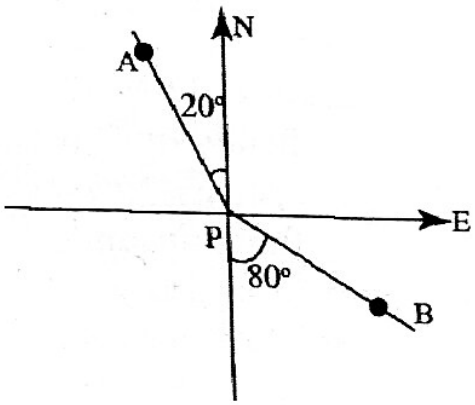
57. Two motor boats A and B leave a port P at the same time. A travels at constant speed of 20km/hr in the direction N20^{0}W and B travels speed of 15km/hr in the direction S 80^{0}E. How far a part are the two boats after 2 hours .....UEE

A. $10\sqrt{31}km$

B. $5\sqrt{37}km$

C. $5\sqrt{13}k$

D. $10\sqrt{37}km$



Solution: In ΔAPB, angle,

$$m(\angle APB) = 20^0 + 90^0 + 10^0 = 120^0$$

$$\Rightarrow \overline{AP} = \left(20\frac{km}{hr}\right)(2hr) = 40km$$

$$\Rightarrow \overline{PB} = \left(15\frac{km}{hr}\right)(2hr) = 30km$$

$$\Rightarrow \left(\overline{AB}\right)^2 = \left(\overline{AP}\right)^2 + \left(\overline{PB}\right)^2 - 2(\overline{AP})(\overline{BP})\cos P$$

$$= (40)^2 + (30)^2 - 2(40)(30)(\cos(120^0))$$

$$\Rightarrow \overline{AB} = \sqrt{1600 + 900 - (2400)\left(\frac{-1}{2}\right)}$$

$$= \sqrt{2500 + 1200} = \sqrt{3700} = 10\sqrt{37}$$

Answer: D

58. A ziza located her two class mates, kiros at a distance of 20 meters N35^{0}W and Guta at 30 meters S 25^0 W away from her position. How far a part are kiros and Guta?